Rank-six fusion rings: categorification criteria and models through multiplicity eight

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Abstract

We study complex categorifiability of the 16612 rank-six fusion rings of multiplicity at most eight in the *Fusion Ring Census*. Arithmetic and sphericalization criteria reduce unrestricted existence to exact finite problems in central induction and module categories. A common framework combines restriction Gram matrices, character and Galois symmetry, pivotal rotation, projection to fusion subcategories, and simultaneous product moments. We apply these necessary conditions automatically and identify positive cases by categorical constructions. The resulting classification has 16552 generally excluded rings, 54 realized rings, and 6 unresolved rings; the exact multiplicity bands $1, 2, 3, 6, 7$ are complete. Every exclusion and model identification has an exact reproducible certificate.

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1 Classification results and scope

A fusion ring is a finite-rank based ring with nonnegative integral structure constants, a distinguished unit, and duality satisfying Frobenius reciprocity. We work with full ordered multiplication tensors; in particular, commutativity is not part of the definition. The multiplicity is the maximum of all structure constants, including the unit coefficients. A categorification means a fusion category over $\mathbb{C}$, without an assumed pivotal, spherical, or unitary structure. We classify existence for a ring, not equivalence classes of categories with that ring.

Vercleyen–Slingerland constructed a substantial low-rank database [42]. Its rank-six portion is complete through multiplicity four. The multiplicity-one categorification problem was settled by Liu–Palcoux–Ren [27]. The rank-four and rank-five projects [12, 11] develop arithmetic, sphericalization, and central-induction certificates. At rank six, however, neither the ring nor a signed quotient of its sphericalization can simply be assumed commutative. This is the principal structural change addressed here.

All positively identified rings and their models appear in Table 3; its structure column is part of the classification statement, not merely a description of a chosen presentation. In particular, Schopieray's category [41, Example 1] realizes $R^{(6)}_{2,124}$ spherically, whereas [41, Theorem 1] excludes *every* pseudo-unitary categorification of that same ring. Thus general or spherical existence cannot be identified with unitary existence, even in rank six and multiplicity two. The electronic ledger records general, pivotal, spherical, pseudo-unitary and unitary existence separately for every tensor. The source identifications are detailed in Section 6.4. The paper is organized by mathematical operation rather than by individual exceptions. Section 2 fixes the source census, its scope, and the tensor conventions used by every automated test. Section 3 reduces arbitrary categorifications to the appropriate arithmetic and spherical cases. Section 4 presents one central-induction framework, with precisely scoped refinements rather than separate ad hoc tests. Section 5 supplies module and Morita constraints. Section 6 treats models by construction, including central-boson quotients, equivariantizations, and relative centers. Section 7 specifies exhaustive certificate checking and the remaining frontier.

In particular, the positive catalogue includes the extended Haagerup category $\mathcal{EH}_1$ [4, 6] and Izumi's $C_5 + 5$ near-group category [23], not just the constructions derived from them. The mixed fermionic quotient of Proposition 6.17, the sign-swap equivariantization of Theorem 6.18, and the relative-center construction of Corollary 6.20 give three additional full-tensor identifications. None is inferred merely from a dimension match or from survival of the necessary criteria.

Theorem 1.1 (Source census [37])**.** *The published* Fusion Ring Census*, at the revision fixed in Section 2, contains exactly* 16612 *based-isomorphism classes of rank-six fusion rings of multiplicity at most eight. The numbers at exact multiplicities* $1, \ldots, 8$ *are*

$$39, \quad 154, \quad 384, \quad 872, \quad 1582, \quad 2768, \quad 3814, \quad 6999.$$

All three possible duality types and all noncommutative cases are included. No categorifiability filter is applied. Completeness of this counting input is the result of the cited repository; the present paper addresses categorification of its tensors.

Theorem 1.2 (Classification through multiplicity three and certified extension)**.** *In the census of Theorem 1.1, the accompanying exact certificates exclude* 16552 *rings from general complex categorification. There are* 54 *positively identified rings:* 53 *have unitary categorifications, whereas* $R^{(6)}_{2,124}$ *has a spherical categorification but no pseudo-unitary categorification. These positive sets and the general exclusions are pairwise disjoint. The remaining* 6 *rings are not decided for general existence by the present criteria and bounded searches. Structure-specific pseudo-unitary exclusions are recorded separately and do not alter that general status. Their distribution is given in Table 1.*

Among the 577 *rings of multiplicity at most three, precisely* 46 *admit general, pivotal, or spherical categorifications; the other* 531 *are generally excluded. Exactly* 45 *of the positive rings admit pseudo-unitary or unitary categorifications; the sole exception is* $R^{(6)}_{2,124}$*. At exact multiplicity three, all* 384 *rings are decided: three have unitary models and* 381 *are generally excluded. In particular, Theorem 5.25 excludes* $R^{(6)}_{3,215}$*, the last residual from that stratum. The completed exact-multiplicity bands are* $1, 2, 3, 6, 7$*. The other tested bands retain the residuals listed in Table 1 and the electronic ledger.*

The low-multiplicity residuals. Every ring at exact multiplicity 2 is decided. Every ring at exact multiplicity 3 is decided. At exact multiplicity 4 the unresolved records are

$$R^{(6)}_{4,22}, \; R^{(6)}_{4,91}, \; R^{(6)}_{4,155}, \; R^{(6)}_{4,629}.$$

Exact multiplicity	Rings	Excluded	Unitary	S, not PU	Undecided
1	39	18	21	0	0
2	154	132	21	1	0
3	384	381	3	0	0
4	872	866	2	0	4
5	1582	1580	1	0	1
6	2768	2765	3	0	0
7	3814	3812	2	0	0
8	6999	6998	0	0	1
Total	16612	16552	53	1	6

Table 1: Complete counting census; categorification is complete at exact multiplicities $1, 2, 3, 6, 7$ and partial in the other displayed bands. The column S, not PU denotes a spherical realization with no pseudo-unitary categorification. All excluded rings are ruled out generally.

The phrase "undecided" is deliberately local to this computation. A search timeout is not an obstruction, and failure to produce a separating functional is not a construction. Nor do we claim that every residual is open throughout the literature. Section 7.2 retains exact feasible points for the now positively identified cyclic-four ring as regression controls. Such points were not used to infer existence; the conformal-embedding construction supplies it.

For reproducible identifiers, retain the unit as the zeroth basis element, use the exported representatives in the generator's fixed coordinate conventions, and sort their full tensors in row-major lexicographic order at each exact multiplicity. This order is a convention on the exported representatives, not a claim that each is the lexicographic minimum under all 120 relabellings. Distinctness up to based isomorphism is supplied by the cited census. The established representatives and identifiers through multiplicity seven are preserved; the multiplicity-eight stratum is sorted by the same row-major convention. The jth tensor at multiplicity m is denoted $R^{(6)}_{m,j}$. The complete tensor, rather than this identifier, is the input to every obstruction. A separate crosswalk records matches to the ancillary database of [42]; its indices play no role in a proof.

All reductions are applied to full tensors. The cheap arithmetic filters are retained because they avoid unnecessary induction searches; the stronger blocks are used only with their stated hypotheses. The rank-four and rank-five results are cited rather than reproved. This is a unified implementation of sufficient exclusion certificates, not a claim that the criterion set is logically minimal.

Completing multiplicity three. Theorem 5.25 generally excludes $R^{(6)}_{3,215}$. Proposition 3.28 first reduces arbitrary categorification to pseudo-unitary categorification by an exhaustive signed sphericalization calculation. The full two-sided Morita-action Gram calculation then forces the involution to interchange the two module structures on each of its fixed simple objects. The resulting local anticommutation relation forces their third Frobenius–Schur indicators to equal -1. Exhaustive exact separation of the central-induction branches contradicts these two requirements. Thus no pivotal or positivity hypothesis remains in the exclusion.

The earlier rank-22 witness survives the joint low-product Gram condition of Section 4.8 and extends to exact modular and fusion data with a positive restriction-ring map. These remain useful records of the limits of the earlier necessary systems. They provide neither a tensor functor nor an algebra structure, and the new local rotation condition excludes them as categorification data for this target. The independent Dong–Schopieray premodular bound and the source-specific cyclic-eight Morita argument retain their stated scopes; neither is used to identify arbitrary modular categories from equal modular data. The higher-multiplicity

exclusions use the spectral module, rank-three descent, and order-six Morita–grading criteria of Section 5, the biquadratic trace and Casimir-parity criterion of Proposition 3.9, and the formal-codegree trace-ideal theorem and ramification criterion of Section 3, the all-orders Gauss obstruction of Proposition 3.12, and the higher-dimensional and quartic stabilizer-trace extensions of Section 5. The completed signed and character-induction searches are recorded in the ledger.

What the exclusion table measures. The sequential table in Section 7.1 applies the available criteria in an explicit order and reports new removals and remaining tensors at each exact multiplicity. Simple structural and arithmetic tests precede finite induction and signed searches. This is a reproducible compromise between cost and yield, not a proof of a universally optimal order. A theorem with a stronger conclusion can be more expensive to apply, and a successful earlier test makes later tests unnecessary for that tensor.

The earliest unresolved multiplicity is the first stratum in which the current automated procedure does not decide every ring. It marks where further work should begin; that work may be a new exclusion criterion, a sharper application of an existing criterion, or a positive model. A zero residual count in a later stratum means that the present methods suffice for that stratum. Even if an entire terminal interval through multiplicity eight has zero residuals, this would not imply that no new criterion is needed at multiplicity nine or higher. Conversely, a bounded search that stops without a certificate does not show that a new theorem is logically necessary.

The same reporting scheme applies at lower ranks: one fixes a complete counting range, runs the necessary criteria and model constructors, and locates the first nonzero residual stratum. Here the detailed automated ledger is rank six. The cited rank-four and rank-five categorification results are used only in their proved ranges; the larger counting ranges of the source repository do not by themselves extend those theorems.

2 Census input and tensor conventions

2.1 One authoritative counting source

All counting data are taken from the public repository *Fusion Ring Census* [37], pinned to revision

`b86eb6dc6eeeafe5fc5f6906d7b074e0322e9726`.

The source files are `results/census.json` and `results/rank6/tables.txt.gz`. They declare a complete, verified rank-six census through multiplicity eight. Its 16612 tensors include all 9613 classes through multiplicity seven and 6999 classes of exact multiplicity eight. The exact-multiplicity definition, inclusion of the unit in the rank, and inclusion of every duality type are unchanged.

The paper trusts this census as its counting input. It supplies neither another census generator nor a claim to have independently enumerated the same domain. The repository contains the enumeration algorithms, coverage arguments, completion records, and tensor checks supporting Theorem 1.1. A valid and duplicate-free tensor list alone would not establish exhaustive coverage; that claim is supplied by the source census, not by the import operation below.

2.2 Frozen data and reproducible identifiers

A copy of the source tensor export is retained with the classification code, so a classification check has no live-network dependency. Its compressed SHA-256 is

`0138c6228dbb06b0861ad38b8dfe67fcd1fc829add80d441c34c09a1afb78675`,

and its uncompressed SHA-256 is

`7898a833dfffd0f645436c1d326740db29ea6a29980de307c67227f0b67122b0`.

These hashes identify the input bytes; they are not substitutes for the source's completeness argument or for categorification certificates.

Every ring is represented by its full ordered tensor $N = (N_{ij}^k)_{0 \le i,j,k < 6}$, with $b_0 = \mathbf{1}$. The multiplicity is $\max_{i,j,k} N_{ij}^k$. Noncommutative inputs are retained without symmetrization. A based-ring isomorphism is a unit-fixing basis permutation preserving every structure constant. Consequently, comparison with a model uses the complete tensor, not only dimensions, a fusion graph, or a numerical fingerprint.

The identifier $R_{m,j}^{(6)}$ is local to this paper's frozen tensor registry. Representatives and identifiers through multiplicity seven are retained. The new exact-multiplicity-eight tensors are placed in row-major lexicographic order using the exported representatives. This order need not give the lexicographically smallest representative under all basis permutations. Import matching records the correspondence between the source representatives and the registry. Every mathematical criterion receives the tensor rather than the identifier, so relabelling cannot alter its conclusion.

2.3 Counting, exclusions and constructions

The source census answers a finite counting question about based rings. A categorification exclusion answers a different question about complex fusion categories. We therefore keep the census fixed while applying necessary criteria and matching models. The resulting ledger covers every imported tensor exactly once; accepted negative certificates, positive constructions and unresolved records are disjoint.

The exact prefix $39,154,384,872$ agrees with the complete rank-six range of [42]. We use no external categorifiability flag as a proof. The classification code may reconstruct arithmetic, tensor or isomorphism data required by a criterion, but it does not repeat the census search. The counting proposal in `oeis/` derives directly from the same source counts and has no dependence on whether a categorification has been found.

3 Arithmetic and categorical reductions

3.1 Conventions and imported criteria

Write N_{ij}^k for the coefficient of b_k in $b_i b_j$ and i^* for duality. The matrices used in the software have entries $(N_i)_{jk} = N_{ij}^k$. Consequently they give the right-handed matrix convention

$$N_i N_j = \sum_k N_{ji}^k N_k.$$

This distinction is harmless for commuting matrices, but essential when checking noncommutative signed tables. Frobenius reciprocity gives $N_{i^*} = N_i^{\mathsf{T}}$. The positive dimension character is denoted $d_i = \mathrm{FPdim}(b_i)$ and $D = \sum_i d_i^2$.

The complexified based ring is a semisimple $*$-algebra. For an irreducible representation E of dimension r_E, let f_E be its formal codegree, in the convention of [35]. The regular Casimir matrix

$$C_R = \sum_i N_i N_{i^*}$$

has eigenvalue $r_E f_E$ with multiplicity r_E^2. It is only in the commutative case that all these multiplicities are one and its characteristic roots are themselves the formal codegrees.

We use the following established necessary conditions. Their precise applications are certified by polynomial, prime, dimension-field, and signed-table witnesses.

Proposition 3.1 (Established inputs)**.** *For a complex fusion categorification, the following hold.*

(i) Every formal codegree is a d-number [35]. If $p(t) = t^s + a_1 t^{s-1} + \cdots + a_s$ is its minimal polynomial, this is expressed by $a_s^i \mid a_i^s$ for $1 \le i \le s$.

(ii) For a commutative Grothendieck ring, character values are cyclotomic integers. The polynomial factorization and finite-field witnesses used here are those of [12, Section 2]; the commutativity hypothesis is checked before this test is used.

(iii) A fusion subring gives a fusion subcategory. Its Frobenius–Perron dimension divides that of the ambient category in the ring of algebraic integers. A subring of rank two with $x^2 = 1 + mx$, $m \ge 2$, is excluded by [34].

(iv) Weakly integral fusion categories are pseudo-unitary [14]. For the two-Frobenius–Perron-dimension reduction we use the Galois-conjugacy result of [39, Lemma 3.1]. It is a reduction to a pseudo-unitary Galois conjugate, not a claim that every originally given spherical structure is positive.

(v) The zero-spectrum obstruction of [27, Theorem 2.14] is a general categorification obstruction. Its finite hypothesis is tested directly on the full ordered tensor.

(vi) The spherical dimension, normalized norm, sign-object moment, adjoint-parity, and low-induction criteria are those of [12, Sections 2,3,6,7]. The finite joint relative-trace criterion is the named theorem of [11].

For completeness, the finite tensor hypothesis in input (v) is as follows. Put $\langle ab, cd\rangle = \sum_k N_{ab}^k N_{cd}^k$. There must be nine indices $a, b, c, u, v, w, x, y, z$ such that

$$N_{ua}^w N_{vu}^b N_{vw}^c N_{xz}^a N_{bx}^y N_{yz}^c > 0, \quad N_{ba}^c = 1, \quad \sum_k N_{ux}^k N_{v^*y}^k N_{wz^*}^k = 0,$$

and each of the two sets

$$\{\langle vu, ca^*\rangle, \langle bu^*, cw^*\rangle, \langle v^*b, wa^*\rangle\}, \qquad \{\langle bx, cz^*\rangle, \langle yx^*, ca^*\rangle, \langle b^*y, az^*\rangle\}$$

contains 1. Theorem 2.14 of [27] excludes any such tensor. The certificate stores the nine indices and the checker reevaluates all these expressions. No associator is assumed to be unitary or pivotal.

Input (v) is especially inexpensive because it needs just a small integer witness. We do not import the negative part of any integral classification table or quadratic-family classification as a list of forbidden labels.

Corollary 3.2 (Certified lower-rank descent)**.** *Suppose a rank-six fusion ring contains a based fusion subring of rank four and multiplicity at most* 32*, or of rank five and multiplicity at most* 17*. If that subring is not based-isomorphic to one of the positive rings in the respective main theorem of [12, 11], the rank-six ring is not categorifiable over $\mathbb{C}$.*

Proof. A hypothetical categorification would contain the fusion subcategory on the prescribed basis subset. Apply the cited complete lower-rank classification. The test enumerates unit-containing dual-closed subsets, checks product closure, and compares the restricted tensor with every positive model under every unit-fixing permutation. No negative labels are read. □

3.2 A certificate of a nonabelian character field

Theorem 3.3 (Noncommuting field automorphisms)**.** *Let R be a commutative fusion ring. Suppose an irreducible factor $f \in \mathbb{Q}[t]$ of the characteristic polynomial of some multiplication matrix admits polynomials $a, b \in \mathbb{Q}[t]$ such that*

$$f(a(t)) \equiv f(b(t)) \equiv 0 \pmod{f}, \qquad a(b(t)) - b(a(t)) \not\equiv 0 \pmod{f}.$$

Then R has no complex fusion categorification.

Proof. In a categorification, the eigenvalues in question are values of one-dimensional characters of $R \otimes \mathbb{C}$ and are cyclotomic integers by Proposition 3.1(ii). Hence $K = \mathbb{Q}[t]/(f)$ embeds into an abelian cyclotomic extension of $\mathbb{Q}$. Every subfield of an abelian Galois extension is Galois with abelian Galois group. Substitution by a and b defines unital endomorphisms of the field K; they are injective, hence surjective by finite dimensionality. The last displayed condition says that these automorphisms do not commute, a contradiction. □

Factoring f over $\mathbb{Q}[t]/(f)$ is allowed in discovery, but is unnecessary in verification: irreducibility, divisibility, two substitutions and one nonzero remainder suffice. In the present census the test excludes $R^{(6)}_{2,119}$, with the degree-six factor

$$f(t) = t^6 - 3t^5 - 2t^4 + 9t^3 - 5t + 1.$$

The two substitution polynomials and their nonzero commutator are stored in the certificate and recomputed from this polynomial. This test is a stronger certificate generator for the existing cyclotomicity condition, not a new categorical hypothesis.

The same automated test excludes $R^{(6)}_{6,1891}$. A multiplication matrix has the irreducible factor

$$t^6 - 11t^5 + 5t^4 + 38t^3 - 5t^2 - 11t - 1,$$

and the certificate supplies two noncommuting automorphisms of its degree-six field. Their defining substitutions and nonzero commutator are checked by exact polynomial reduction. Thus this case is decided by the uniform arithmetic criterion without an induction search.

3.3 Positive dimensions after a tensor-sign change

Lemma 3.4 (Tensor-sign Galois reduction)**.** *Let $\mathcal{C}$ be spherical and let σ be an embedding of its simple-dimension field into $\mathbb{R}$. Put $\eta_i = \operatorname{sgn}(\sigma(\dim X_i))$. If*

$$\eta_0 = 1, \qquad N_{ij}^k > 0 \Longrightarrow \eta_i \eta_j = \eta_k, \tag{3.1}$$

then $\mathcal{C}$ has a pseudo-unitary Galois conjugate after changing its spherical structure by a tensor automorphism of the identity.

Proof. Extend σ to the field of definition and conjugate the categorical structure, as in the Galois reduction of [12]. The scalars η_i define a tensor automorphism of the identity precisely because of (3.1). They agree on duals and square to one, so twisting the pivotal structure preserves sphericality. The resulting dimensions are $|\sigma(\dim X_i)| > 0$. A positive ring dimension character is the Frobenius–Perron character, so this spherical structure is positive and the conjugate is pseudo-unitary. □

This lemma must not be replaced by arbitrary basis sign changes: the support condition (3.1) is essential. The spherical checker now tests it at every real embedding of a remaining character component. Combined with the complete signed-sphericalization tests of Section 3.4, it supplies general exclusion certificates for $R^{(6)}_{2,37}$, $R^{(6)}_{2,57}$ and $R^{(6)}_{3,204}$. The new step concerns spherical character components; the exhaustive signed analysis is still required before a conclusion is promoted to general nonexistence.

3.4 Signed sphericalization at rank six

Let $\mathcal{C}$ be an arbitrary fusion category. Its canonical sphericalization $\widetilde{\mathcal{C}}$ has a sign object ε of dimension -1 and two lifts of every simple object, interchanged by tensoring with ε; see [14, 36]. Choose a unit-preserving set of lifts. Passing to

$$T = K_0(\widetilde{\mathcal{C}})/(\varepsilon + 1)$$

gives integral signed structure constants satisfying

$$|T_{ij}^k| \le N_{ij}^k, \qquad T_{ij}^k \equiv N_{ij}^k \pmod 2. \tag{3.2}$$

They retain associativity, the unit, duality, and Frobenius reciprocity. All these statements concern a necessary quotient of an actual sphericalization; conversely, an admissible signed table need not have a categorification.

The sphericalization arguments in [12, 11] choose the lift of each self-dual object with positive second indicator. We keep this normalization. In particular, the square-channel congruences are modulo four rather than merely modulo two whenever that normalization applies. Non-self-dual lifts are chosen compatibly with duality. These restrictions are part of the cited normalization theorem, not heuristic reductions of a search.

3.4.1 An exhaustive oriented parameterization

Set $t_{abc} = T_{ab}^{c^*}$. In the absence of commutativity, identify only the orbits generated by

$$(a,b,c) \longmapsto (b,c,a), \qquad (a,b,c) \longmapsto (c^*,b^*,a^*). \tag{3.3}$$

These are the cyclic and dual-reflection consequences of reciprocity. An arbitrary transposition is not allowed. Unit and duality coefficients are fixed first. Give each remaining orbit its interval in (3.2), impose the applicable square-channel congruences, and substitute into all ordered associativity equations.

Proposition 3.5 (Exhaustive signed search)**.** *The finite search just described contains the signed quotient of every hypothetical categorification, after the indicated lift normalization. If all its associative tables are rejected by necessary signed sphericalization criteria, the original ring has no complex categorification.*

Proof. The quotient construction gives (3.2) and (3.3). Thus every quotient occurs in the finite product of integer domains. Every pruning operation is a necessary consequence of an equation over those domains. At a leaf the full ordered associativity identities are checked. Exhausting the product therefore exhausts every possible normalized quotient. A genuine category would supply one of the remaining tables and one of its spherical dimension characters, so rejection of all tables and all such characters is contradictory. □

The implementation checks machine-integer bounds before native computation and does not interpret overflow or timeout as exhaustion. The exact verifier reconstructs the parameter domains and the complete list of surviving signed tables; it does not merely recheck a collection of stored rejected examples.

3.4.2 When the signed algebra must be commutative

Let $d = (1, d_1, \dots, d_5)$ be the dimension character of the selected lifts, $K = \mathbb{Q}(d_1, \dots, d_5)$, and $\delta = \sum_i d_i d_{i^*}$. Inflating a representation of R to $K_0(\widetilde{\mathcal{C}})$ doubles its formal codegree. Since the sphericalization has dimension 2δ, central induction gives

$$\delta/f_E \in \mathcal{O}_K, \qquad f_E \in K \quad (E \in \mathrm{Irr}(R \otimes \mathbb{C})). \tag{3.4}$$

Indeed, [36, Theorem 2.13] gives a central simple of dimension δ/f_E, and its restriction is an integral combination of the selected lifts and their negatives. Its dimension is a nonzero algebraic integer in K, proving both assertions.

Proposition 3.6 (Rank-six commutative descent)**.** *Suppose R is commutative of rank six and has a hypothetical categorification. Its signed quotient $T \otimes \mathbb{C}$ is commutative if any of the following tensor-checkable conditions holds:*

(a) all six basis elements are self-dual;

(b) exactly two basis elements are self-dual;

(c) a formal codegree has degree greater than two;

(d) the formal codegrees generate two different quadratic subfields;

(e) zero has multiplicity less than four in $\det(tI - C_R) \bmod 2$.

Proof. The coefficient-of-unit pairing of a signed quotient is still positive definite on the distinguished basis after complex conjugation. Its regular matrices are adjoint under duality. Hence the algebra is semisimple. It has a one-dimensional dimension representation. If it is noncommutative of dimension six, its only possible Wedderburn type is

$$\mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}). \tag{3.5}$$

The Galois orbit of the dimension character has at most two elements, so $[K : \mathbb{Q}] \leq 2$. Equation (3.4) excludes (3.5) in cases (c) and (d).

In case (a), duality is the identity on the signed basis. Being also an anti-automorphism, it forces commutativity. For (b), consider duality as a complex-linear involutive anti-automorphism of (3.5). The spherical dimension character is fixed, so both one-dimensional blocks are fixed. On $M_2(\mathbb{C})$, an involutive anti-automorphism is of orthogonal or symplectic type, with linear trace 2 or -2. The total trace is therefore 4 or 0. In the distinguished basis it is the number of fixed basis elements, namely 2, a contradiction.

Finally the unique two-dimensional representation of (3.5) is Galois invariant. Its formal codegree is a rational algebraic integer, hence an integer. The regular Casimir of T has the even eigenvalue $2f_E$ with multiplicity four. Thus its characteristic polynomial is divisible by t^4 modulo two. But (3.2) implies $C_T \equiv C_R \pmod 2$, contradicting (e). □

This proposition is a reduction, not a noncategorification criterion by itself. Once its hypotheses are verified, the commuting-character methods of [12, 11] apply to every signed branch. Otherwise we use Proposition 3.5. If a noncommutative table of type (3.5) remains, its one-dimensional characters are computed as the common kernel of the matrix commutators. That space has dimension two. The two-dimensional block and its formal codegree are then recovered from the regular Casimir with the multiplicities in (3.5). Each decomposition identity is checked exactly.

3.5 A scoped literature obstruction at the first frontier

The general existence question and the premodular existence question must be kept separate. The following consequence of the complete rank-six premodular fusion-rule catalogue is useful for the latter question.

Proposition 3.7. *If a rank-six fusion ring admits a braided spherical categorification, then its Frobenius–Perron dimension is at most* 168.

Proof. Use the fusion-rule classification of Dong–Schopieray [10, Section 5.3, Figures 13–15, 17 and 19]. The integral representation-ring entries and their listed braided variants have dimension at most 168. The other entries belong to the families below. Write $[j]_\ell = \sin(j\pi/\ell)/\sin(\pi/\ell)$. The inequalities

$$\sin(\pi/\ell) \geq 2/\ell, \qquad |\sin(j\pi/\ell)| \leq 1, \qquad \sqrt{5} < 3, \qquad \sqrt{21} < 5$$

give the following rational upper bounds. Each is at most 168.

Family of fusion rules	Upper bound
Pointed rank three times Fibonacci	12
$\mathrm{Rep}(S_3)$ times Fibonacci	24
Dimension-eight Ising and generalized-Ising rules	8
Ising times Fibonacci	16
Pointed rank two times $\mathcal{C}(A_1, 7)_{\mathrm{ad}}$	51
Fibonacci times $\mathcal{C}(A_1, 7)_{\mathrm{ad}}$	102
$\mathcal{C}(G_2, 21)$	110
Dimension-twenty metaplectic rules	20
$\mathcal{C}(A_1, 13)_{\mathrm{ad}}$	2197/16
$\mathcal{C}(B_2, 9)_{\mathrm{ad}}$	81
$\mathcal{C}(A_1, 12)_{\mathrm{ad}}$	108

For example, the Fibonacci dimension is less than 4, and the sum of squared dimensions of $\mathcal{C}(A_1, 7)_{\mathrm{ad}}$ is at most $1 + 2(7/2)^2$. For the two adjoint A_1 entries in the last three rows, use their dimension formulae $\frac{13}{4}\csc^2(\pi/13)$ and $3\csc^2(\pi/12)$. For B_2, the formula is $9u^2$ with $u = 1 - 2\cos(8\pi/9)$ and $|u| \leq 3$. For G_2, its six dimensions are bounded by $1, 4, 4, 4, 5, 6$. These cover all rows of the cited catalogue. The caveat there about counting braided equivalence classes for two families does not add fusion rules to the list. □

The implementation derives the positive character and its number-field embedding from each tensor. It then decides $\mathrm{FPdim}(R) - 168 > 0$ by exact real-root isolation, recording the defining polynomial, dimension coordinates and isolating interval. No tensor identifier is used to select the outcome. In particular, the former multiplicity-three residual $R^{(6)}_{3,215}$ has the automatically verified positive dimension vector

$$(1, 1, d, d, d-1, d+1), \qquad d = 4 + \sqrt{17}, \qquad \mathrm{FPdim}(R) = 136 + 32\sqrt{17} > 168.$$

It therefore has no premodular categorification. Its general, pivotal, spherical, pseudo-unitary and unitary existence questions are not resolved by this proposition, because none of those hypotheses alone supplies a braiding. They are resolved negatively by the independent stabilizer-rotation argument of Theorem 5.25.

For a separate complete-tensor model audit, the rank-six ancillary data of Ng–Rowell–Wen [29] contain 192 modular records in 20 Galois orbits and 11 based-ring types. The supplied parser reads only literal integer tensors and compares all 120 permutations fixing the unit. Their largest fusion coefficient is 2, so this source supplies no new model at multiplicity eight or for the multiplicity-three frontier. This comparison is a model search within a specified literature class, not an exclusion of arbitrary fusion categories.

3.6 Biquadratic spherical traces

The following test uses sphericalization to obstruct a fusion category without assuming that it itself admits a spherical structure. Throughout this subsection the input ring is commutative, self-dual, and of rank six. Write N_i for its regular multiplication matrices and $T_0 = \mathrm{Tr}(\sum_i N_i^2)$.

Lemma 3.8 (Dimension-field containment)**.** *If the input ring is categorifiable, there is a commutative integral signed based algebra A of rank six, with symmetric multiplication matrices M_i, such that $M_i \equiv N_i \pmod 2$ and $|(M_i)_{jk}| \le (N_i)_{jk}$. It has a dimension character $d = (1, d_1, \dots, d_5)$ in a totally real field K of degree at most six. If $\Delta = \sum_i d_i^2$, every original formal codegree f belongs to K, and Δ/f is an algebraic integer in K. The same statement holds for every formal codegree of A.*

Proof. Sphericalization has two lifts of each simple and a central sign object ϵ of dimension -1. Each lift of a self-dual simple is self-dual: its two possible lifts have opposite nonzero dimensions, whereas a dual has the same dimension. Thus its Grothendieck ring is commutative. Quotienting by $\epsilon + 1$ gives A and the asserted bounds and parity. Its regular matrices are real symmetric, hence simultaneously diagonalizable with real characters. Evaluation at the dimension character is a field quotient of $A \otimes \mathbb{Q}$, proving $[K : \mathbb{Q}] \le 6$. An original character extends to sphericalization with $\epsilon = 1$ and formal codegree $2f$. The induction theorem for formal codegrees [36, Theorem 2.13] produces a central simple object of dimension $2\Delta/(2f) = \Delta/f$. Forgetting it to sphericalization expresses this dimension as an integer combination of the numbers $\pm d_i$. The same argument applies to signed characters with $\epsilon = -1$. □

Suppose the original formal-codegree fields generate $E = \mathbb{Q}(\sqrt{p}, \sqrt{q})$, where $p, q > 1$ are squarefree, $p \equiv 1 \pmod 4$, and their quadratic field discriminants are coprime. The lemma forces $K = E$, since $4 \mid [K : \mathbb{Q}] \le 6$. Write $d_i = (\alpha_i + \beta_i\sqrt{p} + \gamma_i\sqrt{q} + \delta_i\sqrt{pq})/4$. Orthogonality of the four distinct Galois-conjugate characters gives

$$16 + \sum_{i=1}^{5} \alpha_i^2 = p \sum_{i=1}^{5} \beta_i^2 = q \sum_{i=1}^{5} \gamma_i^2 = pq \sum_{i=1}^{5} \delta_i^2.$$

Consequently $\mathrm{Tr}_{E/\mathbb{Q}}(\Delta) \in pq\mathbb{Z}_{>0}$. The two remaining signed formal codegrees are greater than one, since all are positive and their reciprocals sum to one. Since the signed Casimir trace is at most T_0,

$$\mathrm{Tr}_{E/\mathbb{Q}}(\Delta) < T_0 - 2. \tag{3.6}$$

Proposition 3.9 (Finite biquadratic obstruction)**.** *Assume in addition that the Frobenius–Perron formal codegree has minimal polynomial $X^2 - 5nX + 25n$, where $n = s^2 + 4 = k^2\ell$ and $\ell \in \{q, pq\}$. Every categorification supplies an integral element*

$$\theta = \frac{\Delta}{5k\sqrt{\ell}} = \frac{x + y\sqrt{p} + z\sqrt{q} + w\sqrt{pq}}{4}$$

subject to all the following finite tests:

1. *Δ is totally positive, its trace satisfies (3.6), and pq divides that trace. Put $m = z$ if $\ell = q$ and $m = w$ if $\ell = pq$. Then $m > 0$ and $\mathrm{Tr}(\Delta) = 5k\ell m$.*
2. *$\mathrm{Norm}(\theta)$ divides $\ell^2 m^4$.*
3. *Put $1 - \mathrm{Tr}(1/\Delta) = A/B$ in lowest positive terms. For some positive integer j, the last two signed codegrees have sum Aj and product Bj, both roots greater than one, and sum at most $T_0 - \mathrm{Tr}(\Delta)$. If their quadratic polynomial is irreducible, $Bj \mid (Aj)^2$ and its quadratic field embeds into E.*
4. *The product of this quadratic polynomial and $\prod_{\sigma : E \hookrightarrow \mathbb{R}} (X - \sigma\Delta)$ is congruent modulo two to the original Casimir characteristic polynomial.*

If the finite set is empty, no complex fusion category realizes the ring.

Proof. For $u = (s + \sqrt{n})/2$ one has $u^2 - su - 1 = 0$ and $D = 5\sqrt{n}\,u$. Thus $\theta = u\Delta/D$ is integral by Lemma 3.8. The integral basis is the product of the integral bases of the two quadratic fields, because their discriminants are coprime. If $(e, h) = (1, 2)$ for $q \equiv 1 \pmod 4$, and $(e, h) = (0, 1)$ otherwise, its congruences are

$$\begin{aligned} hw &\equiv 0 \pmod 2, & h(z - w) &\equiv 0 \pmod 4, \\ y - ew &\equiv 0 \pmod 2, & x - y - ez + ew &\equiv 0 \pmod 4. \end{aligned}$$

Total positivity supplies the strict coordinate bounds $|w| < m/\sqrt{p}$, $|x| < m\sqrt{q}$, $|y| < m\sqrt{q/p}$ when $\ell = q$; and $|x| < m\sqrt{pq}$, $|y| < m\sqrt{q}$, $|z| < m\sqrt{p}$ when $\ell = pq$. Together with the trace bound these give the finite search. The global dimension is a d-number by [35, Theorem 1.2]. For a nonzero d-number a in a Galois field, $\mathrm{Tr}(a)^4/\,\mathrm{Norm}(a)$ is integral: $\mathrm{Tr}(a)/a$ is integral and one takes its norm. Applying this to Δ gives the norm divisibility in (2). The two complementary codegrees are Galois stable, so their sum and product are integers. The reciprocal identity gives (Aj, Bj); the quadratic d-number test and Lemma 3.8 give (3). Finally $M_i \equiv N_i \pmod 2$ implies the congruence in (4). □

The tensor-only implementation `new_biquadratic_trace.py` excludes `R6_7_3813`, `R6_8_6940`, `R6_8_6973`, and `R6_8_6998`. Their fields are respectively $\mathbb{Q}(\sqrt{5}, \sqrt{193})$, $\mathbb{Q}(\sqrt{5}, \sqrt{34})$, $\mathbb{Q}(\sqrt{17}, \sqrt{257})$, and $\mathbb{Q}(\sqrt{5}, \sqrt{13})$. For `R6_8_6973`, the trace is $2570m$, the trace bound gives $1 \le m \le 2$, and the lattice condition requires $17 \mid m$. For `R6_8_6998`, the only remaining complementary pair has polynomial $X^2 - 2350X + 2500$; its product with the four dimension codegrees is X^6 modulo two, whereas the original Casimir polynomial is $X^6 + X^2$. All four certificates replay with Python optimization enabled. The checker also reproduces the imported multiplicity-thirteen example's 1129 totally positive candidates, rejects altered certificates, respects simultaneous basis permutations, and excludes none of the 54 established positive models.

3.7 Formal-codegree trace ideals

The field-containment lemma admits a useful consequence even when the signed dimension field has not been identified. For an algebraic integer f, write

$$\mathfrak{t}(f) = \mathrm{Tr}_{\mathbb{Q}(f)/\mathbb{Q}}(f\mathcal{O}_{\mathbb{Q}(f)}) = \tau(f)\mathbb{Z}, \qquad \tau(f) > 0.$$

Conjugating f does not change this integer ideal.

Theorem 3.10 (Formal-codegree trace-ideal obstruction)**.** *Let R be a commutative self-dual rank-six fusion ring, with regular matrices N_i and Casimir trace $T_0 = \mathrm{Tr}(\sum_i N_i^2)$. Choose any collection of its formal codegrees f and put $L = \mathrm{lcm}_f\, \tau(f)$. If R is categorifiable, the signed sphericalization has a dimension field K and global dimension Δ satisfying*

$$0 < \mathrm{Tr}_{K/\mathbb{Q}}(\Delta) \in L\mathbb{Z}, \qquad \mathrm{Tr}_{K/\mathbb{Q}}(\Delta) \le T_0.$$

In particular, $L > T_0$ excludes every complex fusion categorification. No pivotality assumption or local-field isolation is required.

Proof. Lemma 3.8 gives $F = \mathbb{Q}(f) \subseteq K$ and $\Delta/f \in \mathcal{O}_K$. The relative trace of an algebraic integer is integral. Therefore

$$\mathrm{Tr}_{K/F}(\Delta) = f\,\mathrm{Tr}_{K/F}(\Delta/f) \in f\mathcal{O}_F.$$

Transitivity of trace gives the asserted divisibility for every selected f. The embeddings of K give distinct characters of the signed algebra, with formal codegrees the corresponding conjugates of Δ. All its formal codegrees are positive. Thus their total, the signed Casimir trace, is at least $\mathrm{Tr}_{K/\mathbb{Q}}(\Delta)$. The signed coefficient bounds give $\mathrm{Tr}(\sum_i M_i^2) = \sum_{i,j,k}(t_{ij}^k)^2 \le \sum_{i,j,k}(N_{ij}^k)^2 = T_0$. □

The implementation uses every rational and quadratic original codegree. For a rational integer f, $\tau(f) = f$. For a quadratic codegree $f = (A + b\sqrt{q})/2$, with $q > 1$ squarefree, the integral basis gives

$$\tau(f) = \begin{cases} \gcd(A, (A + bq)/2), & q \equiv 1 \pmod 4, \\ \gcd(A, bq), & q \equiv 2, 3 \pmod 4. \end{cases}$$

Omitting higher-degree factors weakens the test and creates no exclusion. The calculation depends only on the complete multiplication tensor.

For `R6_8_6997`, the rational formal codegrees include 15, while the FP codegree has minimal polynomial $X^2 - 6145X + 30725$ and trace ideal $6145\mathbb{Z}$. Hence $L = 18435 > T_0 = 6173$, giving a general exclusion. The same test supplies independent exclusions of `R6_7_3812`, `R6_8_6234`, `R6_8_6509`, `R6_8_6940`, and `R6_8_6973`.

Corollary 3.11 (Signed saturation from a trace ideal)**.** *For an unordered triple of nonunit basis labels with coefficient $a \geq 2$, let w be its number of ordered permutations, and put*

$$\eta = \min w(a^2 - (a - 2)^2),$$

where the minimum runs over those triples. If $0 \leq T_0 - L < \eta$, every signed sphericalization coefficient has absolute value equal to its original nonnegative coefficient. All remaining choices are finite sign choices subject to associativity.

Proof. Parity forces any decrease of absolute value to be at least two. Full symmetry forces the same decrease on all w ordered permutations. Such a decrease loses at least η from the Casimir trace, whereas Theorem 3.10 permits loss at most $T_0 - L$. The unit coefficients do not change in the signed quotient. □

For `R6_8_6520`, one has $L = 6145$, $T_0 = 6165$, and $\eta = 24$. Thus its signed quotient is saturated. Exhaustive exact signed associativity leaves one table up to the certified sign gauges; that table is gauge-equivalent to the original ring. The non-FP character orbits have zero entries and cannot be dimension characters. Consequently every categorification would have a pseudo-unitary Galois conjugate. Proposition 3.12 excludes that possibility, completing the general noncategorifiability proof.

The authoritative implementation is `new_trace_ideal.py`. It checks all arithmetic from the tensor, replays signed enumeration when used, and rejects altered certificates and omitted signed tables. Its control report uses the frozen initial forty residual tensors and all fifty-four established positive models; the latter are all admitted by this necessary test.

3.8 An all-orders quadratic Gauss obstruction

The trace-ideal reduction also leads to a new complete exclusion at multiplicity eight. The argument below treats every possible central restriction column and every root-of-unity order; it does not impose a numerical conductor cutoff.

Write $R(m)$ for the commutative self-dual rank-six tensor on $1, a, b, c, d, e$ whose nonunit coefficients equal m, except that the four unordered triples

$$(a, a, b),\ (a, a, c),\ (a, b, b),\ (b, b, c)$$

have coefficient $m - 1$, and

$$(c, d, d),\ (c, e, e),\ (d, d, e),\ (d, e, e)$$

have coefficient $m+1$. Coefficients are invariant under permutations of the three indices, and the unit coefficients are fixed by self-duality. Set

$$n = 25m^2 + 4, \quad U = \frac{5m + \sqrt{n}}{2}, \quad j = (1,1,1,1,1)^t, \quad \delta = (-1,-1,0,1,1)^t.$$

The FP dimensions are $(1, U-1, U-1, U, U+1, U+1)$, and $D = 5(U^2+1) = (5n+25m\sqrt{n})/2$. The other five characters are the quadratic conjugate of the FP character and, with $p = (1+\sqrt{5})/2$, $q = (1-\sqrt{5})/2$,

$$(1,p,q,-1,0,0),\ (1,q,p,-1,0,0),\ (1,0,0,1,-p,-q),\ (1,0,0,1,-q,-p).$$

The corresponding formal codegrees are $D, D', 5, 5, 5, 5$. These identities, character completeness, and the full tensor are checked exactly by `software/new_gauss_family.py`.

Proposition 3.12 (Quadratic Gauss family obstruction)**.** *The tensor $R(7) = R^{(6)}_{8,6520}$ has no pseudo-unitary categorification. Consequently the trace-ideal signed reduction excludes every complex fusion categorification of this tensor. More generally, the same exact verifier gives a necessary criterion for $R(m)$, $m \geq 3$: whenever its exhaustive column enumeration has the families described below, it requires*

$$12m^2 + 2 \ \geq \ \frac{\varphi(f)mb}{2c}, \qquad n = b^2 q_0, \quad f = \operatorname{disc} \mathbb{Q}(\sqrt{q_0}), \quad c = \sqrt{f/q_0},$$

where q_0 is squarefree.

Proof. Assume a pseudo-unitary categorification and use its positive spherical structure. If v_Y is the restriction column of a central simple, then $\sum_Y v_Y v_Y^t = G$, where $G = \sum_i N_i^2$. The unit column of G is $(6, 5mj + \delta)^t$, and its nonunit block is

$$5I + (25m^2+1)jj^t + 5m(j\delta^t + \delta j^t) + \delta\delta^t.$$

By central induction, $I(1)$ contains six distinct simples with dimensions $1, z = U^2, t = U^2+1, t, t, t$ and twists one [36, Theorems 2.5 and 2.13]. The symmetry of the central S-matrix gives, for character-labelled low columns,

$$\dim(Y_\chi)\chi(v_\psi) = \dim(Y_\psi)\psi(v_\chi).$$

Applying this with the quadratic-conjugate character and the four codegree-five characters determines the z column as $(1, mj)^t$: indeed the conjugate dimension of a t column is $t' = t/z$, so reciprocity gives each small character value as $zt'/t = 1$. The complete character table then determines the column uniquely. Write the remaining four columns as $(1, mj + w_\nu)^t$. Their dimensions and the unit column of the Gram identity give

$$j^t w_\nu = 0, \qquad \delta^t w_\nu = 1, \qquad \sum_\nu w_\nu = \delta.$$

On the three-dimensional subspace perpendicular to j and δ, G restricts to $5I$. Taking the trace of its residual Gram inequality therefore gives $\sum_\nu \|w_\nu\|^2 \leq 16$. Each squared norm is at least two, so every w_ν has squared norm at most ten and coordinates in $[-3, 3]$.

The exhaustive enumeration has 32 individual deviations, 59 unordered quadruples satisfying the sum and norm conditions, and 11 satisfying a labelling by the four small characters and modular reciprocity. Exactly three have positive-semidefinite residual Gram matrix. Their

rows of deviations are

$$W_1 = \begin{pmatrix} -1 & 0 & 1 & -1 & 1 \\ -1 & 1 & -1 & 1 & 0 \\ 0 & -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 0 & 1 \end{pmatrix}, \qquad W_2 = \begin{pmatrix} -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 & 0 \end{pmatrix},$$

$$W_3 = \begin{pmatrix} -1 & 0 & 1 & 1 & -1 \\ -1 & 1 & -1 & 0 & 1 \\ 0 & -1 & 1 & -1 & 1 \\ 1 & -1 & -1 & 1 & 0 \end{pmatrix}.$$

All enumeration bounds have just been proved; no low column is selected in advance.

A remaining column is $(0, v)^t$, with $v \in \mathbb{Z}^5_{\geq 0} \setminus \{0\}$. Let Q be the nonunit residual Gram matrix. The necessary conditions are

$$Q - vv^t \succeq 0, \qquad Q - vv^t \geq 0 \text{ entrywise}, \qquad D/\dim(v) \text{ algebraically integral}.$$

The last condition follows because $D^2/\dim(v)^2$ is a formal codegree of the modular center. All three residuals annihilate $(1, 1, -4, 1, 1)^t$. Hence the first, second, fourth, and fifth coordinates determine the third. The bounds $v_i \leq \lfloor \sqrt{Q_{ii}} \rfloor$ and

$$|v_a - v_b| \leq \lfloor \sqrt{Q_{aa} + Q_{bb} - 2Q_{ab}} \rfloor, \qquad |v_d - v_e| \leq \lfloor \sqrt{Q_{dd} + Q_{ee} - 2Q_{de}} \rfloor$$

make the enumeration finite and complete. Gram positivity is checked with exact rational Schur complements, or equivalently by the kernel condition and $v_J^t Q_J^{-1} v_J \leq 1$ on a full-rank principal submatrix. For a quadratic number $x + y\sqrt{n}$, integrality is equivalent to integral trace $2x$ and integral norm $x^2 - ny^2$.

For $m = 7$ the counts at the bounded, Gram-compatible, and integral-quotient stages are respectively

$$(511, 58, 1), \qquad (6147, 282, 9), \qquad (511, 58, 1).$$

The outer branches admit only j, although Q has rank two, and are therefore impossible. The complete middle list is

$$j, \quad j \pm (e_a - e_b), \quad j \pm (e_d - e_e), \quad 7j - e_i + e_k \quad (i \in \{a, b\},\ k \in \{d, e\}).$$

For the general verifier, the accepted list may also include the first five types with j replaced by $2j$ and the last four types with m replaced by $2m$; any additional high type causes the criterion to abstain.

Put $u = e_a - e_b$, $v = e_d - e_e$. Exact Gram identities are

$$\delta^t Q \delta = 32, \quad \delta^t Q j = 80m, \quad u^t Q u = v^t Q v = 8, \quad \delta^t Q u = \delta^t Q v = u^t Q v = 0.$$

If n_m, n_{2m} count the two possible cross-column families, the first two equations give $n_m + n_{2m} = 8$ and $n_m + 2n_{2m} = 8$. Thus exactly eight cross columns occur, all with coefficient m. They saturate both u, v squared forms, excluding all offset types. The three vanishing mixed forms force each of the four columns $B_{ik} = mj - e_i + e_k$ to occur twice. Subtracting them leaves

$$Q - 2\sum_{i,k} B_{ik} B_{ik}^t = (12m^2 + 2) jj^t.$$

Thus the number of uniform columns, with a column $2j$ counted four times, is $L = 12m^2 + 2$.

Let Θ_{ik} be the sum of the two twists belonging to B_{ik}, and let Λ be the sum of uniform twists with these weights. It is a sum of exactly L roots of unity. The first twist moment

of $I(X)$ vanishes for nonunit X [36, Theorem 2.5], so, putting $w = 5U$, the five equations combine as

$$w\Lambda j + t\sum_{i,k}\Theta_{ik}B_{ik} = -mzj - t(4mj + \delta).$$

Pairing with δ gives $\sum\Theta_{ik} = -2$; pairing with j gives

$$w\Lambda = -mz - 2mt, \qquad \Lambda = -\frac{m^2 + m\sqrt{n}}{2}.$$

For completeness the required root-of-unity bound holds for all orders. For $K = \mathbb{Q}(\sqrt{q_0})$ with positive fundamental discriminant f, let $c = \sqrt{f/q_0}$. The coefficient of $\sqrt{q_0}$ in the normalized trace to K of a single root of unity has absolute value at most $c/\varphi(f)$. If the root's cyclotomic field does not contain K, the trace is rational. Otherwise its order is divisible by f. Averaging over the kernel of the primitive quadratic character expresses the irrational part as an induced quadratic Gauss sum divided by $\varphi(N)$. This sum is zero unless the extra factor N/f is squarefree and coprime to f; in the nonzero case its magnitude is $\sqrt{f}$. The formula follows directly by Chinese remaindering: excess exponents at primes of f cancel over lifts, and the coprime factor contributes its Ramanujan sum. As $\varphi(N) \geq \varphi(f)$, the stated bound follows. Normalized trace and the triangle inequality then require

$$L \geq \frac{\varphi(f)}{c}\,|[\sqrt{q_0}]\Lambda| = \frac{\varphi(f)mb}{2c}.$$

For $m = 7$, $n = q_0 = f = 1229$, $b = c = 1$, and $\varphi(f) = 1228$. The forced sum of 590 roots is $-(49 + 7\sqrt{1229})/2$, but its required length is at least $1228 \cdot 7/2 = 4298 > 590$. This contradiction proves the pseudo-unitary exclusion. The separate complete signed trace-ideal certificate supplies a pseudo-unitary Galois conjugate of any hypothetical categorification, proving the general exclusion. □

The executable certificate is tensor-based and invariant under relabelling of the five nonunit objects. Its normal and optimized-Python replay reconstructs every low and high column, rejects missing branches and altered arithmetic, and preserves all 54 realized controls. The frozen 40-input continuation scan applies the criterion to every input; only $R^{(6)}_{7,3582}$ and $R^{(6)}_{8,6520}$ match this tensor family, and both are excluded in the pseudo-unitary scope. The latter is a new general exclusion after the independent trace-ideal reduction. A scan of all 16,612 census tensors also checks the parameter-three through parameter-seven members uniformly:

tensor	forced root count	Gauss lower bound
$R^{(6)}_{4,803}$	110	342
$R^{(6)}_{5,1500}$	194	400
$R^{(6)}_{6,2554}$	302	1440
$R^{(6)}_{7,3582}$	434	1344
$R^{(6)}_{8,6520}$	590	4298

The first three already had exclusions in the incoming classification; this table records independent application of the new necessary criterion.

3.9 Ramification and rational-codegree trace bounds

This criterion applies to arbitrary complex fusion categorifications. It does not assume a pivotal structure and does not enumerate selected dimension characters. Let R be commutative and self-dual of rank six, and write $T_0 = \sum_{i,j,k}(N^k_{ij})^2$.

Proposition 3.13 (Ramified dimension field and Casimir trace)**.** *Suppose the FP global dimension is $D = a + b\sqrt{q}$, where $q > 1$ is squarefree, $q \equiv 2, 3 \pmod 4$, and $a, b \in \mathbb{Z}$, $b > 0$. Suppose also that*

$$R/2R \cong \mathbb{F}_2[z]/(z^2) \times \prod_j \mathbb{F}_{2^{e_j}}.$$

Let $\mathcal{L} \subseteq \mathbb{Z}^2$ be the lattice of pairs (S, B) for which

$$\frac{2qaS - 2qbB}{a^2 - qb^2}, \qquad \frac{2aB - 2qbS}{a^2 - qb^2}, \qquad \frac{2qS}{c}, \qquad \frac{2B}{c} \quad \text{are integers}$$

for every rational original formal codegree c. Let v be the positive generator of the projection of $\mathcal{L}$ onto its first coordinate. If

$$4qv + 5 > T_0,$$

then R has no complex fusion categorification.

Proof. Sphericalize a hypothetical categorification and quotient its Grothendieck ring by the central sign object plus the unit. The result is a rank-six integral signed algebra A with multiplication coefficients t_{ij}^k satisfying

$$t_{ij}^0 = \delta_{ij}, \qquad |t_{ij}^k| \le N_{ij}^k, \qquad t_{ij}^k \equiv N_{ij}^k \pmod 2.$$

The two lifts of each original self-dual simple have opposite nonzero spherical dimensions, so duality cannot interchange them. Consequently the regular matrices M_i of A are real symmetric and commute. The dimension character has values d_i in a totally real field factor K of $A \otimes \mathbb{Q}$, and $\Delta = \sum_i d_i^2$ is the global dimension of the original category.

For every original formal codegree f, its pulled-back character on sphericalization has formal codegree $2f$. The central-induction theorem [36, Theorem 2.13] therefore gives a central simple of dimension Δ/f. Forgetting it expresses this dimension as an integer combination of the $\pm d_i$. Thus $\Delta/f \in \mathcal{O}_K$, and in particular $D \in K$ and $\mathbb{Q}(\sqrt{q}) \subseteq K$. This also recalls the dimension-field argument of Lemma 3.8.

Since $A/2A = R/2R$, lift the displayed primitive idempotents to $A \otimes \mathbb{Z}_2$. A finite free factor with finite-field reduction is finite etale over $\mathbb{Z}_2$, so its generic fiber is unramified. The only remaining factor has rank two. But $\mathbb{Q}_2(\sqrt{q})$ is ramified for $q \equiv 2, 3 \pmod 4$, and every component of $K \otimes \mathbb{Q}_2$ contains this field. No such component can be supported on an unramified factor. Hence $[K : \mathbb{Q}] \le 2$, and therefore $K = \mathbb{Q}(\sqrt{q})$.

Now $\mathcal{O}_K = \mathbb{Z}[\sqrt{q}]$. Write $d_i = \alpha_i + \beta_i\sqrt{q}$ for $1 \le i \le 5$, and set $S = \sum_i \beta_i^2 > 0$ and $B = \sum_i \alpha_i\beta_i$. The two conjugate dimension-character vectors are orthogonal, because the M_i are commuting real symmetric matrices. It follows that

$$1 + \sum_i \alpha_i^2 = qS, \qquad \Delta = 2qS + 2B\sqrt{q}, \qquad \Delta + \Delta' = 4qS.$$

Integrality of Δ/D gives the first two conditions defining $\mathcal{L}$. Integrality of Δ/c gives the other two for every rational codegree c. Therefore $S \ge v$.

The signed Casimir $\sum_i M_i^2$ has the six signed formal codegrees as its eigenvalues. The two dimension characters contribute $4qS$. Each of the other four codegrees is strictly greater than one: equality would force every nonunit character value to vanish, contrary to the unit coefficient in x_i^2. Since the trace is integral,

$$4qv + 5 \le \operatorname{Tr}\Big(\sum_i M_i^2\Big) = \sum_{i,j,k} (t_{ij}^k)^2 \le T_0,$$

contradicting the asserted strict inequality. □

Corollary 3.14. *The rings $R^{(6)}_{7,3812}$, $R^{(6)}_{8,6234}$, and $R^{(6)}_{8,6509}$ have no complex fusion categorifications.*

Proof. For all three rings,

$$q = 226, \qquad D = 2260 + 150\sqrt{226}, \qquad D^2 - 4520D + 22600 = 0.$$

The following table lists their exact local factors and trace data. Here $B_2 = \mathbb{F}_2[z]/(z^2)$ and D' is the conjugate of D.

Ring	Algebra modulo two	Formal codegrees	v	T_0
$R^{(6)}_{7,3812}$	$B_2 \times \mathbb{F}_4 \times \mathbb{F}_2^2$	$D, D', 3, 15, 5, 5$	15	4548
$R^{(6)}_{8,6234}$	$B_2 \times \mathbb{F}_4 \times \mathbb{F}_2^2$	$D, D', 3, 15, 5, 5$	15	4548
$R^{(6)}_{8,6509}$	$B_2 \times \mathbb{F}_2^4$	$D, D', 3, 3, 15, 15$	15	4556

In particular $\Delta/15 \in \mathcal{O}_K$ and $\Delta = 452S + 2B\sqrt{226}$ imply $15 \mid S$, because $\gcd(452, 15) = 1$. Thus the signed Casimir trace is at least $904 \cdot 15 + 5 = 13565$, exceeding each displayed T_0. □

The finite local checks exhaust all 64 elements modulo two, find every primitive idempotent, and verify orthogonality and completeness of the factor decomposition. Every nonzero element of a claimed field factor is checked to have an inverse; the remaining rank-two factor has exactly one nonzero square-zero element. The lattice step is recovered from exact congruences, not floating-point bounds. The three certificates require neither a signed-table search nor a pseudo-unitary exclusion.

The tensor-based implementation is `software/new_ramified_signed.py`. It was applied to the frozen second-round frontier of 40 rings, including all 39 with multiplicity four through eight. Of those 39, the hypotheses fail for 34, one ($R^{(6)}_{7,3582}$) proceeds to a bounded signed reduction, and four are excluded directly: the three above and an independent field-containment exclusion of $R^{(6)}_{8,6940}$. The latter is also covered by the biquadratic criterion, so it is not counted twice. The complete per-input report is `research/round2_higher/ramification_application.json`. All 54 positive models are preserved. Ordinary and optimized Python checks verify the three certificates after nontrivial basis permutations and reject six certificate corruptions and a corrupted tensor.

3.10 Local character fields and quadratic codegree saturation

The next criterion strengthens the signed Casimir bounds by determining the dimension field before enumerating possible formal codegrees. Write A for the signed algebra supplied by Lemma 3.8, K for its dimension character field, and Δ for its dimension codegree. Thus every original or signed formal codegree f lies in K, and Δ/f is an algebraic integer. In particular this statement concerns the sphericalization of an arbitrary categorification; it does not assume a pivotal structure on that category.

Proposition 3.15 (Two local ways to isolate the dimension field)**.** *Let R be commutative and self-dual of rank six, and suppose an original formal codegree generates $E = \mathbb{Q}(\sqrt{q})$, where $q > 1$ is squarefree. Either of the following conditions implies $K = E$ and that the other four signed formal codegrees are rational integers greater than one:*

1. *$q \equiv 5 \pmod 8$, and $R/2R$ is a product of finite fields with exactly one even-degree factor, of degree two;*
2. *$R/2R$ has three local factors: $\mathbb{F}_8$, a rank-two local algebra, and $\mathbb{F}_2$.*

In the second case the conclusion uses cyclotomicity of characters of a categorifiable fusion ring.

Proof. The dimension-field containment lemma gives $E \subset K$. Because the regular matrices of A are symmetric, $A \otimes \mathbb{Q}$ is a product of totally real number fields. Its primitive local idempotents at two lift from $A/2A = R/2R$ to $A \otimes \mathbb{Z}_2$. A finite free local factor with finite-field special fibre is finite étale: its relative differentials vanish by Nakayama, and it is flat. It is therefore the unramified extension with that residue degree. Equivalently one may apply the classification of finite étale algebras over a henselian local ring; see the Stacks Project, Tag 04GK.

In the first case $E \otimes \mathbb{Q}_2$ is the unramified quadratic field. Every completion of a number field containing E has even degree over $\mathbb{Q}_2$. Since K is a field quotient of $A \otimes \mathbb{Q}$, its completions must use only the unique even-degree local factor. That factor has degree two, so $K = E$. If a codegree belonging to another character field F were irrational, it would generate E, because it already belongs to $K = E$. Then F would require another even-degree local factor, which is impossible. All remaining codegrees are therefore rational.

In the second case the lifted rank-three factor has generic fibre the unramified cubic extension of $\mathbb{Q}_2$. Let F be the global character field containing this local factor. A signed character extends to a character of the sphericalization with the sign object sent to -1. Hence its values are cyclotomic by [14, Theorem 8.51]; in particular $F/\mathbb{Q}$ is abelian and Galois. All its completions at two have the same local degree, here three. Thus $[F : \mathbb{Q}]$ is three or six. Degree six would require two degree-three local factors, whereas the remaining lifted blocks have ranks two and one. Hence F is cubic and uses the entire rank-three block. The remaining rational algebra has dimension three. The field K contains the quadratic field E, so it lies in this remainder and has degree two. Thus $K = E$. The other character fields are F and $\mathbb{Q}$; their intersections with E are rational. The remaining codegrees, which lie in these intersections, are rational as claimed.

In both cases rational formal codegrees are algebraic integers. They are positive, and the sum of reciprocals of the six signed codegrees is one. Each is consequently greater than one. □

The second argument does not assert that the whole two-adic order is étale. In particular a rank-two order with dual-number reduction may have split generic fibre. Only its rank and its separation from the cubic block are needed.

Proposition 3.16 (Finite quadratic spectrum bound)**.** *Suppose $K = E = \mathbb{Q}(\sqrt{q})$ and the complementary four codegrees are rational, as in Proposition 3.15. Assume $q \equiv 1 \pmod 4$ and the original Frobenius–Perron codegree has the form $D = c\sqrt{q}\,u$, where c is a positive integer and u is an integral unit of norm -1. Let $T = \operatorname{Tr}(\sum_i N_i^2)$. Every categorification supplies integers x, y and $f_1 \le \cdots \le f_4$ satisfying the following finite system:*

$$\Delta = \frac{c(qy + x\sqrt{q})}{2}, \qquad x \equiv y \pmod 2, \quad y > 0, \quad x^2 < qy^2, \quad cqy \le T - 5,$$

$$n = qy^2 - x^2, \qquad n \mid 4qy^2, \qquad \sum_{i=1}^{4} \frac{1}{f_i} = 1 - \frac{4y}{cn} > 0,$$

$$f_i > 1, \qquad f_i \mid \gcd\left(\frac{c(qy - x)}{2}, cx\right), \qquad cqy + \sum_{i=1}^{4} f_i \le T.$$

In addition Δ/f is integral for every original rational codegree f, and the polynomial

$$\left(z^2 - cqy\,z + \frac{c^2 qn}{4}\right) \prod_{i=1}^{4} (z - f_i)$$

is congruent modulo two to the original Casimir polynomial. If the system is empty, R is not categorifiable. Otherwise its least total trace bounds the possible signed Casimir deficit.

Proof. Since Δ/D is integral, $\theta = \Delta/(c\sqrt{q}) = u\Delta/D$ is integral. The integral basis of E gives $\theta = (x + y\sqrt{q})/2$ with the stated parity. Total positivity of Δ gives $y > 0$ and $x^2 < qy^2$. Its two dimension characters contribute trace cqy; the other four codegrees exceed one. The signed Casimir trace is an integer at most T, giving the displayed finite bound.

The global dimension is a d-number by [35, Theorem 1.2]. Here

$$\mathrm{Tr}(\Delta) = cqy, \quad \mathrm{N}(\Delta) = \frac{c^2qn}{4}, \quad \mathrm{Tr}(1/\Delta) = \frac{4y}{cn}.$$

The quadratic d-number divisibility gives $n \mid 4qy^2$, and the reciprocal-codegree identity gives the reciprocal equation. The integral-basis coordinates of Δ are $c(qy - x)/2$ and cx. Thus Δ/f_i is integral precisely when f_i divides their greatest common divisor. Divisibility by original rational codegrees follows separately from induction. Finally the signed and original matrices are congruent modulo two, so their Casimir characteristic polynomials have the same reduction. Every enumeration range is finite. The case $x = 0$ is retained: K can be quadratic even when Δ itself is rational. □

Theorem 3.17. *The rings $R^{(6)}_{8,6524}$, $R^{(6)}_{8,6525}$ and $R^{(6)}_{8,6999}$ have no complex fusion categorifications.*

Proof. For the first two rings, $q = 149$, $c = 9$,

$$D = \frac{6705 + 549\sqrt{149}}{2}, \qquad T = 6741.$$

Their reductions at two have field degrees $(2, 3, 1)$ and $(2, 1, 1, 1, 1)$, respectively. Proposition 3.15(1) applies. The finite spectrum test has $1 \le y \le 5$ and leaves only

$$y = 5, \qquad x = \pm 61, \qquad (f_1, f_2, f_3, f_4) = (9, 9, 9, 9).$$

Both traces equal 6741.

For $R^{(6)}_{8,6999}$, $q = 41$, $c = 18$,

$$D = 3690 + 576\sqrt{41}, \qquad T = 7416.$$

The local factors are $\mathbb{F}_8$, the dual numbers of rank two, and $\mathbb{F}_2$. Proposition 3.15(2) applies. The bound is $1 \le y \le 10$; the only survivors are

$$y = 10, \qquad x = \pm 64, \qquad (f_1, f_2, f_3, f_4) = (9, 9, 9, 9),$$

again with total trace T.

In all three cases the signed Casimir trace is the sum of the squares of the signed structure constants. Since each absolute value is bounded by its original coefficient, equality of traces forces equality of every absolute value. The remaining unrestricted sign-only associativity systems are exhausted exactly. Every solution is a dual-compatible sign gauge of the original tensor. After this gauge, the lift fusion rules are the original positive ones, and the recorded dimension-character tests leave only the Galois orbit of the positive character. Any categorification would therefore have a pseudo-unitary Galois conjugate.

The pseudo-unitary possibility is excluded by the exact `global_codegree_moment` certificates, checked independently on the full input tensors. For the first two rings the obstruction is visible already in the integral span of simple dimensions. They are $1, u - 1, u, u, u, u$, where $u = (37 + 3\sqrt{149})/2$ and $u^2 - 37u + 7 = 0$. The codegree 9 would require a central simple of dimension

$$D/9 = (61u - 11)/3 \notin \mathbb{Z} + \mathbb{Z}u,$$

which cannot be the dimension of an integral restriction to the original category. This also gives a direct check on the final induction obstruction in those two cases. □

The tensor-only generator and exact verifier are `software/round3_arithmetic.py`; all local algebras, surviving spectra, sign gauges, and complete general certificates are recorded in `research/round3/arithmetic/`. The independent enumeration in `research/round3/models/arithmetic_audit.py` reproduces the first pair of saturated spectra without an optimizer or the generator's recursive divisor enumeration.

3.11 Noncommutative signed spectra

The preceding field-isolation argument has a different replacement when the signed quotient is forced to be noncommutative. This follows either from reduction modulo two or from a separate exhaustive exclusion of the commutative signed sector. It is essential in this case to keep the representation multiplicities.

Proposition 3.18. *Let R have rank six. Suppose either that its reduction modulo two is noncommutative, or that every commutative signed quotient permitted by sphericalization has been excluded by necessary conditions. Suppose its two linear characters form a real quadratic orbit over $E = \mathbb{Q}(\sqrt{q})$, and let D, D' be their formal codegrees and c that of its unique two-dimensional representation. For any categorification there are $\Delta \in E$ and $h \in \mathbb{Z}_{>2}$ satisfying*

$$\frac{1}{\Delta} + \frac{1}{\Delta'} + \frac{2}{h} = 1, \qquad \mathrm{Tr}_{E/\mathbb{Q}}(\Delta) + 8h \le T, \qquad T = \sum_{i,j,k} (N_{ij}^k)^2.$$

Both embeddings of Δ are positive, Δ is a d-number, and

$$\Delta/D, \quad \Delta/D', \quad \Delta/c, \quad \Delta/h \quad \textit{belong to } \mathcal{O}_E.$$

For fixed R these conditions are a finite exact search.

Proof. The negative-sign quotient of sphericalization is now an oriented signed algebra A. Its signed basis is orthonormal for the coefficient-of-unit inner product, and duality gives adjoints of left multiplication. Thus its regular representation is a finite dimensional $*$-algebra with positive inner product and is semisimple. In the first case parity preserves a nonzero commutator; in the second case the commutative alternative is excluded separately. In either case dimension six and the existence of a linear character give

$$A \otimes \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}).$$

There are exactly two linear characters. The spherical dimension character is real and dual invariant, and its field K therefore has degree at most two. Induction applied to the original characters still gives $\Delta/f \in \mathcal{O}_K$. In particular $D \in K$, so $K = E$. The signed linear codegrees are Δ, Δ'.

The unique two-dimensional representation is preserved by Galois action. Its formal codegree h is consequently rational and an algebraic integer, hence an integer. The formal-codegree reciprocal identity weights it by its representation dimension two. On the four-dimensional regular matrix block the Casimir has eigenvalue $2h$, so that block contributes $8h$ to the regular trace. This proves both displayed identities, with the trace inequality following from the coefficient bounds. Positivity of the two linear codegrees and the reciprocal equation imply $h > 2$. Induction from the negative-sign character gives Δ/h; induction from the original characters gives the other quotients. The usual global-dimension d-number theorem applies to Δ.

For example, when $q \equiv 1 \pmod 4$, write $\Delta = (a + b\sqrt{q})/2$, with $a \equiv b \pmod 2$. Then $0 < a \le T - 24$ and $qb^2 < a^2$. These bounds are finite. For each pair (a, b) the reciprocal equation determines

$$h = \frac{2}{1 - a/\,\mathrm{N}(\Delta)}.$$

Testing integrality, positivity, the four quotients, and the quadratic d-number condition completes the exact necessary search. □

Theorem 3.19. *The ring $R^{(6)}_{6,170}$ has no complex fusion categorification.*

Proof. Its original codegrees are $148 \pm 24\sqrt{37}$ and 4, with representation dimensions $1, 1, 2$, and $T = 328$. The preceding finite system leaves exactly

$$\Delta = 148 \pm 24\sqrt{37}, \qquad h = 4, \qquad \mathrm{Tr}(\Delta) + 8h = 328.$$

Thus every signed coefficient has its original absolute value. The complete oriented sign-only associativity enumeration has 1631 search nodes and 16 solutions; all sixteen are dual-compatible sign gauges of R. No commutativity or indicator normalization is imposed. Under each gauge the two possible dimension characters are Galois conjugates of the positive one, so every categorification would have a pseudo-unitary Galois conjugate.

Multiplicity-weighted central induction has 146 candidate restriction columns for the nontrivial linear character and 25 for the two-dimensional representation. Their complete compatibility search yields nineteen low frames. Every frame violates an ordinary exact Gram or moment condition. The pseudo-unitary possibility is therefore impossible, completing the general exclusion. □

The implementation is `software/round3_noncommutative.py`, with arithmetic spectra, all oriented sign tables, the nineteen induction branches, and replays in `research/round3/noncommutative/`. For rings with commutative reduction modulo two, the first hypothesis of Proposition 3.18 is unavailable. Their entire commutative signed sector must be treated before the noncommutative spectrum calculation supplies a general reduction.

Theorem 3.20. *The rings $R^{(6)}_{7,200}$ and $R^{(6)}_{8,491}$ have no complex fusion categorifications.*

Proof. Both reductions modulo two are commutative. First impose commutativity on the unrestricted oriented signed coefficient problem, identifying opposite coefficient variables and intersecting their original parity intervals. If an opposite original coefficient is zero, its identified signed coefficient is forced to zero. This exhausts all commutative signed quotients without an indicator normalization.

The complete enumerations have respectively 176 and 208 signed tables, forming 11 and 13 dual-compatible sign-gauge orbits. Their character algebras have respectively 27 and 53 components. Each component has a zero dimension, unequal dimensions on dual objects, or an exact norm-divisibility obstruction to induction. The latter uses the original formal codegrees, whose quadratic polynomial is $z^2 - 1560z + 1872$, and the matrix-block codegree 12. The regular Casimir eigenvalue of that block is 24; it must not replace the formal codegree in the induction test. The recorded norm obstructions occur at primes 3 or 13. Consequently no categorical signed quotient is commutative.

Proposition 3.18 now applies. For both rings its finite system has exactly

$$\Delta = 780 \pm 216\sqrt{13}, \qquad h = 12, \qquad \mathrm{Tr}(\Delta) + 8h = 1656 = T.$$

The resulting sign-only searches have 4927 and 6463 nodes, respectively, and sixteen solutions each. Every solution is a dual-compatible gauge of the original tensor. Thus any categorification has a pseudo-unitary Galois conjugate. In each ring the multiplicity-weighted induction problem has ten complete low frames, all excluded by exact ordinary Gram or moment conditions. This excludes the pseudo-unitary possibility and proves the claim. □

The separate commutative-sector verifier is `software/round3_commutative_sector.py`; its complete certificate is included in each general certificate in `research/round3/noncommutative/`. Both general replays also reconstruct the noncommutative arithmetic and induction calculations. Removing the sector certificate is explicitly rejected by the verifier.

Scope and controls. Each detector reconstructs its hypotheses from the full tensor. The new family and arithmetic criteria are reapplied to the complete census through multiplicity eight; exclusions of already resolved rings are counted as confirmations. Exact replay disables numerical optimizers and includes altered-certificate rejection and changes of basis. All 54 positively realized rings are controls, with ineligibility distinguished from survival of an applicable system. The C4 family includes the positive parameter-zero member in its applicability scan, although the present irrational-dimension trace engine does not apply to it. Its positive parameter-two member survives the necessary system. These controls check scope; they do not turn a surviving necessary condition into a categorical model.

3.12 Dimension-block Fourier bounds and saturated twist support

Let $K_1 = \mathbb{Q}(d_i)$, $K_0 = \mathbb{Q}(D)$ and $e = [K_1 : K_0]$ for a spherical dimension function. For a character χ of $\mathrm{Gal}(K_0/\mathbb{Q})$, let f_χ be its primitive Dirichlet conductor. Schopieray's dimension-block theorem [50, Theorem 5.1] gives the additional necessary bound

$$\left| \sum_{\sigma \in \mathrm{Gal}(K_0/\mathbb{Q})} \frac{\chi(\sigma)}{\sigma(D)} \right|^2 \leq \frac{1}{e^2 f_\chi}. \tag{3.7}$$

Its equality case supplies a restriction on individual twists, rather than only on their averages. Here S denotes the center's modular S-matrix normalized by D, and T is its matrix of actual twists.

Proposition 3.21 (Saturated conductor support)**.** *Suppose equality holds in* (3.7)*. Inflate χ to $G_1 = \mathrm{Gal}(K_1/\mathbb{Q})$, and let $w_\chi = \sum_{\sigma \in G_1} \chi(\sigma) e_{A_\sigma}$, where A_σ is the low central summand indexed by the dimension character $\sigma(d)$. The cyclic subrepresentation generated by w_χ in the canonical modular representation of $Z(\mathcal{C})$ factors through $\mathrm{SL}_2(\mathbb{Z}/f_\chi\mathbb{Z})$. Consequently, for a central simple with restriction column v,*

$$L_\chi(v) := \sum_{\sigma \in G_1} \chi(\sigma) \sigma\left(\frac{\sum_i v_i d_i}{D} \right) \neq 0 \quad \Longrightarrow \quad \theta_Z^{f_\chi} = 1. \tag{3.8}$$

If several saturated characters give nonzero coefficients, the twist order divides the greatest common divisor of their conductors.

Proof. We extract the equality information from the local compression calculation in [50, Lemma 4.1]. Let $\Gamma = \mathrm{SL}_2(\mathbb{Z}/\ell^a\mathbb{Z})$, let U be its upper unipotent subgroup, and let P project onto U-fixed vectors. For a nontrivial torus character α of conductor ℓ^c, the universal permutation representation on primitive columns realizes $\mathrm{Ind}_U^\Gamma \mathbf{1}$. Its α-space of U-invariants splits into the one-dimensional big-cell space and the small-cell space. The compression PsP maps the big-cell line to the small-cell space and maps the latter back to a line. Both nonzero singular values are $\ell^{-c/2}$. Hence every vector attaining this norm belongs to the span of the big-cell vector and the adjoint image of the opposite-character big-cell vector.

Writing a primitive column as (p, q), these two vectors are, up to nonzero scalar factors,

$$\alpha(q)[q \text{ is a unit}], \qquad \alpha(p)[q \equiv 0 \pmod{\ell^c}].$$

Both are pullbacks of functions on primitive columns modulo ℓ^c. Their entire Γ-cyclic spans therefore factor through reduction modulo ℓ^c. Every irreducible representation with U-invariants embeds equivariantly and isometrically, up to a scalar, into this permutation representation. Thus an irreducible constituent containing a nonzero norm-attaining vector itself factors through that reduction.

For the trivial local character the bound is 1. Equality implies that sw is U-fixed as well as w. The vector w is therefore fixed by both upper and lower unipotents. These generate Γ, so its cyclic representation is trivial. Decompose a general unitary representation orthogonally into irreducibles, and then use the Chinese remainder tensor decomposition at the prime divisors of its level. Equality in the product norm bound forces every occupied local factor to attain its own bound. The preceding conclusions show that a norm-attaining vector generates a representation of level dividing f_χ.

The proof of the dimension-block bound estimates the matrix coefficient $\langle w_{\bar\chi}, PSPw_\chi\rangle$ by $f_\chi^{-1/2}\|w_\chi\|^2$. Equality in the bound forces equality in this compression-norm estimate and in Cauchy–Schwarz. The preceding argument applies to w_χ. In particular $(T^{f_\chi}-I)Sw_\chi = 0$. The known dimension-character rows give $S_{Z,A_\sigma} = \sigma(d_Z/D)$, so the Z-coordinate of Sw_χ is precisely $L_\chi(v)$. This proves (3.8). The greatest-common-divisor assertion follows by intersecting root-of-unity groups. □

The exact audit `research/round5/unification/dimension_block_bound.py` enumerates every dimension-field character for the six unresolved rings. All satisfy the bound. Their nontrivial equality conductors are 20 for $R_{6,4,22}$, 16 for $R_{6,4,91}$, 15 for $R_{6,4,629}$, 12 for $R_{6,5,1204}$, and $12, 24$ for $R_{6,8,1660}$; $R_{6,4,155}$ has strict inequalities. The receipt records all Fourier values, including trivial characters, in exact cyclotomic coordinates. The reusable implementation `software/dimension_block_support.py` computes (3.8) from the dimensions and each restriction column. When the forced roots lie in the retained field, their complete power tuples filter the projected twist types exactly.

Proposition 3.22 (Universal moments of a saturated real character)**.** *In Proposition 3.21, suppose χ is real and put $f = f_\chi$, $\lambda = \sum_{\sigma\in G_1}\chi(\sigma)/\sigma(D)$. Let $M = \operatorname{Ind}_B^{\mathrm{SL}_2(\mathbb{Z}/f\mathbb{Z})}\chi$, where B is the upper triangular subgroup and its character is trivial on unipotents and equals χ on the diagonal torus. Write S_f, T_f for the standard unitary operators on M, and P_χ for projection onto its unipotent-fixed, torus-χ space. Let the columns of K span the subspace on which $P_\chi S_f P_\chi$ has eigenvalue λ. For every exact matrix identity*

$$\sum_{n=0}^{f-1} c_n K^* S_f^* T_f^n S_f K = 0,$$

the following necessary moment equation holds:

$$\sum_{n=0}^{f-1} c_n \sum_Z |L_\chi(v_Z)|^2 \theta_Z^n = 0. \tag{3.9}$$

Proof. By the preceding proposition, the cyclic representation generated by w_χ factors through this finite group. It embeds equivariantly and isometrically into M. Indeed, in an isotypic summand $\pi\otimes\mathbb{C}^m$, the generating vector lies in $\pi^{B,\chi}\otimes\mathbb{C}^m$; cyclicity forces $m \le \dim \pi^{B,\chi}$. Frobenius reciprocity gives exactly this latter multiplicity in M. Complete reducibility and invariant inner products make the embedding isometric.

Since χ is real, equality in Cauchy–Schwarz in the preceding proof says $P_\chi S w_\chi = \lambda w_\chi$. Thus its embedded vector lies in the column space of K. Applying the displayed matrix identity to that vector gives $\sum_n c_n\langle Sw_\chi, T^n Sw_\chi\rangle = 0$. The simple-coordinate expansion of this inner product is exactly (3.9). □

The signed permutation model on the projective line, its projectors, equality eigenspace and complete matrix-relation kernel are constructed by `research/round5/unification/equality_modrep.py`. This uses the full equality eigenspace, so it does not select an unproved

irreducible constituent or a particular vector within it. There is also a direct spectral support test. If E_r is the orthogonal projector onto the ζ_f^r-eigenspace of T_f, compute

$$K^* S_f^* E_r S_f K = \frac{1}{f} \sum_{n=0}^{f-1} \zeta_f^{-rn} K^* S_f^* T_f^n S_f K.$$

If this matrix is zero, positivity implies $E_r S_f K = 0$. Thus $L_\chi(v_Z) \neq 0$ forbids $\theta_Z = \zeta_f^r$. This exact discrete Fourier calculation can remove phase variables before the moment equations are solved.

Corollary 3.23 (Different saturated conductors are orthogonal)**.** *Let χ, ψ be saturated characters with distinct primitive conductors f_χ, f_ψ. Then for every integer n,*

$$\sum_Z \overline{L_\chi(v_Z)} L_\psi(v_Z) \theta_Z^n = 0. \tag{3.10}$$

Only the exponents $0 \leq n < \gcd(f_\chi, f_\psi)$ are needed.

Proof. Every irreducible constituent in the cyclic span of w_χ contains a nonzero unipotent-fixed vector of torus character χ. Its local congruence level is at most the conductor exponent, by Proposition 3.21, and cannot be smaller: otherwise this torus character would factor through a smaller modulus. The constituent therefore has exact conductor level f_χ. Constituents belonging to distinct conductors cannot be isomorphic. Orthogonal complete reducibility of the unitary modular representation now makes the two cyclic subspaces orthogonal. In particular $\langle S w_\chi, T^n S w_\psi \rangle = 0$, which is the displayed formula. Every nonzero summand has both Fourier coefficients nonzero, so its twist order divides both conductors. The expression is therefore periodic with period their greatest common divisor. □

3.13 Different ideals and a split quadratic dimension field

The remaining dense quadratic profile requires a refinement that also controls possible larger signed dimension fields.

Theorem 3.24 (Different-ideal obstruction)**.** *Let K be the dimension-character field of the commutative signed algebra obtained by sphericalization of a commutative self-dual fusion ring. Its global dimension Δ belongs to the different $\mathfrak{D}_{K/\mathbb{Q}}$.*

Proof. Write d_i for the signed basis dimensions, and let $e_K \in K \otimes K$ be the separability idempotent. Orthogonality of distinct Galois-conjugate characters gives

$$(\Delta \otimes 1) e_K = \sum_i d_i \otimes d_i \in \mathcal{O}_K \otimes \mathcal{O}_K.$$

For a in the codifferent, contract the second factor by $x \mapsto \mathrm{Tr}(ax)$. The right side is integral, and the left side is Δa. Thus $\Delta \mathfrak{D}^{-1} \subseteq \mathcal{O}_K$, which is the assertion. □

For a local extension with ramification index e and different exponent d,

$$\mathrm{Tr}(\mathfrak{p}^k) = p^{\lfloor (k+d)/e \rfloor} \mathbb{Z}_p.$$

In particular, every tamely ramified odd prime divides the trace of the different. A cyclic cubic extension of conductor 9 has $e = 3, d = 4$ and contributes 9. We use these standard local trace formulas and the conductor–discriminant formula; see Serre, *Local Fields*, Chapters III and VI, and Milne, *Class Field Theory*, Chapter V, Theorem 3.27.[1]

[1]The latter reference is available at `https://www.jmilne.org/math/CourseNotes/CFT.pdf`.

Proposition 3.25. *A commutative self-dual rank-six fusion ring with formal codegrees*

$$9, 9, 9, 9, \qquad 3690 \pm 576\sqrt{41},$$

Casimir trace 7416*, and reduction* $\mathbb{F}_2^4 \times \mathbb{F}_2[\epsilon]/(\epsilon^2)$ *satisfies the following necessary condition for categorifiability: its signed Casimir trace is* 7416*. Consequently every signed coefficient has its original absolute value.*

Proof. Put $E = \mathbb{Q}(\sqrt{41})$. The dimension-field containment lemma gives $E \subseteq K$, $\Delta/9 \in \mathcal{O}_K$, and

$$\frac{\Delta}{18\sqrt{41}} \in \mathcal{O}_K, \qquad 738 \mid \mathrm{Tr}_{K/\mathbb{Q}}(\Delta), \qquad \mathrm{Tr}_{K/\mathbb{Q}}(\Delta) \leq 7416.$$

Indeed $3690 + 576\sqrt{41} = 18\sqrt{41}(32 + 5\sqrt{41})$, and the last factor is an integral unit of norm -1. Cyclotomicity makes $K/\mathbb{Q}$ abelian Galois, of degree 2, 4, or 6.

If $[K : \mathbb{Q}]$ is 4 or 6, then 2 splits completely in K. The four etale rank-one blocks of the completed signed order have generic fibre $\mathbb{Q}_2$; its only other block has rank two. A Galois number field using a completion of degree two would require at least two such local components, whereas only one rank-two block is available. The same argument excludes ramification at 2. It allows rational idempotents inside the non-etale block and does not assume that block is maximal.

Theorem 3.24 and the trace bound restrict ramified primes outside $\{3, 41\}$ to 5 and 7, and forbid their simultaneous occurrence, which would require trace at least $35 \cdot 738$. The complete list of possible larger fields follows from elementary Dirichlet-character arithmetic:

1. A biquadratic field $E\mathbb{Q}(\sqrt{m})$ requires $m > 1$ squarefree, positive, prime to 41, supported on $\{3, 5, 7\}$, and $m \equiv 1 \pmod 8$. The only such product is 105, already forbidden by its primes 5 and 7.

2. A cyclic quartic character with square the quadratic character of conductor 41 is a quartic character modulo 41 times a quadratic character of positive odd discriminant m. Since $2^{10} \equiv -1 \pmod{41}$, splitting at 2 requires $m \equiv 5 \pmod 8$. The permitted products leave exactly $m = 5, 21$. The conductors are $205, 861$, and the possible dimension traces are respectively $3690, 7380$ and 5166.

3. A degree-six field is EL with L cyclic cubic. Its conductor is 7, 9, or 63: neither 5 nor 41 admits a cubic character, and a nontrivial cubic character at 3 has conductor 9. The prime 2 is inert in the conductor-7 and conductor-9 cubic fields. Only conductor 63 remains, with dimension trace 5166.

There are no additional prime-power conductors: characters of order dividing 4 or 6 are trivial on the higher prime-power unit groups at primes other than 2 or 3; the relevant quadratic odd characters have squarefree conductor.

In the sextic case, coprimality of the discriminants gives $\mathcal{O}_{EL} = \mathcal{O}_E \otimes \mathcal{O}_L$ and a product of differents. Write

$$\Delta = 18\sqrt{41}\,\frac{a + b\sqrt{41}}{2}, \qquad a, b \in \mathcal{O}_L, \quad a \equiv b \pmod 2.$$

Positivity at both embeddings of E over each real embedding of L makes b totally positive, with $\mathrm{Tr}_{L/\mathbb{Q}}(b) = 7$. At 7, the cubic different is $\mathfrak{p}_7^2$, and 18 is a unit. The different theorem therefore forces $b \in \mathfrak{p}_7^2$. Hence

$$49 \leq \mathrm{Norm}(b) \leq (7/3)^3 < 49,$$

a contradiction.

The cyclic quartic cases require a short exact lattice enumeration. For $M = 5, 21$, set

$$\alpha^2 = \frac{M(41 + 5\sqrt{41})}{2}, \qquad \beta^2 = \frac{M(41 - 5\sqrt{41})}{2}, \qquad \alpha\beta = 2M\sqrt{41}.$$

The defining polynomial is $X^4 - 41MX^2 + 164M^2$, and an integral basis is

$$1, \quad \frac{1 + \sqrt{41}}{2}, \quad \frac{1 + \sqrt{41} + 2\alpha}{4}, \quad \frac{1 - \sqrt{41} + 2\beta}{4}.$$

The checker verifies its integral multiplication, cyclic Galois action, complete splitting at 2, and discriminant 41^3M^2. The preceding field list and the conductor–discriminant formula prove maximality. For $\theta = \Delta/(18\sqrt{41}) = \sum_j z_j b_j$ and trace-dual basis b_j^*,

$$z_j = \mathrm{Tr}\left(\Delta \frac{b_j^*}{18\sqrt{41}}\right).$$

At fixed positive trace T, this lies between T times the smallest and largest conjugates of $b_j^*/(18\sqrt{41})$. Exact algebraic sign comparisons certify a finite integer box for every z_j. Imposing the trace and total positivity leaves $28, 216, 16$ candidates at traces $3690, 7380, 5166$, respectively. Every candidate fails a recorded d-number divisibility. The supplementary certificate contains all 260 candidates and the exact coefficient witnessing each failure.

It remains to consider $K = E$. Write $\Delta = 18(41y + x\sqrt{41})/2$. Then $x \equiv y \pmod 2$, $1 \le y \le 10$, $x^2 < 41y^2$, and

$$41y^2 - x^2 \mid 164y^2.$$

The last condition is the quadratic d-number test. The four complementary codegrees form either four rational integers, two rational integers and a quadratic pair, or two quadratic pairs in E. Rational codegrees divide the integral content of Δ. A quadratic pair has polynomial $X^2 - SX + P$, with $S^2 - 4P = 41r^2$, $P \mid \mathrm{Norm}(\Delta)$, $P \mid S^2$, both roots greater than one, and both dividing Δ integrally. The total trace bounds S and r. The reciprocal identity $\sum_f 1/f = 1$ completes the finite test. Exactly seven dimension candidates reach this stage; none admits a quadratic complementary pair. The only complete spectra have $y = 10$, $x = \pm 64$, and four codegrees 9. Their total trace is 7416, proving the proposition. □

The full tensor of `R6_8_6526` satisfies these hypotheses. Exact saturated signed associativity leaves one table up to the certified sign gauges, equivalent to the original tensor. Every non-FP original character orbit fails the arithmetic dimension-character tests. Thus a categorification would have a pseudo-unitary Galois conjugate, which the existing global-codegree moment certificate excludes. The resulting *general* exclusion is implemented in `software/round3_split_arithmetic.py`, with canonical certificate `research/round3/arithmetic/R6_8_6526_general.json`.

3.14 The dimension order and signed central traces

The positive-dimension induction system has two extensions that are useful when an exhaustive sphericalization search leaves a non-positive spherical character. The first sharpens an arithmetic condition without adding any central variables. The second retains the algebraic part of the same trace system, but discards every inequality that depends on positivity of categorical dimensions.

3.14.1 The actual order of simple dimensions

Let $d = (1, d_1, \ldots, d_{r-1})$ be a dimension character under consideration, and let $K = \mathbb{Q}(d_i)$ and $\delta = \sum_i d_i d_{i^*}$. The finitely generated group

$$\mathcal{O}_d = \sum_i \mathbb{Z} d_i \subset K$$

is an order: it contains 1, consists of algebraic integers, and is closed under multiplication because the d_i satisfy the multiplication equations with integer coefficients. It need not be the maximal order of K.

Proposition 3.26 (Dimension-order obstruction)**.** *If a fusion ring R has a spherical categorification with simple dimensions d_i, then for every formal codegree f of R,*

$$f \in K, \qquad \delta/f \in \mathcal{O}_d. \tag{3.11}$$

The same conclusion holds for the dimensions of selected lifts in the canonical sphericalization of an arbitrary categorification of R. Thus the order-membership test may be applied to each dimension-character branch of a complete signed-sphericalization search.

Proof. In the spherical case the central simple A_E attached to an irreducible representation of R with codegree f has dimension δ/f, by [36, Theorem 2.13]. Its forgetful image is an object of the original category. Its dimension is therefore a nonnegative integral linear combination of the d_i, in particular an element of $\mathcal{O}_d$. Since δ is nonzero, this also puts f in K.

For an arbitrary categorification use its sphericalization, of dimension 2δ. An irreducible representation of R lifts through the forgetful quotient to one with formal codegree $2f$; both selected and opposite basis lifts contribute to that codegree. The same central induction theorem gives dimension $(2\delta)/(2f) = \delta/f$. All simple dimensions of the sphericalization are among $\{d_i, -d_i\}$, so forgetting this central object gives an integral linear combination of the selected dimensions. This proves the assertion without assuming that the selected lifts themselves form a fusion subcategory. □

The test is finite and exact. Express the d_i in a rational power basis of K, clear one common denominator, and put their coordinate lattice in Hermite normal form. Membership of δ/f is then an integer linear-algebra question. A negative certificate can equivalently give a rational linear functional $\ell : K \to \mathbb{Q}$ satisfying

$$\ell(d_i) \in \mathbb{Z} \quad \text{for all } i, \qquad \ell(\delta/f) \notin \mathbb{Z}.$$

The checker verifies the defining polynomial of f, reconstructs the lattice and the functional, and checks both displayed conditions. A failure on one character component is not a noncategorification proof unless all possible sphericalization and character branches are closed. In particular a failed character of a positively realized ring cannot be promoted to an exclusion of that ring.

For illustration, the complete certificate for $R^{(6)}_{4,375}$ has four oriented signed tables, all gauges of the original tensor. Its nonzero spherical components have order $\mathbb{Z}[\sqrt{5}]$ and dimensions

$$\delta = 70 + 30\sqrt{5} \quad \text{or} \quad \delta = 30 - 10\sqrt{5}.$$

The codegree 4 gives quotients outside that order in both cases. The supplied functional certificates, not this identifier or this displayed calculation, are the inputs to the automated verification.

3.14.2 The algebraic trace system without positive dimensions

Suppose a spherical character takes values in a real field K of degree at most two, has nonzero simple dimensions, and satisfies $d_i = d_{i^*}$. Write $\delta = \sum_i d_i^2$. The restriction vectors of central simples remain nonnegative integral vectors, even when some categorical dimensions are negative. Consequently the Gram identity (4.2), its finite entry bounds, and the formal-codegree low-column equalities remain valid. They do not use an inequality between categorical dimensions.

For a commutative input let χ_a be a character label of a low simple Z_a and put $\beta_a = \delta/f_a$. If v_W is a candidate restriction vector, define

$$d_W = d \cdot v_W, \qquad H_{aW} = \beta_a \chi_a(v_W), \qquad \Lambda_W = (H_{aW}/d_W)_a.$$

Assume that the full ring character table is defined over K. The character correspondence of Theorem 4.9 gives $H_{aW} = S_{Z_a,W}$, with S unnormalized. The following formulation uses no absolute value of d_W.

Theorem 3.27 (Signed central-induction obstruction)**.** *A spherical categorification with the specified character supplies a nonnegative integral solution of the following finite necessary system. The unknowns count high central restriction vectors and their joint relative twist-trace types. They satisfy the Gram equations, the signed identities*

$$\sum_W (v_W)_i d_W \theta_W^j = \delta \nu_j(b_i), \tag{3.12}$$

and the cyclic-operator bounds on the right-hand indicators. Low columns have twist one and dimension β_a. They admit at least one complete character labelling satisfying modular reciprocity.

All coordinates of Λ_W are algebraic integers. Their joint Galois types have balanced multiplicities, separately for each relative twist-trace type. An absent Galois type forbids the corresponding restriction vector. Finally, if P is the normalized trace from a common cyclotomic field to K, then

$$\sum_W P(\theta_W) H_{aW} H_{bW} = \delta S_{Z_a,Z_b} \tag{3.13}$$

for every pair of low labels.

Exhaustive failure of the system for all bounded low columns, all compatible character labellings, and all permitted second indicators excludes the specified spherical character. Exact rational Farkas certificates suffice for failure of the nonnegative rational relaxation.

Proof. Adjunction proves the Gram identity with nonnegative restriction multiplicities. Central induction supplies the low dimensions and their unit multiplicities exactly as in (4.3); neither statement assumes pseudo-unitarity. The trace formula [30] proves (3.12) with the actual, possibly negative, dimensions. Bounds obtained as traces of finite-order cyclic operators on ordinary complex vector spaces do not assume positivity of categorical dimensions. Only these algebraic bounds, and not the positive low-mass inequalities, are retained here.

The finite relative-trace list is justified by the exponent-prime support theorem [5] and the same relative-trace calculation used in (4.7), with δ in place of D. The high columns are bounded by the positive integral Gram matrix, not by the sign of d_W. Their additional algebraic-integrality and d-number conditions follow from the spherical modular center as in the proof of Theorem 4.3. Columns of dimension zero are impossible.

Each Λ_W is the restriction of the simultaneous character of integer fusion matrices of the center to the labelled low objects. It therefore consists of algebraic integers, and its full Galois orbit occurs with the corresponding multiplicities. For the normalized center representation

$S/\delta, T$, the permutation of columns sends θ_W to $\sigma^2(\theta_W)$; see [5, Theorem 2.9]. A square fixes the real field K of degree at most two, so normalized relative traces of every twist power are unchanged. This proves the asserted balances of joint eigenvalue types. It does not identify d_W with its absolute value, nor impose the positive dimension-orbit rule (4.25).

The center has Gauss sum δ. Its modular relation, together with the twist-one property of low objects, is $\sum_W \theta_W S_{Z_a,W} S_{W,Z_b} = \delta S_{Z_a,Z_b}$. All coefficients other than the twists lie in K. Taking P proves (3.13). The actual central simples give integer counts satisfying all the displayed equations. Expanding them in a rational basis of K and relaxing to nonnegative rational counts can only enlarge the solution set. Farkas' lemma therefore proves the last assertion. □

Exact scope of the implementation. The routine first checks the complete signed character table, its multiplication equations, duality, nonzero dimensions and field degree. It enumerates low columns using Gram bounds and exact dimension equalities, without calling a positive-dimension low-mass test. It then uses signed versions of the inherited indicator rows and adds joint spectrum and low modular rows. In particular it does not reuse the absolute-value dimension balances of Proposition 4.10. Only the second-indicator rows depend on the chosen signs, and each such row depends on one sign. The checker reconstructs the all-minus and all-plus endpoint systems and selects the exact normalized row for each sign choice. Independent mixed-sign reconstructions check this optimization; every separating-vector inequality is still verified. The second indicators of self-dual objects are both allowed unless an independently checked condition forces their sign.

For $R^{(6)}_{4,446}$ the remaining non-positive component is

$$d = (1, 1, -1, -1, 2 - \sqrt{6}, 2 + \sqrt{6}), \qquad \delta = 24.$$

There are 290 low matrices. Reciprocity leaves two, each with two character labellings; all 32 choices of second indicators are retained for each labelling. The 128 exact separation certificates exclude these systems. The positive character component has the preceding character-induction obstruction. The full oriented search has sixteen gauge tables, so the assembled certificate excludes general categorification, not just this displayed spherical structure.

3.14.3 Application and structure-specific conclusions

Component-level rejection and general nonexistence are different conclusions. For example, rejecting one character of $R^{(6)}_{4,174}$ does not exclude that ring. Complete proof trees for the order and signed framework include

$$R^{(6)}_{4,375}, \quad R^{(6)}_{4,446}, \quad R^{(6)}_{5,1507}, \quad R^{(6)}_{5,1508}, \quad R^{(6)}_{5,1582}, \quad R^{(6)}_{6,2767}, \quad R^{(6)}_{7,1907}.$$

For the last five, shorter norm and moment certificates close the exhaustive signed alternatives, and these are retained rather than credited to a stronger criterion unnecessarily. The finite applications are specified by full tensors and exact proof trees.

The shorter pseudo-unitary certificates for $R^{(6)}_{7,3582}$ and $R^{(6)}_{7,3812}$ are intermediate ingredients. Their unrestricted existence branches are closed by the ramified signed reductions and Casimir trace criteria above; both rings are generally excluded.

3.15 Sphericality and Galois reduction at multiplicity three

Proposition 3.28. *Every complex fusion category with based ring $R^{(6)}_{3,215}$ admits a spherical structure and has a pseudo-unitary Galois conjugate. Consequently general categorifiability of this ring is equivalent to pseudo-unitary categorifiability. This reduction alone asserts neither existence nor nonexistence; the final obstruction is proved in Theorem 5.25.*

Proof. Apply the exhaustive oriented signed-sphericalization construction of Section 3.4, using every coefficient in its full parity interval and imposing no positive-indicator or modulo-four normalization. There are 1480 associative signed quotients. The 32 unit-preserving sign gauges partition them into 63 orbits; one orbit consists of the 32 gauges of the original fusion tensor. On representatives of the other 62 orbits there are 271 rational character components. Of these, 228 have a zero simple dimension and are impossible. For each of the other 43, the norm of the prospective global dimension has 17-adic valuation zero. The original ring has a formal codegree with minimal polynomial $t^2 - 272t + 1088$, whose norm has 17-adic valuation one. Thus the signed-sphericalization norm-divisibility condition rejects every remaining component. No nongauge quotient can occur for an actual category.

For completeness, a gauge quotient gives sphericality of the original category, not merely compatibility with spherical dimensions. If $T_{ij}^k = \eta_i\eta_j\eta_k N_{ij}^k$, replace the chosen lift of X_i in the sphericalization by its tensor product with the sign object when $\eta_i = -1$. The new signed quotient has $T = N$. The sum and difference of the multiplicities of the two lifts of X_k in a tensor product are then both N_{ij}^k, so the opposite lift has multiplicity zero. These chosen lifts therefore generate a full tensor subcategory of the sphericalization which projects tensor-equivalently onto the original category. Transport its spherical structure along that equivalence.

The four rational characters of the original ring have a zero entry, so none is a spherical dimension character. The only remaining character orbit is

$$(1, 1, d, d, d-1, d+1), \qquad d^2 - 8d - 1 = 0.$$

One of its two real embeddings is the Frobenius–Perron character, $d = 4 + \sqrt{17}$. Galois conjugation to that embedding gives a spherical category with positive Frobenius–Perron dimensions, hence a pseudo-unitary category. Conversely a pseudo-unitary categorification is a complex fusion categorification.

The complete unrestricted signed enumeration has 274655 search nodes and is frozen in `research/new_frontier/frontier_spherical_reduction.json`. The companion Python script recomputes all 1480 quotients and the complete sign orbits, verifies the original gauge orbit, checks all 271 nongauge character components and their arithmetic obstructions, and reconstructs the original dimension-character orbit using exact arithmetic. The older normalized twelve-table enumeration is retained separately as a regression control; it is not needed for this proof.

□

4 One central-induction certificate framework

The following subsections are blocks of one restriction-and-trace system. The Gram data and representation multiplicities are fixed first. Character labels, reciprocity, Galois orbits, indicators, and modular identities then add equations or remove impossible columns from that same system. Each block states its own scope: Frobenius–Perron positivity is used only in the pseudo-unitary blocks, whereas the signed block retains negative spherical dimensions. A Farkas vector certifies infeasibility only after all low frames, character assignments, and permitted indicator choices have been exhausted. No particular block is mandatory when a shorter inherited certificate already proves the required exclusion.

4.1 Central induction with representation multiplicities

Assume in this section that $\mathcal{C}$ is pseudo-unitary, with its positive spherical structure. Let $F : Z(\mathcal{C}) \to \mathcal{C}$ be forgetful and I its adjoint. Put

$$B_{ji} = [FI(b_j) : b_i] = \sum_{a,k} N_{aj}^{k} N_{k,a^*}^{i}. \tag{4.1}$$

This is the induction Gram matrix. In a commutative ring it agrees with the regular Casimir. In a noncommutative ring one must use (4.1), not silently substitute the Casimir.

For a central simple Z, write $v_Z = ([F(Z) : b_i])_i$. Adjunction gives

$$B = \sum_{Z \in \mathrm{Irr}\, Z(\mathcal{C})} v_Z v_Z^{\mathsf{T}}. \tag{4.2}$$

The summands of $I(\mathbf{1})$ are indexed by irreducible representations E of $R \otimes \mathbb{C}$:

$$I(\mathbf{1}) = \bigoplus_E r_E A_E, \quad (v_{A_E})_0 = r_E, \quad \dim A_E = D/f_E, \quad \theta_{A_E} = 1. \tag{4.3}$$

These are the formal-codegree and induction statements of [36, Theorems 2.5,2.13].

Theorem 4.1 (Multiplicity-weighted low induction)**.** *For each E let $v_E \in \mathbb{Z}^6_{\geq 0}$ satisfy*

$$(v_E)_0 = r_E, \qquad d \cdot v_E = D/f_E, \qquad \sum_E r_E v_E = B_{0,*}. \tag{4.4}$$

A necessary condition for a pseudo-unitary categorification is that at least one such collection has a nonnegative integral Gram completion

$$Q = B - LL^{\mathsf{T}} = \sum_{Z \not\subset I(\mathbf{1})} v_Z v_Z^{\mathsf{T}}, \qquad L = (v_E)_E, \tag{4.5}$$

and satisfies the central-indicator necessary conditions of [12, 11]. In particular, Q is entrywise nonnegative and positive semidefinite. The columns of L occur once each in LL^{T}; they are not repeated r_E times.

For a noncommutative rank-six ring the low unit multiplicities are $1, 1, 2$. Equations (4.4) *and* (4.5)*, with these multiplicities, give a finite exhaustive low-column search.*

Proof. The first two identities of (4.4) are (4.3). Restricting $I(\mathbf{1})$ proves the third. Separating the distinct simples occurring in $I(\mathbf{1})$ from the sum (4.2) proves (4.5). This also explains why r_E weights the row-sum equation but not the number of copies of a column in L.

Finiteness follows from $(v_E)_i^2 \leq B_{ii}$ and $r_E(v_E)_i \leq B_{0i}$. Thus each entry belongs to an explicitly bounded integer interval. All dimension equalities are checked in the exact number field. The usual trace identities involve only restriction multiplicities and categorical dimensions, so their proofs and the finite relative-trace construction do not require commutativity of R once the correct low data and Gram matrix are supplied. Finally a noncommutative semisimple rank-six algebra with a dimension character has type (3.5), giving the stated multiplicities. □

In the implementation the one-dimensional character subspace is computed before selecting a Frobenius–Perron embedding. The unique matrix-block eigenvalue c of the Casimir gives $f_E = c/2$, and the identity

$$\det(tI - C_R) = (t - c)^4 \det(tI - C_R|_{\mathrm{linear}})$$

is checked. A positive gate from Proposition 3.1(iv) is required before a pseudo-unitary exclusion is promoted to a general one. The five higher-multiplicity noncommutative Haagerup-type tensors in the census are excluded in this way. The multiplicity-one Haagerup ring and the group ring of S_3 are retained positive controls.

4.2 Balanced finite-indicator systems

The principal positive-dimension test is a single relative-trace construction. We strengthen it without introducing a new case-specific obstruction. Throughout this section suppose all dimensions lie in $K = \mathbb{Q}$ or $K = \mathbb{Q}(\sqrt{q})$, with $q > 1$ squarefree in the latter case. The low matrix L is one of the possibilities of Theorem 4.1.

4.2.1 The inherited finite system

For a simple b_i, the Ng–Schauenburg identity [30] is

$$\sum_Z (v_Z)_i \dim(Z)\, \theta_Z^k = D\nu_k(b_i). \tag{4.6}$$

The first indicator vanishes for nonunit simples, and the second is zero for non-self-dual simples and ± 1 for self-dual simples. Higher indicators satisfy the bounds used in the finite-indicator theorem of [11]. We use exactly that theorem, including its permitted indicator choices and rational inequalities, rather than assuming unknown indicators have prescribed values.

For a chosen height h, take joint normalized relative traces of $(\theta, \dots, \theta^h)$ to K. The exponent prime support is controlled by the categorical Cauchy theorem [5]. Let P be the prime divisors of the norm of D, and let f be the conductor of K (take $f = 1$ for $K = \mathbb{Q}$). The trace vectors are covered by roots whose orders divide

$$M_h = \prod_{p \in P} p^{\max\{1, v_p(f)\} + \lfloor \log_p h \rfloor}, \tag{4.7}$$

together with the zero vector. As proved in [11], a higher prime-power layer contributes zero to every one of the first h traces. Thus this is not a bound imposed on the actual twist orders. Ramanujan and quadratic Gauss sums compute the finite trace vectors exactly. Extra vectors may be retained as a relaxation; they cannot produce a false exclusion.

An admissible high column has zero unit entry, a nonnegative integral tail v, and $Q - vv^{\mathsf{T}} \succeq 0$. Entrywise Gram bounds and the algebraic-integrality condition $D/\dim(v) \in \mathcal{O}_K$ further restrict it. We also require $\dim(v)$ to be a d-number: the center is braided, so its simple squared norms are d-numbers by [35, Theorem 1.8]; in the positive spherical structure these are the squares of the dimensions, and [35, Corollary 2.9] removes the square. In a quadratic field this is the exact condition that the square of the trace divided by the norm is integral. Denote the finite list by $\mathcal{V}$, and the finite joint trace list by $\mathcal{T}$. Variables $x_{v,\tau} \geq 0$ give the number of high simples with column v and trace type τ; the linear relaxation allows nonintegral values. Gram identities, (4.6), and the cited higher-indicator inequalities give an integer linear system

$$Ax = b, \qquad x \geq 0, \tag{4.8}$$

after adding slack variables and clearing denominators. The code reconstructs this system from the tensor, the low matrix, and the indicator choices.

4.2.2 Duality and Galois balance

Write $v_i^* = v_{i^*}$. Low simples are stable under duality, so the high part is stable as well. Duality preserves the ribbon twist: $\theta_{Z^*} = \theta_Z$, not θ_Z^{-1}. Consequently the actual integer counts satisfy

$$x_{v,\tau} = x_{v^*,\tau} \qquad (v \in \mathcal{V},\ \tau \in \mathcal{T}). \tag{4.9}$$

If a partner column is absent from the list, its variable is zero.

For the quadratic case let $\sigma(\sqrt{q}) = -\sqrt{q}$ and define, for a positive central dimension $\beta \in K$,

$$\gamma(\beta) = \left| \frac{D}{\sigma(D)} \sigma(\beta) \right|, \tag{4.10}$$

where absolute value is at the fixed real embedding. Let c_β be the number of *distinct* low central simples of dimension β. The low multiplicity r_E in $I(\mathbf{1})$ is not included in this count.

Lemma 4.2 (Dimension-count balance)**.** *Galois symmetry permutes the central simples with dimension β onto those with dimension $\gamma(\beta)$. Hence their high counts satisfy*

$$\sum_{\dim(v)=\beta,\tau} x_{v,\tau} - \sum_{\dim(v)=\gamma(\beta),\tau} x_{v,\tau} = c_{\gamma(\beta)} - c_\beta. \tag{4.11}$$

Moreover $\gamma^2(\beta) = \beta$.

Proof. Extend σ to a cyclotomic field of the center. Its normalized S-matrix is $s = S/D$, since the center has global dimension D^2. Signed Galois symmetry gives

$$\sigma(s_{0Z}) = \epsilon_Z s_{0,\widehat{\sigma}(Z)}, \qquad \epsilon_Z \in \{1, -1\}.$$

The entry s_{0Z} equals $\dim(Z)/D$. All central dimensions are positive. Taking absolute values therefore gives $\dim \widehat{\sigma}(Z) = \gamma(\dim Z)$, proving the permutation statement. Subtract the known low counts to obtain (4.11). Finally the map $u \mapsto (D/\sigma D)\sigma(u)$ squares to the identity. Taking absolute values commutes with changing a real sign, so the positive version squares to the identity as well. See also the squared-dimension form of Galois symmetry in [28, Equation (2.4)]. □

The factor $D/\sigma(D)$ and the absolute value in (4.10) are both essential. Omitting either does not give a valid positive dimension permutation in general.

Theorem 4.3 (Balanced finite-indicator criterion)**.** *Append* (4.9) *and, when quadratic,* (4.11) *to the finite system of [11]. If for every admissible low matrix and every allowed second-indicator choice the resulting system has an exact Farkas certificate*

$$A^{\mathsf{T}} y \geq 0, \qquad b^{\mathsf{T}} y < 0, \tag{4.12}$$

then there is no pseudo-unitary categorification. Coupled with a valid general-to-positive reduction, this excludes general complex categorification.

At a fixed height and with the same admissible low and high columns, this criterion contains the former unbalanced finite-indicator test: every old separating functional extends by zero on the appended rows.

Proof. An actual category supplies integral nonnegative counts satisfying the inherited system and both new balances, by duality and Lemma 4.2. But (4.12) would imply $0 > b^{\mathsf{T}} y = x^{\mathsf{T}} A^{\mathsf{T}} y \geq 0$. The last assertion follows by appending zero coordinates to y. □

This containment justifies using the balanced theorem as the common finite-indicator framework. It does not imply that every earlier non-indicator criterion is redundant. In particular, the cheap arithmetic filters and signed reductions remain useful.

4.2.3 An integral refinement and its scope

For a high-column list $v_1, \dots, v_s$, enumerate nonnegative integers m_j with

$$Q = \sum_{j=1}^{s} m_j v_j v_j^{\mathsf{T}}. \tag{4.13}$$

This is finite because each nonzero v_j has a positive diagonal entry. For each frame append $\sum_\tau x_{v_j,\tau} = m_j$ to the balanced system. Exhaustive infeasibility for every frame would be a further valid obstruction. We implement this refinement principally to diagnose the frontier. The feasible point in Section 7.2 survives it. We do not count this unproductive refinement as an independent successful exclusion criterion.

4.3 Cyclic-character compatibility for tensor powers

The preceding indicator system bounds different higher indicators separately. There is additional compatibility: several of them are traces of powers of a single finite-order linear operator. This gives extra linear equations inside the same exact-certificate framework.

Lemma 4.4 (Block rotation)**.** *In a pivotal fusion category, let $E_X^{(n)}$ be cyclic rotation on* $\mathrm{Hom}(1, X^{\otimes n})$*. For every positive divisor r of n,*

$$(E_X^{(n)})^n = \mathrm{id}, \qquad \mathrm{Tr}((E_X^{(n)})^r) = \nu_{n/r}(X^{\otimes r}) = \sum_i [X^{\otimes r} : X_i]\nu_{n/r}(X_i). \tag{4.14}$$

Here the trace is the ordinary trace on the displayed Hom space.

Proof. Pivotal strictification preserves the rotation operators. In the strict pivotal setting, equation (5.4) in [31] states that consecutive rotations of adjacent tensor blocks equal the rotation of their tensor product. Induction gives $(E_X^{(n)})^r = E_{X^{\otimes r}, X^{\otimes(n-r)}}$, exactly the identity used in the proof of [31, Theorem 5.1]. If $r \mid n$, group the factors into n/r consecutive blocks of length r. This identifies the operator with one rotation on the blocks. Taking traces gives the first indicator equality; the final equality is additivity of higher indicators on a direct sum. For $r = n$ it is the identity operator, with trace $[X^{\otimes n} : 1]$. □

Fix the dimension field K, the low columns, and the second indicators as in Section 4.2. For a positive integer n, let $\mathcal{U}_{K,n}$ be the finite set of vectors

$$(P_K(\zeta^r))_{r|n,\ r<n}, \qquad \zeta^n = 1,$$

where P_K is normalized relative trace from a common cyclotomic extension to K. This definition is independent of enlarging the common extension. The vectors are computed by the same Ramanujan and quadratic Gauss sums used for the center twists, but their orders now divide n exactly. In particular no unbounded-order approximation occurs in this auxiliary set.

Theorem 4.5 (Joint tensor-rotation criterion)**.** *In a pseudo-unitary categorification with simple dimensions in K, for each simple X and each $3 \le n \le h$ there are nonnegative integers $y_{X,n,u}$, $u \in \mathcal{U}_{K,n}$, such that*

$$\sum_u y_{X,n,u} = [X^{\otimes n} : 1], \tag{4.15}$$

$$\sum_u y_{X,n,u} u_r = \sum_i [X^{\otimes r} : X_i] P_K(\nu_{n/r}(X_i)) \qquad (r \mid n,\ r < n). \tag{4.16}$$

Together with the center trace identities, these are linear equations in the existing high-center counts and the new y variables after multiplication by D and expansion in a rational basis

of K. Infeasibility of the nonnegative real relaxation is an obstruction to pseudo-unitary categorification. Combined with any proved sphericalization or pseudo-unitarity reduction, it is a general categorification obstruction.

Proof. The operator in Lemma 4.4 is diagonalizable over $\mathbb{C}$ and has nth roots of unity as eigenvalues. Group its eigenvalues, with multiplicity, by their joint relative-trace vectors. This gives nonnegative integer counts satisfying (4.15) and (4.16). Use (4.6) to express each indicator on the right. Its denominator is the known scalar D, while its numerator is the known low contribution plus a linear combination of the high-center variables. Since P_K is K-linear, multiplying by D gives linear equations with coefficients in K. Clearing rational denominators gives an integer system. An integer solution would also be a nonnegative real solution; an exact separating functional therefore proves the claimed obstruction. □

For clarity, let ℓ_i be the dimension of the known low part of $I(X_i)$, with $\ell_0 = D$, and let (v, τ) index a high variable. The equation appended for a divisor r is exactly

$$D\sum_u y_{X,n,u}u_r - \sum_{v,\tau} x_{v,\tau}\Big(\sum_{i>0}[X^{\otimes r} : X_i]v_i\Big)\dim(v)\tau_{n/r} = \sum_i [X^{\otimes r} : X_i]\ell_i. \tag{4.17}$$

This formula specifies the implemented rows without a graphical normalization ambiguity. The unit indicator is one, and unit-free high columns have unit entry zero.

The criterion strengthens the existing system by adding equations and variables with nonnegative constraints; no previously valid inequality is dropped. Old dual certificates extend by zero. Its height-six application excludes $R^{(6)}_{2,55}$, the self-dual Klein-four quadratic ring with two noninvertibles of dimension $2+\sqrt{6}$. The complete ten low-column alternatives and every permitted second-indicator choice are checked. The two-Frobenius–Perron-dimension reduction supplies the general scope. No separate theorem declaring this particular ring impossible is used by the program.

The cyclic-four ring $R^{(6)}_{2,4}$ has the same dimension type but does not satisfy that exclusion certificate. Section 7.2 records an exact integer-count feasible point for the stronger system. Thus neither dimension-type identification nor a failed numerical optimization can replace full-tensor testing.

4.4 Consequences of a proved twist exponent

Suppose the actual twists of $Z(\mathcal{C})$ have order dividing a proved integer N. In this subsection $\mathcal{C}$ is pseudo-unitary, $D = \mathrm{FPdim}(\mathcal{C})$, and all indicators are actual indicators, before relative trace projection. The center formula makes $\nu_k(X)$ periodic in k modulo N. It also gives

$$|\nu_k(X)| \le D^{-1}\sum_Z [F(Z) : X]d_Z = d_X. \tag{4.18}$$

The dimension identity on the right is $\mathrm{FPdim}\, I(X) = Dd_X$.

Proposition 4.6 (Full rotation spectra and prime recovery)**.** *Let $h_n(X) = [X^{\otimes n} : \mathbf{1}]$.*

1. *If $n \mid N$, the entire character of rotation on $\mathrm{Hom}(\mathbf{1}, X^{\otimes n})$ is determined by the actual center variables. Its value at the identity is $h_n(X)$. For $0 < r < n$, put $d = \gcd(n, r)$ and $m = n/d$. Then*

$$t_r = \mathrm{Tr}((E_X^{(n)})^r) = \sigma_{r/d}\Big(\nu_m(X^{\otimes d})\Big), \qquad c_e = \frac{1}{n}\sum_{r=0}^{n-1} t_r\zeta_n^{-er} \in \mathbb{Z}_{\ge 0} \quad (0 \le e < n). \tag{4.19}$$

Here $\sigma_{r/d}$ acts on $\mathbb{Q}(\zeta_m)$; any unit lift modulo N gives the same answer.

2. If $p \nmid N$ is prime, then

$$\nu_p(X) \in \mathbb{Z}, \qquad \nu_p(X) \equiv h_p(X) \pmod{p}, \qquad |\nu_p(X)| \le d_X. \tag{4.20}$$

Consequently, for an integer bound $B_X \ge d_X$ and a prime $p > 2B_X$, the residue $h_p(X) \bmod p$ allows at most one value of $\nu_p(X)$ in $[-B_X, B_X]$. If no such value exists, categorification is impossible. Values obtained from primes in the same residue class modulo N must agree.

Proof. Lemma 4.4 identifies $(E_X^{(n)})^d$ with rotation on m blocks of length d. This operator has mth roots as eigenvalues. Raising it to the power r/d, which is coprime to m, applies the cyclotomic automorphism $\sigma_{r/d}$ to its trace. This proves the first formula. The reduction map from units modulo N to units modulo $m \mid N$ is surjective, and its kernel fixes $\mathbb{Q}(\zeta_m)$, so the choice of lift is immaterial. Fourier inversion gives the actual eigenvalue multiplicities c_e, proving their integral positivity.

For a prime $p \nmid N$, the indicator is both a rotation trace in $\mathbb{Q}(\zeta_p)$ and a center trace in $\mathbb{Q}(\zeta_N)$. The center field statement follows from its canonical congruence representation [47, Theorem II]. These fields intersect in $\mathbb{Q}$, so it is a rational algebraic integer. A rational character of C_p has equal multiplicity b at each primitive pth root. If the multiplicity at one is a, then $h_p = a + (p-1)b$ and $\nu_p = a - b = h_p - pb$. This proves the congruence; (4.18) proves the bound. An interval of length $2B_X < p$ contains at most one integer in a specified residue class. Finally, periodicity modulo N proves the compatibility between primes. All $h_p \bmod p$ are obtained from based-ring matrix powers modulo p; a maximum row sum of a fusion matrix supplies an integer Perron–Frobenius bound B_X. No indicators are guessed. □

The same argument shows why an averaged primitive-root bucket cannot always be treated as an arbitrary integer count: for $p \nmid N$ its total multiplicity is divisible by $p-1$. Nonnegative real relaxations lose this divisibility, but exact modular row certificates retain it.

Proposition 4.7 (Coprime higher Gauss sums of the center)**.** *Write $\tau_n = \sum_Z d_Z^2 \theta_Z^n$. For every n coprime to N,*

$$\tau_n = \frac{D^2}{\sigma_{n^{-1}}(D)}, \tag{4.21}$$

where the inverse is modulo N.

Proof. Use the canonical anomaly-one representation of the center, so that $\tau_1 = D$, $S_{0Z} = d_Z/D$, and $T_{ZZ} = \theta_Z$. The signed Galois permutation π_n satisfies

$$\sigma_n(d_Z)^2 = \frac{\sigma_n(D)^2}{D^2} d^2_{\pi_n(Z)}, \qquad \theta_{\pi_n(Z)} = \theta_Z^{n^2}$$

by the Galois formulas of [47, Theorem II]. Hence

$$\sigma_n(\tau_n) = \frac{\sigma_n(D)^2}{D^2} \sum_Z d^2_{\pi_n(Z)} \theta_{\pi_n(Z)} = \frac{\sigma_n(D)^2}{D}.$$

Applying $\sigma_{n^{-1}}$ gives (4.21). These are exact linear equations in the actual restriction-and-twist counts. They do not require an independent choice of higher central charges. □

4.5 Character-aware central induction

The dimension-only induction matrix forgets an important part of the formal-codegree correspondence: each of its low columns has a specified *character*, not only a specified dimension. Retaining these characters adds exact modular reciprocity and a joint Galois-spectrum condition to the same finite-indicator system. No classification of modular data, numerical eigenvalue decision, or enumeration of center fusion rings is required.

4.5.1 The character attached to a low column

Let $\mathcal{C}$ be pseudo-unitary, let $R = K_0(\mathcal{C})$ be commutative of rank n, and write $d_i = \mathrm{FPdim}(X_i)$ and $D = \sum_i d_i^2$. For a character $\chi : R \to \mathbb{C}$, put

$$f_\chi = \sum_i \chi(X_i)\chi(X_i^*), \qquad \beta_\chi = D/f_\chi.$$

Use the unnormalized modular matrix S of $Z(\mathcal{C})$, so that $S_{1W} = \dim W$. The correspondence preceding [36, Theorem 2.13] identifies the simple summand Z_χ of $I(1)$ by the equality of characters

$$\chi(FW) = \frac{S_{Z_\chi, W}}{\beta_\chi} \qquad (W \in \mathrm{Irr}\, Z(\mathcal{C})). \tag{4.22}$$

Indeed, pull back χ along F. The resulting character of the modular fusion ring is its unique normalized S-row, and [36, Theorem 2.13] identifies the row label, its dimension β_χ, and its multiplicity one in $I(1)$. Thus this statement uses the character-level correspondence, not an arbitrary choice among central simples of equal dimension.

For an integral restriction vector $v = (v_i)$ write $d(v) = \sum_i v_i d_i$ and $\chi(v) = \sum_i v_i \chi(X_i)$. If a_χ is the restriction vector of Z_χ, symmetry of S gives

$$\boxed{\beta_\chi \chi(a_\psi) = \beta_\psi \psi(a_\chi)}. \tag{4.23}$$

The unit column is labelled by the Frobenius–Perron character. When formal codegrees repeat, every dimension-compatible assignment of the remaining distinct characters must be considered. Selecting one assignment without exhausting the others would not be an obstruction.

4.5.2 Joint Galois closure

The next elementary spectral observation is useful well beyond the present induction problem.

Lemma 4.8 (Joint spectral closure)**.** *Let B be a commutative fusion ring and $b_1, \dots, b_s \in B$. The multiset*

$$\{(\varphi(b_1), \dots, \varphi(b_s)) : \varphi \in \mathrm{Irr}(B \otimes \overline{\mathbb{Q}})\}$$

is invariant under every automorphism of $\overline{\mathbb{Q}}$ over $\mathbb{Q}$. Consequently a finite set containing all these tuples must contain every Galois conjugate of each tuple that actually occurs.

Proof. The multiplication equations of B have integral coefficients. Composing a character with a field automorphism gives another character, and the inverse automorphism makes this operation a permutation. Equivalently, the commuting integer multiplication matrices have a Galois-invariant joint spectrum. Restriction to the specified elements preserves the multiset, including repeated restricted tuples. □

For a nonzero prospective restriction vector v define

$$\Lambda(v) = \left(\frac{\beta_\chi \chi(v)}{d(v)} \right)_{\chi \in \mathrm{Irr}\, R}. \tag{4.24}$$

If $v = FW$ for a central simple W, then (4.22) and symmetry give

$$\frac{\beta_\chi \chi(v)}{d(v)} = \frac{S_{W, Z_\chi}}{\dim W}.$$

These are the joint eigenvalues of the integer fusion matrices of Z_χ evaluated at the character labelled by W. In particular, Lemma 4.8 applies to the *whole tuple*. Checking its coordinates separately would be weaker.

Fix a candidate low matrix L. Let $\mathcal{H}_L$ be any finite set containing all possible high restriction columns for that low matrix; the Gram and algebraic-integrality construction in Section 3.1 and the finite-indicator implementation supplies such a set. Let $\mathcal{A}_L$ be its union with the low columns, and set

$$\mathcal{T}_L = \{\Lambda(v) : v \in \mathcal{A}_L\}.$$

Choose a finite Galois field K containing all character values and all d_i. Every coordinate of every tuple belongs to K. A column v is *forbidden* if $\sigma(\Lambda(v)) \notin \mathcal{T}_L$ for some $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. A forbidden low column rejects L; a forbidden high column must have multiplicity zero. Notice that (4.24) uses the fixed character order and D/f_χ, so this test is independent of the permutation used to label equal-dimensional low columns.

Theorem 4.9 (Character-aware finite-indicator obstruction)**.** *For every low matrix of a pseudo-unitary categorification with commutative Grothendieck ring:*

(i) there is a dimension-compatible bijection from its columns to $\mathrm{Irr}\, R$ *satisfying* (4.23)*;*

(ii) none of its low columns is forbidden by the joint Galois test;

(iii) the finite-indicator system remains feasible after all variables supported on forbidden high columns are set to zero.

Therefore the absence of an assignment in (i), failure of (ii), or an exact Farkas obstruction to (iii), for every candidate low matrix and every permitted second-indicator choice, excludes a pseudo-unitary categorification.

Proof. The formal-codegree correspondence proves (i). Every actual central restriction belongs to $\mathcal{A}_L$. Lemma 4.8, applied to the low central simples, proves (ii) and the asserted zero multiplicities. The finite-indicator variables group actual central simples by restriction and joint relative twist trace. Every variable with a forbidden restriction is therefore zero; the remaining actual multiplicities satisfy the inherited equations and inequalities. After slack variables are inserted these have the form $Ax = b$, $x \geq 0$. A rational vector y with $A^{\mathsf{T}}y \geq 0$ and $b^{\mathsf{T}}y < 0$ is impossible, because $b^{\mathsf{T}}y = x^{\mathsf{T}}A^{\mathsf{T}}y \geq 0$. All ordinary low-matrix rejections remain applicable before these additional tests. □

This is a strengthening of the existing finite-indicator framework, not a competing list of special forbidden rings. The implementation uses exact splitting fields of degree at most sixteen generated by quadratic eigenvalues of one or two ring generators. This is an implementation eligibility bound, not a hypothesis of the theorem. Every proposed character is checked against all multiplication equations; the complete set has cardinal n. Field automorphisms are constructed by splitting the primitive-element polynomial over K and checking that all of its distinct roots have been obtained. The verifier reconstructs all labellings, all possible high columns, the forbidden set, and every rational separating vector. It never calls the numerical optimizer used to discover a vector.

4.5.3 General categorification and the multiplicity-two completion

Two illustrative tensors give a compact reproducibility summary:

Ring	Low matrices	Reciprocity survivors	General reduction
$R^{(6)}_{2,20}$	77	2	two FP dimensions
$R^{(6)}_{2,41}$	84	2	two exhaustive signed tables

For $R^{(6)}_{2,20}$, both surviving matrices and both permitted second indicators have exact height-six obstructions. The two-FP-dimension Galois reduction of Proposition 3.1 makes this a general

exclusion. For $R^{(6)}_{2,41}$, one reciprocity survivor is already rejected by the inherited height-six equations. For the other, there are 104 possible high columns; joint Galois closure forbids 66, and the restricted height-six system has an exact separating vector.

For this latter tensor the complete oriented signed enumeration has exactly two tables. Both are sign gauges of the original tensor, with sign vectors $(1,-1,-1,1,1,1)$ and $(1,1,1,1,1,1)$ in census order. The ordinary spherical-character test excludes every character outside the FP Galois orbit by a zero dimension or unequal dimensions on a dual pair. That orbit is excluded by Theorem 4.9. The signed-sphericalization reduction already established in the preceding sections therefore excludes *general* categorification, without assuming the conjectural existence of pivotal structures. These statements are recomputed from the tensors in the certificate, including the completeness of the oriented enumeration.

Table 2 records 23 certified examples of this common character-aware mechanism. Their stored proof trees include the complete sphericalization or Galois reduction required for general nonexistence. In particular, the signed calculation for $R^{(6)}_{3,132}$ is detailed in Section 4.5.4. The certificate controls recompute these applications without a numerical optimizer. This illustrative collection is not a separate allocation of the entire census: the current sequential counts, eligibility outcomes and bounded-search diagnostics appear in Section 7.1 and its machine-readable ledger.

Exact multiplicity	Example certificates
2	2
3	2
4	6
5	3
6	10
Total	23

Table 2: Illustrative character-aware induction certificates for Theorem 4.9, generated from the supplied exact proof examples. These are examples of the mechanism, not a disjoint allocation of the current census.

The multiplicity-three applications are detailed in Section 4.5.4. The complete consecutive classification range now reaches multiplicity three by the stabilizer-rotation argument of Theorem 5.25. The counting bound eight is not presented as a categorification bound.

4.5.4 Modular equations retained inside the indicator system

We describe one further strengthening of Theorem 4.9. It uses the same high-column and relative-trace variables; it does not require enumerating an unknown center fusion ring. Its purpose here is also to specify exactly which stronger necessary conditions the former multiplicity-three residual satisfies before the stabilizer constraint is imposed.

Assume that all ring characters and positive dimensions lie in a real field K of degree at most two, and let P be the normalized relative trace from a common cyclotomic field to K. For a central simple W write $v_W = FW$, $d_W = d(v_W)$, and

$$H_{aW} = S_{Z_a,W} = \beta_a \chi_a(v_W).$$

Here Z_a ranges over the labelled low objects. Thus H_{aW} denotes an entry of the unnormalized matrix S.

Proposition 4.10 (Joint dimension, twist and low modular constraints)**.** *In addition to the preceding induction conditions, every pseudo-unitary categorification satisfies the following statements.*

Every coordinate of $\Lambda(v_W)$ is an algebraic integer. Under a Galois automorphism σ, the pair consisting of a simple dimension and its low fusion-eigenvalue tuple is permuted according to

$$(d_W, \Lambda(v_W)) \longmapsto \left(\left| \frac{D}{\sigma(D)} \sigma(d_W) \right|, \sigma(\Lambda(v_W)) \right). \tag{4.25}$$

This permutation preserves every joint relative twist-trace type

$$(P(\theta_W), P(\theta_W^2), \ldots, P(\theta_W^h)).$$

Finally, for any two low labels a, b,

$$\sum_W P(\theta_W) H_{aW} H_{bW} = D S_{Z_a, Z_b}. \tag{4.26}$$

Thus the finite indicator system may impose equality of multiplicities within each joint orbit in (4.25)*, separately for each trace type, and may impose all equations* (4.26)*. A missing orbit member forbids a candidate column; an exact infeasibility certificate for the enlarged system is a pseudo-unitary obstruction.*

Proof. The entries of $\Lambda(v_W)$ are eigenvalues of integer fusion matrices, so they are algebraic integers. Use the canonical normalized modular representation of the center, namely S/D and T. Its Galois symmetry gives

$$\sigma(S_{0W}/D) = \epsilon_\sigma(W) S_{0,\widehat{\sigma W}}/D, \qquad \Lambda(v_{\widehat{\sigma W}}) = \sigma(\Lambda(v_W)),$$

where $\epsilon_\sigma(W) \in \{1, -1\}$; see [5, equations (2.12)–(2.13)]. Positivity of all simple Frobenius–Perron dimensions gives exactly (4.25), including its absolute value.

For this canonical center normalization, the Galois symmetry of T is $\theta_{\widehat{\sigma W}} = \sigma^2(\theta_W)$, by [5, Theorem 2.9(b)]. A square of a Galois automorphism fixes K, since its degree is at most two. The normalized trace P is invariant under automorphisms over K, and cyclotomic Galois groups are abelian. Consequently $P(\theta_{\widehat{\sigma W}}^j) = P(\theta_W^j)$ for every j. The simultaneous dimension-and-eigenvalue permutation therefore gives multiplicity balances within each trace type, including the low objects in the twist-one type. This statement concerns multiplicities of tuples; it does not assert that a restriction vector itself is Galois invariant.

The positive Gauss sum of the center is D. Its modular relation is

$$D S_{Z_a, Z_b} = \theta_{Z_a} \theta_{Z_b} \sum_W \theta_W S_{Z_a, W} S_{W, Z_b},$$

by [5, equation (2.6)]. Both low twists are one. All the other coefficients in this equation belong to K, by the character formula. Applying P proves (4.26). The actual central simples satisfy the resulting rational equations and nonnegative multiplicity constraints. An exact Farkas obstruction therefore applies as in Theorem 4.9. □

The code reconstructs all finite Galois conjugates, checks algebraic integrality by integer characteristic polynomials, and adds coefficient rows over $\mathbb{Q}$. This is one refinement of the common induction system, not an independently allocated family of forbidden rings. In particular, the final general exclusion of $R_{3,132}^{(6)}$ already has a certificate using Theorem 4.9; no stronger criterion is credited when an inherited one suffices.

4.5.5 Exact multiplicity-three outcome and the higher scan

The general certificate for $R^{(6)}_{3,132}$ has the following finite structure. Its positive type is $(1, 1, d, d+1, d+1, d)$, with $d = 4 + \sqrt{21}$, and its global dimension is $168 + 36\sqrt{21}$. There are 172 low matrices. Modular reciprocity rejects 168; the other four have exact height-six indicator certificates after the joint-spectrum exclusions. The general sphericalization reduction exhausts sixteen oriented signed tables. Eight are sign gauges of the original table; for the other eight, all character branches are rejected by the checked norm or zero-dimension tests. These are certificate statistics reconstructed from the full tensor. They are not premises selected by its identifier.

Corollary 6.12 supplies the unitary model for $R^{(6)}_{3,218}$, and Theorem 5.25 excludes $R^{(6)}_{3,215}$. This completes multiplicity three. The earlier feasible induction and modular data for the latter ring remain regression controls: they demonstrate the strict need for the additional stabilizer-rotation condition, and are not categorical models.

The same framework applies at every tested multiplicity. In particular, non-positive spherical branches are treated by Section 3.14; a positive-dimension obstruction alone would not establish general nonexistence.

4.6 Ambient cyclotomic traces for quartic dimension fields

The relative-trace method does not require the dimension field to be quadratic. One may instead embed it into a cyclotomic field and retain all coefficients of the resulting normalized traces.

Proposition 4.11 (A finite ambient trace domain)**.** *Let $L = \mathbb{Q}(\zeta_F)$ contain all simple dimensions of a pseudo-unitary fusion category, and let S be the prime support supplied by the categorical Cauchy theorem. Suppose every prime dividing F lies in S. For height h, put*

$$B = \prod_{p \in S} p^{\max\{v_p(F),1\} + \lfloor \log_p h \rfloor}.$$

Every joint normalized relative-trace vector

$$\left(\frac{\mathrm{Tr}_{L(\theta)/L}(\theta^j)}{[L(\theta) : L]} \right)_{j=1}^{h}$$

of a root of unity whose order has prime support in S is either zero, or occurs among roots of unity of order dividing B.

Proof. If p^a divides the order with $a > \max\{v_p(F), 1\} + \lfloor \log_p h \rfloor$, then for each $1 \le j \le h$ the p-primary part of θ^j has order strictly larger than $p^{\max\{v_p(F),1\}}$. Summing over the last degree-p cyclotomic layer gives zero: its conjugates differ by the complete set of pth roots of unity. This layer fixes L and the other primary components, so transitivity of trace gives zero in L. Otherwise the order divides B. Since $F \mid B$, all remaining traces can be computed inside $\mathbb{Q}(\zeta_B)$ by averaging the automorphisms $\zeta_B \mapsto \zeta_B^a$ with $a \equiv 1 \pmod F$. □

Use each pair consisting of an admissible high restriction column v and a trace vector as a nonnegative count variable. The Gram equations remain integral. Apply normalized relative trace to the central indicator formula of [30, Theorem 4.1]; the low induction objects have twist one. Expanding in the power basis of L gives rational linear equations for the first three moments. The usual second indicator choices are exhausted. When $3 \nmid F$ and $3 \notin S$, the third indicator lies both in $\mathbb{Q}(\zeta_3)$ and in a cyclotomic field of conductor coprime to three, hence is rational. Its cyclic multiplicities give the necessary description

$$\nu_3(Y) = a - b/2, \qquad a, b \ge 0, \qquad a + b = \dim \mathrm{Hom}(\mathbf{1}, Y^3).$$

The count b includes both primitive third-root eigenspaces. For a four-dimensional cubic space with the anticommuting stabilizer of Proposition 5.21, impose in addition $\nu_3(Y) \le 1$. These conditions are linear after adding nonnegative slack variables. Allowing rational multiplicities enlarges the necessary system and is therefore legitimate for exclusion.

4.7 Full-character traces and low Verlinde compatibility

The full character field can contain useful information that disappears under projection to the Frobenius–Perron dimension field. We retain this information by applying the ambient trace construction of Proposition 4.11 to a cyclotomic field containing all characters. The following formula makes this extension inexpensive.

Lemma 4.12 (Direct cyclotomic relative trace)**.** *Let $K = \mathbb{Q}(\zeta_F)$ and let $\theta = \zeta_m^b$ have exact order m. Put $d = \gcd(m, F)$ and $r = m/d$. If $\gcd(r, F) > 1$, the normalized relative trace of θ to K is zero. Otherwise it is*

$$\frac{\mathrm{Tr}_{K(\theta)/K}(\theta)}{[K(\theta) : K]} = \frac{\mu(r)}{\varphi(r)}\,\zeta_d^{br^{-1}},$$

where the inverse is taken modulo d, and the last factor is 1 when $d = 1$.

Proof. If a prime divides both r and F, its exponent in m exceeds its exponent in F. The last cyclotomic layer at this prime fixes K and the other primary components; the corresponding primitive root sum is zero. Transitivity of normalized trace proves the first assertion. Otherwise r is coprime to F, and hence to d. The Chinese remainder identity writes

$$\zeta_m^b = \zeta_d^{br^{-1}}\zeta_r^{bd^{-1}}.$$

The first factor lies in K. Averaging the second factor over the cyclotomic Galois group gives the primitive root sum $\mu(r)$ divided by $\varphi(r)$. Coprimality ensures that adjoining K does not change this average. □

Apply the lemma separately to the powers θ^j. With the finite bound B of Proposition 4.11, this computes the complete joint trace domain directly in K, without constructing $\mathbb{Q}(\zeta_B)$. An independent implementation compares this formula with explicit Galois averaging in compositum fields: all 1056 exact comparisons, covering 176 conductor–order pairs, agree.

Proposition 4.13 (Full-character moment compatibility)**.** *Suppose a commutative fusion ring has a pseudo-unitary categorification. Let D be its global dimension and let $K = \mathbb{Q}(\zeta_F)$ contain its full character table. For every possible low induction frame, the nonnegative counts of high restriction columns and joint twist traces satisfy the following conditions:*

(i) *the induction Gram identities and the central indicator formulas;*

(ii) *for every simple X, every tested order n, and every proper divisor r of n, the cyclic rotation character identity*

$$\mathrm{Tr}(E_{X,n}^r) = \nu_{n/r}(X^r), \qquad \dim \mathrm{Hom}(\mathbf{1}, X^n) = [X^n : \mathbf{1}];$$

(iii) *the low-pair modular equations and the integrality and Galois compatibility of the joint fusion eigenvalues;*

(iv) *equality of the multiplicities of paired types (λ, τ) and $(\sigma\lambda, \sigma^2\tau)$, where λ is the joint low fusion eigenvalue vector and τ the joint normalized twist-trace vector.*

Thus exact infeasibility of all these finite systems excludes a pseudo-unitary categorification.

Proof. The Gram identities and the indicator formula are the established induction constraints; the latter are applied coefficientwise in K using [30, Theorem 4.1]. The cyclic rotation operator has nth roots of unity as eigenvalues, and the divisor-power identity follows by grouping the tensor factors into blocks of length r; see [31]. Introduce nonnegative variables for these rotation eigenvalue multiplicities and apply normalized relative trace. The resulting equations are rational after expansion in a power basis.

For low central objects A, B, both twists are one, so the center identity $STS = DT^{-1}ST^{-1}$ gives the low-pair equations. Their joint eigenvalues are algebraic integers and occur with Galois-invariant multiplicities. Under a Galois automorphism the fusion eigenvalues transform by σ, whereas the canonical center twists transform by σ^2. Normalized relative trace commutes with these automorphisms, proving the paired-type equality. If an ambient automorphism requires a lift to a compositum, the resulting normalized trace is independent of that lift. The twist-one low objects contribute the known constant terms. Relaxing integer multiplicities to nonnegative real variables enlarges the set of possible solutions, so an exact separating functional proves the claimed necessary obstruction. □

There is a complementary refinement using low fusion products. Write v_Z for the restriction column of a central simple Z and $d(v) = \sum_i v_i \,\mathrm{FPdim}(b_i)$. For a low central simple A_a associated to a character χ_a of formal codegree f_a, put

$$t_a(v) = \frac{D}{f_a}\frac{\chi_a(v)}{d(v)}.$$

Then $t_a(v_Z) = S_{A_a,Z}/d_Z$.

Corollary 4.14 (Self-dual low-product constraints)**.** *For self-dual input, all low central simples are self-dual. The general word and restriction formulas in Section 4.8 therefore apply with $a^* = a$ and real $\lambda_a(Z) = t_a(v_Z)$. In particular the cubic moments are the low fusion coefficients, and summing the nonnegative high coefficients gives the restriction-product equations and their coordinatewise upper bounds.*

Proof. The input character values are real. Hence the corresponding low S-rows are real, and $S_{a^*,Z} = \overline{S_{a,Z}}$ together with invertibility of S forces $a^* = a$. The assertion follows from the general formulas, retaining the same bounded high coefficient variables. □

The implementation may retain integer variables and test congruences. Only rows with integer-count variables are used for a congruence witness. In the applications below, however, nonnegative linear infeasibility already suffices: each stored ray y satisfies $y^T A \geq 0$ and $y^T b < 0$ over the integers. No numerical optimizer is used by the final verifier.

Theorem 4.15. *The rings $R^{(6)}_{4,211}$ and $R^{(6)}_{4,243}$ have no complex fusion categorification.*

Proof. For $R^{(6)}_{4,211}$, the simple dimensions are $(1, 1, d, 2d+1, 2d+1, d)$, where $d = 2+\sqrt{6}$, and $D = 120 + 48\sqrt{6}$. Its full character field is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, so take $F = 24$. At height three the Cauchy prime support $\{2, 3\}$ gives $B = 144$ and 53 joint trace types. The exhaustive low enumeration has 130 frames. Only two frames have relevant character assignments: each has two assignments and four second-indicator choices. All 16 systems are excluded by exact rays from Proposition 4.13. Here even the weaker equality of total joint-type Galois multiplicities suffices; the coupled trace refinement is not needed.

For $R^{(6)}_{4,243}$, the dimensions are $(1, 1, d, d+1, d+1, d)$ with $d = 7+\sqrt{57}$ and $D = 456+60\sqrt{57}$. The full character field is quadratic. The height-six systems use Proposition 4.14, including restriction-product bounds. There are 210 low frames. Four relevant frames contribute

respectively two, one, one, and two character assignments, each with eight second-indicator choices. The resulting 48 systems all have exact separating rays.

Finally, general scope is supplied independently by unrestricted signed sphericalization. For the two rings the complete signed enumerations have respectively 128 and 1584 tables, in 4 and 53 sign-gauge orbits. Exactly one orbit in each case is gauge isomorphic to the original tensor; the remaining character branches are rejected by respectively 8 and 176 exact arithmetic obstructions. The surviving nonzero dimension characters belong to the Frobenius–Perron Galois orbit. Thus any complex categorification would admit a pseudo-unitary Galois conjugate, contradicting the preceding exclusions. □

The full certificates and exact replay reports are retained in `research/round3/m4`. The verifiers `software/round3_m4.py` and `software/round3_verlinde.py` reconstruct the complete branch domains from the input tensors. Both new necessary systems are also applied to the entire frozen 22-ring frontier and all 54 positive model controls, with explicit eligibility and time-budget outcomes; a timeout is inconclusive. In particular, $R^{(6)}_{4,155}$ and $R^{(6)}_{4,629}$ survive the stronger full-character height-six systems with coupled Galois spectrum/twist counts. These surviving systems are not asserted to be modular data or categorifications.

4.8 Joint restriction and word Gram realization

The cubic Verlinde, higher-word and integer-square conditions are different projections of one necessary integral Gram factorization. Retaining the restriction coordinates in that factorization makes it substantially stronger without introducing new categorical assumptions.

Lemma 4.16 (Simple restriction support)**.** *Let $F : \mathcal{B} \to \mathcal{C}$ be a tensor functor between fusion categories. Suppose a proposed complete list of restriction columns for simples of $\mathcal{B}$ is available. Call a column low if its tensor-unit coordinate is positive, and let v be the restriction column of a simple object Z. Suppose $\mathbf{1}_{\mathcal{B}}$ is the only simple object whose restriction column has positive tensor-unit coordinate and is componentwise bounded by the based-ring product vv^*. Then the tensor-unit coordinate of vv^* is one. Moreover, there is at most one simple of $\mathcal{B}$ having restriction column v.*

Proof. Simplicity gives $[ZZ^* : \mathbf{1}_{\mathcal{B}}] = 1$. Every simple constituent of ZZ^* has restriction componentwise bounded by vv^*. Only the tensor unit can contribute to the unit coordinate of $F(ZZ^*)$, so that coordinate is one. If distinct simples Z, W had restriction v, then $[ZW^* : \mathbf{1}_{\mathcal{B}}] = 0$, whereas the unit coordinate of $F(ZW^*)$ would be positive. A nonunit low constituent would be required, contrary to the hypothesis. □

Let $\mathcal{C}$ be pseudo-unitary, put $D = \mathrm{FPdim}(\mathcal{C})$, and write $F : Z(\mathcal{C}) \to \mathcal{C}$ for the forgetful functor. Fix distinct low central simples $A_0 = \mathbf{1}, A_1, \dots, A_{\ell-1}$, their duality permutation $a \mapsto a^*$, and their restriction columns $u_a = [F(A_a)]$. We assume that their fusion eigenvalues

$$\lambda_a(Z) = \frac{S_{A_a,Z}}{S_{\mathbf{1},Z}}$$

are known from the restriction column $v_Z = [F(Z)]$. Here S is normalized and $S_{\mathbf{1},Z} = d(v_Z)/D$, where $d(v) = \sum_i v_i \,\mathrm{FPdim}(X_i)$. For a commutative input ring and the character-labelled low simples of canonical induction, the existing character formula gives

$$\lambda_a(v) = \frac{D}{f_a}\frac{\chi_a(v)}{d(v)},$$

where χ_a has formal codegree f_a. The low simples are distinguished objects even when their restriction columns coincide. Fix a complete high restriction profile: for every admissible nonzero column v, its multiplicity is m_v.

For a word $P = A_{a_1} \cdots A_{a_k}$, set $\lambda_P(v) = \prod_j \lambda_{a_j}(v)$. The empty word is the unit. Dualizing a word dualizes each index; the order is immaterial in the commutative central Grothendieck ring.

Theorem 4.17 (Joint restriction–word Gram condition)**.** *Choose finitely many words $P_1, \dots, P_s$ in the low simples. Their pairings and their low-simple coefficients are determined by the profile:*

$$H_{pq} = \sum_{Z \in \operatorname{Irr} Z(\mathcal{C})} \frac{d(v_Z)^2}{D^2} \overline{\lambda_{P_p}(v_Z)} \lambda_{P_q}(v_Z), \tag{4.27}$$

$$L_{pa} = \sum_{Z} \frac{d(v_Z)^2}{D^2} \overline{\lambda_a(v_Z)} \lambda_{P_p}(v_Z). \tag{4.28}$$

These numbers are nonnegative integers. Let r_p be the restriction of P_p after its low constituents are removed, and put

$$G = H - LL^{\mathsf{T}}, \qquad Q = \sum_v m_v v v^{\mathsf{T}}, \qquad B_{pi} = (r_p)_i.$$

There are nonnegative integral vectors $x_Z = ([P_p : Z])_{p=1}^s$ for the high central simples such that

$$\begin{pmatrix} Q & B^{\mathsf{T}} \\ B & G \end{pmatrix} = \sum_{Z \text{ high}} \begin{pmatrix} v_Z \\ x_Z \end{pmatrix} \begin{pmatrix} v_Z \\ x_Z \end{pmatrix}^{\mathsf{T}}. \tag{4.29}$$

In particular G and the joint block matrix are positive semidefinite, have nonnegative integral entries, and each x_Z satisfies

$$x_Z \in \operatorname{im}_{\mathbb{Q}} G, \qquad 0 \le (x_Z)_p \le \min \left\{ \lfloor \sqrt{G_{pp}} \rfloor, \min_{i:(v_Z)_i > 0} \left\lfloor \frac{(r_p)_i}{(v_Z)_i} \right\rfloor \right\}.$$

Proof. In a commutative based ring the matrices of a simple and its dual are transposes and commute. They are therefore normal and simultaneously diagonalizable, with $\lambda_{a^*}(Z) = \overline{\lambda_a(Z)}$. The Verlinde formula in the modular center, whose global dimension is D^2, gives

$$[P : \mathbf{1}] = \sum_Z \frac{d(v_Z)^2}{D^2} \lambda_P(v_Z).$$

Apply this identity to $P_p^* P_q$ and to $A_a^* P_p$. Rigidity identifies the resulting numbers with $\dim \operatorname{Hom}(P_p, P_q)$ and $[P_p : A_a]$, respectively. This proves (4.27) and (4.28), including their integral positivity. It also explains why complex conjugation cannot be omitted when the low simples are not self-dual.

Decompose each P_p into its low and high constituents. The orthonormality of the simple basis for the Frobenius pairing gives $G_{pq} = \sum_{Z \text{ high}} [P_p : Z][P_q : Z]$. Applying the tensor functor F to the same decomposition gives $r_p = \sum_{Z \text{ high}} [P_p : Z] v_Z$. These are the lower-right and mixed blocks of (4.29); its upper-left block is the fixed high induction profile. Thus the displayed matrix is an integral Gram matrix.

For a real matrix X, the images of X and XX^{T} agree. Apply this to the matrix of coefficient vectors x_Z. Since all entries are rational, each x_Z lies in the rational image of G. A diagonal entry of G is a sum of their squared coordinates, giving the square-root bound. Each coordinate of r_p is a sum of nonnegative integers $(x_Z)_p (v_Z)_i$, giving the restriction bound. □

Equivalently, writing X for the matrix with columns x_Z, the lower and mixed blocks are

$$G = XX^{\mathsf{T}}, \qquad \sum_{Z \text{ high}} (x_Z)_p v_Z = r_p. \tag{4.30}$$

All terms in the theorem are computed from the input tensor, a complete profile, a character labelling and the chosen words. Thus a nonintegral pairing, a negative entry or an impossible finite Gram factorization rejects that specified profile. No hand choice of central fusion coefficients is part of the criterion.

4.8.1 Exact finite domains and conditional ellipsoids

The coordinate bounds already make the coefficient-vector search finite. A stronger bound exploits the whole Gram matrix, including its mixed block.

Lemma 4.18 (Exact rank-one Gram domains)**.** *Let M be a rational positive-semidefinite matrix, and choose a principal submatrix M_{JJ} of size* $\operatorname{rank} M$ *that is invertible. For a rational vector z, the condition $M - zz^{\mathsf{T}} \succeq 0$ is equivalent to*

$$z = M_{\bullet J} M_{JJ}^{-1} z_J, \qquad z_J^{\mathsf{T}} M_{JJ}^{-1} z_J \le 1.$$

Consequently every column in a Gram factorization belongs to this exact ellipsoid in the rational image of M.

Proof. Positivity of $M - zz^{\mathsf{T}}$ forces z to be orthogonal to $\ker M$, hence to lie in $\operatorname{im} M$. On that image, use the independent coordinates indexed by J. Congruence reduces the assertion to positive definiteness of M_{JJ} and positivity of $M_{JJ} - z_J z_J^{\mathsf{T}}$. Conjugating by the positive square root of M_{JJ}^{-1} gives $I - ww^{\mathsf{T}} \succeq 0$, equivalently $w^{\mathsf{T}} w \le 1$, proving the formula. All matrices required for checking the final formula are rational; the square root is only used in this proof. □

For the joint matrix in (4.29), fix the restriction column v. Choose independent principal coordinates J of Q, and put

$$\begin{aligned} c_v &= B_{\bullet J} Q_{JJ}^{-1} v_J, \\ C &= G - B_{\bullet J} Q_{JJ}^{-1} B_{\bullet J}^{\mathsf{T}}, \\ \rho_v &= 1 - v_J^{\mathsf{T}} Q_{JJ}^{-1} v_J. \end{aligned} \tag{4.31}$$

The necessary image identities for v and B^{T} are checked first. Block Gaussian congruence then turns the joint matrix into $\operatorname{diag}(Q, C)$ and its column (v, x) into $(v, x - c_v)$. Lemma 4.18 therefore gives

$$x - c_v \in \operatorname{im}_{\mathbb{Q}} C, \qquad (x - c_v)_K^{\mathsf{T}} C_{KK}^{-1} (x - c_v)_K \le \rho_v,$$

where K is any independent principal coordinate set for C. If $C = 0$, this simply forces $x = c_v$. A negative ρ_v is impossible. This calculation uses the inverse of an explicit nonsingular rational matrix, not a numerical pseudoinverse.

To enumerate the domain, select independent coordinates of x itself. The image condition expresses all other coordinates as rational affine functions of these, and the theorem bounds each free coordinate in a finite integer interval. During enumeration, interval bounds for the unassigned coordinates discard a branch only when some reconstructed coordinate cannot lie in its prescribed interval. In addition, any assigned prefix of independent coordinates satisfies its own principal ellipsoid inequality: it is obtained by minimizing over the remaining real coordinates. A prefix violating that exact rational inequality can also be discarded. At each leaf check integrality, every coordinate bound, the full ellipsoid inequality, and the componentwise

inequality $zz^{\mathsf{T}} \le M$. Every admissible integer column follows a unique unpruned path, and every retained leaf satisfies all stated constraints. A node or time limit interrupts this enumeration and supplies no completeness claim.

Let $\mathcal{Q}_v$ be the resulting complete finite domain for restriction type v. Its zero vector is retained whenever allowed: a high central simple may occur in none of the selected words. Introduce nonnegative integer multiplicities $z_{v,x}$, for $x \in \mathcal{Q}_v$. The remaining necessary conditions are the integer linear equations

$$\sum_{x\in\mathcal{Q}_v} z_{v,x} = m_v, \qquad \sum_{v,x} z_{v,x} x x^{\mathsf{T}} = G, \qquad \sum_{v,x} z_{v,x} x_p v = r_p. \tag{4.32}$$

The upper-left block follows from the first equations and the fixed profile, so these enforce the complete joint Gram factorization. Exact rational separation or an exact integer obstruction rejects a profile. A solution supplies compatible coefficients for the selected words, and makes no categorical existence assertion.

The optimizer-free implementation is `software/round5_unified.py`. It uses the same theorem for quadratic, cubic or longer words, retains complex duality, and records incomplete domains separately. Exact unimodular changes of coordinates may shorten the ellipsoid enumeration without changing its integral points.

Categorical controls independently derive the centers of Vec_{C_2}, Vec_{C_3} and $\mathrm{Rep}(S_3)$. For the last, finite centralizer character sums give its eight-simple center and exact Verlinde gives all 512 fusion coefficients. Every selected quadratic or cubic nonunit low word has a nonzero high part. Its actual integral assignments satisfy the joint Gram and coupled balancing systems; replacing all twists by one violates two product-balancing rows. The complex pointed control separately detects omission of the dual index. These optimizer-free controls accompany `verify_domains.py`, `pointed_center_control.py` and `s3_center_control.py` in `research/round5/unification/`.

4.8.2 Completing the action of the low simples

Proposition 4.19 (Finite word-action completion)**.** *Expand an integral solution of* (4.32) *into individual proposed central simples. Let V be their restriction matrix, including the low columns, and let $u(P)$ be the complete coefficient vector of each retained word. If these data come from a categorification, there exist nonnegative integral matrices N_a, with $(N_a)_{ZY} = [A_aY : Z]$, satisfying*

$$N_a u(P) = u(A_aP), \qquad N_a e_0 = e_a, \qquad N_{a^*} = N_a^{\mathsf{T}}, \qquad N_aN_b = N_bN_a,$$
$$VN_a = N_{F(A_a)}V,$$

where the word equation is imposed whenever both words are retained. Each entry is bounded by the restriction product:

$$0 \le (N_a)_{ZY} \le \min_{i:(v_Z)_i>0} \left\lfloor \frac{([F(A_a)]v_Y)_i}{(v_Z)_i} \right\rfloor.$$

For every $k \ge 1$, one also has the unweighted spectral identity

$$\mathrm{Tr}(N_a^k) = \sum_{Z\in \mathrm{Irr}\, Z(\mathcal{C})} \lambda_a(v_Z)^k. \tag{4.33}$$

Equivalently, if the proposed center has rank r, the first r trace identities specify the characteristic polynomial

$$\det(tI - N_a) = \prod_Z (t - \lambda_a(v_Z)).$$

If duality on the proposed simple objects has been selected, rigidity also gives $(N_a)_{ZY} = (N_a)_{Y^*Z^*}$. *In particular every* N_a *is normal, since it commutes with* $N_{a^*} = N_a^{\mathsf{T}}$. *Consequently completion of the selected low fusion actions is a finite exact problem for each integral word-coefficient assignment.*

Proof. Multiplication in the central Grothendieck ring gives the word equation and the unit column. Rigidity gives $N_{a^*} = N_a^{\mathsf{T}}$; braidedness gives commutativity. Applying the forgetful tensor functor to each product gives the restriction equation. Its terms are nonnegative, so the contribution of any single constituent is bounded in every positive coordinate of its nonzero restriction column. This proves the finite entry bound.

The regular fusion representation is diagonalized by the modular S-matrix. Its eigenvalues for multiplication by A_a are $\lambda_a(v_Z)$, one for every central simple Z, counted with the multiplicities in the complete profile. Taking traces proves (4.33); no dimension-square weight occurs here. Newton's identities give the equivalent characteristic polynomial. The further entry identity is the Frobenius reciprocity equality $[A_aY : Z] = [A_aZ^* : Y^*]$. The entry bounds make the matrix domain finite. All displayed identities can therefore be checked exactly after exhaustive bounded enumeration. □

A contradiction excludes only the coefficient assignment being completed; to exclude its restriction profile, all integral coefficient assignments must be covered. Conversely, compatible low actions need not specify products of every pair of high simples. Even a full based ring and modular datum still require a categorical construction before an existence claim.

Corollary 4.20 (Unit-column saturation)**.** *Let* $\mathcal{A} \subset \mathrm{Mat}_r(\mathbb{Q})$ *be the rational algebra generated by proposed known fusion actions and the identity. If they extend to a fusion ring on the proposed simple basis, then* $\mathrm{ev}_0(M) = Me_0$ *is injective on* $\mathcal{A}$. *Moreover,*

$$Me_0 \in \mathbb{Z}^r \Longrightarrow M \in \mathrm{Mat}_r(\mathbb{Z}), \qquad Me_0 \in \mathbb{Q}_{\geq 0}^r \Longrightarrow M \in \mathrm{Mat}_r(\mathbb{Q}_{\geq 0}).$$

Both conditions admit finite exact tests before unknown fusion actions are reconstructed.

Proof. Every M in the generated algebra is multiplication by the element z whose coefficient vector is Me_0. Thus $M = \sum_i z_i N_i$. This gives injectivity and both implications. Compute a rational basis of $\mathcal{A}$ and its unit-column image. Injectivity is a rank test. A basis of the saturated lattice $\mathrm{ev}_0(\mathcal{A}) \cap \mathbb{Z}^r$, obtained by exact integer normal forms, suffices to test integrality of its lifted matrices. The nonnegative cone of unit columns is rational polyhedral; its finitely many generating rays suffice to test entrywise positivity. Equivalently, each possible negative matrix entry gives a rational linear feasibility problem in that cone. □

4.8.3 The earlier moment conditions as projections

For $P = A_aA_b$, formula (4.28) is precisely $N_{A_a,A_b}^{A_c}$, with the factor λ_{c^*} required in the complex case. The corresponding diagonal entry of H is

$$\dim \mathrm{Hom}(A_aA_b, A_aA_b) = \sum_Z (N_{A_a,A_b}^Z)^2. \tag{4.34}$$

In the earlier cubic notation this says

$$t_{abc} = \frac{1}{D^2} \sum_Z d(v_Z)^2 \lambda_a(v_Z)\lambda_b(v_Z)\lambda_c(v_Z), \qquad N_{A_a,A_b}^{A_c} = t_{abc^*}. \tag{4.35}$$

Writing $q_{ab,v} = \sum_{Z:\, v_Z = v} N_{A_a,A_b}^Z$ for high types, the restriction identity is

$$u_au_b = \delta_{a,b^*}u_0 + \sum_{c \neq 0} t_{abc^*}u_c + \sum_{v \text{ high}} q_{ab,v}v. \tag{4.36}$$

The off-diagonal entries couple different products, so retaining only these individual squared norms loses necessary constraints. For integers $0 \le n \le M$ and $k \ge 0$,

$$n^2 - (2k+1)n + k(k+1) = (n-k)(n-k-1) \ge 0, \qquad n^2 \le Mn.$$

Summing these inequalities over a restriction type recovers the integer-square cuts. They are valid relaxations of (4.32), rather than separate categorical principles. The vectorized full Morita-action Gram factorization uses the same finite-domain lemma after its own categorical Gram identity has been established.

There is also a useful exact variance bound, omitting zero-multiplicity types, when the type totals $T_{pv} = \sum_{Z:\, v_Z = v} (x_Z)_p$ are known. Write $T_{pv} = m_v q_{pv} + r_{pv}$ with $0 \le r_{pv} < m_v$. Then

$$G_{pp} \ge \sum_v (m_v q_{pv}^2 + r_{pv}(2q_{pv}+1)). \tag{4.37}$$

Indeed, among m_v nonnegative integers with fixed sum, the smallest sum of squares occurs when their values differ by at most one: replacing a, b with $a-1, b+1$ decreases the sum whenever $a \ge b+2$. Subtracting the type averages also proves $G - T\,\mathrm{diag}(m_v^{-1})T^{\mathsf{T}} \succeq 0$. If this matrix is zero, every coefficient vector in a restriction type equals that type's average; their integrality is then forced.

If an additional word has zero Schur complement against the previously selected words, its high coefficient vector is a forced rational linear combination of their vectors. Checking its nonnegative integrality therefore strengthens the finite domains without adding a free coordinate. This follows directly from the image identity in Lemma 4.18, applied after adjoining that word.

4.8.4 Coupling product coefficients to projected twists

The same coefficient vectors can be coupled to the finite twist-trace types, imposing balancing on individual products rather than only aggregate modular moments. Care with duality is essential here. Our normalized S follows the modular-representation convention of [5, equation (2.3)]; its balancing identity is

$$D\theta_a\theta_b S_{a,b} = \sum_Z N^Z_{A_a^*, A_b}\, \theta_Z d(v_Z).$$

Thus, for low simples, whose twists are one, the product $A_a A_b$ has right-hand side $DS_{a^*,b}$, not $DS_{a,b}$ in general.

Proposition 4.21 (Projected product balancing)**.** *Let K contain the dimensions and the known low S-entries, and let P be normalized relative trace from a common cyclotomic field to K. Refine the coefficient counts in* (4.32) *to $z_{v,x,\tau}$, where $\tau = (P(\theta_Z), \dots, P(\theta_Z^h))$ is a permitted joint trace type. If the selected words include $P_p = A_a A_b$, then*

$$\sum_{v,x,\tau} z_{v,x,\tau}\, x_p\, d(v)\tau_1 = DS_{a^*,b} - \sum_{c \text{ low}} L_{pc} d(u_c). \tag{4.38}$$

These equations, the Gram and restriction equations, and every indicator or Galois-multiplicity equation must use the same count variables, with their respective marginals.

Proof. Apply the balancing identity to A_a^*, A_b; the double dual has the same simple label as A_a. Separate the low constituents, whose twists are one. The other coefficients and the dimensions belong to K, so normalized relative trace replaces only θ_Z by τ_1. Grouping the central simples by (v, x, τ) proves the formula. Summing over τ recovers the coefficient counts, while summing over x recovers the original restriction/twist counts. Consequently all earlier necessary equations hold for these marginals. Using independent variables for the two systems would forget the compatibility asserted here. □

Equation (4.38) concerns a product of two specified simple objects with known twists. It does not attach a scalar twist to an arbitrary longer word. Nor is a normalized trace type silently treated as an actual root of unity.

Lemma 4.22 (Actual twists from an extremal projected trace)**.** *Let τ_k be the normalized relative trace of θ_Z^k to a retained cyclotomic field. If $\tau_k^m = 1$, then $\theta_Z^k = \tau_k$ and $\theta_Z^{km} = 1$. Consequently, if every occupied projected twist type has such an extremal coordinate, all actual lifts of that projected assignment can be enumerated in a finite root-of-unity group.*

Proof. At the chosen complex embedding the normalized trace is the average of the conjugates of θ_Z^k, all on the unit circle. Since $|\tau_k| = 1$, equality in the triangle inequality forces every conjugate to equal τ_k, including θ_Z^k itself. The root bound follows. For several occupied types, take the least common multiple of their resulting exponents and check every power-trace coordinate of every root in that group. Integer compositions of each type's multiplicity enumerate all multisets of actual lifts. □

Before combining actual roots into projected trace types, retain their individual dimension capacities. In the pseudo-unitary setting, put

$$M_X(\lambda) = \frac{D_{\mathcal{C}}}{D_{\mathcal{D}}} \sum_{\substack{W \in \operatorname{Irr} Z(\mathcal{D}) \\ \theta_W = \lambda}} d_W [F_{\mathcal{D}}(W) : X].$$

Corollary 4.31 and positivity give, for every central simple of restriction v and twist λ,

$$d(v) v_X \le M_X(\lambda) \qquad (X \in \operatorname{Irr} \mathcal{D}). \tag{4.39}$$

A projected column/type pair is therefore impossible if none of its allowed actual roots satisfies all these inequalities. Coincident relative traces do not add the capacities available to a single object. More generally, the individual-root mass equations can be imposed as an exact integral allocation problem for the proposed columns. These are direct consequences of the same projection identity, independent of any choice of simultaneous word coefficients.

4.9 Lifted central subcategories and adjunction Gram matrices

Projection to a fusion subcategory constrains each central column and its twist. When the induction data additionally force an actual nondegenerate fusion subcategory of the center, its entire action can be retained.

Proposition 4.23 (Central packets and restricted adjunction)**.** *Let $\mathcal{B} \subset Z(\mathcal{C})$ be a nondegenerate braided fusion subcategory and let $\mathcal{E} = \mathcal{B}'$ be its centralizer. For $b \in \operatorname{Irr} \mathcal{B}$ and $Y \in \operatorname{Irr} \mathcal{E}$, put $w_b = [F(b)]$ and $v_Y = [F(Y)]$. The complete central restriction profile is the disjoint union of the packets*

$$\{ w_b v_Y : b \in \operatorname{Irr} \mathcal{B} \}, \qquad Y \in \operatorname{Irr} \mathcal{E}.$$

Their twists are $\theta_b \theta_Y$. Let $T = F|_{\mathcal{E}}$, let I be its right adjoint, and set $A = I(\mathbf{1})$. With restriction matrix $V_{iY} = [T(Y) : X_i]$ and fusion convention $(N_A)_{XY} = [Y \otimes A : X]$, one has

$$V^{\mathsf{T}} V = N_A. \tag{4.40}$$

Neither this identity nor its proof requires T to be dominant.

Proof. The nondegenerate centralizer decomposition gives a braided equivalence $\mathcal{B} \boxtimes \mathcal{E} \simeq Z(\mathcal{C})$; see [15]. Its simple objects are precisely the pairs (b, Y). Tensoriality of F gives their restriction $w_b v_Y$, and the ribbon product gives their twist. Coincident restriction columns do not identify the corresponding simple pairs.

Semisimplicity gives the right adjoint I. Rigidity, tensoriality of T and adjunction yield, naturally in $X, Y \in \mathcal{E}$,

$$\begin{aligned}\mathrm{Hom}_{\mathcal{E}}(X, I(T(Y))) &\cong \mathrm{Hom}_{\mathcal{C}}(T(X), T(Y)) \\ &\cong \mathrm{Hom}_{\mathcal{C}}(T(Y^* \otimes X), \mathbf{1}) \\ &\cong \mathrm{Hom}_{\mathcal{E}}(Y^* \otimes X, A) \\ &\cong \mathrm{Hom}_{\mathcal{E}}(X, Y \otimes A).\end{aligned}$$

Thus $I(T(Y)) \cong Y \otimes A$. For simple X, Y, the dimension of the second space is the scalar product of their restriction columns, while the dimension of the last space is $(N_A)_{XY}$, proving (4.40). □

For a forced Ising subcategory with simples $1, g, \sigma$, a packet has columns $v, gv, \sigma v$ and twists $\lambda, -\lambda, z\lambda$, with the appropriate primitive sixteenth root z. In particular the rank of the center is divisible by three. All packet orientations and all inherited Ising choices must be covered. This factors the center; it does not assert a tensor-product decomposition of $\mathcal{C}$.

The algebra A is the $\mathcal{E}$-part of canonical unit induction: for simple $Y \in \mathcal{E}$, $[A : Y] = \dim \mathrm{Hom}_{\mathcal{C}}(F(Y), \mathbf{1})$. Consequently, when the packet assignments force $A = \mathbf{1} \oplus c$, the full fusion matrix of c is already known: $N_c = V^{\mathsf{T}}V - I$. Every other fusion matrix must commute with it, have its prescribed unit column, and satisfy $N_{Y^*} = N_Y^{\mathsf{T}}$. Since $\mathcal{E}$ is braided, these matrices are normal and mutually commuting. Together with the restriction identities, these give finite exact generator constraints before any modular-data completion. No converse identity for VV^{T} is asserted: it does not generally equal multiplication by $F(A)$.

Corollary 4.24 (Grading separates the adjunction summands)**.** *Suppose $\mathcal{E}$ has a known group grading and $I(\mathbf{1}) = \mathbf{1} \oplus \bigoplus_a A_a$, where the nonunit simple summands A_a have pairwise distinct grading degrees. Write γ_Y for the degree of a simple Y. Then*

$$(N_{A_a})_{ZY} = (V^{\mathsf{T}}V - I)_{ZY}[\gamma_Z = \gamma_{A_a}\gamma_Y].$$

Thus the individual fusion matrices are fixed by the restriction matrix and a proposed assignment of degrees to simples.

Proof. The adjunction identity gives $V^{\mathsf{T}}V - I = \sum_a N_{A_a}$. Grading homogeneity permits a contribution in position (Z, Y) only from the summand with degree $\gamma_Z\gamma_Y^{-1}$; by hypothesis there is at most one such summand. This proves the formula. □

Pointed monodromy supplies such gradings. For two summands of opposite $\{+1, -1\}$ charges, the two masks are respectively the equal-charge and opposite-charge entries. Assignments on repeated restriction columns can be enumerated by their charge counts, up to permutations of copies.

Full faithfulness of $F|_{\mathcal{B}}$ does not imply a tensor factorization of $\mathcal{C}$, even when $\mathcal{B}$ is semion. For example, take $\mathcal{C} = \mathrm{Vec}^{\omega}_{\mathbb{Z}/4}$ with $\omega(a, b, c) = i^{a\lfloor (b+c)/4 \rfloor}$ for representatives $0 \le a, b, c < 4$. The element 2 lifts centrally with half-braiding $\gamma(x) = e^{3\pi i x/4}$; its square lift is the unit and its self-braiding is $-i$. Thus its semion subcategory forgets fully faithfully, whereas a tensor factorization would identify the fusion group $\mathbb{Z}/4$ with $(\mathbb{Z}/2)^2$. The exact replay `central_semion_counterexample.py` checks all 256 cocycle pentagons, 16 half-braiding hexagons and four square-lift identities by arithmetic modulo eight.

For a pointed central subgroup G, put $q(g) = \theta_g$ and $b(g, k) = q(gk)/(q(g)q(k))$. Write $\epsilon_g(Z)$ for the scalar monodromy on $g \otimes Z$ when Z is simple. Since g is invertible, this scalar is nonzero.

Lemma 4.25 (Monodromy charges of word constituents)**.** *For a low word $P = A_{a_1} \cdots A_{a_r}$ define $\epsilon_g(P) = \prod_j \epsilon_g(A_{a_j})$. Every simple constituent Z of P has $\epsilon_g(Z) = \epsilon_g(P)$ for every $g \in G$. Consequently the words on which a coefficient vector x_Z is positive must all have the same charge character of G. Moreover,*

$$\epsilon_k(g \otimes Z) = b(k, g)\epsilon_k(Z).$$

For the character-labelled induction lows in our S convention,

$$\epsilon_g(A_a) = \frac{DS_{g^*,A_a}}{d(A_a)} = \chi_a(F(g^*)) = \chi_a(F(g))^{-1}.$$

For involutions, the last two values agree.

Proof. The braiding hexagon expresses monodromy with a tensor product as the product of the monodromies with its factors, whenever these are scalar. Thus monodromy on $g \otimes P$ is the displayed scalar product. Naturality with respect to each simple inclusion $Z \to P$ gives the constituent assertion. Taking one factor to be an invertible g gives the transport formula, since its monodromy with k is $b(k, g)$. The S-matrix equality uses the same dual-index convention as Proposition 4.21; substituting the low character formula gives the final identities. □

In particular the pairings of differently charged words vanish, and no nonnegative Gram column may mix their supports. This gives a cheap exact sieve before word-action completion or twist coupling. A zero coefficient vector imposes no charge choice and remains admissible.

Proposition 4.26 (Pointed stabilizer divisibility)**.** *Let* $\mathrm{rad}(b) = \{g : b(g, k) = 1 \text{ for all } k \in G\}$. *For a restriction column v, set $K_v = \{g \in G : [F(g)]v = v\}$ and let m_v count the central simples with that column. Then*

$$\frac{|K_v|}{|K_v \cap \mathrm{rad}(b)|} \mid m_v.$$

Proof. If $g \otimes Z \cong Z$, the transport formula in Lemma 4.25 forces $b(k, g) = 1$ for every k, hence $g \in \mathrm{rad}(b)$. The group K_v acts on the finite set of simples with restriction v. Every stabilizer H is contained in $K_v \cap \mathrm{rad}(b)$, so each orbit size $|K_v|/|H|$ is a multiple of the displayed divisor. Summing orbit sizes proves the assertion. The forgetful functor need not be fully faithful on G. □

4.10 Modular-word closure and finite representation completion

Once a complete twist profile is fixed, the known low rows also determine part of the modular representation by exact linear algebra. This is a separate use of the same central data: no fusion coefficients are guessed. Write S, T for the canonical normalized matrices of $Z(\mathcal{C})$, $N = \mathrm{ord}(T)$, and $C = S^2$ for charge conjugation. The canonical representation factors through $SL_2(\mathbb{Z}/N\mathbb{Z})$. For a coprime to N, let Q_a be the signed permutation defined by $\sigma_a(S) = SQ_a$, where $\sigma_a(\zeta_N) = \zeta_N^a$. The congruence and Galois formulas of [47, Theorem II] give, for $ab \equiv 1 \pmod N$,

$$ST^bS = T^{-a}SQ_aT^{-a}. \tag{4.41}$$

In particular, if all twist-one simples are among the known lows, then Q_a preserves that set. Its signed action can be determined from the known low block whenever the matching signed columns are unique.

Proposition 4.27 (Exact modular closure and aggregate Verlinde test)**.** *Suppose the action of S is known on a subspace W of the C-fixed space, and T is known on the entire simple basis. The following operations are necessary and exact.*

1. *Known pairs (v, Sv) satisfy every linear relation among their first coordinates. A zero first coordinate with nonzero proposed image is an obstruction.*

2. *If $u, v \in W$ satisfy $u = Tv$, then*

$$S(TSu) = T^{-1}Sv.$$

Both this pair and its reverse may be adjoined to the known action.

3. *If P, Q, R are nonnegative integral sums of central simples whose S-images are known, then*

$$\sum_j \frac{(SP)_j (SQ)_j (SR)_j}{S_{0j}} = [PQR : \mathbf{1}] \in \mathbb{Z}_{\geq 0}. \tag{4.42}$$

Thus this test applies to an entire class of indistinguishable restriction and twist columns even before its individual rows are known.

Proof. The first assertion is linearity. For the second, use the anomaly-one relation $STS = T^{-1}ST^{-1}$ and $u = Tv$:

$$STSu = T^{-1}ST^{-1}u = T^{-1}Sv.$$

Charge conjugation commutes with both S and T, so the new vectors are still C-fixed. On this space $S^2 = 1$, which justifies reversing the pair. The starting low basis must therefore be verified self-dual; self-duality of all high simples is not assumed. Finally, the trilinear Verlinde formula gives (4.42) for simples. Additivity proves the statement for nonnegative sums, with no assumption on duality within a class. □

The process terminates after at most the central rank many increases of the known span. A constant rank is a statement about this closure process, not by itself a proof that the unknown matrix is unique. If its orthogonal complement is explicitly verified to be T-stable, its remaining representation can instead be enumerated in the finite quotient. For any proved exponent bound N, the Chinese remainder theorem gives

$$SL_2(\mathbb{Z}/N\mathbb{Z}) \cong \prod_{p^a \| N} SL_2(\mathbb{Z}/p^a\mathbb{Z}).$$

The representation factors through this quotient even when the actual twist order is a proper divisor of N. Enumerating each finite group, its conjugacy classes and its class-algebra multiplication determines all irreducible characters: the simultaneous primitive eigenspaces of the class algebra give the central characters, and character orthogonality determines their degrees. Tensor products then give all irreducibles of the finite quotient. Comparing their exact T-spectra with the remaining spectrum is a finite integral multiplicity problem. Applied to the entire simple basis, this gives the necessary condition that its unweighted twist-multiplicity vector be a nonnegative integral sum of irreducible T-spectra. Applied to a proved invariant complement, it gives the corresponding condition on that complement. The action of $-I$ must also agree with an allowable duality permutation of the original simple basis. On each twist eigenspace the multiplicity of its $+1$ eigenvalue is at least that of its -1 eigenvalue: a dual pair contributes one of each, and a self-dual simple contributes only $+1$. The difference is at least the number of known self-dual simples of that twist. These inequalities apply to the full representation, or after subtracting the known contribution when a complement is used.

If the remaining T-eigenvalues are distinct and their representation is unique, all intertwiners preserving T are diagonal. Symmetry of S fixes their relative squares; when these ratios are nonzero it leaves only finitely many sign choices, modulo common scalar. Each resulting matrix is then checked for the full modular relations, the prescribed known rows, and every Verlinde coefficient. Passing these checks supplies necessary fusion and modular data; it does not construct an associator or establish categorifiability.

4.11 Subcategory projection and exact twist support

Information about the center of a proper fusion subcategory imposes constraints on individual restriction columns of the ambient center. It does not require a braiding on the ambient fusion category, or an extension of a chosen subcategory braiding.

Proposition 4.28. *Let $\mathcal{D} \subset \mathcal{C}$ be a full fusion subcategory of a pivotal fusion category, with the inherited pivotal structure. Let $P(V)$ be the sum of all simple summands of V belonging to $\mathcal{D}$. If (V, c) is an object of $Z(\mathcal{C})$, then $P(V)$ has a canonical $\mathcal{D}$-half-braiding. If V is central simple with twist λ, every simple constituent of $P(V)$ in $Z(\mathcal{D})$ has twist λ. Consequently, writing $F_{\mathcal{D}} : Z(\mathcal{D}) \to \mathcal{D}$ for the forgetful functor, the $\mathcal{D}$-part of a central restriction column belongs to the nonnegative integral span of the columns $[F_{\mathcal{D}}(W)]$ with $\theta_W = \lambda$.*

Proof. The complement of $\mathcal{D}$ is stable under tensoring on either side by objects of $\mathcal{D}$. Indeed, if a simple $S \notin \mathcal{D}$ and $X, T \in \mathcal{D}$ satisfy $\mathrm{Hom}(T, X \otimes S) \neq 0$, rigidity makes S a constituent of $X^* \otimes T \in \mathcal{D}$, a contradiction. Hence P is a projection of $\mathcal{D}$-bimodule categories. Put $W = P(V)$ and choose $i : W \to V$, $p : V \to W$ with $pi = 1_W$. For $X \in \mathcal{D}$ the maps

$$c_X^W = (1_X \otimes p) c_X (i \otimes 1_X)$$

are isomorphisms. Naturality and the half-braiding identity restrict to this summand, so they make W an object of $Z(\mathcal{D})$.

It remains to check the twist, rather than merely the half-braiding. Suppress associators and use the Drinfeld isomorphism

$$u_V = (\mathrm{ev}_V \otimes 1_{V^{**}})(c_{V^*} \otimes 1_{V^{**}})(1_V \otimes \mathrm{coev}_{V^*}) : V \longrightarrow V^{**}.$$

The duality identities give

$$(1_{V^*} \otimes p^{**}) \,\mathrm{coev}_{V^*} = (p^* \otimes 1_{W^{**}}) \,\mathrm{coev}_{W^*},$$
$$\mathrm{ev}_V (p^* \otimes 1_V) = \mathrm{ev}_W (1_{W^*} \otimes p).$$

Using naturality of c in its second argument between these two identities gives $p^{**} u_V i = u_W$. If j denotes the pivotal isomorphism, then $p^{**} j_V i = j_W$. With the convention $\theta = u^{-1} j$, central simplicity gives $u_V = \lambda^{-1} j_V$, and therefore $u_W = \lambda^{-1} j_W$. Thus $\theta_W = \lambda 1_W$. Decomposing W into central simples proves the integral-span assertion. The opposite twist convention gives the same conclusion when used in both centers. □

Proposition 4.29 (Modular subcategory projection)**.** *Suppose $\mathcal{C}$ is pseudo-unitary and $\mathcal{D} \subset \mathcal{C}$ is a full fusion subcategory, both with their positive spherical structures. Let $\Pi : Z(\mathcal{C}) \to Z(\mathcal{D})$ be the additive central projection constructed above, and put*

$$B_{WZ} = \dim \mathrm{Hom}_{Z(\mathcal{D})}(W, \Pi Z).$$

For the canonical normalized modular matrices of the two centers,

$$S_{\mathcal{D}} B = B S_{\mathcal{C}}, \qquad T_{\mathcal{D}} B = B T_{\mathcal{C}}, \qquad B_{W,\mathbf{1}} = \delta_{W,\mathbf{1}}. \tag{4.43}$$

In particular, each individual simple $W \in Z(\mathcal{D})$ has the mass and capacity bounds

$$\sum_Z B_{WZ} d_Z = \frac{D_{\mathcal{C}}}{D_{\mathcal{D}}} d_W, \qquad B_{WZ} d_Z \leq \frac{D_{\mathcal{C}}}{D_{\mathcal{D}}} d_W. \tag{4.44}$$

The second inequality concerns one ambient central simple and one subcategory central simple, even if several W have the same twist and forgetful image.

Proof. Regard $\mathcal{M} = \mathcal{C}$ as a $\mathcal{D}$–$\mathcal{C}$ bimodule by tensor multiplication. It is indecomposable, since its right $\mathcal{C}$ action alone generates every simple from the unit. The full center of this module category is a Lagrangian algebra

$$L \in Z(\mathcal{D}) \boxtimes Z(\mathcal{C})^{\mathrm{rev}};$$

see the module-category full-center construction [51, Section 6, Theorem 6.2] and the correspondence [52, Proposition 4.8]. We identify its multiplicities explicitly. For $W \in Z(\mathcal{D})$ and $Z \in Z(\mathcal{C})$, define bimodule endofunctors on $\mathcal{M}$ by

$$L_W(X) = F_{\mathcal{D}}(W) \otimes X, \qquad R_Z(X) = X \otimes F_{\mathcal{C}}(Z).$$

Their left and right module structures use the respective half-braidings. The defining full-center adjunction identifies the coefficient of $W \boxtimes Z^*$ with $\dim \mathrm{Nat}_{\mathcal{D}-\mathcal{C}}(L_W, R_Z)$. A right $\mathcal{C}$-module natural transformation is determined by its component at the unit, a map $f : F_{\mathcal{D}}(W) \to F_{\mathcal{C}}(Z)$. Its left $\mathcal{D}$-linearity says precisely

$$c^Z_X(f \otimes 1_X) = (1_X \otimes f)c^W_X \qquad (X \in \mathcal{D}).$$

Because its source belongs to $\mathcal{D}$, f factors uniquely through $P(F_{\mathcal{C}}(Z))$. The displayed condition then says that this factorization is a morphism $W \to \Pi Z$ in $Z(\mathcal{D})$. Consequently

$$L \cong \bigoplus_{W,Z} B_{WZ}\, W \boxtimes Z^*.$$

This fixes the dual convention: when $\mathcal{D} = \mathcal{C}$, B is the identity and L is the usual diagonal Lagrangian algebra.

The class of a Lagrangian algebra is fixed by the canonical normalized S and T. Indeed, the Morita description identifies it with induction of the unit in a center, where this is [36, Theorem 2.5, Proposition 2.8]; the positive spherical structures make the center equivalence ribbon. Let $C_{\mathcal{C}}$ be the duality permutation of the ambient center. The coefficient matrix of L is $BC_{\mathcal{C}}$, and the reversed center has matrix $\overline{S_{\mathcal{C}}} = S_{\mathcal{C}}^{-1}$ and inverse twists. Thus invariance is

$$S_{\mathcal{D}} B C_{\mathcal{C}} S_{\mathcal{C}}^{-1} = B C_{\mathcal{C}}, \qquad T_{\mathcal{D}} B C_{\mathcal{C}} T_{\mathcal{C}}^{-1} = B C_{\mathcal{C}}.$$

Duality commutes with both matrices, so these equalities give the first two identities in (4.43). Projection sends the ambient central unit to the subcategory central unit, proving the last identity. Finally compare the ambient-unit column of $S_{\mathcal{D}} B = B S_{\mathcal{C}}$ and use $S_{\mathcal{D},W0} = d_W/D_{\mathcal{D}}$ and $S_{\mathcal{C},Z0} = d_Z/D_{\mathcal{C}}$. Positivity of all summands proves the individual capacity inequality. □

Corollary 4.30 (Disjoint central projection rows)**.** *In the setting of Proposition 4.29, suppose every ambient central simple projects either to zero or to a single simple of $Z(\mathcal{D})$ with multiplicity one. Then, for a positive integer r,*

$$BB^{\mathsf{T}} = rI.$$

Thus $B^{\mathsf{T}}/\sqrt{r}$ is an isometric intertwining embedding of the entire canonical modular representation of $Z(\mathcal{D})$ into that of $Z(\mathcal{C})$. Its T-spectrum can be subtracted before the finite-quotient representation test. Any signed charge-conjugation bound for the complement is obtained by subtracting this known subrepresentation's signed trace from the corresponding full-space bound; it need not be nonnegative.

Proof. The hypothesis says each column of B is zero or a standard basis vector. Hence BB^{T} is diagonal, with entries equal to the numbers of simples in the projection fibers. Each entry is positive by (4.44). The normalized modular representations are unitary, so the intertwining identities imply that BB^{T} commutes with $S_{\mathcal{D}}$ and $T_{\mathcal{D}}$. Comparing its $W, \mathbf{1}$ entries and using

$S_{\mathcal{D},W\mathbf{1}} = d_W/D_{\mathcal{D}} > 0$ shows that all diagonal entries equal the same positive integer r. The embedding and the subtraction statements follow by taking adjoints. The orthogonal complement is a representation space, but is not required to have a simple basis permuted by charge conjugation. □

The disjoint-column hypothesis must be verified. The exact categorical control `research/round5/unification/subcategory_intertwiner_control.py` derives S, T and B from finite group character sums for $\mathrm{Vec}(C_2), \mathrm{Vec}(C_3) \subset \mathrm{Vec}(S_3)$. Both rectangular modular identities and all individual masses hold. The second projection has rank five although its target center has rank nine, illustrating why a general subcategory inclusion alone does not justify subtracting the entire target representation. Mutating one restriction multiplicity violates the checked S-identity.

For an Ising subcategory, the two twist-minus-one simples of its center are $\mathbf{1} \boxtimes g$ and $g \boxtimes \mathbf{1}$, each of dimension one. Hence every high ambient central simple whose restriction contains g has dimension at most $D_{\mathcal{C}}/4$. This is stronger than the combined twist-minus-one mass $D_{\mathcal{C}}/2$. For $\mathcal{D} = \mathrm{Vec}$, the same proposition says that the multiplicity vector of $I_{\mathcal{C}}(\mathbf{1})$ is fixed by S and T; thus the full modular representation contains a trivial constituent.

The projection also gives exact, twist-resolved dimension identities. Here and below spherical structures are inherited, so the centers carry their standard ribbon structures.

Corollary 4.31. *Let $\mathcal{C}$ be spherical and $\mathcal{D} \subset \mathcal{C}$ a full fusion subcategory. Put $D_{\mathcal{C}} = \dim(\mathcal{C})$ and $D_{\mathcal{D}} = \dim(\mathcal{D})$. For a simple $X \in \mathcal{D}$ and a root of unity λ,*

$$\sum_{\substack{Z \in \operatorname{Irr} Z(\mathcal{C}) \\ \theta_Z = \lambda}} \dim(Z)[F_{\mathcal{C}}(Z) : X] = \frac{D_{\mathcal{C}}}{D_{\mathcal{D}}} \sum_{\substack{W \in \operatorname{Irr} Z(\mathcal{D}) \\ \theta_W = \lambda}} \dim(W)[F_{\mathcal{D}}(W) : X].$$

An empty right-hand side is zero. In the pseudo-unitary case all weights are positive and dimensions can be replaced by Frobenius–Perron dimensions.

Proof. Choose N divisible by the orders of all twists in $Z(\mathcal{D})$. Proposition 4.28 shows that every central simple contributing to the left-hand side has twist in μ_N. The indicator formula [30] gives, for $1 \le n < N$,

$$D_{\mathcal{C}} \nu_n(X) = \sum_Z \theta_Z^n \dim(Z)[F_{\mathcal{C}}(Z) : X].$$

The same formula holds in $\mathcal{D}$, and the indicators agree: their defining pivotal rotations act on the same spaces $\mathrm{Hom}(1, X^{\otimes n})$. For $n = 0$, the corresponding equality is $\dim I_{\mathcal{C}}(X) = D_{\mathcal{C}} \dim(X)$, and similarly in $\mathcal{D}$. Fourier inversion on μ_N proves the assertion. □

Corollary 4.32. *In the pseudo-unitary setting the following subcategory constraints hold.*

1. *For a Fibonacci subcategory with basis $(1, a)$, $a^2 = 1+a$, write $\varphi = \mathrm{FPdim}(a) = (1+\sqrt{5})/2$. A nonzero projected column (r, s) has either twist 1 and $r \ge s$, or $r = 0$ and twist ζ_5^2 or ζ_5^{-2}. For each of the latter twists, the a-mass in Corollary 4.31 is $D_{\mathcal{C}}\varphi/(\varphi+2)$; the twist-one a-mass is $D_{\mathcal{C}}\varphi^2/(\varphi+2)$.*
2. *For an Ising subcategory with basis $(1, g, \sigma)$, a nonzero projected column (r, s, t) has one of the following forms: twist 1 with $t = 0$ and $r \ge s$; twist -1 with $r = t = 0$; or one of four Ising twists $\{z, -z, z^{-1}, -z^{-1}\}$ with $r = s = 0$, where z is an appropriate primitive sixteenth root. The g-masses at 1 and -1 are each $D_{\mathcal{C}}/2$. The σ-mass at each of its four twists is $D_{\mathcal{C}}\sqrt{2}/4$.*

In particular, a high column, whose unit coordinate is zero, cannot combine g and σ in its Ising projection, or contain either a, g or σ at twist one in the respective cases.

Proof. The rank-two and Ising classifications [34, 15] provide nondegenerate braidings on these subcategories. For either such category $\mathcal{D}$, its center is $\mathcal{D} \boxtimes \mathcal{D}^{\mathrm{rev}}$ and the restriction of (U, V) is $U \otimes V$. For Fibonacci the four columns are

$$(1,0), \quad (0,1), \quad (0,1), \quad (1,1),$$

with twists $1, z, z^{-1}, 1$. Positivity gives the usual Fibonacci modular S-matrix; balancing gives $z^2 + \varphi z + 1 = 0$, hence $z = \zeta_5^{\pm 2}$. The center dimensions are respectively $1, \varphi, \varphi, \varphi^2$. For Ising, the twist-one columns are $(1,0,0)$ twice and $(1,1,0)$ once; the twist--1 columns are $(0,1,0)$ twice; and each of $z, -z, z^{-1}, -z^{-1}$ has the column $(0,0,1)$ once. Their dimensions are $1, 2, \sqrt{2}$ as determined by these columns. Taking their additive spans and applying the preceding corollary gives every assertion. All inherited Ising spherical choices must be retained. If the second indicator of σ is fixed, the identity $z^2 + z^{-2} = \nu_2(\sigma)\sqrt{2}$ selects the appropriate set of four twists. □

Computational use. The criteria first restrict the allowed high-column/twist pairs and the twist-one low columns. They then add the exact mass identities to the integer linear equations. When using normalized relative twist traces, one projects the identities to the chosen field, combining coincident trace types; one must not confuse a projected trace with an actual twist. These are necessary conditions only. General exclusions still require the separately checked general-to-pseudo-unitary reduction.

4.12 Fibonacci projection over a biquadratic character field

The projection obstruction of Proposition 4.28 can be combined with full-character induction without computing in an unnecessarily large cyclotomic field. We record the exact reduction used here.

Proposition 4.33 (Projected twist traces)**.** *Let $K \subseteq L = \mathbb{Q}(\zeta_F)$ be a Galois field containing all the characters of a commutative fusion ring, and suppose that* $\mathrm{Gal}(K/\mathbb{Q})$ *has exponent at most two. Let P be the allowed prime support of the Frobenius–Schur exponent and $h \geq 2$. For a root of unity θ, put*

$$T_K(\theta) = \left(\frac{\mathrm{Tr}_{K(\theta)/K}(\theta^j)}{[K(\theta) : K]} \right)_{1 \leq j \leq h}.$$

Every such vector with root order supported on P is either zero or is obtained from a root whose order divides

$$B = \prod_{p \in P} p^{\max(1, v_p(F)) + \lfloor \log_p h \rfloor}.$$

These vectors are computed exactly by first taking normalized trace to L and then averaging over $\mathrm{Gal}(L/K)$*. The induction Gram, rotation, and low-character equations remain necessary after this projection. Their Galois multiplicity equalities may be imposed separately for each projected twist vector.*

Proof. The tower identity for normalized traces gives the stated computation. The conductor bound follows from Lemma 4.12: if an order has a prime exponent larger than the displayed bound, every power $1 \leq j \leq h$ still has a nontrivial final cyclotomic layer over L, and its relative trace vanishes. Averaging once more preserves zero. Otherwise the order divides B.

All coefficients other than roots of unity in the induction and low-character equations lie in K. The normalized trace is K-linear, so applying it preserves these equations. The variables counting central simple objects and cyclic eigenspaces remain nonnegative; identifying equal projected vectors merely sums these variables and thus weakens the necessary system.

For Galois balance, a modular Galois automorphism acts on joint character eigenvalues by σ and on the canonical twist by σ^2. The latter automorphism restricts to the identity on K, by the exponent-two hypothesis. Normalized trace to K is invariant under this automorphism. Consequently the projected twist vector is unchanged along the corresponding permutation of central simples. This proves the separate multiplicity equalities. No integrality of the projected trace, or of a solution to the relaxed linear system, is asserted. □

For the ring $R_{6,8,3664}$, the Fibonacci basis object is b_1: $b_1^2 = 1 + b_1$. Its Frobenius–Perron character field and global dimension are

$$K = \mathbb{Q}(\sqrt{5}, \sqrt{6}), \qquad D = 300 + 132\sqrt{5} + 120\sqrt{6} + 54\sqrt{30}.$$

The formal codegree polynomials are

$$t^2 - 15t + 45, \qquad t^4 - 1200t^3 + 18000t^2 - 86400t + 129600,$$

each with multiplicity one. Thus one may use $F = 120$, $h = 3$, and $P = \{2, 3, 5\}$. The bound is $B = 720$, and there are exactly 43 distinct projected vectors. Arithmetic takes place in degree four, whereas the ambient cyclotomic field has degree 32.

By Corollary 4.32, each low column c necessarily satisfies $c_1 \leq c_0$. For a high column, $c_0 = 0$; if $c_1 > 0$, its twist must be ζ_5^2 or ζ_5^{-2}. The two roots have the same normalized trace to the totally real field K, so this condition leaves one projected twist vector for such a column. Columns with $c_1 = 0$ retain the full necessary trace domain.

Theorem 4.34. *The fusion ring $R_{6,8,3664}$ has no categorification over $\mathbb{C}$.*

Proof. Unrestricted signed sphericalization gives 768 signed quotients and 24 sign orbits. One orbit is gauge equivalent to the original ring; the other orbits are eliminated by 36 exact dimension-character checks. The unique surviving nonzero dimension-character component is the Frobenius–Perron Galois orbit. The general-to-pseudo-unitary reduction therefore applies.

For the pseudo-unitary case, exhaustive low induction enumeration gives 13 frames. Four fail an ordinary moment inequality; three fail the Fibonacci twist-one projection inequality; four have no compatible low-character labelling. The remaining two frames each have one labelling and one allowed second-indicator tuple. Their necessary systems have, respectively, 708 equations in 2184 variables and 663 equations in 1792 variables. Both systems have certified rational separating vectors. The verifier reconstructs every field, frame, character assignment, indicator branch, and rational equation from the full input tensor, and checks $A^T y \geq 0$, $b^T y < 0$ exactly. The complete certificate is `research/round4/m8/R6_8_3664_certificate.json`, replayed by `software/round4_m8_subfield.py`. □

The criterion was screened on all ten inputs to this continuation and all 54 known realizations. The target is the only remaining input containing a Fibonacci basis subcategory. Eight realized rings contain one; three lie in the implemented full-character subfield domain and survive the necessary system, while the other five are explicitly outside that domain. The remaining 46 controls have no Fibonacci basis subcategory. The structural census screen finds 443 Fibonacci subring cases among all 16612 tensors; it records eligibility only, not an additional 443 exclusion replays. Normal and optimized Python replays of the new general certificate disable both numerical optimizers. Mutation checks separately reject omitted branches, corrupted domain data, false projection witnesses, and altered separating vectors.

4.13 Ising twist masses at multiplicity eight

The two multiplicity-eight rings $R_{6,8,311}$ and $R_{6,8,1660}$ have formal codegrees with characteristic polynomial

$$(t - 4)^2(t^4 - 576t^3 + 11520t^2 - 73728t + 147456).$$

They contain Ising basis subcategories, with labels $(g, \sigma) = (b_4, b_3)$ and (b_1, b_4), respectively. Their different duality involutions remain part of the full tensor input; the common polynomial does not identify the rings.

Here the entire character field embeds in $\mathbb{Q}(\zeta_{24})$. A pseudo-unitary Ising category has twists $1, -1, \xi$, where ξ is a primitive sixteenth root. Its center has four σ-restriction objects, with twists $\xi, -\xi, \xi^{-1}, -\xi^{-1}$. On taking normalized trace to $\mathbb{Q}(\zeta_{24})$, their height-three vectors become

$$(0, \eta, 0), \qquad (0, \eta^{-1}, 0), \qquad \eta = \xi^2,$$

with each vector occurring for two of the four objects. The sign $\epsilon = \nu_2(\sigma)$ determines the unordered pair:

$$\eta + \eta^{-1} = \epsilon\sqrt{2}.$$

Both possible signs are retained whenever permitted by the second indicator conditions.

Corollaries 4.32 and 4.31 now give the following simultaneous constraints. Every low column satisfies $c_\sigma = 0$ and $c_g \le c_0$; moreover

$$\sum_{\text{low } Z} d_Z c_g(Z) = D/2.$$

A high column with $c_g > 0$ has twist -1 and $c_\sigma = 0$. A high column with $c_\sigma > 0$ has $c_g = 0$ and one of the two displayed projected vectors. For each of those vectors the exact dimension mass is $D\sqrt{2}/2$. These equations are imposed together with the complete induction Gram, full-character reciprocity, Galois balance, and cyclic rotation equations. They do not assume that the Ising braiding extends to the ambient category.

Theorem 4.35. *The fusion ring $R_{6,8,311}$ has no categorification over $\mathbb{C}$.*

Proof. The unrestricted signed reduction has 96 quotients and eight sign orbits. One orbit is gauge equivalent to the original ring, and 13 exact dimension-character checks reject all other possible categorical dimension components. The surviving component is the Frobenius–Perron Galois orbit, so it suffices to exclude a pseudo-unitary categorification.

The complete low induction domain has two frames. Each has two character labellings and four allowed second-indicator tuples. With $F = 24$ and height three, the complete order bound is 144 and there are 53 normalized trace vectors. The first frame gives systems of 387 equations in 802 variables; the second gives 1017 equations in 1896 variables. Each of the 16 branches has an exact rational separating vector. The direct certificate is `research/round4/m8/R6_8_311_certificate.json`; the full-tensor verifier is `software/round4_m8_ising.py`. □

The same criterion is implemented uniformly and screened on all ten frontier inputs and all 54 realized controls. Three frontier inputs contain an Ising basis subcategory: the two rings just discussed and $R_{6,4,91}$, treated in its own campaign. Six realized rings contain one. Five are outside the present full-character-field implementation, and the applicable control $R_{6,2,94}$ survives the necessary system.

For $R_{6,8,1660}$, the unrestricted signed reduction has 640 quotients in 24 sign orbits and 58 nongauge dimension-character rejections. It proves sphericality and a pseudo-unitary Galois reduction. Morita reconstruction over $1 + g$ returns the same based ring, so it supplies no transfer to the excluded non-self-dual partner. The complete current profile enumeration and actual-root twist constraints are given in the frontier analysis below, replacing the earlier bounded searches.

4.14 Twist-resolved subcategory constraints at higher multiplicity

The projection in Proposition 4.28 can be used before enumerating a complete modular datum. In particular, it couples a restriction column to its twist; checking the column and the twist separately loses information. We record two explicit source tables which make this necessary condition fully effective.

The half-E_6 subcategory. Suppose that the based subring on $\{1, g, x\}$ satisfies $g^2 = 1$, $gx = xg = x$, and $x^2 = 1 + g + 2x$. Put $d = 1 + \sqrt{3}$ and $D_0 = 2 + d^2 = 6 + 2\sqrt{3}$. Hagge–Hong classify its four categorifications [45, Theorem 1 and Sections 4.7–4.8]. The relations imply that every tensor automorphism of the identity is trivial: its scalar on x satisfies $\eta_x^2 = \eta_x$, and then $\eta_g = 1$. Hence a pivotal structure, when present, is unique. Two Galois conjugates have $\dim(x) = 1 - \sqrt{3} < 0$ and cannot be pseudo-unitary. The two remaining cases are complex conjugate. Their center restriction columns and twists are given by Hong–Rowell–Wang [46, Section 2.1]. Writing $\zeta = \exp(2\pi i/12)$, one table is

Z	1	Y	X_1	X_2	X_3	X_4	X_5	U	V	W
$F(Z)$	1	g	x	x	x	x	x	$1+x$	$g+x$	$1+g+x$
θ_Z	1	-1	ζ^{-3}	ζ^{5}	ζ^{5}	ζ^{2}	ζ^{-4}	1	-1	$1.$

The other table conjugates every twist. Complex conjugation of the entire putative pseudo-unitary category preserves its based ring and its positive dimensions, and exchanges these two subcategory choices. Thus exclusion for the displayed choice excludes both choices; this is a reduction of existence, not an omission of a categorical branch.

For a high induction column c, the unit coordinate is zero. Consequently, $c_g > 0$ forces $\theta = -1$ and $c_g \geq c_x$; if $c_g = 0 < c_x$, the twist must be one of the four displayed X-twists. A low column has twist one, and its coordinates must satisfy $0 \leq c_g \leq c_x \leq c_1$. These statements concern all structure constants, with the subring located by its multiplication table, rather than by census labels.

The twist masses give additional equalities. If the ambient category has FP dimension D, Corollary 4.31 implies

$$\sum_{Z:\,\theta_Z=\lambda} c_{x,Z}\,\mathrm{FPdim}(Z) = \frac{D}{D_0} M_\lambda,$$

where the successive values of M_λ at $\lambda = \zeta^{-3}, \zeta^{5}, \zeta^{2}, \zeta^{-4}, -1, 1$ are

$$d, \quad 2d, \quad d, \quad d, \quad d+1, \quad 2d+3.$$

Known low columns are subtracted from the right-hand sides. The existence reduction by conjugating the entire category, given above, covers the conjugate source table as well. The implementation combines it with the full induction Gram equations, the integrality of joint character spectra, and equality of multiplicities along Galois orbits. Optional refinements add the low Verlinde coefficients and restriction products.

Cyclic pointed subcategories of order three. For a generator g of a pointed C_3 subcategory, all associator classes are represented by $s \in \{0, 1, 2\}$. A standard cocycle representative is

$$\omega_s(a, b, c) = \exp\!\left(\frac{2\pi i s a}{3}\left\lfloor \frac{b+c}{3} \right\rfloor\right), \qquad 0 \leq a, b, c < 3.$$

Solving the one-dimensional half-braiding equations gives, over the underlying object g^a, the three twists

$$\exp\!\left(2\pi i\left(\frac{sa^2}{9} + \frac{ab}{3}\right)\right), \qquad b \in \{0, 1, 2\}.$$

Thus a central object whose restriction contains g or g^2 has $\theta^3 = \exp(2\pi is/3)$. For the ambient field $\mathbb{Q}(\zeta_{24})$, the three height-three relative-trace types when $s = 0$ are $(\lambda, \lambda^2, 1)$, $\lambda^3 = 1$. For $s = 1, 2$, the three possible ninth roots have the same normalized relative-trace type $(0, 0, \exp(2\pi is/3))$. All three associator classes must be retained. No triviality assumption on the pointed associator is used.

Exact arithmetic and scope. The supplementary implementations retain rational equations and integer separating rays. For the half-E_6 input with FP field $\mathbb{Q}(\sqrt{3}, \sqrt{5})$, normalized traces from $\mathbb{Q}(\zeta_{60})$ to the real FP field reduce the height-three twist domain from 131 to 40 types. A field pullback is accepted only after exact reconstruction of every coordinate. The Galois group of this real field has exponent two, so the square action on these relative twist traces is trivial; joint-spectrum multiplicity equalities can therefore be imposed separately for each relative-trace type. Additional negative-stabilizer rotation conditions are imposed only after exhaustive reconstruction of the relevant involution module. Surviving systems remain necessary relaxations and do not provide categorifications or modular data.

Certified applications and the remaining multiplicity-five input. For $R_{6,6,49}$ the complete low domain has 353 frames, of which only one remains after the ordinary and character-assignment checks. Its four assignments, three pointed associator classes, and two second-indicator choices give 24 exact separating rays. The established two-FP-value reduction gives unrestricted categorical scope. For $R_{6,6,1035}$ there are 12 low frames, of which two admit a character assignment. Each has two second-indicator choices; all four full cyclotomic projection systems have exact separating rays. The unrestricted signed reduction exhausts 448 signed tables in 14 sign orbits and reduces every possible categorification to the pseudo-unitary problem. These two exclusions complete the exact multiplicity-six stratum. The final half-E_6 certificate uses the full conductor-60 system; its validity does not depend on the exploratory real-field MILP infeasibility results or on additional Clifford rotation constraints.

The multiplicity-five input $R_{6,5,1204}$ remains unresolved. Its 39 low frames admit eight character assignments across two frames. The height-three full-character system, and the subsequent low-Verlinde and restriction-product refinement, exclude the two assignments of the first frame but retain a necessary relaxation on the second. Neither numerical feasibility nor the failure of particular Morita model searches is a categorification or a general exclusion.

4.15 Integer-square linearization and central-invertible transport

The complex low-product and fourth-moment identities are special cases of Section 4.8. They retain dual indices even when the original ring is not self-dual. The following linear consequences are useful when a complete integral Gram decomposition is too large. For $P = A_a A_b$, put $q_{ab,v} = \sum_{F(Z)=v}[P : Z]$, let x_v count that restriction type, and let $B_{ab,v}$ be the coordinatewise restriction upper bound from the unified theorem. Write $Q_{ab} = [P : P]$.

Introduce $z_{ab,v} = \sum_{F(Z)=v}(N^Z_{A_a,A_b})^2$ and corresponding square variables for the known low coefficients. The integer-square inequalities in Section 4.8 apply coefficientwise. Summing them over a restriction type gives

$$(2k+1)q_{ab,v} - k(k+1)x_v \le z_{ab,v} \le B_{ab,v}q_{ab,v}. \tag{4.45}$$

For a low simple, replace x_v by 1 and $q_{ab,v}$ by its individual Verlinde multiplicity. Equation (4.34) becomes the sum of these square variables, together with the unit contribution δ_{a,b^*}. Any finite selection of the lower inequalities is valid; the implementation supports both all $0 \le k < B$ and a stated cutoff. This gives an exact linear strengthening, with integrality retained in bounded discovery experiments. Failure of a mixed-integer optimizer to finish is recorded only as an incomplete search.

4.15.1 Transport by central invertibles

Suppose the low frame has exactly one column equal to e_0. A central invertible restricting to the unit must occur in $I(\mathbf{1})$, so the forgetful homomorphism on invertible objects has trivial kernel. Its image is therefore a subgroup K of the based-ring invertible group, and each $g \in K$ has exactly one invertible lift. For any such lift L, tensoring by L permutes all central simples and sends a restriction vector v to the coefficient vector of gv. In particular, the total counts of these restriction types agree, including low objects.

There is a stronger constraint when $g^2 = 1$ and $\nu_2(g) = -1$. The lift has order two and twist $\theta_L = \pm i$. Its monodromy with a simple Z is a scalar in $\{1, -1\}$: monodromy with invertibles is multiplicative, and $L^2 = \mathbf{1}$. The balancing identity therefore gives

$$\theta_{LZ} = \pm i \theta_Z.$$

In particular $LZ \ncong Z$, so this permutation is a fixed-point-free involution. When the ambient cyclotomic field contains i, normalized relative traces transform by $(t_j)_j \mapsto ((\pm i)^j t_j)_j$. One can encode the resulting matching with nonnegative integral edge variables between allowed pairs of restriction/trace types. An edge joining distinct types contributes one to each marginal; a loop in the *type* graph contributes two, because it still represents two distinct simples. This last convention is necessary when relative trace loses the underlying twist information. These matching equations are necessary only; for several lifts they do not impose all simultaneous group-action relations.

4.16 Exhaustive induction profiles at the quartic frontier

The two quartic tensors admit a substantially smaller finite search space than the unrestricted central systems suggest. We enumerate complete induction multiplicity profiles before searching for additional twists or fusion coefficients. A profile specifies the multiplicity of every possible restriction column; it does not specify a fusion category.

Fix a low induction frame L and let $v_1, \dots, v_t$ be the remaining admissible high restriction columns. If m_j is the number of central simples with column v_j, then

$$\sum_{j=1}^{t} m_j v_j v_j^{\mathsf{T}} = G, \qquad m_j \in \mathbb{Z}_{\geq 0}, \qquad m_j \leq \min_{i:(v_j)_i > 0} \left\lfloor \frac{G_{ii}}{(v_j)_i^2} \right\rfloor,$$

where G is the induction Gram matrix after subtracting LL^{T}. The joint character spectrum supplies additional linear equations equating Galois multiplicities. All these equations have integer coefficients.

Proposition 4.36 (Exact bounded affine enumeration)**.** *For an integer system $Am = b$ with prescribed finite bounds $0 \leq m_j \leq u_j$, row reduction over $\mathbb{Q}$, followed by the bounded search below, enumerates every integral solution exactly once.*

Proof. Choose the nonpivot coordinates as free variables. Each pivot coordinate is an affine rational function of them. Clear its denominator separately. Enumerate the free coordinates in any fixed order within their given integer bounds. At a partial assignment, the unused coordinates give an explicit lower and upper bound for each pivot numerator: for a coefficient a and remaining variable in $[0, u]$, its contribution belongs to $[\min(0, au), \max(0, au)]$. Discard a node only if one such interval misses the allowed interval for that numerator. At a leaf check divisibility by each cleared denominator and the original equations. Every integral solution follows a unique unpruned path; every retained leaf satisfies the original system and all bounds. A node or time budget interrupts this argument and is therefore recorded as incomplete, never as an exclusion. □

There is a useful inexpensive sieve after enumeration. Write D for the global dimension of the original pseudo-unitary category and d_Z for a central dimension. If $A_1, \dots, A_s$ are the nonunit low central simples, put $\lambda_a(Z) = S_{A_a,Z}/S_{\mathbf{1},Z}$, with the normalized modular matrix S. The character and restriction data determine these numbers. For every word $w = (a_1, \dots, a_k)$, simultaneous diagonalization of the fusion matrices gives

$$\dim \mathrm{Hom}(\mathbf{1}, A_{a_1} \otimes \cdots \otimes A_{a_k}) = \sum_Z \frac{d_Z^2}{D^2} \prod_{j=1}^{k} \lambda_{a_j}(Z) \in \mathbb{Z}_{\geq 0}. \tag{4.46}$$

No duality assumption on the A_a is needed. In degree three these are precisely the low fusion multiplicities after the appropriate dual label is inserted. Their restrictions must leave a nonnegative remainder in every low product. Both tests are exact linear evaluations of a fixed profile. We subsequently check all nonunit low words through degree six.

Proposition 4.37 (Intermediate quartic induction profiles)**.** *Before full fusion completion, each of the two surviving low-character labellings has the following complete domain of necessary induction profiles:*

tensor	*Gram/Galois*	*degree three*	*words through six*	*refined*	*center ranks*
*R*6_4_22	93	3	2	2	30, 36
*R*6_4_91	524	6	6	2	36, 42

*Any categorification of R*6_4_22 *has exactly one central simple whose restriction is its nontrivial invertible object. For R*6_4_91*, both simples g, σ of its Ising subcategory have exactly one central lift. These lifts generate an Ising fusion subcategory of its center. Theorems 7.3 and 7.1 refine these intermediate domains to center ranks* 36 *and* 42*, respectively.*

Proof. The general sphericalization and Galois-to-pseudo-unitary reductions are replayed in the accompanying records, so it suffices to impose the pseudo-unitary necessary conditions. All low frames and all second indicator choices are covered. The previous exact modular trace rays remove the other frames and indicator branches; their rational dual inequalities are replayed here, rather than inferred from numerical solver status.

For the remaining low frame, Proposition 4.36 visits respectively 20791 and 4295 exact search nodes, yielding the first column of counts. Equation (4.46) and restriction products give the next two columns. One of the three intermediate profiles for *R*6_4_22 has multiplicity 9/2 for the word $(1, 1, 1, 1)$ and is discarded.

We then impose the pointed or Ising subcategory projection, the coupled Galois–twist equations, and the low fusion and restriction equations. For *R*6_4_91, four of the six profiles fail an exact congruence modulo 2, on both labellings. Each certificate is a specified linear combination of integer-count equations: its coefficient vector vanishes modulo 2, while its right side is 1. The two remaining profiles have exact nonnegative witnesses with the required integral multiplicities. For *R*6_4_22, the rank-36 profile has such a witness. At this stage the rank-30 profile is retained without a feasibility assertion. The joint word–twist balancing criterion of Proposition 4.21 subsequently excludes it for both character labellings. Likewise the centralizer analysis in Section 7.5 excludes the rank-36 profile of *R*6_4_91. The displayed table records the complete intermediate domains used by those stronger proofs.

Finally, inspection of the exhaustively verified profiles gives multiplicity one for each stated pure restriction column. In the Ising case write the lifts as G, Σ. The unique pure unit column is the central unit. Uniqueness makes Σ self-dual, so rigidity supplies the unit summand of Σ^2; the remaining forgotten object is g. Duality and restriction therefore give $G^2 = \mathbf{1}$, $G\Sigma = \Sigma$ and $\Sigma^2 = \mathbf{1} + G$, proving the last assertion. These statements descend from a pseudo-unitary Galois conjugate because the fusion coefficients and restriction multiplicities are unchanged. □

The canonical records are `R6_4_22_frontier_certificate.json` and `R6_4_91_frontier_certificate.json`, in `research/round4/quartic_front/`. The modules `round4_quartic_profiles.py` and `round4_quartic_twists.py` reconstruct the finite domains and check all equalities, congruences, rays and retained witnesses without a numerical optimizer. The current stronger outcomes and their exact verification are given in Section 7.5.

4.17 Odd cyclotomic orbit sizes for induction columns

The relative twist traces can be strengthened by retaining their orbit sizes. This refinement is a necessary condition for a pseudo-unitary categorification; a feasible system is not a construction of a fusion category.

Proposition 4.38. *Let $\mathcal{C}$ be pseudo-unitary with commutative Grothendieck ring, and let $K = \mathbb{Q}(\zeta_F)$ contain its complete character table and its Frobenius–Perron dimensions. Assume that F is odd and that every prime divisor of the Frobenius–Schur exponent divides F. Fix an induction restriction column v, and let m_v be the number of simple objects of $\mathcal{Z}(\mathcal{C})$ having this column. For a twist of order n, put*

$$e(n) = [\mathbb{Q}(\zeta_{\mathrm{lcm}(F,n)}) : K] = \prod_{p \mid F} p^{\max(0, v_p(n) - v_p(F))}.$$

The multiset of twists over v is a disjoint union of full relative Galois orbits. In particular an orbit of twists of order n contributes $e(n)$ objects, so $e(n) \le m_v$. Its contribution to the first h twist moments is

$$e(n)\left(\frac{\mathrm{Tr}_{K(\zeta_n)/K}(\zeta_n^a)}{e(n)}, \dots, \frac{\mathrm{Tr}_{K(\zeta_n)/K}(\zeta_n^{ah})}{e(n)}\right).$$

Proof. Write $D = \mathrm{FPdim}(\mathcal{C})$ and let A_χ denote the multiplicity-one simple summand of $I(\mathbf{1})$ associated to a character χ, of formal codegree f_χ. With normalized S-matrix,

$$S_{A_\chi, Z} = \frac{\chi(FZ)}{f_\chi}, \qquad S_{\mathbf{1}, Z} = \frac{\mathrm{FPdim}(FZ)}{D}.$$

Thus all these entries belong to K. A modular Galois automorphism fixing K fixes the unit row and every A_χ row: its possible sign is positive, as seen in the positive unit column, and distinct simple objects have distinct S-matrix rows. It also fixes the entries in these rows when permuting any column Z. Invertibility of the ordinary character table therefore shows that FZ is unchanged. This proves invariance of each restriction-column class.

For the canonical ribbon structure on a Drinfeld center, the modular Galois permutation transforms a twist by $\theta \mapsto \sigma^2(\theta)$, as in the modular Galois symmetry of [5]; there is no anomaly factor. The group $\mathrm{Gal}(K(\zeta_n)/K)$ has the displayed order $e(n)$, which is odd, and the square map is an automorphism of this abelian group. Hence its square action on primitive nth roots has the same orbits as its ordinary Galois action. Each such orbit has $e(n)$ elements, with equal multiplicities in the fixed restriction-column class. Summing the powers of these roots proves the formula. □

Here is the finite form used in the supplementary implementation. If G is the residual induction Gram matrix after the low columns have been removed, a nonzero high column v has multiplicity at most

$$B_v = \min_{i : v_i \neq 0} \left\lfloor \frac{G_{ii}}{v_i^2} \right\rfloor.$$

Consequently only orders n with $e(n) \le B_v$ are relevant. They are exhaustively enumerated by $0 \le v_p(n) \le v_p(F) + \lfloor \log_p B_v \rfloor$. For each normalized trace type, the count must belong to

the additive semigroup generated by the corresponding orbit sizes. Removing a size which is a positive multiple of a smaller size does not change this semigroup. The code uses a separate nonnegative integer variable for each retained size, multiplying the corresponding moment column by that size. No divisibility assertion is made when the relative Galois group has even order.

4.18 The conductor-fifteen multiplicity-four frontier

For $R^{(6)}_{4,629}$, the formal-codegree polynomial is

$$(t-5)^2(t^4-360t^3+5850t^2-30375t+50625).$$

The unrestricted signed enumeration has exactly 32 tables in one sign orbit, all gauge-equivalent to the original table. The exact dimension character test leaves only the Frobenius–Perron Galois orbit. Thus every categorification is spherical and has a pseudo-unitary Galois conjugate; the following necessary-data reduction therefore applies to general categorifications as well.

There are seven low induction frames. Joint-spectrum integrality eliminates six. The remaining frame has two possible character labellings and 85 admissible high restriction vectors. Applying Proposition 4.36 to its Gram and Galois equations gives exactly 784 nonnegative integral column-multiplicity profiles. Two independent implementations agree: the native enumerator visits 22 641 594 nodes, while the optimizer-free Python replay reconstructs the same complete set. The low-word tests through degree six, including nonnegativity of the restricted low-product remainders, leave exactly five profiles for each character labelling, of ranks

$$24, \quad 30, \quad 32, \quad 35, \quad 38.$$

These are finite necessary induction data, not fusion categories or modular data. In particular, the ring remains unresolved.

The height-six cyclotomic system, strengthened by Proposition 4.38, eliminates all five profiles for one labelling and the rank-24 and rank-32 profiles for the other. Each elimination is witnessed by an exact integral Farkas functional. At this intermediate stage the retained induction-profile ranks are 30, 35, and 38, with one character labelling. The rank-35 and rank-38 systems have explicit nonnegative integral witnesses. A stronger rank-35 witness also satisfies the low-Verlinde and restriction-product constraints. These witnesses verify only the stated necessary equations. The subsequent exact fusion and modular-data completion in the current frontier analysis eliminates ranks 30 and 38 and determines the remaining rank-35 modular candidates; it still does not assert categorical existence.

5 Module categories and Morita constraints

These criteria use module categories forced by algebra objects and complement the central-induction system. They do not impose a pivotal structure on the original category. Field-automorphism and tensor-sign reductions are kept with the arithmetic reductions in Section 3.

5.1 An involution-module obstruction

Theorem 5.1 (Antiinvariant module representation)**.** *Let R be a commutative fusion ring and let $g \neq 1$ be an invertible basis element with $g^2 = 1$. Suppose the set $J = \{x : gx = x\}$ is nonempty. If R has a complex fusion categorification, there are integer matrices $T_i \in M_{|J|}(\mathbb{Z})$, indexed by all basis elements, satisfying*

$$T_0 = I, \qquad T_iT_j = \sum_k N_{ij}^k T_k, \qquad T_{i^*} = T_i^{\mathsf{T}}, \tag{5.1}$$

$$|(T_i)_{xy}| \le N_{ix}^{y}, \qquad (T_i)_{xy} \equiv N_{ix}^{y} \pmod 2 \qquad (x, y \in J). \tag{5.2}$$

Failure of this finite system is a general categorification obstruction.

Proof. Let $\mathcal{C}$ be a hypothetical categorification. The additive category generated by a fixed simple x is a rank-one module category over the pointed subcategory generated by g. Choose isomorphisms $g \otimes x \simeq x$. The scalar module-associativity equation says that the pointed associator three-cocycle is a coboundary. Thus that subcategory is tensor equivalent to Vec_{C_2}, and $A = 1 \oplus g$ admits the regular connected separable algebra structure. This is a use of module associativity, not an assumption that g has a central lift. Standard separable-algebra module theory is as in [15, Chapter 7].

Let $\mathcal{M} = \mathcal{C}_A$ be the category of right modules. Its free and forgetful functors I_A, F_A satisfy $F_A I_A(X) = X \oplus (X \otimes g)$. Adjunction shows that a two-element g-orbit gives one simple module, whose restriction is the sum of the two simples in the orbit. A fixed simple x gives two nonisomorphic simples M_x^+, M_x^-, each restricting to x. Indeed, $\dim \mathrm{End}(I_A(x)) = 2$, so the squared multiplicities in its semisimple decomposition sum to two. The mixed Gram entries exclude constituents from other orbits. Every simple module occurs in a free module because A is separable.

The algebra automorphism α acting as $+1$ on 1 and -1 on g induces an involution Q of $\mathcal{M}$ by twisting the right action. It fixes the module attached to a two-element orbit and interchanges M_x^+ and M_x^-. The latter assertion follows directly because the nonzero action map $x \otimes g \to x$ changes sign, whereas an endomorphism of the simple underlying object is scalar. This twisting functor commutes with the left action of $\mathcal{C}$.

Consequently the (-1)-eigenspace of Q on $K_0(\mathcal{M}) \otimes \mathbb{Q}$ is a representation of R, with basis $e_x = [M_x^+] - [M_x^-]$, $x \in J$. Between two fixed pairs the action matrix has a block $\left(\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right)$. Forgetting modules gives $a + b = N_{ix}^{y}$, while its action on the antiinvariant basis has coefficient $a - b$. Nonnegativity and integrality of a, b give (5.2). The module associator gives the multiplication identity in (5.1). Finally module Frobenius reciprocity makes the action of i^* the transpose of the action of i. All e_x have the same squared length two, so the restricted matrices obey the same transpose identity. No spherical or unitary structure has been used. □

Proposition 5.2 (Exact generator enumeration)**.** *The conditions of Theorem 5.1 can be decided by a finite integer search. If a selected set of basis elements generates $R \otimes \mathbb{Q}$ as an algebra, it suffices to enumerate their matrices, reconstruct the other matrices by rational polynomial expressions, and check* (5.1)–(5.2)*.*

Proof. Every entry has an explicit finite interval and parity. Choose monomials $w_0 = 1, w_1, \dots, w_{r-1}$ forming a rational basis of the rank-r algebra, and let V have as its rows their coefficients in the based-ring basis. The expression for b_i is row i of V^{-1} applied to the monomials. Thus any representation is uniquely reconstructed from the selected generator matrices. Testing all the ring relations, transpose identities, and entry bounds after reconstruction accepts every possible solution and nothing outside the stated necessary system. □

The direct enumeration implementation treats $|J| \le 2$ and chooses one or two generators when such a set is found. Applying it to the census gives nine complete exclusion certificates, including $R_{3,79}^{(6)}$ and $R_{3,216}^{(6)}$. The respective generator searches contain 4608 and 1536 assignments. These are automatic applications of one theorem; their indices are not inputs to the test. The spectral refinement in Section 5.2 also treats larger fixed-point sets and supplies two further exclusions with $|J| = 4$.

5.2 Spectral pruning of the antiinvariant module

The matrix system in Theorem 5.1 is useful beyond the two-dimensional antiinvariant space. Its entry bounds impose an additional spectral restriction that makes larger fixed-point sets accessible.

Proposition 5.3 (Perron-factor exclusion)**.** *Retain the notation of Theorem 5.1, and let $i = i^*$. Suppose the nonnegative regular fusion matrix N_i is irreducible. Then no T_i can have an eigenvalue whose minimal polynomial contains* $\mathrm{FPdim}(i)$ *as a root. Moreover, if for every $x \in J$ there are $a, y \in J$ with N_{ax}^y odd, then $T_g = I$.*

Proof. Write $B_i = (N_{ix}^y)_{x,y\in J}$. Since J is a proper subset of the basis, the strict Perron–Frobenius inequality for a proper principal submatrix of an irreducible nonnegative matrix gives

$$\rho(T_i) \le \rho(|T_i|) \le \rho(B_i) < \rho(N_i) = \mathrm{FPdim}(i).$$

The characteristic polynomial of the integer matrix T_i is rational; thus an irreducible factor containing $\mathrm{FPdim}(i)$ cannot divide it. The bounds on T_g make it diagonal with entries ± 1. For $a \in J$, the identity $ga = a$ gives $T_g T_a = T_a$. A diagonal entry -1 would therefore force the corresponding row of every such T_a to vanish, contradicting the indicated odd entry. □

If i generates $R \otimes \mathbb{Q}$, express each basis element as a rational polynomial $p_a(i)$. The preceding proposition removes the Perron factor from the possible minimal polynomial of T_i; when $T_g = I$, every remaining factor must also divide $p_g(t) - 1$. Since $T_i \equiv B_i \pmod 2$, its characteristic polynomial must reduce to that of B_i modulo two. Enumerate all products of the remaining irreducible factors of total degree $|J|$, allowing repetitions. Each product fixes both $\mathrm{Tr}(T_i)$ and $\mathrm{Tr}(T_i^2) = \sum_{x,y}(T_i)_{xy}^2$. These equalities reduce the finite entry search substantially. Reconstruction by the p_a and verification of the entire system (5.1)–(5.2) finish the test. The search is exhaustive because every possible module representation has one of the enumerated rational characteristic polynomials.

Proposition 5.4. *The rings $R_{8,2945}^{(6)}$ and $R_{8,3071}^{(6)}$ have no categorification as complex fusion categories.*

Proof. In both rings the involution is $g = b_1$ and $J = \{b_2, b_3, b_4, b_5\}$; all basis elements are self-dual. All tensor identities used below are checked automatically from the census tensors in the supplementary certificates.

For $R_{8,2945}^{(6)}$, the element b_3 generates the rational algebra, its regular matrix is irreducible, and

$$\det(tI - N_3) = t(t-2)(t^2 - 23t - 1)(t^2 + t - 1).$$

The Perron factor is $t^2 - 23t - 1$. The parity criterion gives $T_g = I$, and the character with b_3-value zero has g-value -1, so this character cannot occur. Thus only $t - 2$ and $t^2 + t - 1$ remain. The fixed block has characteristic polynomial $t^4 + t^2 + 1$ modulo two. Of the three degree-four products of the two remaining factors, only $(t^2 + t - 1)^2$ has this reduction. Hence

$$\mathrm{Tr}(T_3) = -2, \qquad \mathrm{Tr}(T_3^2) = 6, \qquad T_3^2 + T_3 - I = 0.$$

In the ordered basis J, the required odd entries of T_3 are the last two diagonal entries and the symmetric pairs $(1,4), (2,3)$. Their squared contributions already sum to six. Consequently

$$T_3 = \begin{pmatrix} 0 & 0 & 0 & \varepsilon \\ 0 & 0 & \delta & 0 \\ 0 & \delta & -1 & 0 \\ \varepsilon & 0 & 0 & -1 \end{pmatrix}, \qquad \varepsilon, \delta \in \{1, -1\}.$$

Rational reconstruction, reduced modulo t^2+t-1, gives $T_2=-3T_3-4I$. Its first diagonal entry is -4, contradicting the bound $|(T_2)_{11}|\le N_{b_2b_2}^{b_2}=2$.

For $R^{(6)}_{8,3071}$, the regular matrix N_3 is irreducible and

$$\det(tI-N_3)=t^3(t+2)(t^2-26t+24).$$

The last factor is the Perron factor. After removing it, the symmetric matrix T_3 has eigenvalues in $\{0,-2\}$, so $T_3^2=-2T_3$. Every diagonal entry of its fixed block is six, and therefore every diagonal entry d of T_3 is even. The diagonal of the matrix identity gives

$$d^2+\sum_{y\neq x}(T_3)_{xy}^2=-2d.$$

It follows that $d\in[-2,0]\cap 2\mathbb{Z}=\{-2,0\}$, and either choice forces all off-diagonal entries in that row to vanish. This contradicts $N_{b_3b_2}^{b_3}=7$, which requires the corresponding off-diagonal entry to be odd. □

The exact generator search checks respectively four and zero trace-and-norm-compatible matrices for these two records. The second search uses b_5 as a rational algebra generator; the displayed direct argument uses b_3 to make the contradiction shorter. Both certificates are general categorification obstructions, with no pivotal assumption.

5.3 Descent to a rank-three near-group subring

Theorem 5.5 (Rank-three subring obstruction)**.** *Suppose the Grothendieck ring of a complex fusion category contains three distinct basis elements $1,g,x$ satisfying*

$$g^2=1,\qquad gx=xg=x,\qquad x^2=1+g+mx.$$

Then $m\le 2$.

Proof. The three basis elements are closed under products and duals, so their simple objects generate a fusion subcategory with Grothendieck ring $K(1,0,m,0)$. By [36, Theorem 4.1], categorifiability of this ring implies $m\le 2$. The cited theorem uses the Galois-pseudounitary reduction for this near-group family and does not impose a pivotality hypothesis on the ambient category. □

In $R^{(6)}_{4,162}$, the basis elements (b_0,b_3,b_4) have precisely these products with $m=4$. The exact certificate checks all coefficients in their nine products, including the vanishing coefficients outside this subring. Theorem 5.5 therefore excludes this ring from general complex categorification. The same automatic scan is applied to the entire census through multiplicity eight.

5.4 Small-dimension Morita obstructions

The next criteria use categorical Morita equivalence rather than additional indicator-height searches. They apply to general complex fusion categories. In particular, no chosen pivotal structure enters their hypotheses.

5.4.1 A cyclic-index bound and dimension thirty

For group-theoretical data (G, ω, H, ψ), the simple objects of $\mathcal{C}(G, \omega, H, \psi)$ are indexed by a double coset HgH and an irreducible projective representation π of $S_g = H \cap gHg^{-1}$. Their Frobenius–Perron dimensions are

$$[H : S_g] \dim \pi. \tag{5.3}$$

The objects supported on H form a fusion subcategory equivalent to $\mathrm{Rep}(H)$. One can see this after replacing the data by an equivalent adapted-cocycle presentation with $\psi = 1$ and $\omega|_{G\times G\times H} = 1$; see [38, Proposition 2.5]. These descriptions also follow from [20].

Lemma 5.6 (Cyclic-index bound)**.** *Suppose a finite group G has a cyclic normal subgroup B of index at most two. Every simple object of every group-theoretical category $\mathcal{C}(G, \omega, H, \psi)$ has Frobenius–Perron dimension at most two.*

Proof. Put $K = H \cap B$. Since every subgroup of a cyclic group is characteristic, K is normal in G. Thus $K \subseteq S_g \subseteq H$ and $[H : K] \leq 2$. If $[H : S_g] = 2$, then $S_g = K$ is cyclic. Every irreducible projective representation of a cyclic group over $\mathbb{C}$ has dimension one, so (5.3) is two.

If $S_g = H$, it suffices to bound the dimensions of projective representations of H. In the twisted group algebra of H, the subalgebra on K is commutative and semisimple: its cyclic generator satisfies $u^{|K|} = c \neq 0$, a polynomial with distinct roots. The remaining coset, when present, permutes its weight spaces in orbits of size at most two. In an irreducible module there is one such orbit. For an orbit of size one, a coset generator squares to a nonzero scalar and hence has a one-dimensional invariant subspace. For an orbit of size two, a line in one weight space and its translate span an invariant two-dimensional subspace. Irreducibility therefore gives dimension at most two. This argument includes arbitrary projective factor sets. □

Proposition 5.7 (Integral dimension thirty)**.** *Every simple object of an integral fusion category of Frobenius–Perron dimension* 30 *has dimension one or two.*

Proof. The category is group-theoretical by [16, Theorem 9.2]. Its presenting group has order thirty. Every group of order thirty is solvable: the possible numbers of Sylow five- and three-subgroups are respectively $1, 6$ and $1, 10$; if neither is one, the corresponding nonidentity elements already number $24 + 20 > 29$. A normal Sylow subgroup and the solvable quotient prove the claim. Hall's theorem now supplies a subgroup of order fifteen. It is cyclic by the Sylow theorems and normal because it has index two. Lemma 5.6 applies. □

The adjective *integral* is essential: the statement does not claim the same bound for nonintegral categories of total dimension thirty. The certificate checks a positive integral dimension character, not just an integer value of the total dimension.

5.4.2 A Morita dichotomy in dimension sixty

Lemma 5.8. *A fusion category of Frobenius–Perron dimension sixty is either solvable or categorically Morita equivalent to $\mathrm{Vec}^{\omega}_{A_5}$ for some three-cocycle ω. If it is integral and solvable, has exactly two invertible simples, and satisfies $\mathcal{C}_{\mathrm{ad}} = \mathcal{C}$, every simple has dimension at most four.*

Proof. By [16, Theorem 9.16], the category is weakly group-theoretical. It is therefore Morita equivalent to a nilpotent category of the same dimension, obtained from vector spaces by successive extensions by finite groups whose orders multiply to sixty. If all these groups are solvable, refining their normal series into cyclic prime-order factors shows that the category is solvable. Solvability is invariant under categorical Morita equivalence; see [16, Section 4.2].

Otherwise an extension group has nonsolvable order dividing sixty. All proper divisors involve at most two primes, apart from thirty, which was handled above. Burnside's theorem and the fact that the unique nonabelian simple group of order sixty is A_5 imply that this group is A_5 itself. Its order uses the entire dimension. The nilpotent category is consequently pointed, of the form $\mathrm{Vec}^{\omega}_{A_5}$.

For the second assertion, a nontrivial last step of a solvable construction is a cyclic prime-order extension or equivariantization. An extension would give a nontrivial grading, contrary to $\mathcal{C}_{\mathrm{ad}} = \mathcal{C}$. In an equivariantization $\mathcal{C} = \mathcal{D}^{C_p}$, the canonical $\mathrm{Rep}(C_p)$ is a fusion subcategory. Exactly two invertible simples force $p = 2$, and $\mathrm{FPdim}(\mathcal{D}) = 30$.

The simples of a C_2-equivariantization are indexed by orbits of simples of $\mathcal{D}$ and projective representations of their stabilizers. Those stabilizers are trivial or cyclic of order two, so the projective representations have dimension one. A simple dimension in $\mathcal{C}$ is therefore either d or $2d$, with d a simple dimension in $\mathcal{D}$. Since $\mathcal{C}$ is integral, d is rational; as an algebraic integer it is an integer. Thus $\mathcal{D}$ is integral. Proposition 5.7 gives $d \leq 2$, proving the bound four. □

5.4.3 Cocycle-independent involution returns

For a normalized three-cocycle on a group of order two, its value on the triple of nonidentity elements is unchanged by a normalized coboundary. In particular, triviality of its restriction class implies that this value is one.

Theorem 5.9 (Involution-return indicator)**.** *Let $\mathcal{C}(G, \omega, H, \psi)$ be group-theoretical, let $g \in G$, and put $S = H \cap gHg^{-1}$. Suppose every element in*

$$T_g = \{r \in gH : r^2 \in S\}$$

is an involution conjugate in G to an element of H. For the simple object $M(g, \pi)$ associated to an irreducible projective representation π of S, one has

$$\nu_2(M(g, \pi)) = \frac{|T_g| \dim \pi}{|S|}. \tag{5.4}$$

In particular, if $T_g \neq \varnothing$, these simples are self-dual. This conclusion holds for every admissible ω, ψ.

Proof. Use the equivalent adapted presentation supplied by [38, Proposition 2.5]; its cocycle restricts to one on H^3. If r is an involution conjugate into H, the restriction class on $\langle r \rangle$ is trivial. The preceding coboundary observation gives $\omega(r, r, r) = 1$ in this adapted presentation, independently of any remaining values of the cocycle.

The second-indicator specialization of [38, Theorem 4.2, equation (4.6)] is

$$\nu_2(M(g, \pi)) = \frac{1}{|S|} \sum_{r \in T_g} \pi_{-2}(r) \chi_\pi(r^{-2}).$$

For an involution, the scalar $\pi_{-2}(r)$ in that formula is $\omega(r, r, r)^{-1} = 1$, whereas $\chi_\pi(r^{-2}) = \chi_\pi(1) = \dim \pi$. This proves (5.4). Group-theoretical categories have their positive spherical structures; a simple with nonzero second indicator is self-dual. Self-duality is preserved under tensor equivalence, so the conclusion is independent of the presentation. □

Corollary 5.10 (Dihedral supports in A_5)**.** *In every category $\mathcal{C}(A_5, \omega, D_{2q}, \psi)$ with $q \in \{3, 5\}$, every simple of dimension q has second indicator one. In particular all such simples are self-dual.*

Proof. The dihedral subgroup has order $2q$. Its ordinary irreducible representations have dimensions one or two, so a simple of dimension $q > 2$ is not supported on it. Formula (5.3) then forces stabilizer order two and a one-dimensional projective representation. For these double cosets the following exact group data hold:

$\lvert H\rvert$	double-coset sizes	$\lvert S\rvert$	$\lvert T_g\rvert$ when $\lvert S\rvert = 2$
6	$6, 18, 36$	2	2
10	$10, 50$	2	2

Every element in T_g is an involution. All fifteen involutions in A_5 are conjugate, and H contains an involution. Thus Theorem 5.9 gives $\nu_2 = 2/2 = 1$.

These group assertions are certified independently from the sixty even permutations of five letters. The verifier generates every order-$2q$ subgroup using an element of order q and an involution, checks that they form one conjugacy class, and enumerates every right coset and every square-return element. It does not test only a selected coset or an assumed cocycle. The coset lists are included in each certificate. □

Theorem 5.11 (Tensor-checkable small-dimension obstruction)**.** *Let R be an integral fusion ring. Each of the following conditions excludes a general complex categorification.*

(a) $\mathrm{FPdim}(R) = 30$ *and some basis dimension exceeds two.*

(b) $\mathrm{FPdim}(R) = 60$*, the adjoint based subring is R, and exactly two basis elements are invertible. Some basis dimension exceeds four. For a prime $q \in \{3, 5\}$ the complete set of based-subring dimensions is $\{1, 2, 2q, 60\}$, and a basis element of dimension q is not self-dual.*

Proof. Part (a) is Proposition 5.7. For (b), Lemma 5.8 excludes solvability and gives a presentation $\mathcal{C}(A_5, \omega, H, \psi)$. The canonical $\mathrm{Rep}(H)$ subcategory forces $|H| \in \{1, 2, 2q, 60\}$. If $|H| \leq 2$, every simple dimension is at most two, contradicting the stated dimension exceeding four. If $H = A_5$, the category is $\mathrm{Rep}(A_5)$ and has only one invertible simple, because A_5 is perfect. Thus H is an order-$2q$ subgroup, necessarily a conjugate of D_{2q}. Corollary 5.10 contradicts the non-self-dual dimension-q basis element. □

5.4.4 Uniform census application

The implementation supplies the same tensor test at every multiplicity. It first checks a positive integral dimension character; it then computes the adjoint closure, all based subrings, duality, and the number of invertibles. Illustrative applications are

ring	sorted dimension type	branch	q
$R^{(6)}_{2,51}$	$(1, 1, 2, 2, 5, 5)$	(b)	5
$R^{(6)}_{3,15}$	$(1, 1, 2, 3, 3, 6)$	(b)	3
$R^{(6)}_{4,19}$	$(1, 1, 1, 1, 1, 5)$	(a)	–

The program finds these applications from the tensor, without selecting labels in advance. The current census scan and its exact dimension checks are recorded by `software/morita_controls.py`. The criterion also excludes $R^{(6)}_{2,25}$ and $R^{(6)}_{2,29}$. Overlap with other criteria is counted only once in the sequential exclusion table.

5.5 Two-sided module Gram constraints

Let $\mathcal{C}$ be a fusion category, let A be a connected separable algebra, and let $\mathcal{M} = \mathcal{C}_A$. Write M_i for the action matrix of a simple $X_i \in \mathcal{C}$, and $\mathcal{D} = \mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{M})$ for the dual category, with a

fixed choice of opposite tensor convention. All dimension statements here use Frobenius–Perron dimensions, not a positive pivotal structure.

Proposition 5.12 (Two-sided module Gram identities)**.** *For every simple $m \in \mathcal{M}$, let B_m be the matrix of the evaluation functor $\mathcal{D} \to \mathcal{M}$, $Y \mapsto Y(m)$. Then*

$$B_m B_m^{\mathsf{T}} = \sum_i (M_i)_{mm} M_i. \tag{5.5}$$

The tensor unit column of B_m is e_m. If d_n are the positive module dimensions in any fixed normalization, each column v belonging to a simple $Y \in \mathcal{D}$ satisfies

$$\sum_n v_n d_n = d_m \operatorname{FPdim}(Y). \tag{5.6}$$

Proof. The internal endomorphism algebra of m in $\mathcal{C}$ is $A_m = \bigoplus_i (M_i)_{mm} X_i$, up to the harmless replacement of every index by its dual; these multiplicities agree by module reciprocity. The module category is equivalently $\mathcal{C}_{A_m}$, sending m to the free rank-one module. The dual category is the category of A_m-bimodules. Evaluation becomes the functor forgetting the left action, whose adjoint is free left multiplication by A_m. Their composite on $\mathcal{M}$ is therefore the action of A_m. Adjunction gives (5.5). Evaluation of the tensor unit is m, giving the specified column. The commuting action of the dual category on the positive module dimension vector has eigenvalue $\operatorname{FPdim}(Y)$; this gives (5.6). These assertions are standard consequences of separable-algebra Morita theory; see [15, Chapter 7]. □

5.5.1 The split-involution input and a finite system

Suppose $g^2 = 1$ is an invertible object fixing a simple object of $\mathcal{C}$. The module-associativity argument in the main manuscript supplies the algebra $A = 1 \oplus g$. Free two-element g-orbits give one simple of $\mathcal{M}$; fixed simples give two. The sign automorphism of A interchanges the two actions on every fixed simple. Thus the action matrices M_i can be reconstructed from the full tensor and an anti-invariant representation satisfying the parity, transpose and entry bounds of Theorem 5.1.

Normalize the dimension of the free module A to one. An orbit module has dimension $\operatorname{FPdim}(X)$ and a split fixed module has dimension $\operatorname{FPdim}(X)/2$. These module dimensions need not be algebraic integers. At $m = A$, the Gram matrix in (5.5) is $I + M_g$. Since M_g is an involutive permutation, its nonnegative integral Gram columns are forced: a fixed point has two unit columns, and a transposed pair has one column supported on that pair. Hence (5.6) determines the multiset of simple dimensions of $\mathcal{D}$. One of the unit-dimension columns is its tensor unit.

For another simple m, remove its known tensor-unit column. For every remaining prescribed dual dimension e, enumerate nonzero integer vectors $v \geq 0$ with

$$v_i^2 \leq Q_{ii}, \quad v_i v_j \leq Q_{ij}, \quad \sum_i v_i d_i = e d_m, \qquad Q = \sum_i (M_i)_{mm} M_i - e_m e_m^{\mathsf{T}}.$$

The integer multiplicities of these columns must add their outer products to Q and respect the prescribed number of columns of each dual dimension. Relaxing the multiplicities to nonnegative rationals gives a necessary rational linear system. An exact Farkas vector proves infeasibility. A general exclusion requires infeasibility for every admissible anti-invariant representation; selecting only one is invalid.

The executable implementation treats at most two fixed simples and simple-dimension fields of degree at most four. It enumerates all admissible generator matrices and reconstructs the complete representation. The vector search uses exact field coordinates and explicit integer

bounds. A domain-budget limit or an ineligible fixed-point count gives no conclusion. Negative checking, when a certificate is proposed, reconstructs the full domain and checks only exact arithmetic.

The implementation in `software/module_dual_gram.py` is a uniform necessary-condition test. It does not assign categorical existence when the relaxed system is feasible. Its successful certificates, if any, must enter the same exact-checking and disjoint allocation process as every other obstruction.

5.6 Coupling all Morita action matrices

The two-sided module test can be strengthened by retaining the complete associativity pairing, rather than its restrictions to one module simple at a time. The strengthening uses Frobenius–Perron positivity only to recognize module categories; it makes no pivotal or spherical assumption.

Theorem 5.13 (Full Morita action Gram identity)**.** *Let ${}_{\mathcal{C}}\mathcal{M}_{\mathcal{D}}$ be an invertible bimodule category, with simple objects $m_0, \dots, m_{s-1}$. Write*

$$L_i(a,b) = [X_i \otimes m_a : m_b], \qquad R_j(a,b) = [m_a \otimes Y_j : m_b]$$

for its left and right action matrices. Then

$$\sum_j R_j(a,d)R_j(b,c) = \sum_i L_i(b,a)L_i(c,d) \qquad (0 \le a,b,c,d < s). \tag{5.7}$$

Consequently the matrix

$$G_{(a,d),(b,c)} = \sum_i L_i(b,a)L_i(c,d)$$

has a nonnegative integral Gram factorization whose columns are the vectorized matrices R_j. Each such column belongs to $\operatorname{im}_{\mathbb{Q}} G$, and its coordinate indexed by (a,d) is at most $\lfloor\sqrt{G_{(a,d),(a,d)}}\rfloor$. The corresponding matrices commute with every L_i, include the identity, and are permuted by transposition.

Proof. Use duals in the Morita bicategory. The multiplicity of Y_j in $m_b^* m_c$ is $R_j(b,c)$, by Frobenius reciprocity. Thus the multiplicity of m_d in $m_a(m_b^* m_c)$ is the left-hand side of (5.7). The multiplicity of X_i in $m_a m_b^*$ is $L_i(b,a)$, so associativity identifies the same multiplicity with its right-hand side. If W is the matrix of vectorized right actions, this is $G = WW^{\mathsf{T}}$. Hence $\operatorname{im}_{\mathbb{R}} W = \operatorname{im}_{\mathbb{R}} G$; since all matrices are rational, each column of W belongs to the stated rational image. A diagonal entry of G is the sum of the squares of the corresponding entries of W, which proves the bound. The commuting left and right actions, the tensor unit of $\mathcal{D}$, and $R_{j^*} = R_j^{\mathsf{T}}$ prove the remaining assertions. □

Suppose now that $A = \mathbf{1} \oplus g$ is the split involution algebra and $m_0 = A$ in $\mathcal{C}_A$. The existence argument for this algebra and the exhaustive module-action construction are those of the preceding module criterion. Since the internal end of m_0 is A, setting $c = d = 0$ in (5.7) gives

$$\sum_j R_j(\,\cdot\,,0)R_j(\,\cdot\,,0)^{\mathsf{T}} = I + L_g. \tag{5.8}$$

Here L_g is an involutive permutation matrix. Its right-hand side has a unique nonnegative integral Gram factorization with nonzero columns, up to permutation of columns: a fixed point contributes two copies of its unit vector, and a transposition contributes the sum of its two unit vectors. Indeed, zero off-diagonal entries prevent columns from meeting different orbits. On a fixed point the squared entries sum to two; on a transposition the diagonal entries

and the off-diagonal entry are all one. These observations prove the assertion orbit by orbit. Transposition supplies the same finite list for the possible first rows of each right action matrix.

The implementation intersects the rational image of G with these prescribed first rows and columns. Choosing free coordinates among the actual matrix entries, the diagonal bounds in Theorem 5.13 give a finite integer box. Every candidate is checked for nonnegativity, the entrywise necessary bounds $vv^{\mathsf{T}} \leq G$, and commutation with the left actions. If the candidates are $v_1, \ldots, v_t$, the unknown multiplicities must satisfy

$$\sum_{h=1}^{t} z_h v_h v_h^{\mathsf{T}} = G - \operatorname{vec}(I)\operatorname{vec}(I)^{\mathsf{T}}, \qquad z_h \in \mathbb{Z}_{\geq 0},$$

together with the first-column multiplicities from (5.8), after removing the unit action. An exact Farkas certificate for the relaxation $z_h \geq 0$ excludes that module-action branch. As before, exclusion of the fusion ring requires exhausting every eligible branch. The program `software/full_morita_gram.py` reconstructs the finite domains and checks separating vectors using integer arithmetic.

Proposition 5.14 (The involution dual of the first residual)**.** *If $\mathcal{C}$ categorifies $R^{(6)}_{3,215}$ and g is its nontrivial invertible simple, then $\mathbf{1} \oplus g$ admits a connected separable algebra structure and*

$$K_0({}_A\mathcal{C}_A) \cong K_0(\mathcal{C}) \quad \textit{as based rings}, \qquad A = \mathbf{1} \oplus g.$$

This assertion does not identify the two fusion categories up to tensor equivalence.

Proof. The involution fixes two simples, so the previously established split involution argument applies. Exhaustive reconstruction gives sixteen module-action branches. Two violate the rank bound $\operatorname{rank} G \leq \operatorname{rank} K_0({}_A\mathcal{C}_A)$, and twelve have exact Farkas certificates for the full Gram system. Each of the remaining two has Gram rank six and six dual simples. Therefore its six right action matrices are linearly independent; in particular they cannot repeat. Exhaustive selection from the finite candidate lists gives exactly one Gram frame for each branch, up to permuting its columns. Multiplying those six matrices reconstructs a nonnegative integral fusion tensor, and a recorded based permutation identifies all its coefficients with $R^{(6)}_{3,215}$.

The complete sixteen-branch record is `research/new_frontier/frontier_full_morita.json`. The script `research/new_frontier/verify_frontier_full_morita.py` reconstructs the module actions and candidate domains, checks all rank and Farkas obstructions, proves uniqueness of each surviving frame by finite exact enumeration, and verifies every matrix product and the based-ring identification. Verification does not call an optimizer. □

The two surviving frames are based bimodule data, not associators or categorifications. In particular the proposition does not supply an algebra $\mathbf{1} \oplus X$ for a noninvertible simple X. The uniform full Gram scan covers all 47 residual inputs through multiplicity eight: 21 are eligible for the implemented involution search and 26 are ineligible. It yields no additional whole-ring exclusion. All 54 supplied positive models pass the control scan. These outcomes are retained separately from the exclusion ledger.

5.7 Integral multiplicities in the Morita Gram identity

The full Morita action Gram identity of Theorem 5.13 has an elementary integral refinement. The unknown attached to a possible right-action matrix counts simple objects of the dual category; it is an integer, even when distinct simple objects have the same action matrix. Retaining this integrality detects obstructions invisible in the rational nonnegative relaxation.

Proposition 5.15 (Morita Gram congruence obstruction)**.** *Fix one of the exhaustively enumerated module actions for a forced split involution algebra* $\mathbf{1} \oplus g$*. Let* $Ax = b$ *be its full Morita Gram system, including the prescribed first-column multiplicities, with all possible right-action matrices retained. If there exist* $q > 1$ *and an integral row vector* y *such that*

$$yA \equiv 0 \pmod{q}, \qquad yb \not\equiv 0 \pmod{q},$$

then this module action cannot occur in a complex fusion category.

Proof. Every categorification would give a vector x of nonnegative integral multiplicities satisfying $Ax = b$. Multiplication by y would give $yb = yAx \equiv 0 \pmod{q}$, a contradiction. All entries of A and b come from the integral fusion and module actions. The argument therefore requires no pivotal or spherical structure and no positivity assumption on categorical dimensions. □

For discovery, Gaussian elimination over $\mathbb{F}_2$ produces the row vector y. Verification only multiplies integer vectors and checks the two congruences. Repeated equality rows may be removed by a deterministic rule. The completeness of the right-action candidate list is exactly the finite Gram enumeration already used in Section 5.6; no new search cutoff is introduced.

Proposition 5.16. *The rings* `R6_5_638` *and* `R6_7_1225` *admit no categorification by a fusion category over* $\mathbb{C}$*.*

Proof. The unrestricted signed reduction is verified separately for each full fusion tensor. There are respectively 11 800 and 46 856 signed tables, partitioned into 418 and 1 560 sign orbits. Each has one original gauge orbit. Every other orbit is excluded by its exact character obstructions; within the original orbit the only admissible nonzero dimension characters lie in the Frobenius–Perron Galois orbit. It therefore suffices to exclude pseudo-unitary categorification.

In each case the involution-module enumeration has sixteen branches. Two fail the full Morita rank bound. Twelve fail Proposition 5.15, with $q = 2$. Both remaining branches force negative stabilizer commutators on the two fixed simple objects, indexed 4 and 5 in the stored tensor. Their cubic rotation spaces have common dimension $n = 4$ for `R6_5_638` and $n = 6$ for `R6_7_1225`. The higher-dimensional Clifford rotation criterion of Section 5.10 applies. The exponent prime supports are respectively $\{2, 5, 13\}$ and $\{2, 5, 29\}$, so the third indicators belong respectively to

$$\{-2, 1\} \qquad \text{and} \qquad \{-3, 0\}.$$

The actual separating certificates already hold on the weaker intervals $[-2, 1]$ and $[-3, 3/2]$; they do not rely on selecting a discrete indicator value.

For either tensor the complete low-induction enumeration gives 91 frames, of which 31 have a nonempty character-labelling set. The 50 labellings and four second-indicator choices yield 200 exact height-six systems. Every system has a verified integral separating functional. The two dimension fields are $\mathbb{Q}(\sqrt{65})$ and $\mathbb{Q}(\sqrt{145})$; their low-column counts are respectively $(217, 225, 225, 61, 61)$ and $(469, 481, 481, 127, 127)$. All possible twist orders are covered by the inherited relative-trace construction. Thus the two remaining module branches also fail. □

The exact verifier is `software/round3_integral_morita.py`, with certificate kind `general_integral_morita_rotation`. The full finite domains, congruence vectors, separating vectors, and optimizer-free replay reports are stored in `research/round3/higher/`. Applying the new congruence test to all 22 inherited residuals and all 54 inherited realized controls completes 76 runs without a timeout or false exclusion. Of the realized controls, 27 admit the tested module domain and 27 are ineligible. The modulo-two test alone leaves a surviving module branch on every inherited residual; the two new exclusions use its combination with the forced Clifford indicators.

5.8 An order-six Morita–grading obstruction

The dimension reconstruction above can force a dual category whose based ring has a short exhaustive parametrization. This supplies two further exclusions without assuming a pivotal structure on the original category.

Theorem 5.17. *Let $\mathcal{C}$ be a fusion category, and let g be an invertible involution fixing a simple object. Set $A = \mathbf{1} \oplus g$. Suppose the simple Frobenius–Perron dimensions of $\mathcal{D} = {}_A\mathcal{C}_A$ are*

$$\underbrace{1, \dots, 1}_{6}, \qquad \underbrace{d, \dots, d}_{6}, \qquad d = 2 + \sqrt{5}.$$

Then $\mathcal{C}$ has a faithful C_3 grading. In particular, a fusion ring with trivial universal grading and the stated forced dual dimension profile cannot be categorified. This excludes $R^{(6)}_{5,100}$ and $R^{(6)}_{5,408}$ over $\mathbb{C}$.

Proof. The algebra A exists by the fixed-object argument of Theorem 5.1. We first exhaust the possible based rings of $\mathcal{D}$. Its invertible objects form a group G of order six. The multiplicity of an invertible h in $\rho\rho^*$, for a noninvertible simple ρ, is one precisely when $h\rho \cong \rho$. The identity $d^2 = 1 + 4d$ and irrationality of d therefore imply that the stabilizer of ρ is trivial. Thus the noninvertibles are exactly $h\rho$, $h \in G$. The right action is free for the same reason, so

$$\rho h = \phi(h)\rho, \qquad \rho^2 = u + \sum_{h \in G} a_h h\rho, \qquad \sum_{h \in G} a_h = 4, \tag{5.9}$$

where $\phi \in \mathrm{Aut}(G)$, $u \in G$, and $a_h \in \mathbb{Z}_{\geq 0}$. Associativity gives $\phi^2 = \mathrm{Ad}(u)$ and $\phi(u) = u$; comparison of $(\rho^2)h$ with $\rho(\rho h)$ also makes the coefficient function invariant under

$$k \longmapsto \phi^2(h) k \phi(h)^{-1}.$$

Only C_6 and S_3 can occur as G.

The cyclic case. The automorphism ϕ is either the identity or inversion. Inversion would make (a_h) constant on cosets of the square subgroup of order three, contradicting $\sum a_h = 4$. Hence ϕ is the identity. Use additive labels $0, \dots, 5$ for C_6. Duality gives $\rho^* = (-u)\rho$ and

$$u \in \mathbb{Z}/6\mathbb{Z}, \qquad a_k \geq 0, \qquad \sum_{k=0}^{5} a_k = 4, \qquad a_k = a_{u-k}. \tag{5.10}$$

There are exactly 60 pairs (u, a) satisfying these conditions. Each supplies the entire candidate tensor by

$$h \cdot k = h + k, \qquad h \cdot (k\rho) = (h + k)\rho, \qquad (h\rho)(k\rho) = h + k + u + \sum_{j} a_j (h + k + j)\rho.$$

An exact exhaustive calculation partitions the 60 pairs as follows:

Number	Exact consequence
42	A formal-codegree norm obstruction
6	A closed based subring isomorphic to $R^{(6)}_{2,20}$
12	A faithful C_3 grading with neutral invertible involution

The first row is a general obstruction here: by [39, Lemma 3.1], a category with these two distinct Frobenius–Perron dimensions is Galois conjugate to a pseudo-unitary category with the same based ring. In that category $\mathrm{FPdim}(\mathcal{D})/f$ must be an algebraic integer for every

formal codegree f. The cyclic candidate ring is commutative, so the Casimir characteristic roots used in the check are its formal codegrees, with no representation dimension correction. Its global dimension is

$$D = 6 + 6d^2 = 60 + 24\sqrt{5}, \qquad p_D(t) = t^2 - 120t + 720.$$

For each of the 42 cases, the certificate gives the exact minimal polynomial q of a formal codegree and a prime p for which

$$\frac{v_p(|p_D(0)|)}{\deg p_D} < \frac{v_p(|q(0)|)}{\deg q}. \tag{5.11}$$

This contradicts integrality of D/f: take the norm in any common number field containing D and f. The second row uses the previously established general exclusion for $R_{2,20}^{(6)}$, whose original exact certificate is replayed, not just its allocated status. For each of the remaining 12 pairs the grading is explicitly

$$\deg(h) = h \pmod 3, \qquad \deg(h\rho) = h + 2u \pmod 3. \tag{5.12}$$

Every nonzero structure constant respects this degree assignment. The unique nonidentity invertible involution, namely $3 \in C_6$, has degree zero.

The nonabelian case. Every automorphism of S_3 is inner. Replacing ρ by an invertible multiple makes ϕ the identity. Then u is central, so $u = 1$, and a is constant on conjugacy classes. Its values (a_e, a_t, a_c) on the identity, transpositions and three-cycles satisfy

$$a_e + 3a_t + 2a_c = 4.$$

The complete domain and its obstructions are

(a_e, a_t, a_c)	Obstruction
$(4, 0, 0)$	The rank-two subring $\rho^2 = 1 + 4\rho$
$(2, 0, 1)$	The closed subring $R_{2,20}^{(6)}$
$(0, 0, 2)$	The closed subring $R_{2,21}^{(6)}$
$(1, 1, 0)$	A one-dimensional formal-codegree norm obstruction

The first row is excluded by [34]. In the next two rows the subring is generated by C_3 and ρ; its based isomorphism and the previously established general certificate are checked exactly. For the last row, extend the sign representation of S_3 by assigning ρ a root λ of $\lambda^2 + 2\lambda - 1 = 0$. Equation (5.9) verifies this as a one-dimensional representation of the full ring. Its genuine formal codegree is

$$f = \sum_{h \in S_3} 1 + \sum_{h \in S_3} \lambda^2 = 6(1 + \lambda^2) = 24 \pm 12\sqrt{2}, \qquad q(t) = t^2 - 48t + 288.$$

Here $\rho^* = \rho$. Since $v_2(720) = 4$ and $v_2(288) = 5$, condition (5.11) holds. The two-dimension reduction again makes this a general categorical obstruction. In particular, this step does not identify an unweighted Casimir eigenvalue from a higher-dimensional irreducible representation with a formal codegree. All nonabelian possibilities have now been excluded.

Returning through Morita duality. Consequently $\mathcal{D}$ has one of the faithful gradings (5.12). Regard A as the distinguished simple right module in $\mathcal{M} = \mathcal{C}_A$, and let $B = \underline{\mathrm{End}}_{\mathcal{D}}(A)$. Double Morita duality identifies $\mathcal{C}$ with ${}_B\mathcal{D}_B$, up to the opposite tensor convention; see [15, Chapter 7]. For a simple A-bimodule Y, its multiplicity in B is

$$\dim \mathrm{Hom}_{\mathcal{M}}(A \otimes_A Y, A) = \dim \mathrm{Hom}_{\mathcal{M}}(Y, A).$$

There are exactly two simple bimodules restricting to the right module A: the regular bimodule and its twist by the nontrivial character of C_2. Each has multiplicity one. This can also be read directly from the two unit columns at A in the Gram matrix $I + M_g$ of Proposition 5.12; off-diagonal entries in that row vanish. All other simple bimodules have zero multiplicity. Thus $B = \mathbf{1} \oplus b$, where b is a nontrivial invertible involution. In the cyclic case b is necessarily $3 \in C_6$, hence B lies in the neutral component.

A connected separable algebra in the neutral component of a faithful group grading transfers that grading to its bimodule category. Indeed, the left and right algebra actions preserve each homogeneous support, and tensor product over the algebra multiplies the degrees. Every component is nonzero: for a nonzero homogeneous object Y, the free bimodule $B \otimes Y \otimes B$ is nonzero and has the same degree. It follows that $\mathcal{C}$ is faithfully C_3-graded.

Finally, for $R^{(6)}_{5,100}$ the involution is basis label 4, the fixed labels are $3, 5$, and the exhaustive anti-invariant generator search has 16 tuples. For $R^{(6)}_{5,408}$ these data are respectively 1, $4, 5$, and 8. Each search has exactly one admissible representation, in which the involution acts as $+I$ on the two anti-invariant coordinates. The Gram reconstruction forces exactly the dimension profile in the statement. In both original tensors the multiplicative closure of the supports of all XX^* is the whole basis. Thus the adjoint subcategory would be all of $\mathcal{C}$, which rules out any nontrivial faithful grading. This proves the two claimed exclusions. □

Exact finite certificate. The implementation is `software/new_morita_order6.py`, with audit output in `research/model_continuation/morita_order6_certificates.json`. The input certificate stores the involution, fixed labels, generator search size, every admissible anti-invariant representation, and the reconstructed dual dimensions in exact field coordinates. Verification reconstructs the exhaustive representation domain, rather than trusting a proposed list of branches. The finite dual routine enumerates every composition of 4 into six nonnegative parts, applies exactly (5.10), and checks that there are 60 cases. It constructs each full tensor, factors its Casimir polynomial exactly, and checks the recorded norm witness, the full grading identities, or the closed-subring support and unit-preserving based isomorphism. The four S_3 class triples are independently exhausted. The inherited certificates for $R^{(6)}_{2,20}$ and $R^{(6)}_{2,21}$ are replayed before they are used. Thus the audit output is a reproducible record of a complete finite necessary-condition calculation, with no unresolved branch silently discarded.

A separate replay, `research/model_continuation/replay_morita_order6.py`, checks equality with the saved finite audit and rejects mutations of the involution, branch list, generator count, representation matrices, dual dimension counts, and input basis labels. Its checks use explicit exceptions and remain active under `python -O`.

5.9 Forced pointed associators and the multiplicity-four frontier

The next obstruction uses the associator of a pointed subcategory to restrict central twists. It is a tensor criterion and applies to an entire family.

Proposition 5.18. *Let $m \geq 0$, and suppose a fusion category has invertible objects $1, t, t^2, t^3$, with t of order four, and two self-dual noninvertible objects X, Y satisfying*

$$tX = Y = Xt, \qquad tY = X = Yt,$$
$$X^2 = Y^2 = 1 + t^2 + m(X + Y), \qquad XY = t + t^3 + m(X + Y).$$

Then the pointed associator has class 2 in $H^3(C_4, \mathbb{C}^\times) \cong \mathbb{Z}/4\mathbb{Z}$. In a pseudo-unitary realization, every simple central summand of $I(t)$ or $I(t^3)$ has twist satisfying $\theta^4 = -1$.

Proof. Put $H = \langle t^2 \rangle$. Because t^2 fixes a simple object, the associator restricts trivially to H: the associativity scalars of the action on that object give a two-cochain trivializing the restriction. Thus its class in $H^3(C_4, \mathbb{C}^\times)$ is either 0 or 2.

Suppose first that the class is zero. The regular pointed algebra $A = 1 \oplus t \oplus t^2 \oplus t^3$ exists. Its right-module category has three simple objects. The invertible orbit contributes A; the orbit $\{X, Y\}$ contributes two simples M_+, M_-, each with underlying object $X \oplus Y$. Indeed its stabilizer is C_2, whose projective irreducible representations are the two ordinary one-dimensional representations, because $H^2(C_2, \mathbb{C}^\times) = 0$. This is the usual orbit–stabilizer description of modules over a regular pointed algebra (equivalently, the classification of modules over its action groupoid algebra).

Let $d = \mathrm{FPdim}(X) = \mathrm{FPdim}(Y)$, so $d^2 = 2 + 2md$ and $d = m + \sqrt{m^2 + 2}$. Normalize module dimensions by $\mathrm{FPdim}(A) = 4$; their vector is $(1, d/2, d/2)$. Every invertible fixes the simple module A and permutes M_+, M_-. On their difference line its action is a character $\chi : C_4 \to \{\pm 1\}$, hence $\chi(t^2) = 1$. The action matrix of X is symmetric, since X is self-dual. Its first row and column are $(0, 1, 1)$ by free–forgetful adjunction. The dimension eigenvector therefore forces its remaining block to be

$$\begin{pmatrix} a & 2m - a \\ 2m - a & a \end{pmatrix} \quad (a \in \mathbb{Z}).$$

Thus its eigenvalue on the difference line is the integer $\lambda = 2a - 2m$. Applying the fusion relation for X^2 on this line gives

$$\lambda^2 = 2 + m(1 + \chi(t))\lambda.$$

For $\chi(t) = -1$ this says $\lambda^2 = 2$; for $\chi(t) = 1$ it says $(\lambda - m)^2 = m^2 + 2$. Both are impossible for integers: $m^2 + 2$ lies strictly between consecutive squares for $m \geq 1$, and equals 2 for $m = 0$. Hence class 0 is impossible.

For completeness use representatives $a, b, c \in \{0, 1, 2, 3\}$ and standard cyclic cocycles

$$\omega_k(a, b, c) = \exp\left(\frac{2\pi i k}{4} a \left\lfloor \frac{b + c}{4} \right\rfloor\right).$$

Their restrictions have $\omega_k(2, 2, 2) = (-1)^k$, confirming that the restriction kernel is $\{0, 2\}$. The canonical pseudo-unitary indicator formula gives $\nu_4(t) = i^k$ and $\nu_4(t^3) = i^{9k}$; for $k = 2$ both equal -1. Reversing the cocycle convention inverts these values and has no effect on this conclusion.

The center indicator formula reads

$$\mathrm{FPdim}(\mathcal{C})\nu_4(t) = \sum_{Z \subset I(t)} [Z : I(t)] \, \mathrm{FPdim}(Z)\theta_Z^4.$$

The positive coefficients on the right sum to $\mathrm{FPdim}(I(t)) = \mathrm{FPdim}(\mathcal{C})$. Equality in the triangle inequality forces every $\theta_Z^4 = -1$. The same argument applies to t^3; compare [39, Proposition 3.8]. □

Lemma 5.19. *Under the hypotheses of Proposition 5.18, m is even. For the algebra $1 \oplus t^2$, its anti-invariant module action on the two fixed-object coordinates is, up to sign gauge,*

$$T_t = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T_{t^2} = -I, \qquad T_X = T_Y = 0.$$

In particular both noninvertibles have the anticommuting stabilizer used in Proposition 5.21.

Proof. The involution-module construction gives $T_t = \left(\begin{smallmatrix} 0 & \epsilon \\ \delta & 0 \end{smallmatrix}\right)$, where $\epsilon, \delta \in \{\pm 1\}$, and $T_{t^2} = \epsilon\delta I$. Since X, Y are self-dual, T_X and T_Y are symmetric integer matrices; they commute with T_t and satisfy $T_Y = T_t T_X$. If $\epsilon\delta = 1$, commutation makes the diagonal entries of T_X equal. Its

eigenvalue λ on the $+1$ eigenspace of T_t is therefore an integer. But the relation for X^2 gives $\lambda^2 = 2 + 2m\lambda$, which is impossible as above. Consequently $\epsilon\delta = -1$. A symmetric matrix commuting with this skew-symmetric T_t is scalar, so $T_X = aI$. The symmetry of $T_Y = aT_t$ forces $a = 0$. Finally every entry of T_X is congruent modulo two to the corresponding coefficient of multiplication by X on $\{X, Y\}$, namely m. Hence m is even. Changing the sign of one fixed-object coordinate gives the displayed sign gauge. □

Theorem 5.20. *The fusion ring $R_{4,6}^{(6)}$ has no categorification over $\mathbb{C}$.*

Proof. Its full tensor has the family in Proposition 5.18 with $m = 4$ and $(t, t^2, t^3, X, Y) = (b_1, b_5, b_2, b_3, b_4)$. All simple Frobenius–Perron dimensions belong to $\{1, 4 + 3\sqrt{2}\}$. The two-dimension Galois-pseudo-unitarity theorem therefore reduces general existence to pseudo-unitary existence, without a pivotality assumption on the original category.

Lemma 5.19 gives exactly one anti-invariant module representation for the split involution algebra $1 \oplus t^2$, after sign gauge. The finite module checker independently reconstructs it. The action of t^2 is $-I$ on both fixed-object coordinates. The cubic spaces of X and Y each have dimension four; Proposition 5.21 therefore forces

$$-2 \le \operatorname{Re}\nu_3(X), \operatorname{Re}\nu_3(Y) \le 1.$$

The exact central-induction calculation has ten low frames and four second-indicator assignments per frame. All low summands have twist one. Every high restriction column containing t or t^3 is restricted by Proposition 5.18 to $\theta^4 = -1$. In the normalized relative-trace domain over $\mathbb{Q}(\sqrt{2})$, this means its fourth coordinate is exactly $(-1, 0)$. Keeping only those trace types is necessary; actual roots with $\theta^4 = -1$ have precisely that coordinate under every relative Galois embedding. The order-four trace system includes all root orders with the allowed prime support $\{2, 3\}$, including arbitrarily large orders through their zero trace type.

Each of the resulting forty rational nonnegative systems has a recorded integral separating vector. Verification reconstructs the low frames, the unique module branch, the entire joint trace domain, the four second-indicator assignments, and the forced-twist variable deletion. For every system $Az = b$, its vector y satisfies $A^T y \ge 0$ and $b^T y < 0$. Therefore no feasible central induction data, and hence no pseudo-unitary categorification, exists. This contradicts the general-to-pseudo-unitary reduction. □

Certificate and implementation. The tensor-based implementation is `software/round3_smallest.py`. Its verifier never calls an optimizer; the discovery routine merely proposes exact rays. The certificate, proof audit, all-census applicability scan, positive controls and optimized replay are retained in `research/round3/smallest/`.

5.10 A stabilizer obstruction to scalar pivotal rotation

Proposition 5.21 (Anticommuting stabilizers and third rotation)**.** *Let $\mathcal{C}$ be a spherical fusion category, let g be an invertible of order two for which $A = \mathbf{1} \oplus g$ is an algebra, and let Y be a simple object with $gY \cong Y \cong Yg$. Suppose that left multiplication by g exchanges the two right A-module structures on Y. Put $m = \dim \operatorname{Hom}(\mathbf{1}, Y^{\otimes 3})$. Then m is even. If $m > 0$, the pivotal third-rotation operator on this space is not scalar. More precisely,*

$$\nu_3(Y) = -(a + b\zeta_3 + c\zeta_3^2), \qquad a, b, c \in \mathbb{Z}_{\ge 0}, \quad a + b + c = m/2.$$

If 3 is coprime to the Frobenius–Schur exponent, this becomes

$$\nu_3(Y) \in \{-m/2 + 3b : 0 \le b \le \lfloor m/4 \rfloor\}.$$

In particular, if $m = 2$ under this coprimality hypothesis, then

$$\nu_3(Y) = -1.$$

Proof. Use strict monoidal notation and an associative identification $q : g^2 \xrightarrow{\sim} \mathbf{1}$ supplied by the algebra A. Choose isomorphisms $l : gY \to Y$ and $r : Yg \to Y$, normalized by

$$l(\mathrm{id}_g \otimes l) = q \otimes \mathrm{id}_Y, \qquad r(r \otimes \mathrm{id}_g) = \mathrm{id}_Y \otimes q.$$

The two normalized right module structures are r and $-r$. Transporting the right action on gY through l gives

$$r' = l(\mathrm{id}_g \otimes r)(l^{-1} \otimes \mathrm{id}_g).$$

The hypothesis says $r' = -r$, or equivalently

$$r(l \otimes \mathrm{id}_g) = -l(\mathrm{id}_g \otimes r). \tag{5.13}$$

This also shows directly how the hypothesis is read from the anti-invariant module matrices: it is $T_g[Y, Y] = -1$.

Define an invertible endomorphism of Y^2 by

$$a = (r \otimes l)(\mathrm{id}_Y \otimes q^{-1} \otimes \mathrm{id}_Y),$$

and put $P = a \otimes \mathrm{id}_Y$, $Q = \mathrm{id}_Y \otimes a$ on Y^3. Expanding either composite inserts two copies of q^{-1}, with common intermediate tensor word $YggYggY$. The endpoint maps agree. On the middle gYg, the two composites use the two sides of (5.13). Consequently

$$PQ = -QP.$$

The restrictions to $V = \mathrm{Hom}(\mathbf{1}, Y^3)$ remain invertible. Taking determinants shows that $\dim V$ is even.

Let E be pivotal cyclic rotation moving the last Y to the first position. The canonical cyclic map

$$\mathrm{Hom}(\mathbf{1}, B \otimes Y) \longrightarrow \mathrm{Hom}(\mathbf{1}, Y \otimes B)$$

is natural in the entire object B. Applying this naturality to $B = Y^2$ and $a \in \mathrm{End}(B)$ gives the exact identity

$$EP = QE.$$

The construction and its naturality are those of [31, Definition 3.1, Lemma 3.4 and Theorem 5.1]. No braiding or self-duality sign is used. If E were scalar on nonzero V, then $P = Q$, contradicting $PQ = -QP$ and invertibility.

The module-action normalizations also give $a^2 = I$: after two pair insertions, the first endpoint applies $r(r \otimes \mathrm{id}_g)$, the second applies $l(\mathrm{id}_g \otimes l)$, and the two resulting copies of q cancel the inserted copies of q^{-1} by associativity. Set $A_j = E^j P E^{-j}$ for $j \in \mathbb{Z}/3\mathbb{Z}$. These are involutions, and the anticommutation of $A_0 = P$ and $A_1 = Q$, together with cyclic conjugation, makes them pairwise anticommuting. They therefore give a representation of the complex Clifford algebra $\mathrm{Cl}_3 \cong M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$. In that algebra the element

$$U = \frac{-I + A_0A_1 + A_1A_2 + A_2A_0}{2}$$

satisfies $U^3 = I$, $UA_jU^{-1} = A_{j+1}$, and has trace -1 on each two-dimensional simple module. These identities also follow by direct multiplication using $A_i^2 = I$ and $A_iA_j = -A_jA_i$. The eight-dimensional universal algebra calculation is reproduced with rational coefficients by `research/round2_audit/clifford_rotation.py`; it verifies $U^3 = I$ and $UA_j = A_{j+1}U$. Its scalar coefficient is $-1/2$, while the three bivectors are traceless on either two-dimensional simple, giving trace -1 without a matrix-choice assumption. Thus $S = U^{-1}E$ commutes with the Clifford action and satisfies $S^3 = I$. On its multiplicity spaces, of total dimension $m/2$, let

a, b, c be the multiplicities of $1, \zeta_3, \zeta_3^2$ for S. Taking the trace of $E = US$ proves the asserted formula. Reversing the cyclic convention exchanges b and c, so the same value set applies to the standard indicator convention.

On the other hand, the central-twist formula puts $\nu_3(Y)$ in $\mathbb{Q}(\zeta_N)$, where N is the Frobenius–Schur exponent. If $3 \nmid N$, then $\mathbb{Q}(\zeta_N) \cap \mathbb{Q}(\zeta_3) = \mathbb{Q}$. Hence $b = c$, and $a + 2b = m/2$ gives the rational-value list. □

Remark 5.22 (No self-duality requirement). The stabilizer-rotation proposition applies equally to a non-self-dual simple Y. Its hypotheses and proof use isomorphisms $gY \cong Y \cong Yg$, not an isomorphism $Y \cong Y^*$. In particular, rigidity gives

$$m = \dim \mathrm{Hom}(\mathbf{1}, Y^{\otimes 3}) = N_{Y,Y}^{Y^*};$$

one must not replace the last index by Y unless Y is self-dual. The pair-insertion operators act on Y^3 itself, and naturality of pivotal rotation applies to Y^2 and its endomorphism a without dualizing Y. Thus the same Clifford multiplicity formula applies to each member of a dual pair separately.

Corollary 5.23 (Projected third-indicator spectrum)**.** *Under the hypotheses of Proposition 5.21,*

$$-m/2 \le \mathrm{Re}\, \nu_3(Y) \le m/4, \qquad \mathrm{Re}\, \nu_3(Y) \in \{m/4 - 3a/2 : a = 0, 1, \ldots, m/2\}.$$

Let K be any number field satisfying $K \cap \mathbb{Q}(\zeta_3) = \mathbb{Q}$. The normalized relative trace of $\nu_3(Y)$ to K equals its real part and therefore belongs to the displayed finite set. This includes every real number field and the complex cyclotomic field $\mathbb{Q}(\zeta_8)$. If instead K contains ζ_3, the unprojected possibilities are the full set

$$\{-(a + b\zeta_3 + c\zeta_3^2) : a, b, c \in \mathbb{Z}_{\ge 0},\ a + b + c = m/2\}.$$

Proof. The preceding proposition gives $\nu_3(Y) = -(a + b\zeta_3 + c\zeta_3^2)$ with $a + b + c = m/2$. Its real part is $-a + (b + c)/2 = m/4 - 3a/2$, which gives the stated set and bounds. Under the intersection hypothesis, $K(\zeta_3)/K$ is quadratic and its nontrivial automorphism exchanges ζ_3 and ζ_3^2. Consequently its normalized relative trace sends both roots to $-1/2$, and sends the indicator to $-a + (b + c)/2$. Transitivity gives the same value when the trace is taken from any larger common field. The quadratic field $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$ does not embed into $\mathbb{Q}(\zeta_8)$: the latter's three quadratic subfields are $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$. This proves the stated complex-field example. When K contains ζ_3, normalized trace fixes the entire indicator, so the original Clifford multiplicity set must be retained. □

Corollary 5.24. *Every pseudo-unitary categorification of $R_{3,215}^{(6)}$, in the basis of dimensions $(1, 1, d, d, d - 1, d + 1)$, must satisfy*

$$\nu_3(b_4) = \nu_3(b_5) = -1.$$

Proof. The exact full Morita-action calculation for $A = 1 + g$ leaves two branches, both with $T_g = -I$ on the two fixed simples b_4, b_5. They satisfy $N_{44}^4 = N_{55}^5 = 2$, and both are self-dual. Their third-rotation spaces therefore have dimension two. The global dimension is $136 + 32\sqrt{17}$, whose norm is $1088 = 2^6 \cdot 17$. The Cauchy theorem [5, Theorem 3.9] excludes the prime 3 from the Frobenius–Schur exponent. Apply Proposition 5.21. □

5.11 Exact certificate and exhaustive coverage

The new obstruction is implemented without a distinguished census label in `software/stabilizer_rotation.py`. Its input is the multiplication tensor. The certificate first reconstructs the complete list of anti-invariant actions for the forced algebra $\mathbf{1} \oplus g$. For the multiplicity-three frontier this list has sixteen entries. Twelve are rejected by exact separating functionals for the full Morita-action Gram system, and two by its rank bound. In each of the remaining two entries, left multiplication by g exchanges both pairs of module structures on the fixed simples b_4, b_5. Equivalently, the diagonal entries of T_g on their anti-invariant lines are both -1. Their cubic invariant spaces have dimension two. The pair-insertion theorem therefore forces $\nu_3(b_4) = \nu_3(b_5) = -1$, since the Cauchy prime support is $\{2, 17\}$.

The central calculation is exhaustive; it does not assume the rank-twenty-two modular datum retained elsewhere as a relaxation witness. All ninety-one low induction frames are reconstructed from the tensor and its formal codegrees. Thirty-one frames admit character labellings; there are fifty labellings in total and four possible second-indicator assignments. The certificate contains an exact Farkas vector for each of these two hundred systems. Each system includes the induction Gram equations, relative twist traces through order three, cyclic-rotation character constraints, joint Galois compatibility of dimensions and eigenvalues, and the low-row modular relations. It additionally imposes the two forced third indicators. If $Ax = b$ is such a system, its recorded integral vector y is checked to satisfy

$$A^{\mathsf{T}} y \geq 0, \qquad b^{\mathsf{T}} y < 0.$$

Every feasible central decomposition supplies $x \geq 0$, so these inequalities exclude every branch. The variables allow all roots of unity whose prime support is $\{2, 17\}$ via their finite joint relative-trace types; no finite upper bound on the actual exponent or central rank is imposed. For clarity, the order-three relative-trace domain has exactly ten joint patterns. Write a root's order as $2^a 17^b$. If $b \geq 2$ or $a \geq 3$, its normalized traces at powers $1, 2, 3$ all vanish. Otherwise $a \in \{0, 1, 2\}$ and $b \in \{0, 1\}$ give the remaining nine distinct patterns, with the two quadratic signs when 17 divides the order. For primitive seventeenth roots the first trace is $(-1 \pm \sqrt{17})/16$; the quadratic character of 2 is $+1$ and that of 3 is -1. This elementary enumeration explicitly includes arbitrarily large exponent orders through the all-zero pattern. Integral high-column decompositions are not needed for this final exclusion.

The general-scope wrapper is `software/galois_pseudo_reduction.py`. It reconstructs the full unrestricted signed-sphericalization enumeration, including all 1480 signed quotient tables and all sign orbits. Every nongauge character component is rejected by its exact arithmetic certificate, and the only nonzero dimension-character orbit of the original ring is checked directly to be its Frobenius–Perron orbit. Consequently every hypothetical complex categorification has a pseudo-unitary Galois conjugate to which the preceding obstruction applies. No positivity normalization is imposed during the signed enumeration. The combined certificate is stored in `certificates/stabilizer_rotation_215.json`; the public replay entry point is `stabilizer_rotation.verify_general(R,c)`.

Theorem 5.25 (Resolution of the multiplicity-three frontier)**.** *The based ring $R^{(6)}_{3,215}$ has no complex fusion categorification.*

Proof. The unrestricted signed-sphericalization calculation of Proposition 3.28, independently reconstructed by the general certificate above, shows that any hypothetical categorification has a pseudo-unitary Galois conjugate. Work with that conjugate.

The complete involution-module domain has sixteen branches. The full Morita-action Gram identities exclude fourteen, and either remaining branch satisfies the hypotheses of Corollary 5.24. Consequently both $\nu_3(b_4)$ and $\nu_3(b_5)$ equal -1.

The exhaustive central-induction calculation in Section 5.11 excludes those two equalities for every permitted low frame, character assignment and second-indicator choice. Each of its 200 integral separating vectors is checked against the reconstructed rational system; no numerical infeasibility output is used as a certificate. Thus no pseudo-unitary categorification exists, contradicting the reduction. □

This proof uses no identification of a hypothetical Drinfeld center with a known braided category. The compatible cyclic-eight modular datum from the previous continuation remains a useful test of the older necessary conditions, but the local stabilizer and rotation constraints prevent it, and every other central-induction branch, from arising here.

5.12 Further stabilizer exclusions at multiplicity four

Corollary 5.26. *The ring $R^{(6)}_{4,416}$ has no complex fusion categorification.*

Proof. The same local theorem excludes `R6_4_416` without changing its hypotheses to fit the input. The unique relevant invertible involution is b_1, with fixed simples b_4, b_5. Both exhaustive anti-invariant module actions have negative stabilizer sign on both fixed objects. Each cubic invariant space has dimension four. The Clifford-rotation trace therefore lies in $\{-2, 1\}$, since the Cauchy prime support is $\{2, 5\}$. In particular its upper bound is one.

The quadratic central-induction calculation reconstructs all 895 low frames. After character reciprocity and joint Galois pruning, two relevant character labellings remain. All sixteen second-indicator choices for each labelling are excluded by exact Farkas rays, using the interval bound $-2 \le \nu_3(b_4), \nu_3(b_5) \le 1$. Thus only 32 separating vectors are needed, and no discrete guess for the third indicators is required.

For general scope, the unrestricted signed enumeration has 2376 quotient tables in 83 sign orbits. All nongauge branches are rejected by 222 exact character obstructions. Of the original nonzero dimension-character components, the non-Frobenius–Perron component has global dimension polynomial $x^2 - 32x + 128$ and fails the norm condition at prime 5, required by the Frobenius–Perron global dimension polynomial $x^2 - 800x + 3200$. Consequently the same general-to-Galois-pseudo-unitary reduction applies. The combined general-scope certificate is `certificates/stabilizer_rotation_416.json`; it has been replayed under optimized Python with all numerical optimizer entry points disabled.

□

The ring $R^{(6)}_{4,174}$. The four invertible basis elements have Frobenius–Perron dimension 1; the remaining two have dimension $d = 4 + 3\sqrt{2}$, with $d^2 = 2 + 8d$. Thus the global dimension is $72 + 48\sqrt{2}$, whose norm is 576. The involution-module calculation has a single antiinvariant branch, and its stabilizer commutator is negative on both noninvertible simples. Each cubic invariant space has dimension four. Since the global-dimension prime support contains 3, the exact relative traces of their third indicators are allowed to be $-2, -1/2$, or 1. The computation first uses the weaker interval $[-2, 1]$.

At trace height six, there are ten low induction frames. Eight character labellings survive the earlier exact conditions; their full second-indicator assignments give 128 systems. Every system admits an exact separating ray, already with the interval constraint. Hence no pseudo-unitary categorification exists. The two distinct Frobenius–Perron dimensions supply the previously established general-to-Galois-pseudo-unitary reduction, so the exclusion holds for arbitrary complex fusion categories. The composite certificate uses the general higher-stabilizer verifier, with the reduction payload `two_fp_values`; all matrix and branch data are reconstructed from the input tensor.

The ring $R^{(6)}_{4,427}$. Write $d = 7+5\sqrt{2}$, so that $d^2 = 1+14d$. The Frobenius–Perron dimensions are $(1, 1, d, d, d-1, d+1)$ and the global dimension is $400+280\sqrt{2}$, of norm 3200. There is one antiinvariant involution-module branch. Its stabilizer commutator is negative on the last two simples, whose cubic invariant spaces both have dimension four. The prime 3 does not divide the global-dimension norm; the Clifford criterion therefore restricts both third indicators to $\{-2, 1\}$.

The trace-height-six computation enumerates 315 low induction frames. The surviving character assignments comprise eight labellings and 32 second-indicator systems. Exact separating rays exclude every system even using the weaker interval $[-2, 1]$. This proves pseudo-unitary nonexistence. The independent unrestricted signed-quotient reduction has 2608 tables, forming 87 sign-gauge orbits, with 239 nongauge character rejections. Its exact character certificates reduce every categorification to the Galois-pseudo-unitary case. Combining the two verified components gives general nonexistence through the same `general_stabilizer_rotation_higher` verifier.

Theorem 5.27 (Quartic stabilizer-trace exclusions)**.** *The rings $R^{(6)}_{4,258}$, $R^{(6)}_{4,325}$, $R^{(6)}_{8,570}$ and $R^{(6)}_{8,2203}$ have no complex categorification.*

Proof. The tensor-only anti-invariant module enumeration has one possible representation for each ring. The order-two invertible has mixed stabilizer sign -1 on the self-dual simple b_5 in the two multiplicity-four rings, and on b_4 in the two multiplicity-eight rings. Each of these cubic invariant spaces has dimension four. Thus the preceding bound applies in every pseudo-unitary categorification.

For $R^{(6)}_{4,258}$, set $u = 2\cos(2\pi/20)$. The exact dimension-field embedding gives

$$D = 12u^3 + 20u^2 - 20u - 20, \qquad p_D(t) = t^4 - 120t^3 + 2400t^2 - 16000t + 32000.$$

The Cauchy prime support is $\{2, 5\}$. At height three, Proposition 4.11 gives $B = 40$ and 31 distinct trace vectors, including zero. The complete low-induction search has three frames and four second-indicator choices per frame. Every one of these twelve systems has an exact separating ray.

For $R^{(6)}_{4,325}$, set $u = 2\cos(2\pi/16)$. Here

$$D = 8u^3 + 16u^2, \qquad p_D(t) = t^4 - 128t^3 + 2560t^2 - 16384t + 32768.$$

The prime support is $\{2\}$, the height-three order bound is 32, and there are 25 trace vectors. The two low frames and four second-indicator choices per frame give eight exact separating rays. For both $R^{(6)}_{8,570}$ and $R^{(6)}_{8,2203}$, take $u = 2\cos(2\pi/20)$. The shared dimension-field data are

$$D = 160u^3 + 300u^2 - 220u - 400, \qquad p_D(t) = t^4 - 1400t^3 + 44000t^2 - 400000t + 800000.$$

Again $S = \{2, 5\}$, $B = 40$, and the trace domain has 31 vectors. Each ring has seven low frames. There is one second-indicator choice for $R^{(6)}_{8,570}$ and four for $R^{(6)}_{8,2203}$, giving respectively seven and twenty-eight exact separating rays. These computations establish pseudo-unitary nonexistence in all four cases.

The general certificates also reconstruct all signed sphericalization quotients and their character obstructions. For $R^{(6)}_{4,258}$ there are 32 quotients in one sign-gauge orbit. For $R^{(6)}_{4,325}$ there are 96 quotients in four orbits; the three nongauge orbits require nine character rejections. The two multiplicity-eight rings give respectively 96 quotients in six orbits with ten nongauge character rejections, and 1472 quotients in 46 orbits with 111 such rejections. The surviving orbit in each case is gauge isomorphic to the original ring, and the nonpositive original

characters are excluded by the recorded exact tests. Consequently any complex categorification would have a Galois conjugate with the positive spherical dimensions just excluded.

The canonical implementation is `software/cyclotomic_stabilizer_trace.py`. Its `verify_general` entry point checks both the sphericalization reduction and the complete pseudo-unitary certificate. It reconstructs the module domain, field embedding, trace domain, low frames, high columns and second-indicator alternatives from the input tensor; no census identifier enters a mathematical test. Every retained high column satisfies the exact integral dimension quotient and central simple d-number conditions. All separating inequalities are checked with integers. The certificate files are `certificates/cyclotomic_stabilizer_R6_4_258.json`, `certificates/cyclotomic_stabilizer_R6_4_325.json`, `certificates/cyclotomic_stabilizer_R6_8_570.json` and `certificates/cyclotomic_stabilizer_R6_8_2203.json`. □

The same ambient-field computation applies to the other quartic inputs recorded in `research/round2_literature/quartic/`. Its exploratory modular refinement keeps global Galois multiplicity balance; it does not copy the quadratic assertion that Galois squares preserve every relative twist-trace type. A surviving linear system is retained as a scoped necessary relaxation and does not supply a category.

6 Categorical models and exact tensor reconstruction

Positive records are established by tensor matches to constructions, not by lack of an obstruction. We use Deligne products of the standard rank-two and rank-three unitary categories, the TT_3 category of [13, Definition 2.5], and the positive quadratic constructions of [28, Proposition 5.1, Corollary 5.8]. We use the latter results only for existence, not as a negative classification oracle. Every displayed tensor is reconstructed from its defining rules and compared under unit-preserving permutations with the census.

6.1 A central fermion and a cyclic-four quadratic ring

Let $\mathcal{D}$ be the unitary near-group category $\mathrm{Ad}(E_6)$ with basis $1, h, X$ and rules

$$h^2 = 1, \qquad hX = Xh = X, \qquad X^2 = 1 + h + 2X. \tag{6.1}$$

Its noninvertible dimension is $d = 1 + \sqrt{3}$, and $D = \dim \mathcal{D} = 4 + 2d$.

Lemma 6.1 (Central fermion)**.** *There is an invertible $z \in Z(\mathcal{D})$ with $F(z) = h$, $z^2 = 1$, and $\theta_z = -1$. The full subcategory generated by z is the unitary symmetric category* sVec.

Proof. The formal codegrees are D, $D' = 6 - 2\sqrt{3}$, and 2. Thus $I(1) = 1 \oplus A \oplus B$, with dimensions $1, 1 + d, 2 + d$. Since $FI(1) = 3 \cdot 1 + h + 2X$, comparison of the unit multiplicities and of the coefficients of $1, d$ gives

$$F(A) = 1 + X, \qquad F(B) = 1 + h + X.$$

All three low twists are one.

The induction Gram matrix in the basis $(1, h, X)$ is

$$\begin{pmatrix} 3 & 1 & 2 \\ 1 & 3 & 2 \\ 2 & 2 & 8 \end{pmatrix}.$$

Only B among the low objects occurs in $I(h)$, once. The remaining endomorphism dimension is $3 - 1 = 2$, so the remainder consists of two distinct simples, both with multiplicity one. Its

restriction is $FI(h) - F(B) = 2h + X$. Consequently these two objects are z, W with $F(z) = h$ and $F(W) = h + X$. The object z has dimension one and is invertible. Its square restricts to the unit; the only central simple restricting to the unit is the unit itself, as the decomposition of $I(1)$ shows. Therefore $z^2 = 1$.

The pointed associator class on $\langle h \rangle$ is trivial: the rank-one module category on X supplies a two-cochain whose coboundary is the restricted associator. In the positive spherical structure, $\nu_2(h) = 1$. Equation (4.6) at $k = 2$ implies that every twist in $I(h)$ squares to one, because all dimensions are positive and their total is D. At $k = 1$ it gives

$$0 = (2 + d) + \theta_z + (1 + d)\theta_W.$$

The known positive summand has exactly the total dimension of the other two. Equality in the triangle inequality forces $\theta_z = \theta_W = -1$. Finally an order-two invertible in a unitary ribbon category with twist -1 has self-braiding -1 in its canonical positive structure. Its generated pointed braided category is sVec. □

Proposition 6.2 (Fermionic zesting)**.** *The ring $R^{(6)}_{2,2}$ has a unitary categorification. More generally, if a unitary fusion category $\mathcal{D}$ has a central fermion z of order two, the rules*

$$(U, a) \star (V, b) = (U \otimes V \otimes F(z)^{ab}, a + b), \qquad a, b \in \mathbb{Z}/2, \tag{6.2}$$

on $\mathcal{D} \boxtimes \mathrm{Vec}_{\mathbb{Z}/2}$ admit a unitary associator.

Proof. Use the central two-cocycle $\lambda(a, b) = z^{ab}$ and normalized associativity scalar $\tau(a, b, c) = i^{abc}$, where the exponents use representatives $0, 1$. The only central interchange obstruction is $c_{\lambda(a,b),\lambda(c,d)} = (-1)^{abcd}$. A direct calculation gives

$$\begin{aligned} bcd + a(b \oplus c)d + abc - (a \oplus b)cd - ab(c \oplus d) \\ \equiv 2abcd \pmod 4, \end{aligned}$$

where $\oplus$ is addition modulo two. Thus the coboundary of τ is precisely that interchange obstruction. The associative-zesting construction of [9, Definition 3.2, Proposition 3.4], in the relative-center formulation also recalled in [3, Definition 2.1], therefore supplies the pentagon and rigidity for (6.2). All structural maps used are unitary, including the scalar τ, so the resulting fusion category is unitary. This use of a known construction does not claim to have independently solved its individual F-symbol equations.

Apply Lemma 6.1. Let $g = (1, 1)$, identify $g^2 = (h, 0)$, and write $Y = (X, 0)$ and $X_1 = (X, 1)$. Then the invertibles form C_4,

$$gY = X_1, \quad gX_1 = Y, \quad Y^2 = X_1^2 = 1 + g^2 + 2Y, \quad X_1 Y = g + g^3 + 2X_1.$$

These rules reconstruct the full tensor that is based-isomorphic to $R^{(6)}_{2,2}$. The supplement checks that equality and all sixteen scalar coboundary identities exactly. □

The proposition is a positive construction, not an assertion that this fusion category or its ring has not appeared in another presentation. It is particularly useful here because a ring that survived the first obstruction pass is thereby removed from the residual list for the correct reason: existence rather than an attempted exclusion.

6.2 Finite-group and group-theoretical models

For $G = (C_3 \times C_3) \rtimes C_4$, with the order-four rotation $(x, y) \mapsto (-y, x)$, the character ring is reconstructed from four linear characters and two induced characters. Exact character inner products give all fusion coefficients and identify $R^{(6)}_{2,11}$.

We also compute the untwisted group-theoretical categories $\mathcal{C}(G, 1, H, 1)$ for

$$(G, H) = (A_4, C_2), \quad (A_5, D_{10}), \quad (A_5, S_3).$$

Here D_{10} has order ten. The finite-group construction is standard; we recall its explicit tensor calculation to distinguish it from a mere comparison of dimension types. Realize the category as H-bimodules in Vec_G. Its simples are supported on double cosets HgH and indexed by irreducible representations of $H \cap gHg^{-1}$. The tensor product is relative tensor product over the group algebra of H. Equivalently it is convolution of the corresponding equivariant vector bundles. The character of a convolution fiber is computed by summing over the middle H-orbits and taking invariants under the stabilizer; averaging the finite stabilizer action gives its multiplicities. This is a finite exact character-inner-product calculation. The code lists the groups as permutations, computes the double cosets and stabilizers, and uses their exact character values. It then checks the resulting complete fusion tensors.

The three outputs are respectively $R^{(6)}_{2,36}$, $R^{(6)}_{2,105}$, and $R^{(6)}_{3,141}$. The last model is also discussed in [2, Section 6.5]. Finite groups, ordinary unitary representations, and relative tensor product over a finite-dimensional C^*-algebra give unitary fusion categories in these untwisted cases.

6.3 ADE modules, cyclic Izumi models, and a finite group

The following constructions add five realizations at multiplicity two and one at multiplicity three. Each constructor builds its fusion tensor without reading the census, then searches for a full based-ring match.

Proposition 6.3 (ADE modules and their zesting)**.** *The rings $R^{(6)}_{2,75}$, $R^{(6)}_{2,14}$, $R^{(6)}_{2,128}$ and $R^{(6)}_{2,140}$ admit unitary categorifications. They are, respectively, the E_6 module category over $SU(2)_{10}$, its fermionic grading zest, the D_6 module category over $SU(2)_8$, and the local D_{10} module category over $SU(2)_{16}$.*

Proof. Use the unitary $SU(2)_k$ category with simples $V_0, \dots, V_k$ and fusion coefficients

$$N^l_{ij} = 1 \quad \Longleftrightarrow \quad |i - j| \le l \le \min(i + j, 2k - i - j), \quad i + j + l \equiv 0 \pmod 2.$$

The E_6 algebra is $V_0 \oplus V_6$ at level ten. The D-type algebra is $V_0 \oplus V_k$ at levels divisible by four. Their existence, separability, module categories and the ADE module graphs are the Kirillov–Ostrik construction [26, Theorem 6.1 and Section 7]. These algebras and their module tensor products have the standard unitary structures.

For E_6, label the graph vertices $0, \dots, 5$ and its edges

$$01, \quad 12, \quad 23, \quad 34, \quad 25,$$

with unit vertex zero. Multiplication by the free image of V_1 is the adjacency matrix B. The rows $(B^j)_{0,*}$, $0 \le j \le 5$, form an invertible matrix V. Centrality of the free image of V_1 and the cyclicity of the unit vector imply that all multiplication matrices are uniquely determined by $N_i = \sum_j (V^{-1})_{ij} B^j$. The exact constructor evaluates these matrices, checks nonnegative integral entries and all tensor axioms, and matches $R^{(6)}_{2,75}$.

The parity of the original $SU(2)$ labels descends to the module category; in these coordinates it is $(0, 1, 0, 1, 0, 1)$. The free image of V_{10} is the degree-zero invertible at vertex four. The canonical central free functor sends the order-two fermion V_{10} to a central fermion g in the module category. Hence associative zesting may multiply the product of two odd objects by g. Its obstruction is the fermionic sign $(-1)^{abcd}$, trivialized by the scalar cochain i^{abc}, exactly as in Proposition 6.2. All sixteen cochain identities are checked. This constructs a unitary fusion

category and its tensor is $R^{(6)}_{2,14}$. The construction uses the actual grading and central fermion, not just a ring automorphism.

For the D cases put $m = k/2$. The orbit modules are O_j, $0 \le j < m$, and the fixed object V_m splits as $U_+ \oplus U_-$. The free image of V_j is $O_{\min(j,k-j)}$ unless $j = m$. Products of two orbit modules are obtained by applying this induction rule to the displayed $SU(2)$ coefficients. For an orbit module times $U_+ \oplus U_-$ use the same rule; the difference $U_+ - U_-$ has eigenvalue zero for odd j and $(-1)^{j/2}$ for even j. At the two levels used here, $k \equiv 0 \pmod 8$, the resolved objects are self-dual. The fixed-point resolution formulas of [26, Section 7, D_{2m} fusion table] give the remaining products. Equivalently, in the local sector of this congruence class,

$$U_\pm^2 = U_\pm + O_0 + O_4 + \cdots + O_{m-4}, \qquad U_+U_- = O_2 + O_6 + \cdots + O_{m-2}.$$

Together with Frobenius reciprocity these formulas are the constructor's complete specification. At $k = 8$ use all six modules; at $k = 16$ use the local modules $O_0, O_2, O_4, O_6, U_+, U_-$. Their tensors match $R^{(6)}_{2,128}$ and $R^{(6)}_{2,140}$ respectively. □

Proposition 6.4 (The Izumi cyclic-five model)**.** *The ring $R^{(6)}_{2,143}$ is the Grothendieck ring of the commutative even part of the unitary Izumi C_5 subfactor.*

Proof. The existence of the cyclic-five Izumi subfactor and the relevant even part are established in the literature summarized in [21, Section 6.6]. Its commutative fusion rules are those in [17, Theorem 7]. To specify the full tensor used for the match, write $1, a, b_1, b_2, c_1, c_2$, identify indices i and $-i$ modulo five, and put $b_0 = c_0 = 0$, $\Sigma = a + b_1 + b_2 + c_1 + c_2$. The products are

$$\begin{aligned} a^2 = 1 + \Sigma, \qquad ab_i = \Sigma - b_i, \qquad ac_i = \Sigma + c_i, \qquad b_ic_j = \Sigma, \\ b_ib_j = \delta_{ij}1 + (1 - \delta_{ij})a + \Sigma - a - b_{i+j} - b_{i-j}, \\ c_ic_j = \delta_{ij}1 + (1 + \delta_{ij})a + \Sigma - a + c_{i+j} + c_{i-j}. \end{aligned}$$

All products commute and all simples are self-dual. The independent constructor specializes these formulas at five and verifies all 216 coefficients of the matched tensor. Existence is the cited subfactor construction, not a consequence of the based-ring axioms alone. □

Proposition 6.5 (An exact finite-group model)**.** *The ring $R^{(6)}_{3,47}$ is categorified by* $\mathrm{Rep}(GL(3,\mathbb{F}_2)) \simeq \mathrm{Rep}(PSL(2,7))$.

Proof. Generate the group as all invertible 3×3 binary matrices, with actual matrix multiplication. It has 168 elements and six conjugacy classes of sizes $1, 21, 56, 42, 24, 24$, ordered by element orders $1, 2, 3, 4, 7, 7$. Put $a = (-1 + \sqrt{-7})/2$ and $b = \overline{a}$. The candidate character table is

	1	2	3	4	7	7
1	1	1	1	1	1	1
3	3	−1	0	1	a	b
$\overline{3}$	3	−1	0	1	b	a
6	6	2	0	0	−1	−1
7	7	−1	1	−1	0	0
8	8	0	−1	0	1	1

.

We verify its representation-theoretic meaning, not just orthogonality. The program computes the multiplication constants of the six class sums in $\mathbb{C}[GL(3,\mathbb{F}_2)]$. For each row χ it checks that $|C|\chi(C)/\chi(1)$ is a unital character of this class algebra. The six characters are distinct and therefore exhaust its primitive central characters. By semisimplicity they are the normalized central characters of the actual irreducible representations. Row orthogonality and positivity of the displayed degrees fix the normalization, so the rows are the actual irreducible characters.

Taking their product inner products gives nonnegative integer structure constants, evaluated exactly in $\mathbb{Q}(\sqrt{-7})$. The resulting full tensor is based-isomorphic to $R^{(6)}_{3,47}$. A finite-group representation category has its usual unitary realization. □

6.4 Categorical structures, extensions, and central quotients

The following determinations use categorical constructions, followed by exact identification of their full fusion tensors. A based-ring isomorphism transports both existence and a proved obstruction to a specified categorical structure. It does not turn spherical existence into pseudo-unitary existence. All identifications in this section are recomputed by `software/literature_models.py`; no census label is an input to the construction.

6.4.1 Schopieray's spherical model

Schopieray constructs the rank-six modular category $\mathcal{C}(\mathfrak{so}_5, 3/2)_{\mathrm{ad}}$ in [41, Example 1, equations (1)–(2)]. Its displayed fusion matrices, in the order $(X_0, \dots, X_5)$, give our tensor $R^{(6)}_{2,124}$ after the permutation

$$(0, 1, 5, 4, 2, 3).$$

Here, and in the electronic catalogue, a permutation p means $N_{ij}^{\mathrm{census}\,k} = N_{p(i),p(j)}^{\mathrm{source}\;p(k)}$. There can be more than one such permutation; the catalogue stores the lexicographically first one. The six matrices are independently transcribed in the constructor and all 216 coefficients are compared. Schopieray's Theorem 1 proves that this based ring admits no pseudo-unitary categorification. Its exact existence record is therefore

	general	pivotal	spherical	pseudo-unitary	unitary
$R^{(6)}_{2,124}$	yes	yes	yes	no	no.

The last entry follows because a unitary fusion category is pseudo-unitary. This is a ring-level nonexistence theorem, not only the failure of a particular spherical structure to have positive dimensions. It does not contribute to the count of *general* exclusions.

6.4.2 A nilpotent extension and an Izumi even part

Write H_8 for the Kac–Paljutkin Hopf algebra of dimension eight. Choi–Lu–Sun give extensions of $\mathrm{Rep}(H_8)$ with one additional simple object in [7, equation (1.4), Section 7.1 and Tables 3–4]. With basis $(1, a, b, ab, N, D)$ and $a^2 = b^2 = 1$, their fusion rules are

$$\begin{aligned} gN = N, \quad gD = D, \quad N^2 = \sum_{g \in C_2^2} g, \\ ND = DN = 2D, \qquad D^2 = \sum_{g \in C_2^2} g + 2N. \end{aligned} \tag{6.3}$$

The relation $gN = N$ is meant for each $g \in C_2^2$, as is $gD = D$. The source provides existence; it is not inferred from associativity of these integer rules.

For a separate categorical justification of unitarizability, the successive adjoint subcategories determined by (6.3) have bases

$$\{1, a, b, ab, N, D\} \supset \{1, a, b, ab, N\} \supset \{1, a, b, ab\} \supset \{1\}.$$

Thus every categorification with these rules is nilpotent. A nilpotent fusion category is weakly group-theoretical and is unitarizable by [18, Theorem 5.20]. The program recomputes this adjoint series by support closure. The source tensor matches $R^{(6)}_{2,69}$ with the identity

permutation. This proves unitary *existence*; neither nilpotency alone nor a tensor-axiom check is used to assert existence.

The dual even part of the Izumi C_4 subfactor is a unitary fusion category. Its multiplication table in [24, p. 27], in the source order $(1, \kappa, \beta\kappa, \chi, \sigma, \beta)$, matches $R^{(6)}_{2,98}$ after the permutation

$$(0, 5, 1, 2, 4, 3).$$

The constructor recovers every entry of that table before making the comparison. The equality is with $R^{(6)}_{2,98}$, not with the dimension-matching tensor $R^{(6)}_{2,41}$. Equality of dimension lists would not establish this identification.

6.4.3 A free central-boson quotient

A *central boson* here means an invertible object $\widetilde{g}$ in the unitary Drinfeld center $Z(\mathcal{D})$, generating a symmetric subcategory equivalent to $\mathrm{Rep}(C_2)$, whose underlying object g in $\mathcal{D}$ is nontrivial. This includes the half-braiding and the Tannakian structure. Centrality of the basis element $[g]$ in $K_0(\mathcal{D})$ is not a substitute for this hypothesis.

Proposition 6.6 (Free central-boson quotient)**.** *Let $\mathcal{D}$ be a unitary fusion category and let $\widetilde{g}$ be a central boson as above. Suppose tensoring with g has no fixed point on* $\mathrm{Irr}(\mathcal{D})$*. Then the right modules over the central algebra $A = 1 \oplus g$ form a unitary fusion category $\mathcal{D}_A$. Its simples are indexed by the two-element g-orbits. For orbit representatives i, j, k its fusion coefficients are*

$$\widetilde{N}^{[k]}_{[i],[j]} = N^k_{ij} + N^{gk}_{ij}. \tag{6.4}$$

Proof. The algebra A is the image of the regular function algebra of C_2 under the central Tannakian embedding. It is connected and separable, and is commutative as a central algebra. The half-braiding makes right A-modules into a tensor category with product $\otimes_A$; see [15, Theorem 8.23.3]. Use the standard dagger Frobenius structure on the regular algebra. Work first with its unitary modules. Every algebraic module is a summand of a free module by separability; an idempotent in the finite-dimensional C^*-endomorphism algebra of a free unitary module is conjugate to an orthogonal projection. Thus these unitary modules represent the whole algebraic module category up to equivalence. Relative tensor products are the images of the self-adjoint separability projections. The unitary associativity maps of $\mathcal{D}$ descend to unitary maps between their images, and the module morphism spaces inherit the adjoint operation. Thus $\mathcal{D}_A$ is a unitary fusion category.

The free-module functor $F(i) = i \otimes A$ is a tensor functor. By adjunction,

$$\mathrm{Hom}_A(F(i), F(j)) \cong \mathrm{Hom}_{\mathcal{D}}(i, j \oplus jg).$$

For simple i, freeness of the g-action makes the endomorphism algebra of $F(i)$ one-dimensional, so $F(i)$ is simple. The same formula shows $F(i) \cong F(j)$ precisely when $j = i$ or $j = gi$. Every simple module is a summand of a free module, since A is separable, and hence is one of these simple free modules. Finally

$$F(i) \otimes_A F(j) \cong F(i \otimes j).$$

Collecting the two constituents k, gk in each orbit gives (6.4). It also proves independence of the chosen representatives. □

Apply the proposition to

$$\mathcal{D} = \mathrm{Ising} \boxtimes PSU(2)_6, \qquad g = (\psi, V_6).$$

In the principal unitary braidings, ψ and V_6 are order-two fermions. Their product has self-braiding $+1$ and generates a Tannakian $\mathrm{Rep}(C_2)$, with central lift supplied by the product

braiding. The simple labels of $PSU(2)_6$ are V_0, V_2, V_4, V_6; tensoring by V_6 exchanges V_0, V_6 and V_2, V_4. Therefore the diagonal action on the twelve simples of $\mathcal{D}$ is free. Formula (6.4) gives a rank-six unitary category, whose tensor is $R^{(6)}_{2,94}$. The checker reconstructs the parent tensor, checks all twelve action labels, computes the six orbits and verifies the displayed formula and its independence of representatives. Its orbit order is specified in `data/diagonal_fermion_quotient.json`; the final census permutation is $(0, 3, 1, 2, 4, 5)$.

The construction uses *all* right A-modules with the tensor product supplied by the central lift. It does not identify them with the local modules of a braided algebra and does not assert that the quotient inherits the ambient braiding. In particular, no transparency hypothesis has been silently added or inferred from the fusion ring.

6.5 A diagonal fermion with a fixed point

The free-action formula of Section 6.4 does not apply to a fixed simple object. Its two resolutions can have different duality depending on the local rotation data. We give a calculation which resolves the fixed point and determines every fusion coefficient.

We use positive spherical structures in unitary categories. If $f^2 = 1$, $x^* = x$, and $t : f \otimes x \longrightarrow x$ is an isomorphism, denote by $t^{\mathsf{T}} : x \otimes f \longrightarrow x$ its pivotal transpose, using the specified self-dual identifications of f and x. Equivalently it is obtained from t by inserting a coevaluation for x on the right and evaluating the left pair. Associators are suppressed in this notation. Normalize t so that the action of two copies of f is the identity.

Lemma 6.7 (Duality of a resolved fixed point)**.** *Let (f, c) be a central boson of order two, and let V be a self-dual simple fixed by f. Suppose $t : fV \to V$ is a normalized action and*

$$t^{\mathsf{T}} c_{f,V} = \delta t, \qquad \delta \in \{1, -1\}. \tag{6.5}$$

For the central algebra $A = 1 \oplus f$, the two simple left A-modules with underlying object V have actions t and $-t$. Their duality is $V_\epsilon^* = V_{\delta\epsilon}$, where $\epsilon = \pm 1$.

Proof. The two actions satisfy the module equation because t is normalized. They are simple and inequivalent since V is simple and a scalar intertwiner cannot change the sign of its nonzero action. Their sum is the free module on V: the commuting algebra of that free module is the two-dimensional semisimple algebra of its stabilizer.

The dual left action is obtained by transposing the given action and then moving f from the left to the right of V^* with its specified half-braiding. This is exactly the composite in (6.5). The normalization $f^2 = 1$ makes the inverse action equal to the action, and the symmetric Frobenius structure of A supplies the ordinary pivotal transpose. Thus its sign is $\delta\epsilon$. Alternatively, on the one-dimensional ordinary vacuum channel of $V \otimes V$, the projector for the relative tensor product of signs ϵ, η has scalar $(1 + \epsilon\eta\delta)/2$. The unit occurs precisely when $\eta = \delta\epsilon$, giving the same duality rule. □

Theorem 6.8 (Equal-phase fermionic product)**.** *Let $\mathcal{E}_m$ and $\mathcal{E}_n$ be unitary fusion categories with respective self-dual bases $1, f, x$ and $1, f', y$ and rules*

$$f^2 = 1, \quad fx = xf = x, \quad x^2 = 1 + f + mx, \qquad (f')^2 = 1, \quad f'y = yf' = y, \quad y^2 = 1 + f' + ny.$$

Assume specified central fermions f, f' and normalized actions t, s satisfy

$$t^{\mathsf{T}} c_{f,x} = -it, \qquad s^{\mathsf{T}} c_{f',y} = -is. \tag{6.6}$$

The categories have specified positive pivotal identifications used in these equations. Then m, n are even, and the category of modules over the central algebra

$$A = 1 \oplus (f \boxtimes f') \quad \text{in} \quad \mathcal{E}_m \boxtimes \mathcal{E}_n$$

is a unitary fusion category of rank six. With basis $1, g, a, b, u, v$*, its duality fixes* $1, g, a, b$ *and interchanges* u, v*. Its complete multiplication is determined by commutativity and*

$$g^2 = 1, \quad ga = a, \quad gb = b, \quad gu = v, \quad gv = u, \tag{6.7}$$

$$a^2 = 1 + g + ma, \quad b^2 = 1 + g + nb, \quad ab = u + v, \tag{6.8}$$

$$au = av = b + \tfrac{m}{2}(u + v), \qquad bu = bv = a + \tfrac{n}{2}(u + v), \tag{6.9}$$

$$\begin{aligned} u^2 = v^2 = g + \tfrac{m}{2}a + \tfrac{n}{2}b + \tfrac{mn}{4}(u + v), \\ uv = 1 + \tfrac{m}{2}a + \tfrac{n}{2}b + \tfrac{mn}{4}(u + v). \end{aligned} \tag{6.10}$$

If $d_m = \mathrm{FPdim}(x)$ *and* $d_n = \mathrm{FPdim}(y)$*, its dimensions are* $1, 1, d_m, d_n, d_m d_n/2, d_m d_n/2$*.*

Proof. The product of the specified fermions is a central boson: its self-braiding is $(-1)(-1) = 1$. Its central regular algebra A is connected and separable. Modules over this algebra have relative tensor product, with tensor unit A, and form a unitary fusion category. This uses all modules over a *central* algebra; no braiding on the quotient is asserted. Write Q for the tensor free-module functor.

There are four free orbits and one fixed point in the nine simples of the Deligne product. The corresponding modules are

$$1 = Q(1,1), \quad g = Q(f,1), \quad a = Q(x,1), \quad b = Q(1,y), \qquad Q(x,y) = u \oplus v.$$

On the fixed point $V = x \boxtimes y$, use the action $t \boxtimes s$. Its square is one. The transpose and half-braiding in a Deligne product factor, so (6.6) gives

$$(t \boxtimes s)^{\mathsf{T}} c_{f \boxtimes f', V} = (-i)^2 (t \boxtimes s) = -(t \boxtimes s).$$

Lemma 6.7 proves $u^* = v$. The other four objects are self-dual because they are tensor free modules on self-dual objects. The module dimension equals the underlying dimension divided by $\dim A = 2$, giving the asserted dimensions and rank.

Tensoring a resolved fixed point by g changes its action sign. Indeed, to form the relative product with $Q(f,1)$ the diagonal algebra line is moved past $(f,1)$; the scalar is $c_{f,f} = -1$. Identifying $(f,1)V$ with V by $t \boxtimes \mathrm{id}$ leaves no other scalar, by the normalized square of t. Hence $gu = v$, $gv = u$. Taking duals gives $ug = v$, $vg = u$, so g is central on these objects. The free-module tensor identity gives $g^2 = 1$, $ga = ag = a$, $gb = bg = b$ and (6.8).

Since g fixes a, the two products au, av are equal, and likewise $ua = va$. Their respective sums, computed by the tensor functor Q, both equal $2b + m(u + v)$. This proves the first half of (6.9) on both sides and also forces m even. The second half and the evenness of n follow similarly. Thus all products involving a, b, g commute. Since $v = gu$ and g is central, u and v commute as well.

Now $u^2 = v^2$ and $(u^2)^* = v^2 = u^2$. The coefficient of 1 in u^2 is zero and that of g is one. Frobenius reciprocity and (6.9) give coefficients $m/2, n/2$ on a, b. Self-duality of u^2 makes the coefficients of u, v equal. Put $e = d_m d_n/2$ and call the common coefficient k. From $d_m^2 = 2 + m d_m$ and $d_n^2 = 2 + n d_n$ one obtains

$$e^2 = 1 + \tfrac{m}{2} d_m + \tfrac{n}{2} d_n + \tfrac{mn}{2} e.$$

Dimension comparison gives $k = mn/4$ and proves the square formula. Multiplication by g then gives the formula for uv. □

Corollary 6.9 (Three tensor identifications)**.** *The cases* $(m,n) = (0,0), (0,2), (2,2)$ *of Theorem 6.8 have unitary realizations and identify, respectively, the census rings*

$$R^{(6)}_{1,3}, \qquad R^{(6)}_{2,14}, \qquad R^{(6)}_{2,40}.$$

In particular, $R^{(6)}_{2,40}$ *is the diagonal-fermion quotient of two* identical *unitary half-*E_6 *categories.*

Proof. For $m = 0$ use the Ising category with positive dimension $\sqrt{2}$ and Frobenius–Schur sign $+1$. In the conventions of [1, Section 2.1, equations (9)–(13)], choose $A = e^{3\pi i/8}$; then $d = -A^2 - A^{-2} = \sqrt{2}$ and $A^4 = -i$. The displayed fermion half-braiding and pivotal normalizations give (6.6).

For $m = 2$ use unitary half-E_6, the category with (6.1). The local data in [1, Section 7.1, equation (235), Section 7.2, equation (240), Appendix D.1, equation (467)] give the normalized action square one, pivotal rotation scalar one, and the half-braiding scalar $-i$ on the noninvertible object. They therefore give exactly the second input needed in (6.6). Categorical existence and unitarity are those of the Ising and half-E_6 constructions, not inferences from these few scalar identities alone.

The code reconstructs (6.7)–(6.10) for the three pairs, checks every fusion-ring axiom, and matches all 216 coefficients under unit-fixing permutations. For $(2, 2)$ the census basis is $(1, u, v, g, a, b)$, giving the stated identification. □

The two half-E_6 factors must not be replaced without comment by a category and its complex conjugate. Conjugation changes $-i$ to $+i$; the product of the rotation phases is then $+1$, and Lemma 6.7 makes the two resolutions self-dual. The relative-product description of the 3311 category in [25, Section 5.1] uses the opposite/conjugate factors. Our non-self-dual resolution is therefore not a dimension-only reidentification of that model. The construction uses central boson condensation with a fixed point, not the earlier free-orbit formula.

6.6 The exceptional G_2 level-four model

The cyclic-four tensor $R^{(6)}_{2,4}$ has a unitary realization. We identify it with the exceptional quantum subgroup $\mathcal{E}_4(G_2)$ associated with the conformal embedding

$$G_{2,4} \ \subset \ \mathrm{Spin}(14)_1. \tag{6.11}$$

The embedding, vacuum algebra, annular matrix and branching matrix used below are given in [8, Section 4.5, pp. 29–32]. We use existence of this conformal-embedding algebra, not existence inferred merely from a commuting modular-invariant matrix.

Let $\mathcal{B}$ be the unitary WZW category of G_2 at level four, with the long fundamental weight first, as in [8]. The vacuum algebra is

$$A = V_{(0,0)} \oplus V_{(0,3)}, \qquad \dim A = 6 + 2\sqrt{6}.$$

It is the connected commutative separable algebra, with its unitary structure, supplied by (6.11). Thus $\mathcal{B}_A$, the category of all right modules with tensor product over A, is an ordinary unitary fusion category. Its subcategory of local modules is the level-one category of Spin(14). This distinction between all modules and local modules matters: the former has six simples, the latter only four.

Proposition 6.10. *The Grothendieck ring of $\mathcal{B}_A$ is based-isomorphic to $R^{(6)}_{2,4}$. In particular this ring has a unitary categorification.*

Proof. In the source ordering $(1, X, Y, h, g, g^{-1})$, the basic annular matrix and branching matrix

are

$$N_X = \begin{pmatrix} 0&1&0&0&0&0\\ 1&2&2&1&0&0\\ 0&2&2&0&1&1\\ 0&1&0&0&0&0\\ 0&0&1&0&0&0\\ 0&0&1&0&0&0 \end{pmatrix}, \qquad E = \begin{pmatrix} 1&0&0&0&0&0\\ 0&1&0&0&0&0\\ 0&1&1&0&0&0\\ 0&0&1&1&0&0\\ 1&1&1&0&0&0\\ 0&1&1&0&1&1\\ 0&1&0&1&0&0\\ 0&1&1&0&0&0\\ 0&0&1&0&0&0 \end{pmatrix}.$$

The source weights, indexing the rows of E, are

$$(0,0),(0,1),(0,2),(1,0),(0,3),(1,1),(0,4),(1,2),(2,0).$$

The free-module image of $V_{(0,1)}$ is the simple X, by the second row of E, so its annular action really is multiplication by X. The free-module functor is central, hence X commutes with every object. The local simples are $1, h, g, g^{-1}$, corresponding to the vacuum, vector, and two spinor representations. The level-one fusion group of D_7 is $P/Q \simeq C_4$; equivalently its vector is the square of either spinor. Thus $g^2 = h$ and $g^4 = 1$. These four objects remain distinct and invertible in $\mathcal{B}_A$, since the local subcategory is a full tensor subcategory.

Reading the displayed matrix gives

$$X^2 = 1 + h + 2X + 2Y, \qquad hX = X, \qquad gX = g^{-1}X = Y.$$

The image X is self-dual, since $V_{(0,1)}$ is self-dual. Moreover $Y = gX$, so $Y^* = Xg^{-1} = Y$. The category is tensor generated by the central object X and the cyclic local group, so its ring is commutative. Associativity now determines all remaining products:

$$\begin{aligned} g^4 = 1, \quad gX = Y, \quad gY = X, \\ X^2 = Y^2 = 1 + g^2 + 2X + 2Y, \qquad XY = g + g^3 + 2X + 2Y. \end{aligned} \tag{6.12}$$

These are the required full fusion rules, with census order $(1, g, g^{-1}, X, Y, h)$; in source coordinates the permutation is $(0, 4, 5, 1, 2, 3)$. The resulting 216 coefficients, including the nontrivial duality $g^* = g^{-1}$ and the self-duality of X, Y, agree with $R^{(6)}_{2,4}$. □

The mathematical inputs to this identification and its arithmetic checks are separated. Existence and unitarity come from the conformal embedding. The source annular and branching matrices determine the module action; the cyclic local group then determines its full fusion ring. Independently, the constructor calculates all nine $G_{2,4}$ fusion matrices by Weyl alternants in $\mathbb{Q}(\zeta_{24})$, and verifies

$$N_i^{\mathcal{B}} E = E\Big(\sum_a E_{ia} N_a^{\mathcal{B}_A}\Big), \qquad EE^{\mathsf{T}} = I + N^{\mathcal{B}}_{(0,3)}.$$

The latter is free-module adjunction for the stated vacuum algebra. All entries, all unit-fixing identifications with the census, and the cyclic group action are exact. A dimension match alone would not prove the proposition: it would fail to distinguish the cyclic-four ring from the Klein-four rings with the same dimension list.

The dimensions are $(1, 1, 1, 1, 2+\sqrt{6}, 2+\sqrt{6})$ and $\mathrm{FPdim}(\mathcal{B}_A) = 24 + 8\sqrt{6}$. The published construction and its exact tensor identification provide a positive control for the finite necessary systems in Section 7.2.

6.7 Inversion equivariantization of an odd near-group category

The positive case at multiplicity three is an existing unitary model, not a proposed solution of new pentagon equations. We use the coherent order-two action constructed in [23, Section 13.3]. The following formulation separates that categorical input from the reconstruction of the entire fusion tensor.

Proposition 6.11 (An inversion-equivariantization constructor)**.** *Let n be odd, and let $\mathcal{N}$ be a unitary near-group category with invertibles α_j, $j \in \mathbb{Z}/n\mathbb{Z}$, and one further simple ρ, with*

$$\alpha_j\rho = \rho\alpha_j = \rho, \qquad \rho^2 = \bigoplus_j \alpha_j \oplus n\rho.$$

Assume a coherent unitary C_2-action fixes ρ, takes α_j to α_{-j}, fixes the unit channel of ρ^2, and acts on a chosen basis $(T_j)_j$ of $\mathrm{Hom}(\rho, \rho^2)$ by $T_j \mapsto T_{-j}$. Assume moreover that the action on the pointed part has the standard inversion equivariantization $\mathrm{Rep}(\widehat{C}_n \rtimes C_2)$. Then $\mathcal{N}^{C_2}$ is a unitary fusion category of rank $(n+7)/2$. Its simples are

$$1, g, x_+, x_-, v_1, \ldots, v_{(n-1)/2}.$$

Writing $v_0 = 1 + g$ and $v_j = v_{-j}$, with indices modulo n, its products are determined by

$$g^2 = 1, \qquad gx_+ = x_-, \qquad gx_- = x_+, \qquad gv_i = v_i.$$

$$\begin{aligned} v_iv_j &= v_{i+j} + v_{i-j}, \\ v_ix_\pm = x_\pm v_i &= x_+ + x_-, \\ x_+^2 = x_-^2 &= 1 + \sum_i v_i + \frac{n+1}{2}x_+ + \frac{n-1}{2}x_-, \\ x_+x_- = x_-x_+ &= g + \sum_i v_i + \frac{n-1}{2}x_+ + \frac{n+1}{2}x_-. \end{aligned}$$

All its simples are self-dual. The dimensions are $1, 1, d, d, 2, \ldots, 2$, where $d > 0$ satisfies $d^2 = n + nd$.

Proof. For an action on a semisimple category, equivariant simple objects are obtained from orbits of simple objects and irreducible projective representations of their stabilizers. In the present explicitly given action, the fixed simples $1, \rho$ have their indicated equivariant structures and each admits the two sign choices. Denote these by $1, g$ and x_+, x_-. Each nontrivial inversion orbit has trivial stabilizer and yields a single v_i with underlying object $\alpha_i \oplus \alpha_{-i}$. This proves the rank and dimension statements. Tensoring with g changes the sign of an equivariant structure and therefore exchanges x_+, x_- while fixing every v_i.

The pointed equivariantization hypothesis gives the displayed products of the v_i. For $v_i\rho$, the two summands of the underlying object are exchanged by the action. Its $+1$ and -1 eigenspaces are both one-dimensional, proving $v_ix_\pm = x_+ + x_-$, and likewise for products in the opposite order.

The unit channel in ρ^2 contributes 1 to x_+^2. The other invertible channels are paired and contribute the v_i. In the ρ-multiplicity space, inversion fixes exactly T_0 and pairs the other $n-1$ basis vectors. Its eigenspaces have dimensions $(n+1)/2$ and $(n-1)/2$. This proves the formula for x_+^2. Multiplying by g proves the mixed product, and multiplying twice proves $x_-^2 = x_+^2$. The unit coefficients in these formulas show self-duality. A coherent unitary action has a unitary equivariantization; thus the construction uses actual associativity data, not merely an automorphism of a based ring. □

Corollary 6.12. *The ring $R^{(6)}_{3,218}$ admits a unitary categorification.*

Proof. Izumi's $C_5 + 5$ category has the inversion symmetry of [23, Example 13.8]. Section 13.3 constructs an actual order-two automorphism commuting with ρ, not just an outer automorphism class. Its action on the isometries is $S_j \mapsto S_{-j}$ and $T_j \mapsto T_{-j}$. Theorems 13.17–13.18 of that source establish precisely the hypotheses and product decompositions used above, including the pointed representation subcategory.

The source scalar symmetries can also be checked directly: with $\zeta = \exp(2\pi i/5)$, the quadratic phases are $a(j) = \zeta^{2j^2}$ and the nonzero b-phases have exponents $(4, 1, 1, 4)$ at $j = 1, 2, 3, 4$, each divided by $\sqrt{5}$. Both lists are inversion invariant. The dimension used here is derived from the fusion equation,

$$d = \frac{5 + 3\sqrt{5}}{2}, \qquad d^2 = 5 + 5d, \qquad \dim(\mathcal{N}^{C_2}) = 45 + 15\sqrt{5}.$$

Thus no independently copied numerical dimension enters the constructor. The source supplies existence and the coherent action; the finite checks validate the specified permutations and scalar symmetries.

In the proposition's order $(1, g, x_+, x_-, v_1, v_2)$, the census order is $(1, g, x_-, x_+, v_1, v_2)$. The function `inversion_rules(5)` constructs every product and the recorded permutation $(0, 1, 3, 2, 4, 5)$ identifies all 216 structure constants with $R^{(6)}_{3,218}$. This is a based-ring identification, not a match of Frobenius–Perron dimensions alone. □

The generic formula is not a claim that the categorical input exists for every odd n. The positive registry supplies only the cited $n = 5$ input. This constructor and its full based isomorphism are included in Table 3 and in the electronic model catalogue.

6.8 Higher subfactor and affine-group models

6.8.1 The six-object Asaeda–Haagerup category

The unitary category $\mathcal{AH}_1$ has the six-object fusion table in [22, Table 1]. In its source order $(1, \psi, \chi, \sigma, \zeta, \tau)$, the permutation $(0, 1, 3, 2, 5, 4)$ identifies all 216 structure constants with $R^{(6)}_{6,1764}$. The supplement encodes the source table separately from the census and checks this complete equality. Existence comes from the published subfactor category, not from the arithmetic of that table. This is a known unitary model. Its global dimension coinciding with that of another tensor, in particular $R^{(6)}_{3,215}$, proves neither a based-ring isomorphism nor a Morita equivalence.

6.8.2 A regular-orbit representation-ring constructor

Proposition 6.13. *Let V be a nontrivial finite abelian group and suppose that H acts freely and transitively on $\widehat{V} \setminus \{1\}$. Put $h = |H| = |V| - 1$. The irreducibles of $G = V \rtimes H$ consist of the inflations of* $\mathrm{Irr}(H)$ *and one additional self-dual representation ρ of dimension h. Their products satisfy*

$$\sigma\rho = (\dim \sigma)\rho, \qquad \rho^2 = \sum_{\sigma \in \mathrm{Irr}(H)} (\dim \sigma)\sigma + (h - 1)\rho. \tag{6.13}$$

Consequently these rules, together with $\mathrm{Rep}(H)$*, define a unitary categorical model, whenever the stated group action is supplied.*

Proof. The irreducibles of a semidirect product by an abelian normal subgroup are obtained from H-orbits in $\widehat{V}$ and representations of their stabilizers. The trivial orbit gives the inflated

representations of H. The single nontrivial orbit has trivial stabilizer and gives $\rho = \mathrm{Ind}_V^G \lambda$, for any nontrivial λ. Inversion preserves this orbit, so ρ is self-dual. Restriction of an inflated σ to V is $(\dim \sigma)1$, and induction gives the first equation. Its reciprocity then gives the multiplicity $\dim \sigma$ of each inflation in ρ^2. If the remaining multiplicity of ρ is a, comparing dimensions yields $h^2 = \sum_\sigma (\dim \sigma)^2 + ah = h + ah$, so $a = h - 1$. The representation category of a finite group is unitary. □

Take $V = \mathbb{F}_3^2$ and let H be generated by

$$I = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \qquad J = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The identities $I^2 = J^2 = -1$ and $IJ = -JI$ identify H with Q_8. Its orbit on the eight nonzero vectors is free and transitive; the contragredient action has the same property. Thus Proposition 6.13 gives a six-object category of dimension 72, with simple dimensions $(1, 1, 1, 1, 2, 8)$ and multiplicity seven. It realizes $R^{(6)}_{7,572}$.

The constructor generates the actual 72-element semidirect product, checks its multiplication and action, and computes the four linear characters, the inflated two-dimensional quaternion character, and the induced eight-dimensional character. It verifies orthogonality, completeness by the sum of squares, and all tensor coefficients by character inner products. The resulting tensor is then compared to the census. This test is an application of the proposition to explicit group data, not a rule accepting all abstract rings with these dimensions.

6.9 The complete catalogue and group-theoretical constructions

The positive part of the classification is recorded as explicitly as the negative part. Table 3 contains one row for every one of the 54 positive rings, including all twenty-one rings of multiplicity one. The electronic catalogue records every constructor used, not merely one representative: there are 61 constructor records, with an explicit unit-fixing basis permutation from each output to its census representative. These permutations are verified entry by entry. The names in the table are presentation labels, not hypotheses in a categorification test. The symbols U and S distinguish unitary existence from spherical existence with proved nonexistence of a pseudo-unitary categorification. The single S entry is not counted among the unitary models.

6.9.1 Small constructions and conventions

Here $PSU(2)_k$ means the even subcategory of the principal unitary $SU(2)_k$ category, and D_9 has order eighteen. The indicated C_2 action on $C_3 \times C_3$ is inversion. In $C_3 \rtimes C_4$, a generator of C_4 acts by inversion on C_3; the action therefore factors through C_2. The positive Ising-type rings in the table are denoted $\mathcal{I}(G, r)$, where $|G| = 4$, a subgroup H of order two is fixed, and $r \in \{0, 1\}$. Their basis consists of G and σ_0, σ_1. An element of $G/H \cong C_2$ translates the subscripts, and

$$\sigma_i \sigma_j = \sum_{a \in G:\, \bar{a} = i+j+r} a.$$

The unitary realizations are the Ising zestings constructed in [27, Section 4.3]; the metaplectic category $SO(5)_2$ and its listed zest are treated there as well. The program reconstructs these formulas, ordinary finite-group character products, the Haagerup tensor, and the $SU(2)$ fusion rules before matching the resulting tensors. Thus the multiplicity-one positives do not depend on a cardinality-complement argument.

Table 3: All positively identified rings in the census. U means a unitary realization; S means a spherical realization and no pseudo-unitary categorification (hence no unitary one). The numbers of types U and S are given in Table 1. Here $R_{m,j} = R^{(6)}_{m,j}$. Every constructor and its based isomorphism are recorded electronically.

Ring	Model or construction	Type	Source
$R_{1,1}$	$\mathcal{I}(C_4, 1)$: cyclic, non-self-dual Ising zest	U	[27]
$R_{1,3}$	$\mathcal{I}(C_4, 0)$: cyclic, self-dual Ising zest	U	[27]
$R_{1,4}$	$\mathrm{Rep}(C_3 \rtimes C_4)$	U	[27]
$R_{1,6}$	$\mathrm{Vec}_{C_2} \boxtimes \mathrm{Vec}_{C_3} \simeq \mathrm{Vec}_{C_6}$	U	[27]
$R_{1,7}$	Tambara–Yamagami category for C_5	U	[27]
$R_{1,9}$	$\mathcal{I}(C_2^2, 1)$: non-self-dual Ising zest	U	[27]
$R_{1,10}$	Zested $SO(5)_2$ category	U	[27]
$R_{1,11}$	$\mathrm{Fib} \boxtimes \mathrm{Vec}_{C_3}$	U	[27]
$R_{1,14}$	Vec_{S_3}	U	[27]
$R_{1,15}$	Haagerup category $\mathcal{H}_3$	U	[21]
$R_{1,20}$	$\mathrm{Vec}_{C_2} \boxtimes$Ising; also $R(0,0)$	U	[27, 28]
$R_{1,21}$	$\mathrm{Vec}_{C_2} \boxtimes \mathrm{Rep}(S_3)$; also $R(0,1)$	U	[27, 28]
$R_{1,23}$	$\mathrm{Vec}_{C_2} \boxtimes PSU(2)_5$	U	[27]
$R_{1,24}$	$PSU(2)_{10}$ (the other 3311 even-part ring)	U	[27, 25]
$R_{1,26}$	$SO(5)_2$	U	[27]
$R_{1,30}$	$\mathrm{Rep}(D_9)$, with $\lvert D_9 \rvert = 18$	U	[27]
$R_{1,32}$	$\mathrm{Rep}((C_3 \times C_3) \rtimes C_2)$, inversion action	U	[27]
$R_{1,34}$	$PSU(2)_{11}$	U	[27]
$R_{1,36}$	$\mathrm{Fib} \boxtimes \mathrm{Ising}$	U	[27]
$R_{1,37}$	$\mathrm{Fib} \boxtimes \mathrm{Rep}(S_3)$	U	[27]
$R_{1,38}$	$\mathrm{Fib} \boxtimes PSU(2)_5$	U	[27]
$R_{2,2}$	Fermionic C_2-zest of $\mathrm{Ad}(E_6) \boxtimes \mathrm{Vec}_{C_2}$	U	[9]
$R_{2,4}$	$\mathcal{E}_4(G_2)$: modules for $G_{2,4} \subset \mathrm{Spin}(14)_1$	U	[8]
$R_{2,11}$	$\mathrm{Rep}((C_3 \times C_3) \rtimes C_4)$	U	[15]
$R_{2,14}$	Fermionic grading zest of the E_6 category	U	[9]
$R_{2,34}$	TT_3	U	[13]
$R_{2,36}$	$\mathcal{C}(A_4, 1, C_2, 1)$	U	[2]
$R_{2,40}$	Diagonal-fermion quotient of two identical half-E_6 categories	U	[1]
$R_{2,49}$	Quadratic category $R'(2)$	U	[28]
$R_{2,53}$	$\mathrm{Vec}_{C_2} \boxtimes \mathrm{Ad}(E_6)$; also $R(0,2)$	U	[28]
$R_{2,69}$	C_2-extension of $\mathrm{Rep}(H_8)$	U	[7]
$R_{2,71}$	$\mathcal{C}(S_4, \omega, C_2^2, \psi)$ of Theorem 6.16	U	[20]
$R_{2,75}$	E_6 modules over $SU(2)_{10}$	U	[26]
$R_{2,94}$	Diagonal-fermion quotient of $\mathrm{Ising} \boxtimes PSU(2)_6$	U	[15]
$R_{2,97}$	Dual Goodman–de la Harpe–Jones 3311 even part	U	[25]
$R_{2,98}$	Dual even part of the Izumi C_4 subfactor	U	[24]
$R_{2,105}$	$\mathcal{C}(A_5, 1, D_{10}, 1)$	U	[2]
$R_{2,124}$	Schopieray: $\mathcal{C}(\mathfrak{so}_5, 3/2)_{\mathrm{ad}}$	S	[41]
$R_{2,127}$	$\mathrm{Fib} \boxtimes \mathrm{Ad}(E_6)$	U	[26]
$R_{2,128}$	D_6 modules over $SU(2)_8$	U	[26]
$R_{2,140}$	Local D_{10} modules over $SU(2)_{16}$	U	[26]
$R_{2,143}$	Commutative even part of the Izumi C_5 subfactor	U	[17]

Continued on the next page

Table 3 continued

Ring	Model or construction	Type	Source
$R_{2,144}$	Principal G_2 level-three category	U	[40]
$R_{3,47}$	$\mathrm{Rep}(GL(3,\mathbb{F}_2)) \simeq \mathrm{Rep}(PSL(2,7))$	U	[2]
$R_{3,141}$	$\mathcal{C}(A_5,1,S_3,1)$	U	[2]
$R_{3,218}$	Inversion equivariantization of Izumi's C_5+5 near-group category	U	[23]
$R_{4,249}$	Sign-swap equivariantization of the equal half-E_6 quotient	U	§6.12
$R_{4,322}$	Diagonal-fermion quotient of half-$E_6 \boxtimes PSU(2)_6$	U	§6.11
$R_{5,17}$	Izumi's C_5+5 near-group category	U	[23]
$R_{6,345}$	$\mathcal{C}_{4,8,4}$, with ring $R(4,6)$	U	[28]
$R_{6,1712}$	Relative Fibonacci center $Z_{\mathrm{Fib}}(TT_3)$	U	§6.13
$R_{6,1764}$	Asaeda–Haagerup category $\mathcal{AH}_1$	U	[22]
$R_{7,572}$	$\mathrm{Rep}((C_3 \times C_3) \rtimes Q_8)$	U	[15]
$R_{7,2274}$	Extended Haagerup category $\mathcal{EH}_1$	U	[4, 6]

6.9.2 Exact Weyl-character reconstruction

We use the principal unitary categories $\mathcal{C}(\mathfrak{g},k)$ at positive integral level. Their construction and the Weyl-alcove and character formulas are reviewed in [40]; these are categorical inputs, not consequences of a numerically fitted modular matrix. Normalize long roots to have squared length two, let W be the Weyl group, and put $\rho=\sum_i \lambda_i$. For dominant weights in the level-k alcove define

$$C_{\lambda\mu}=\frac{\sum_{w\in W}\det(w)\exp\!\big(-2\pi i(w(\lambda+\rho),\mu+\rho)/(k+h^\vee)\big)}{\sum_{w\in W}\det(w)\exp\!\big(-2\pi i(w\rho,\mu+\rho)/(k+h^\vee)\big)}. \tag{6.14}$$

These are the fusion characters. Their products determine the fusion coefficients by

$$C_{\lambda\mu}C_{\nu\mu}=\sum_{\tau}N_{\lambda\nu}^{\tau}C_{\tau\mu}.$$

All denominators and the required character-matrix rank are checked in the cyclotomic number field. Exact Gaussian elimination determines the coefficients, which are then checked to be nonnegative integers and to satisfy every character equation, including columns not used to solve the linear system. This also works for the root-lattice subcategory of type A_2, using a full-rank submatrix of its restricted characters.

For G_2 at level three, the short fundamental weight is first, and the input data are

$$((\lambda_i,\lambda_j))=\begin{pmatrix}2/3 & 1\\ 1 & 2\end{pmatrix},\qquad h^\vee=4,\qquad \Lambda_3=\{(0,0),(1,0),(2,0),(3,0),(0,1),(1,1)\}.$$

The calculation takes place in $\mathbb{Q}(\zeta_{21})$, and produces a rank-six tensor of multiplicity two.

Proposition 6.14. *The tensor obtained from* (6.14) *for* $(\mathfrak{g},k)=(G_2,3)$ *is based-isomorphic to* $R^{(6)}_{2,144}$. *In particular this ring admits a unitary modular categorification.*

Proof. The Weyl construction supplies the category and the displayed character formula supplies its fusion rules. The exact calculation just described is implemented independently of the census in `software/quantum_rank2.py`. The resulting 216 coefficients and a basis permutation are recorded in the model catalogue; the checker recomputes them and verifies their equality to the census tensor. This is a full-tensor identification, not an identification by dimension type. □

6.9.3 The dual 3311 even part

The dual even-part identification is

$$R_{2,97}^{(6)} \cong K_0((A_{11}^{\mathrm{even}})_{E_6^{\mathrm{even}}}^{*}). \tag{6.15}$$

The category on the right is the dual even part of the Goodman–de la Harpe–Jones 3311 subfactor and is unitary. Its complete multiplication table is Figure 2 of [25, Section 5.1]. The independent constructor uses the basis $(1, B, C, C', D, g)$ of that figure; in particular

$$C^2 = 1 + g + 2C, \quad (C')^2 = 1 + g + 2C', \quad CC' = B + D, \quad gC = C, \quad gC' = C', \quad gB = D.$$

The other products are reconstructed from the same published table and all tensor axioms and the based isomorphism in (6.15) are checked. The other 3311 even-part ring is already $R_{1,24}^{(6)}$. Neither identification implies that another ring with the same Frobenius–Perron dimensions is categorifiable; in particular it does not settle the non-self-dual ring $R_{2,40}^{(6)}$.

6.9.4 A three-double-coset realization criterion

For finite G, a subgroup H, and unit-modulus normalized cochains $\omega \in Z^3(G, \mathbb{C}^\times)$ and $\psi \in C^2(H, \mathbb{C}^\times)$ with $\omega|_H = \delta\psi$, write $\mathcal{C}(G, \omega, H, \psi)$ for the category of ψ-twisted H-bimodules in Vec_G^ω, with the conventions of [20]. For $u \in G$ put $H^u = H \cap uHu^{-1}$. Simple objects over HuH correspond to irreducible projective representations of H^u with factor

$$\begin{aligned}\psi^u(a,b) = \psi(a,b)\psi(u^{-1}b^{-1}u, u^{-1}a^{-1}u) \\ \cdot \frac{\omega(a,b,u)\omega(a,bu,u^{-1}b^{-1}u)}{\omega(abu,u^{-1}b^{-1}u,u^{-1}a^{-1}u)}.\end{aligned} \tag{6.16}$$

Such a simple has Frobenius–Perron dimension $[H : H^u] \dim \pi$. These statements, and support of tensor products on products of double cosets, are [20, Theorems 4.1 and 5.1]. We call a factor on an abelian group nondegenerate when its twisted group algebra has one irreducible representation; equivalently its commutator bicharacter has trivial radical.

Theorem 6.15 (Three-double-coset realization)**.** *Suppose H is abelian of order $h = s^2 > 1$ and*

$$G = H \sqcup tH \sqcup HuH, \qquad N_G(H) = H \sqcup tH, \qquad H^u = \{1\}.$$

If the factor ψ^t in (6.16) *is nondegenerate, then $\mathcal{C}(G, \omega, H, \psi)$ is a unitary categorification of the commutative fusion ring with basis $\widehat{H} \sqcup \{X, Y\}$ and rules*

$$\begin{aligned}\chi X = X\chi = X, \qquad \chi Y = Y\chi = Y, \qquad XY = YX = sY, \\ X^2 = \sum_{\chi \in \widehat{H}} \chi, \qquad Y^2 = \sum_{\chi \in \widehat{H}} \chi + sX + (h-2)Y.\end{aligned} \tag{6.17}$$

Its dimension type is $(1^h, s, h)$.

Proof. The component over H is $\mathrm{Rep}(H)$, so it consists of the h invertibles indexed by $\widehat{H}$ with their ordinary group multiplication. Nondegeneracy of ψ^t gives one simple X of dimension s over tH. The trivial stabilizer of HuH gives one simple Y of dimension h. Hence these are all the simple objects.

The normalizer is a subgroup with quotient $N_G(H)/H$ of order two. Consequently tH is closed under inversion. Its complement HuH in G is also closed under inversion and is stable under multiplication on either side by $N_G(H)$. Since X and Y are unique on their supports, both are self-dual and all the invertibles fix them on both sides. Support forces X^2 into $\mathrm{Rep}(H)$.

Frobenius reciprocity gives $[X^2 : \chi] = [\chi X : X] = 1$, proving the first square formula. Similarly, XY and YX are supported on HuH, and dimensions give $XY = YX = sY$. Reciprocity gives $[Y^2 : \chi] = 1$ and $[Y^2 : X] = s$. The remaining coefficient is therefore determined by $h^2 = h + s^2 + h[Y^2 : Y]$, namely $[Y^2 : Y] = h - 2$.

Finally Vec_G^ω is unitary for unit-modulus ω. The admissible twisted regular algebra of H, with its normalized regular inner product, is a connected separable C^*-algebra object. Its bimodule category is unitary. This is the standard unitary form of the group-theoretical construction, and proves existence rather than merely associativity of the displayed ring. □

6.9.5 An explicit twisted S_4 model

The following cochain certificate makes the realization criterion applicable to a residual that is not realized by the corresponding untwisted construction.

Realize $G = PGL(2, \mathbb{F}_3) \cong S_4$ by its action on $\mathbb{P}^1(\mathbb{F}_3) = \{(1,0), (0,1), (1,1), (1,2)\}$. In each projective class choose the matrix $s(g)$ whose first nonzero entry in row-major order is one. Define

$$s(g)s(k) = (-I)^{\beta(g,k)} s(gk), \qquad \alpha(g) = \begin{cases} 0, & \det s(g) = 1, \\ 1, & \det s(g) = 2, \end{cases} \qquad \omega(g,k,l) = (-1)^{\alpha(g)\beta(k,l)}.$$

Here β is a normalized $\mathbb{F}_2$-valued 2-cocycle from the central extension $GL(2, \mathbb{F}_3)$, and α is a homomorphism. The cup-product identity therefore proves that ω is a normalized 3-cocycle. Take $H = \{1, a, b, c\}$ represented, respectively, by

$$I, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Its action is the subgroup generated by the two disjoint transpositions on the displayed ordered projective line, so $H \cong C_2^2$. Put $\psi(x,y) = i^{q(x,y)}$, with $q(1,x) = q(x,1) = 0$ and

q	a	b	c
a	0	2	0
b	3	0	1
c	1	0	1

in $\mathbb{Z}/4\mathbb{Z}$. (6.18)

For every $x, y, z \in H$ the admissibility identity is

$$q(y,z) + q(x,yz) - q(xy,z) - q(x,y) \equiv 2\alpha(x)\beta(y,z) \pmod 4.$$

It follows either by substitution in the nine-entry table or by the complete 4^3-triple check in the supplement.

Theorem 6.16. *For the cochains just defined, $\mathcal{C}(S_4, \omega, C_2^2, \psi)$ is a unitary categorification of $R_{2,71}^{(6)}$.*

Proof. Choose

$$t = \left[\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \right], \qquad u = \left[\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \right].$$

Matrix multiplication over $\mathbb{F}_3$ gives exactly three double cosets, of sizes $4, 4, 16$, represented by $1, t, u$. The first two form $N_G(H)$, and H^u is trivial. Substitution in (6.16) gives the following exponent matrix for ψ^t, in the order $(1, a, b, c)$:

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 3 & 0 & 1 \\ 0 & 2 & 3 & 1 \end{pmatrix} \pmod 4.$$

Its commutator bicharacter has trivial radical: each nonidentity row has an entry different from the corresponding transposed entry. Thus its twisted group algebra is $M_2(\mathbb{C})$, and Theorem 6.15 applies with $h = 4$, $s = 2$.

The constructor regenerates the twenty-four projective matrices, their faithful permutation action, the subgroup, all double cosets, and every stabilizer factor. It checks the admissibility identity and, in addition to the cup-product proof, all 24^4 cocycle identities. It then uses (6.17) to reconstruct the full tensor and checks its based isomorphism with $R^{(6)}_{2,71}$. The complete data are in `data/twisted_s4_certificate.json`. Thus no inference from Frobenius–Perron dimensions alone is used. □

The three additions in this section are positive determinations, not new exclusions. In particular, searching only untwisted group-theoretical models would miss the last construction. All allocated negative certificates remain applications of the unified obstruction framework stated earlier in the paper.

6.10 Exceptional and near-group source identifications

Two source tensors must be included independently of any obstruction search. The principal even part $\mathcal{EH}_1$ of the extended Haagerup subfactor is unitary by its subfactor construction [4]. Its six fusion matrices are explicitly displayed in [6, Section 5.1]. In that source order, the permutation $(0, 1, 3, 4, 5, 2)$ gives $R^{(6)}_{7,2274}$. The largest coefficient is seven. The constructor transcribes the published matrices and checks all 216 coefficients, not merely the dimension field or formal codegrees.

The unitary $C_5 + 5$ near-group category in [23, Example 13.8] has basis $\alpha_0, \dots, \alpha_4, \rho$ and products

$$\alpha_i\alpha_j = \alpha_{i+j}, \qquad \alpha_i\rho = \rho\alpha_i = \rho, \qquad \rho^2 = \sum_{i=0}^{4} \alpha_i + 5\rho.$$

Its tensor identifies $R^{(6)}_{5,17}$ with permutation $(0, 1, 4, 3, 2, 5)$. The positive root of the dimension equation is $(5 + 3\sqrt{5})/2$. This is the near-group input itself, distinct from the inversion equivariantization of Corollary 6.12. Both source identifications are independent of the census and both are matched under all unit-preserving permutations.

6.11 A mixed diagonal-fermion quotient

Let $\mathcal{E}$ be the unitary half-E_6 category of (6.1), and let $\mathcal{P} = PSU(2)_6$. The object h of $\mathcal{E}$ has the central fermion of Lemma 6.1. In the standard unitary $SU(2)_6$ category, V_6 is an order-two fermion; restricting its half-braiding gives a central fermion in $\mathcal{P}$. Thus (h, V_6) is a central boson in $\mathcal{E} \boxtimes \mathcal{P}$.

Proposition 6.17. *The category of modules over $1 \oplus (h, V_6)$ in $\mathcal{E} \boxtimes \mathcal{P}$ is a unitary fusion category with Grothendieck ring $R^{(6)}_{4,322}$.*

Proof. The second factor has basis (V_0, V_2, V_4, V_6), with V_6 interchanging V_0, V_6 and V_2, V_4. The diagonal action is therefore free, even on objects whose first component is X. Proposition 6.6 applies without a fixed-point choice. In the product order $(1, h, X) \times (V_0, V_2, V_4, V_6)$, its permutation of the twelve simples is

$$(7, 6, 5, 4, 3, 2, 1, 0, 11, 10, 9, 8).$$

Take orbit representatives $(0, 1, 2, 3, 8, 9)$. The quotient formula of Proposition 6.6, applied to the independently constructed half-E_6 and $SU(2)_6$ tensors, gives the census tensor after

permutation $(0,3,1,2,4,5)$. The implementation checks every product and its independence of the chosen representative. Existence and unitarity come from the central separable algebra; the tensor comparison identifies its Grothendieck ring. □

6.12 A sign-twisted swap on a fermionic product

The next construction has a categorical hypothesis. Associativity of the integer tensor for a formal parameter does not provide its input category.

Theorem 6.18 (Sign-swap equivariantization)**.** *Let $\mathcal{E}$ be a unitary fusion category with simples $1, f, x$, a specified central fermion f, and rules*

$$f^2 = 1, \quad fx = xf = x, \quad x^2 = 1 + f + mx.$$

Suppose $d = \mathrm{FPdim}(x)$ is irrational. There is a unitary fusion category with invertible group $G = C_2 \times C_2$ and two self-dual simples X, Y, fixed by all invertibles, with

$$\begin{aligned} X^2 &= \sum_{g \in G} g + mX + 2Y, \\ XY = YX &= 2X + 2mY, \\ Y^2 &= \sum_{g \in G} g + 2mX + m^2 Y. \end{aligned} \tag{6.19}$$

Their dimensions are $2d$ and d^2. The Ising input $m = 0$ gives $R^{(6)}_{2,69}$, and the half-E_6 input $m = 2$ gives $R^{(6)}_{4,249}$.

Proof. In $\mathcal{E} \boxtimes \mathcal{E}$ put $b = f \boxtimes f$ and $A = 1 \oplus b$. This is the regular algebra of a central Tannakian subcategory. The factor-swap tensor functor S preserves the central lift of b. Equip A with the algebra isomorphism $\phi : S(A) \to A$ that is $+1$ on 1 and -1 on b. It satisfies $\phi S(\phi) = \mathrm{id}$. Transport of module actions through S and ϕ defines a coherent unitary C_2 action T on $\mathcal{Q} = (\mathcal{E} \boxtimes \mathcal{E})_A$. The category we use is the ordinary equivariantization $\mathcal{Q}^{C_2}$.

The free b-orbits give four simples of $\mathcal{Q}$: the unit, $g = (f \boxtimes 1)A$, and $a = (x \boxtimes 1)A$, $a' = (1 \boxtimes x)A$. The fixed object $x \boxtimes x$ gives two simples u_+, u_-, each of dimension $d^2/2$. To see the action explicitly, choose the right f-action $t : x \otimes f \to x$ with square one, using the module associativity constraint. Such a normalization exists because $1 \oplus f$ is the regular separable algebra and its two actions on x differ by sign. The right b-actions on the fixed object are $\pm(t \boxtimes t)$. Swap preserves this tensor, while ϕ changes its sign. Hence T interchanges u_+, u_-; it also interchanges a, a' and fixes $1, g$.

An orbit of size two has trivial stabilizer and yields one simple in the equivariantization; a fixed simple has two equivariant structures, since projective representations of C_2 over $\mathbb{C}$ can be linearized. Thus there are four invertibles, one simple X over $a \oplus a'$ and one simple Y over $u_+ \oplus u_-$. Their dimensions are $2d, d^2$. Their supports are invariant under duality, so X, Y are self-dual. They are fixed by all invertibles: tensoring preserves their distinct dimensions and there is only one simple of either dimension.

We check the group law on the four invertibles; dimensions alone would not distinguish C_4 from $C_2 \times C_2$. Work in the pointed part $\mathrm{sVec} \boxtimes \mathrm{sVec}$, with trivial associator and the specified self-braidings -1. Let e_0, e_1 be the two homogeneous components of $g = (f, 1)A$, in degrees $(f,1), (1,f)$. An isomorphism $g \otimes_A g \to A$ has the normalized coefficients

$$\mu_{00} = 1, \qquad \mu_{01} = b, \qquad \mu_{10} = -b, \qquad \mu_{11} = -1.$$

The minus sign in balancing is forced by the half-braiding of b with $(f,1)$; it is not a choice of a ring sign. Give g the equivariance $T(e_0) \mapsto ie_1$, $T(e_1) \mapsto -ie_0$. Its square is the identity. On

each of the four pairs (e_r, e_s), the displayed μ intertwines this tensor-square equivariance with ϕ, which acts on A as $(1, -1)$. Therefore this lift of g squares to the tensor unit. Together with the independent sign representation of C_2, it generates $C_2 \times C_2$. The six scalar equalities (two for coherence, four for multiplication) are also checked exactly over $\mathbb{Q}(i)$.

Frobenius reciprocity now puts each invertible once in X^2, Y^2 and none in XY. Write $d^2 = 2+md$. Comparing coefficients of $1, d$ in $(2d)^2$, $(2d)d^2$ and d^4 forces respectively the coefficients in (6.19); their uniqueness uses the irrationality of d. For example $X^2 = \sum_g g + aX + cY$ gives $8 + 4md = 4 + 2c + (2a + cm)d$, so $c = 2$ and $a = m$. For $XY = aX + cY$, comparison gives $c = 2m, a = 2$; for $Y^2 = \sum_g g + aX + cY$, it gives $c = m^2, a = 2m$. All operations used preserve unitarity.

The Ising and half-E_6 inputs and their central fermions are supplied by standard Ising theory and Lemma 6.1 (see also [1]). Their tensors, constructed from (6.19), match the census with permutations $(0, 1, 2, 3, 5, 4)$ and the identity. These two categorical inputs, not arbitrary values of m, are admitted by the model registry. □

6.13 Reconstruction from a relative center

For a full fusion subcategory $\mathcal{D} \subset \mathcal{C}$, the relative center $Z_{\mathcal{D}}(\mathcal{C})$ consists of objects of $\mathcal{C}$ equipped with a coherent half-braiding against objects of $\mathcal{D}$. It is not the Müger centralizer in a braided category. It is a fusion category, is Morita equivalent to $\mathcal{D} \boxtimes \mathcal{C}^{\mathrm{rev}}$, and has dimension $\mathrm{FPdim}(\mathcal{D})\,\mathrm{FPdim}(\mathcal{C})$ [19, Remark 2.4]. For unitary input, its realization as the dual of the natural unitary module category gives a unitary fusion category.

Proposition 6.19 (Forced relative-center reconstruction)**.** *Suppose $\mathcal{D} = \langle a \rangle \subset \mathcal{C}$ is a Fibonacci subcategory, so $a^2 = 1 + a$. In the simple basis of $\mathcal{C}$, let L_a, R_a be the matrices of left and right multiplication by a, and put*

$$B = I + L_a R_a, \qquad L = (e_0, e_a, e_a, e_0 + e_a), \qquad Q = B - LL^{\mathsf{T}}.$$

Starting from Q, whenever a diagonal entry is one, its column is a forced nonnegative integral Gram column. Subtract its outer product and repeat. If this process reaches zero, it determines the entire restriction matrix $F = (L, H)$ of $Z_{\mathcal{D}}(\mathcal{C})$.

Assume additionally that the high columns are pairwise distinct, distinct from the low columns, and individually invariant under duality in $\mathcal{C}$. Then the relative center is self-dual. Impose the known $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$ *multiplication on its first four simples and the ring-homomorphism equations for F. If these linear equations in the symmetric nonunit structure constants have a unique solution, that solution is its complete Grothendieck tensor.*

Proof. For the forgetful functor and its adjoint, relative central induction has underlying object $FI(V) = V \oplus aVa$; equivalently it is the finite semisimple coend $\bigoplus_{U \in \mathrm{Irr}\,\mathcal{D}} U \otimes V \otimes U^*$. Adjunction gives the Gram identity $FF^{\mathsf{T}} = B$. The full inclusion $\mathcal{D} \subset \mathcal{C}$ induces a fully faithful tensor inclusion $Z(\mathcal{D}) \subset Z_{\mathcal{D}}(\mathcal{C})$: the half-braiding condition and all morphism spaces between objects of $\mathcal{D}$ are unchanged. The standard nondegenerate Fibonacci braiding identifies $Z(\mathcal{D})$ with $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$, whose four restriction columns are exactly L.

If $Q_{ii} = 1$, exactly one remaining simple has restriction coefficient one in row i, and all the others have coefficient zero. Its full restriction column is therefore $Q_{\bullet i}$. Subtracting its outer product is mandatory. At zero residual no further nonzero restriction column is possible. This proves the asserted uniqueness without a numerical factorization or a conjectural central rank.

The low four simples are self-dual. The additional hypotheses force every high simple to be self-dual, because duality permutes restriction columns and no other simple has the same column. A based ring all of whose simples are self-dual is commutative. Frobenius reciprocity makes its nonunit coefficients symmetric in all three indices. Its forgetful functor is a ring

homomorphism and the low multiplication is already known. Hence its actual tensor solves the stated linear system. Uniqueness determines that tensor; categorical existence was supplied before solving the system. If a pivot or uniqueness test fails, the recipe is inconclusive, not an exclusion or a claimed construction from arbitrary ring data. □

Corollary 6.20. *For the unitary category TT_3 and either of its Fibonacci generators, $Z_{\mathrm{Fib}}(TT_3)$ has Grothendieck ring $R^{(6)}_{6,1712}$.*

Proof. Use the source basis $(1, f^{(2)}, \rho, \sigma, \bar{\sigma}, \mu)$ of [13, Definition 2.5] and $a = \rho$. Its dimension vector is $(1, \varphi^3, \varphi, \varphi^2, \varphi^2, \varphi)$, where $\varphi = (1+\sqrt{5})/2$. The source provides the unitary category and its complete multiplication table. The Gram procedure of Proposition 6.19 gives

$$F = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Indeed, after removal of the low columns the row of σ is a diagonal-one pivot; after its forced subtraction the row of $f^{(2)}$ is another such pivot. Their removal leaves zero. Denote the columns by $(1, p, q, pq, X, Y)$. Then $p^2 = 1 + p$, $q^2 = 1 + q$, and the complete remaining rules are

$$\begin{aligned} pX = qX &= Y, & pY = qY &= X + Y, \\ (pq)X &= X + Y, & (pq)Y &= X + 2Y, \\ X^2 &= 1 + pq + 2X + 2Y, \\ XY &= p + q + pq + 2X + 4Y, \\ Y^2 &= 1 + p + q + 2pq + 4X + 6Y. \end{aligned}$$

For detail on uniqueness, there are $\binom{7}{3} = 35$ symmetric nonunit unknowns. The low products and all 36 restriction-product identities give 312 scalar linear equations of exact rank 35. The verifier reconstructs this integer system and its unique nonnegative integral solution, and independently checks every based-ring axiom. The displayed rules also permit direct substitution into every identity.

The full tensor matches $R^{(6)}_{6,1712}$ under $(0, 1, 2, 4, 5, 3)$. Repeating the calculation with $a = \mu$ gives a based-isomorphic result. Applying the same recipe to $\mathcal{C} = \mathcal{D} = \mathrm{Fib}$ recovers $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$ as an independent positive control. Thus neither the rank-six answer nor its census identification is an input to the reconstruction. □

7 Certificates, research coverage, and the remaining frontier

7.1 Automated allocation and verification

The computation has three separate layers: the frozen source census, a discovery pipeline, and an exact categorification-certificate checker. Discovery is permitted to time out or to use numerical linear programming to propose a functional. Only the checker can validate a negative record. The final ledger contains one of four disjoint reporting classes: generally excluded, unitarily realized, spherically realized with no pseudo-unitary categorification, or unresolved. Five separate existence columns record the categorical structures, with no implication from spherical to pseudo-unitary existence. The pipeline rejects any intersection of its positive and negative sets.

Criterion	$m = 1$	$m = 2$	$m = 3$	$m = 4$	Total removed	Seconds
Input	39	154	384	872	—	—
Rank-two subring	0/39	4/150	4/380	49/823	631	0.9
d-number	10/29	87/63	312/68	687/136	14753	11.6
Zero spectrum	2/27	0/63	1/67	0/136	4	0.6
Lower-rank subring	0/27	0/63	6/61	16/120	144	0.4
Involution module	4/23	3/60	19/42	13/107	233	52.0
Dimension quotient	0/23	1/59	2/40	2/105	20	6.5
Cyclotomicity	0/23	0/59	2/38	1/104	14	2.2
Morita descent	0/23	2/57	1/37	1/103	4	0.6
Field automorphisms	0/23	1/56	0/37	0/103	2	5.8
Weighted induction	0/23	1/55	1/36	1/102	7	5.5
Sphericalization/induction	2/21	33/22	32/4	85/17	693	857.6
Dimension order	0/21	0/22	0/4	0/17	1	184.1
Character induction	0/21	0/22	0/4	0/17	4	98.7
Tensor rotation	0/21	0/22	0/4	0/17	0	81.8
Modular induction	0/21	0/22	0/4	0/17	0	65.2
Deferred exact checks	0/21	0/22	0/4	0/17	1	157.2
Added: rank-three subring	0/21	0/22	0/4	1/16	1	—
Added: spectral module	0/21	0/22	0/4	0/16	2	—
Added: order-six Morita	0/21	0/22	0/4	0/16	2	—
Added: character induction	0/21	0/22	0/4	2/14	2	—
Added: stabilizer rotation	0/21	0/22	1/3	5/9	8	—
Added: ramified dimension field	0/21	0/22	0/3	0/9	4	—
Added: biquadratic norm and trace	0/21	0/22	0/3	0/9	4	—
Added: formal-codegree trace ideal	0/21	0/22	0/3	0/9	2	—
Added: forced cyclic associator	0/21	0/22	0/3	1/8	1	—
Added: local signed spectra	0/21	0/22	0/3	0/8	4	—
Added: noncommutative signed spectra	0/21	0/22	0/3	0/8	3	—
Added: integral Morita constraints	0/21	0/22	0/3	0/8	2	—
Added: ambient character rotations	0/21	0/22	0/3	2/6	2	—
Added: subcategory center projection	0/21	0/22	0/3	0/6	4	—
Constructed models	21	22	3	2	—	—
Unresolved	0	0	0	4	—	—

Table 4: Executed ordered sieve and verified continuation, multiplicities one to four. Criterion cells are newly excluded / remaining; the total column covers all eight multiplicities. Seconds are original pass wall times with 4 or 8 workers. Continuation dashes indicate that no comparable pass wall time is assigned. Timeouts remove nothing.

Criterion	$m = 5$	$m = 6$	$m = 7$	$m = 8$
Input	1582	2768	3814	6999
Rank-two subring	81/1501	121/2647	113/3701	259/6740
d-number	1356/145	2442/205	3473/228	6386/354
Zero spectrum	0/145	1/204	0/228	0/354
Lower-rank subring	25/120	28/176	22/206	47/307
Involution module	49/71	23/153	78/128	44/263
Dimension quotient	6/65	2/151	4/124	3/260
Cyclotomicity	1/64	3/148	0/124	7/253
Morita descent	0/64	0/148	0/124	0/253
Field automorphisms	0/64	1/147	0/124	0/253
Weighted induction	1/63	1/146	1/123	1/252
Sphericalization/induction	58/5	140/6	115/8	228/24
Dimension order	0/5	0/6	0/8	1/23
Character induction	0/5	0/6	0/8	4/19
Tensor rotation	0/5	0/6	0/8	0/19
Modular induction	0/5	0/6	0/8	0/19
Deferred exact checks	0/5	0/6	1/7	0/19
Added: rank-three subring	0/5	0/6	0/7	0/19
Added: spectral module	0/5	0/6	0/7	2/17
Added: order-six Morita	2/3	0/6	0/7	0/17
Added: character induction	0/3	0/6	0/7	0/17
Added: stabilizer rotation	0/3	0/6	0/7	2/15
Added: ramified dimension field	0/3	0/6	2/5	2/13
Added: biquadratic norm and trace	0/3	0/6	1/4	3/10
Added: formal-codegree trace ideal	0/3	0/6	0/4	2/8
Added: forced cyclic associator	0/3	0/6	0/4	0/8
Added: local signed spectra	0/3	0/6	0/4	4/4
Added: noncommutative signed spectra	0/3	1/5	1/3	1/3
Added: integral Morita constraints	1/2	0/5	1/2	0/3
Added: ambient character rotations	0/2	0/5	0/2	0/3
Added: subcategory center projection	0/2	2/3	0/2	2/1
Constructed models	1	3	2	0
Unresolved	1	0	0	1

Table 5: The same ordered sieve, multiplicities five to eight. Sphericalization/induction checks retained proof trees for $m \leq 7$ and discovers proofs at $m = 8$. Trees can use nested advanced filters; unfinished retained checks appear in the deferred row. Original advanced passes target $m \in \{3, 8\}$. Continuation rows apply additional complete certificates to the original survivors. This practical order is not claimed optimal.

Tables 4 and 5 give a sequential view: a tensor removed at one stage is not counted as a new removal at any later stage. The order places inexpensive structural and arithmetic obstructions before larger finite branch searches. Effectiveness means the number of *additional* tensors decided at that point, rather than the number that a criterion could reject when run in isolation. The table is a transparent operational order, not a comparison of logical strength or a hardware-independent timing ranking. Ineligible hypotheses, an unfinished search and a feasible necessary relaxation all leave a ring undecided unless another accepted certificate or model resolves it.

The allocation in Table 6 is disjoint. It records the certificate family stored for each exclusion, while Tables 4 and 5 make the sequential accounting explicit. Neither table asserts mathematical independence of the criteria. The signed and positive-dimension records call the same low-induction and finite-indicator framework internally. When an older unbalanced relative-trace certificate suffices, it is interpreted as a certificate for Theorem 4.3 with zero coefficients on the added balance equations. Likewise every inherited Farkas vector remains valid after the Galois-forbidden columns of Theorem 4.9 are removed: the inequalities on retained columns are unchanged. The stronger framework therefore retains the old exact witnesses rather than treating their special cases as unrelated obstructions.

Every certificate is verified from the original ordered tensor. The checker reconstructs number fields, verifies separating elements and factorizations, recomputes low-column and signed-table possibilities, and checks all integer inequalities. In particular it does not assume that a stored list of branches is exhaustive. For Farkas records it checks (4.12) with Python integers. During certificate verification the floating-point linear-programming entry point is replaced by a function that raises an error if called.

The source revision and tensor hashes fix the counting input. No census is regenerated for the categorification analysis. The positive check reconstructs the listed products, the zesting tensor and scalar coboundary, finite-group character rings, and group-theoretical convolution tensors, including the twisted S_4 cochains and their projective stabilizer factors. Exact Weyl characters reconstruct the G_2 model. Every positive entry, including every multiplicity-one entry, is matched by an explicit unit-fixing basis permutation; the complete 61-record constructor catalogue is checked for omissions. Regression tests compare the native integer routines with straightforward exact implementations, and deliberately corrupted certificates must be rejected.

This is an exact computer-assisted verification, not a formalization in a proof assistant. The categorical input theorems, their hypotheses, and the mathematical completeness argument remain part of the written proof. A checksum proves file integrity, not a mathematical theorem.

Proof of Theorem 1.2. Each negative record is an application of Theorems 5.1 and 3.3, or of Proposition 3.1, Corollary 3.2, or of an exhaustive signed/positive reduction followed by Theorem 4.1 or Theorem 4.3 and the cited rank-four or rank-five tests, with the tensor-sign reduction of Lemma 3.4 and the extension of Theorem 4.5 where recorded. The stronger character-aware system of Theorem 4.9, dimension-order test of Proposition 3.26, and signed refinement of Theorem 3.27 are used where recorded. The spectral refinement of Proposition 5.3, rank-three subring descent, Theorem 5.17, and Proposition 3.9 supply the additional continuation certificates. The second continuation also uses the ramification criterion of Proposition 3.13, Theorem 3.10, the signed saturation corollary, and Proposition 3.12. The higher stabilizer and ambient cyclotomic refinements use Corollary 5.23 and Theorem 5.27, with a complete general-to-pseudo-unitary reduction whenever their inner certificate has only pseudo-unitary scope. Theorem 5.25 supplies the final general multiplicity-three exclusion: its sphericalization reduction, full Morita-action calculation, local rotation lemma and complete exact indicator separation are all required. The exact checker validates all hypotheses and exhausts the required finite alternatives. Thus no excluded tensor has a complex categorification.

Section 6 constructs and tensor-matches all claimed positive examples, with the additional models in Sections 6.4 and 6.6, the higher models of Section 6.8, and complete catalogue in Section 6.9. In particular the spherical existence and pseudo-unitary nonexistence for $R^{(6)}_{2,124}$ use the full-tensor identification and the two cited results of Schopieray. No complement argument is used to assign the positive records. The census and disjoint ledger establish the counts in Table 1. The other records are left undecided, with no assertion of nonexistence or realizability. □

7.2 Regression controls

The positively realized ring $R^{(6)}_{2,4}$ supplies independent controls for the balanced and rotation systems. Their exact witnesses are `data/frontier_feasible.json` (formal rank 22) and `data/frontier_rotation_feasible.json` (formal rank 27). Each is checked only against its own reconstructed necessary system; the different formal ranks do not describe a single center. Existence comes from Proposition 6.10.

Conversely, the generally excluded ring $R^{(6)}_{3,215}$ has an exact rank-22 induction, fusion and modular-data relaxation. The stored replay programs in `research/m8_frontier/` preserve this negative control for overinterpreting feasible data. The additional stabilizer-rotation obstruction, not a failure of those weaker systems, proves its nonexistence. These controls are kept outside the main classification argument.

7.3 Reproduction and counting data

The top-level command

```
python verify.py -jobs 2
```

checks the frozen input correspondence, positive models, and allocated negative certificates. The command with `python -O` checks that correctness does not depend on Python assertions. Regression controls exercise exact arithmetic, corrupted certificates and the frontier witnesses. Precise dependencies, commands, report scope and search-stop behavior are given in the accompanying verification guide. These are classification checks, not a new enumeration of fusion rings.

The `oeis/` directory contains an eight-term b-file, a CSV, and proposed definition and provenance text. The definition is the number of based-isomorphism classes of rank-six fusion rings of *exact* multiplicity m, without assuming commutativity, self-duality, integrality or categorifiability. The terms are copied from the pinned census. No accession number is assumed and no online submission is claimed. Counting farther and classifying farther are separate achievements.

7.4 Research coverage and the next unresolved stratum

The residual list is derived from the complete decision ledger, not from a hand-selected sample. Multiplicities one through three are completely decided. The smallest unresolved multiplicity and the exact residual tensors are reported in Section 1; the first such tensor is the priority for further theoretical and automated work. The stage table also shows whether difficulties reappear at larger multiplicities. A criterion applied successfully at one bound remains the same mathematical criterion at the next bound: additional computation alone is not presented as a new theorem.

The module-Gram refinements and relative-center constructions are necessary tests or model constructions with explicit hypotheses. A failed eligibility check does not exclude a ring. An inconclusive Gram factor does not supply a model. For the relative Fibonacci center of TT_3,

the categorical construction and exact tensor reconstruction supply the positive conclusion; feasible Gram data by themselves do not.

Literature comparisons distinguish exact fusion-rule matches from shared dimensions. Unitary-only classifications do not exclude arbitrary complex fusion categories, and candidate modular data are not categorifications unless the source supplies a construction. The same principles govern new models and new necessary criteria: a model must have a justified categorical source and an exact full-tensor match; an obstruction must state its hypotheses and exhaust all branches needed for the claimed categorical structure.

The certificate files and immutable campaign manifests preserve provenance without entering the mathematical criteria as lists of forbidden IDs. The authoritative ledger is rebuilt from their disjoint union. Alternative proofs can be retained, but a tensor receives only one exclusion allocation. `software/continuation_controls.py` independently checks accepted continuation records with numerical optimizers disabled; `verify.py` replays the full current ledger. Discovery logs and incomplete searches are not verification receipts.

The present frontier is described by complete tensor-derived necessary data in the current research report. Its counts are not predictions about categorifiability beyond multiplicity eight. A surviving integral profile, fusion tensor or modular datum is assigned categorical existence only when a categorical construction and exact tensor identification are supplied.

7.5 Exact completion of the quartic centralizer data

The full tensor of $R_{6,4,91}$ contains the Ising based subcategory with $g = b_4$ and $\sigma = b_3$. The preceding signed and induction reductions leave two possible center ranks, 36 and 42. Each profile has exactly one central simple restricting to g and exactly one restricting to σ. Their products, determined by restriction and simplicity, give an Ising fusion subcategory of the center. Every braided Ising category is nondegenerate. Proposition 4.23 therefore applies to its centralizer $\mathcal{E}$.

Theorem 7.1 (Complete centralizer fusion and modular data for $R_{6,4,91}$)**.** *If $R_{6,4,91}$ has a pseudo-unitary categorification $\mathcal{C}$, then $Z(\mathcal{C}) \simeq \mathcal{I} \boxtimes \mathcal{E}$, where $\mathcal{I}$ is Ising and $\mathcal{E}$ has rank 14. The center-rank-36 profile is impossible. For the rank-42 profile, the complete compatible fusion ring of $\mathcal{E}$ has one based-ring isomorphism class. With its restriction columns labelled as in the certificate, it has eight fusion tensors, all related by permutations of repeated restriction columns. There are exactly two compatible pairs of centralizer modular data; they are complex conjugate. These assertions specify necessary central data and do not assert that either pair is categorifiable.*

Proof. Write the low summands of canonical unit induction as $A_0, \dots, A_5$. Their known character eigenvalues at the pure g and σ columns identify their Ising coordinates without choosing a braiding on $\mathcal{C}$:

$$A_0 = 1 \boxtimes 1, \quad A_1 = g \boxtimes a, \quad A_2 = g \boxtimes b, \quad A_3 = 1 \boxtimes c, \quad A_4 = \sigma \boxtimes u, \quad A_5 = \sigma \boxtimes u^*.$$

Both inherited low character labellings give these same coordinates and the same duality permutation (4 5) on low labels. Thus the right adjoint of $F|_{\mathcal{E}}$ at the tensor unit is $1 \oplus c$.

The complete central profile must be a union of packets $(v, gv, \sigma v)$. The nonnegative integral packet-cover equations have one solution for each of the two profiles. Each admits 144 endpoint orientations up to permutations of equal columns. Of these, 135 violate the displayed low Ising coordinates. For a remaining orientation, let V have the proposed centralizer restriction columns. The adjunction identity fixes

$$N_c = V^{\mathsf{T}} V - I.$$

Nonnegative entries, the unit column, and the tensor-functor equation $VN_c = N_{F(c)}V$ reject eight further orientations. Exactly one restriction multiset survives for each rank. This is a decomposition of the center; no tensor-product decomposition of $\mathcal{C}$ is used.

We describe the entire finite fusion search. For every possible duality involution consistent with the restriction columns and the forced pair u, u^*, use the symmetric variables

$$x_{ijk} = N_{ij}^{k^*}, \qquad x_{ijk} = x_{\pi(i,j,k)}.$$

The unit and c products are fixed. Restriction of every product gives linear equations and finite upper bounds

$$N_{ij}^k \le \min_{h:(v_k)_h>0} \left\lfloor \frac{(v_i v_j)_h}{(v_k)_h} \right\rfloor.$$

Duality and commutation with N_c give further exact linear equations. Project this system onto multiplication by the simple whose restriction is b_1, eliminating the complementary variables by rational row reduction. Every actual fusion matrix is normal, since multiplication by its dual is its transpose and the centralizer ring is commutative.

The rank-12 centralizer has one possible duality. Its projected affine system has eight free coordinates and no bounded integral solution; complete enumeration visits 11 nodes. For rank 14, there are four possible dualities. Three give 64 projected matrices each, none normal. The remaining duality gives 512 projected matrices, of which 16 are normal. For each such matrix, its commutation with all the other fusion matrices is now linear. Eight completions are inconsistent; each of the other eight has one bounded integral completion. Direct multiplication checks the unit, duality, nonnegative integral coefficients and every associativity identity. Swapping the three pairs of identical restriction columns identifies these eight tensors in one based-ring orbit.

For each orbit representative the exact character table is obtained by factoring a separating multiplication matrix over $\mathbb{Q}(\zeta_{16})$. All character products and all roots are checked exactly. In any modular labelling, the character attached to object j has formal codegree $\mathrm{FPdim}(\mathcal{E})/d_j^2$, and $\widehat{S}_{ij} = d_j\chi_j(i)$ must be symmetric. Exhaustive assignment of the characters subject to these requirements gives exactly two $\widehat{S}$ matrices.

Put $z = \theta_\sigma$. The inherited second indicator is $\nu_2(\sigma) = -1$, so $z = \zeta_{16}^e$ with $e \in \{3, 5, 11, 13\}$. The low summands have twist one, hence

$$\theta_a = \theta_b = -1, \qquad \theta_c = 1, \qquad \theta_u = \theta_{u^*} = z^{-1}.$$

For each S and each of these four values of z, apply balancing with one of these known-twist objects in the first coordinate:

$$\theta_i\theta_j\widehat{S}_{ij} = \sum_k N_{i^*,j}^k d_k\theta_k.$$

These are linear equations for all fourteen twists. Every branch has rank fourteen. Exactly two are consistent, with Ising exponents 5 and 11, respectively. The unique solutions have twists in μ_{16} and are complex conjugate. In particular, no a priori bound on the unknown twist orders is being substituted for an exhaustive search: the linear equations determine the twists themselves. Each solution passes every balancing equation, $\widehat{S}^2 = \mathrm{FPdim}(\mathcal{E})C$, the full modular relation, and the requirement that the product central charge of $\mathcal{I} \boxtimes \mathcal{E}$ is one.

The record is `research/round5/quartic_pair/R6_4_91_center_reconstruction.json`. The reconstruction and optimizer-free replay are implemented in `software/round5_quartic_pair.py`. They recompute every packet cover, orientation, duality, affine projection, normality test, fusion completion, based-ring orbit, character labelling and twist solution from the frozen full-tensor frontier. Replay also checks the inherited signed reduction and its exhaustive profiles when requested. □

The surviving centralizer has no nontrivial invertible object. Its formal codegree polynomial is

$$(t-8)^2(t^2-128t+2048)^2(t^2-64t+512)^2(t^4-2816t^3+151552t^2-1572864t+4194304).$$

For the representative with $z=\zeta_{16}^5$, in the certificate's unit-first order, the twist exponents are

$$(0,10,12,8,14,14,6,6,4,2,8,0,11,11).$$

The other representative negates these exponents modulo sixteen. An associator, compatible braiding and the requisite induction algebra are not supplied by fusion coefficients and modular matrices alone. Thus the theorem eliminates one inherited profile and completely fixes the remaining fusion and modular data, while leaving the categorical realizability question explicit.

7.6 Full regular-action completion for $R_{6,4,22}$

The remaining quartic ring $R_{6,4,22}$ is treated by the same restriction–word Gram and regular-action criteria, without a centralizer factorization. The inherited center ranks are 30 and 36.

Proposition 7.2 (The rank-30 branch)**.** *The center-rank-*30 *induction profile of* $R_{6,4,22}$ *is impossible.*

Proof. Apply the quadratic and cubic word Gram criterion and couple its coefficient counts to the projected twist types by Proposition 4.21. For each of the two inherited character labellings the simultaneous system has 2950 equations and 4132 nonnegative variables. Each has an exact rational separating vector. The certificate `research/round5/fermion/R6_4_22_certificate.json` and `software/round5_22.py` reconstruct the full domains and check both separating vectors over the rationals. No integrality or numerical infeasibility assertion is needed for these two exclusions. □

Theorem 7.3 (Complete necessary central data for $R_{6,4,22}$)**.** *If* $R_{6,4,22}$ *has a pseudo-unitary categorification, its center has rank* 36*. Its complete central fusion ring, with the forgetful restriction map, has one based-ring isomorphism class among all the inherited possibilities. Each of the two conjugate low character labellings has exactly one compatible modular datum. The two modular data are complex conjugate and have twist order* 20*.*

Proof. For the rank-36 profile, retain all 50 quadratic and cubic words in the five nonunit low central simples. The complete conditional Gram ellipsoids of Theorem 4.17 give 155 possible pairs consisting of a restriction column and its word coefficient vector. Their simultaneous count system has 902 equations. Exact bounded affine enumeration has rank 88, visits 136389 nodes, and gives precisely five nonnegative integral count vectors. The two low labellings give identical word data, which is verified directly; thus this count enumeration covers both.

Expand each count vector into the individual central simples. Duality must dualize its restriction column and permute its word coordinates by the known low duality. In each of the five patterns this leaves two possible involutions, corresponding to the two indistinguishable copies of one column. Apply Proposition 4.19 to multiplication by A_1. In matrix row convention, its entries satisfy

$$(N_{A_1})_{jk}=(N_{A_1})_{k^*,j^*},\qquad U_wN_{A_1}=U_{wA_1},$$

whenever both words have been retained. Tensoriality fixes their restrictions, and the prescribed eigenvalues fix the full characteristic polynomial. All variables have finite integral bounds from restriction products. Exact interval propagation and affine enumeration reject both dualities in four patterns, and one duality in the fifth pattern.

The last case has 1184 affine matrices after the prescribed trace is imposed. Exactly two have the required characteristic polynomial. Adjoining the actions of A_2, A_3, A_4 successively, with commutation against all previously fixed actions, gives one continuation of each; $N_{A_5} = N_{A_4}^{\mathsf{T}}$. The resulting two complete low-action systems differ only by swapping the indistinguishable central simples. This equivalence is checked on every matrix entry.

There is a unique central simple whose restriction is b_1. Its multiplication matrix is now determined by restriction, Frobenius reciprocity and commutation with the fixed low actions: the exact linear system has 465 variables, rank 465, and one nonnegative integral solution. The central invertible and the other simple actions are then completed in the same way. Each step has one solution. The finished tensor is checked directly for the unit, duality, commutativity, nonnegative integral coefficients and all associativity identities. Hence the procedure has completed the entire central ring, not only its products with low simples.

The exact character table of this ring is computed over $\mathbb{Q}(\zeta_{20})$ from a separating multiplication matrix. Formal codegrees, symmetry and the prescribed low character eigenvalues leave one S-labelling for each inherited labelling. Set the six low twists equal to one. Balancing with these known-twist objects is a linear system of rank 36 in the 36 unknown twists. It has a unique solution for each labelling; the solutions are complex conjugate and belong to μ_{20}. The unique nonunit central invertible is forced to have twist -1. Every balancing equation, the full modular relation and central charge one are checked exactly.

The complete reconstruction is `software/round5_regular_completion.py`, using the generic regular-action engine `software/round5_low_actions.py`. The record is `research/round5/quartic_pair/R6_4_22_center_reconstruction.json`. Replay rebuilds the conditional Gram domains, all five count vectors, all ten duality cases, every surviving matrix, the based-ring identifications and both modular completions from the frozen full-tensor frontier. It uses no numerical optimizer and remains valid with Python assertions disabled. □

The two quartic rings have therefore reached complete necessary central fusion and modular data. This is stronger than an integral induction or Gram witness, but it still leaves categorical realization: a compatible associator and braiding, together with the required canonical induction algebra, have not been constructed. The current records retain this remaining question explicitly.

7.6.1 Finite checks after modular completion

The completed matrices allow the same rotation criterion to be applied with every central simple as a source, rather than only the central unit. Write S, T for the normalized, anomaly-one center matrices and $W_{Zi} = [F(Z) : X_i]$. The generalized indicator vector is

$$\left(\nu^Z_{m,1}(X_i)\right)_{Z,i} = ST^mW.$$

The corresponding rotation on $\mathrm{Hom}(F(Z), X_i^{\otimes n})$ has nth power θ_Z^{-1}. By [48, Propositions 3.3–3.7], for $g = \gcd(n,k)$ its normalized power trace is

$$\theta_Z^{k/n}\nu^Z_{n,k}(X_i) = \alpha_{k/g,n/g}\Big(\theta_Z^{g/n}\nu^Z_{n/g,1}(X_i^{\otimes g})\Big),$$

where $\alpha_{a,m}(\zeta_m) = \zeta_m^a$. The expression to which this Galois automorphism is applied must lie in $\mathbb{Q}(\zeta_{n/g})$. Exact Fourier inversion must give nonnegative integral eigenspace dimensions whose sum is $\dim \mathrm{Hom}(F(Z), X_i^{\otimes n})$.

For both conjugate modular data of each quartic ring, these field membership and Fourier conditions hold for every original simple, every central simple, and every $2 \le n \le 16$. The full

center also satisfies signed-permutation Galois covariance and the twist-square compatibility. Finally,

$$\tau_n = \sum_Z d_Z^2 \theta_Z^n > 0 \qquad (\gcd(n, \operatorname{ord} T) = 1),$$

for all eight coprime residue classes of each exponent. Thus every coprime higher central charge is one, as required of a Drinfeld center by [49, Theorem 4.4]. Positivity is checked by exact real-root isolation in the real quartic field. The equality $\tau_n = \mathrm{FPdim}(\mathcal{C})$ is not assumed for general coprime n.

These additional surviving conditions are reconstructed in `software/round5_modular_checks.py`; its record is `research/round5/quartic_pair/modular_checks.json`. They strengthen the verified necessary data and do not resolve the remaining categorical realizability problem.

The saturated dimension-field characters of both quartic rings have order four and are nonreal. Their conductors are respectively 20 and 16. The real-character specialization of Proposition 3.22 is therefore not applied to them. Instead, the exact audit `research/round5/quartic_pair/conductor_intertwiner.py` constructs an equivariant quotient of $\mathrm{Ind}_B^{\mathrm{SL}_2(\mathbb{Z}/f\mathbb{Z})} \chi$ onto the cyclic span of w_χ in each completed modular datum. It checks every S, T generator edge and $P_U S w_\chi = \lambda w_{\bar{\chi}}$, where P_U projects onto T-fixed vectors. The induced and cyclic dimensions are $(36, 18)$ for $R_{6,4,22}$ and $(24, 24)$ for $R_{6,4,91}$, for both saturated characters and both modular variants. Complete reducibility and polar decomposition give an isometric equivariant section of each quotient. Thus every valid universal conductor moment or spectral identity already holds on these cyclic spans, and adds no obstruction to the completed data.

The full central projection of Proposition 4.29 also survives. For the pointed rank-two subcategory of $R_{6,4,22}$ and the Ising subcategory of $R_{6,4,91}$, twist and forgetful compatibility leave respectively five and seven integral splitting variables in B. The exact equations $S_{\mathcal{D}} B = B S_{\mathcal{C}}$ have ranks five and seven and force one nonnegative integral matrix in each conjugate modular variant. Every S, T, forgetful and unit equation is rechecked in `research/round5/quartic_pair/subcategory_projection.py`. Thus the full projection condition adds no quartic obstruction.

7.7 The finite centralizer frontier for $R'(4)$

The ring $R6_4_155$ remains unresolved. The new certificate replaces its previous necessary-data witness by a complete finite list of six induction profiles, each with a forced pair of low fusion matrices in a semion centralizer. This is a reduction of the categorification problem, not an existence claim for any profile.

Write the basis as $\mathbf{1}, X, X^*, h, g, hg$, where the invertibles form C_2^2, h fixes X, X^*, and g exchanges them. The exact dimension and products are

$$d = 4 + 3\sqrt{2}, \qquad D = 72 + 48\sqrt{2}, \qquad X^2 = g + hg + 4X + 4X^*, \quad XX^* = \mathbf{1} + h + 4X + 4X^*.$$

The previously certified sphericalization reduction applies to every categorification of this ring. We therefore use its pseudo-unitary Galois representative throughout the necessary-data computation.

First apply the pointed version of Proposition 4.28 to all three involutions. If a central simple restricts with a nonzero coefficient of an involution k, its twist has square $\nu_2(k)$. Of the ten ordinary induction frames, only one survives the resulting exact systems. Its six low columns are

$$(1,0,0,0,0,0), \quad (1,2,2,0,0,0), \quad (1,2,2,1,0,0) \text{ twice}, \quad (1,1,1,0,0,0) \text{ twice}.$$

Its four character labellings differ by permutations of equal columns; the verifier checks the permutations explicitly. The surviving second indicators, in the displayed basis order after the unit, are $(0, 0, 1, -1, -1)$. This first reduction uses twenty-eight exact Farkas certificates.

The exact right-$(\mathbf{1} \oplus h)$ module enumeration has one anti-invariant representation, in which h exchanges the two module structures on both X and X^*. Proposition 5.21 applies to these non-self-dual objects: its hypothesis concerns left and right stabilizers and does not require self-duality. Since $[X^3 : \mathbf{1}] = [(X^*)^3 : \mathbf{1}] = 4$, the third indicators have the form $-(a+b\zeta_3+c\zeta_3^2)$ with $a + b + c = 2$. Their normalized relative traces to $\mathbb{Q}(\zeta_8)$ belong to $\{-2, -\frac{1}{2}, 1\}$. Duality gives the same value for X and X^*. Eight further exact Farkas certificates exclude the first two values in all four labellings.

The remaining bounded Gram–Galois lattice has exactly 1066 integral profiles; an independent optimizer-free recursion enumerates all of them. The unique central object restricting to the unit makes the image of central invertibles injective. Subgroup closure and their tensor permutation of restriction types leave 874 profiles. The unified joint restriction–word Gram criterion, using all fifteen quadratic and thirty-five cubic low words, excludes 772 profiles by an empty exact coefficient domain and four by exact Farkas inequalities. Coupling the remaining coefficient vectors to the same twist-trace variables and imposing ribbon balancing excludes another 39 profiles. For the non-self-dual low pair the balancing target is $S_{A_a^*, A_b}$, as in Proposition 4.21.

All 59 remaining profiles force unique central lifts of h, g, hg. Their twists satisfy $q(g), q(hg) \in \{i, -i\}$. The lift of h is fermionic: a bosonic lift would allow de-equivariantization by $\mathbf{1} \oplus h$, and the fixed simple X would split into objects of dimension $d/2$; its norm is $-1/2$, contradicting algebraic integrality. If $q(g) = q(hg)$, the pointed monodromy form has trivial radical; otherwise its radical is $\langle h \rangle$. The exact divisibility test in Proposition 4.26 leaves respectively seven and twenty-five profiles, with the former contained in the latter.

Finally the central lift of g generates a nondegenerate semion subcategory, so the *center* decomposes as its product with a modular centralizer $\mathcal{E}$. Restriction columns occur in pairs (v, gv). The low monodromies select the low objects $\mathbf{1}, A_1, A_2$ in $\mathcal{E}$, hence the restricted adjoint algebra is $\mathbf{1} \oplus A_1 \oplus A_2$. For every endpoint orientation the restricted adjunction identity is

$$V^{\mathsf{T}} V = I + N_{A_1} + N_{A_2}.$$

Exhaustive endpoint enumeration, closure under duality, and restriction compatibility leave fifteen profiles and thirty orientations. The possible center ranks are

$$30, 32, 36, 38, 40, 42, 44, 46, 48, 52, 54, 60.$$

No decomposition of the original unbraided category is asserted.

The remaining adjunction matrix splits by Corollary 4.24. The monodromy charges of A_1, A_2 against h are respectively $+1, -1$. For any assignment $\epsilon_i \in \{\pm 1\}$ to the centralizer simples, N_{A_1} is the same-charge block of $V^{\mathsf{T}} V - I$, and N_{A_2} is the complementary block. Equal restriction columns are permuted freely, so the verifier enumerates charge counts within each type rather than labelled sign vectors. It enforces the known low charges, duality, restriction products, and commutation. In the opposite quadratic form the centralizer contains the fermion h, which preserves charges and cannot fix a charge-$+1$ simple. In the equal quadratic form it contains the semion hg, whose tensor action reverses charges. These exact finite tests leave six profiles and nine low-matrix pairs.

Complex conjugation of the entire pseudo-unitary category preserves its positive dimensions and restriction columns and conjugates all twists. The checked low-labelling permutations therefore allow the normalization $q(g) = i$. Three exact simultaneous-pointed-action and ribbon-balancing rays eliminate the remaining equal-form pairs. The final six profiles have center ranks $44, 46, 48, 52, 54, 60$, and each has one forced pair (N_{A_1}, N_{A_2}) in the opposite

quadratic form. The surviving profile indices in the independently reconstructed enumeration are $275, 277, 279, 296, 298, 306$.

The compact certificate is `research/round5/m4_155/frontier_certificate.json`; its independent reconstruction is `replay_frontier.py`. The final packet reconstruction is `replay_packets.py`, with output `packet_frontier.json`. The packet replay also reconstructs the charge splittings and the three final rays. Both replay programs disable numerical optimizers and use explicit checks that remain active under `python -O`. Full fusion associativity and modular completion of this finite frontier remain necessary before any categorification can be claimed.

7.8 The multiplicity-four ring with a rank-35 central datum

The following reduction determines the remaining modular and fusion data for $R_{6,4,629}$. It does not construct its associators or a category. All indices in the certificate files are zero based.

Theorem 7.4 (Complete necessary modular reduction for $R_{6,4,629}$)**.** *Every complex categorification of $R_{6,4,629}$ admits a spherical structure and has a pseudo-unitary Galois conjugate. For that conjugate, the center has rank 35 and twist exponent 15. Its restriction matrix, normalized modular matrices and fusion tensor belong to one explicitly computed class, up to complex conjugation and relabelling of simples having identical restriction columns and twists. The previously retained induction profiles of ranks 30 and 38 are impossible. The rank-35 datum satisfies nonnegative integral Verlinde multiplicities, but categorifiability remains unresolved.*

Proof. The unrestricted signed-sphericalization enumeration has 32 associative quotients, all in the sign-gauge orbit of the original tensor. Changing the chosen lifts by that gauge makes every opposite-lift multiplicity zero, so the chosen lifts form a tensor subcategory equivalent to the original category. Its spherical structure transports to the original category. Of the two rational character components of the original ring, the quadratic component has zero simple dimensions and cannot be the spherical dimension character. The other component is one quartic Galois orbit containing the Frobenius–Perron character. Conjugating to that embedding gives pseudo-unitarity. The unrestricted enumeration and the character calculation are exactly replayed by `research/round4/m4_629/general_reduction.json` and its optimized replay.

The complete induction-profile reduction reconstructs all seven low frames. Six fail the spectral condition. The remaining frame has two compatible character labellings and 784 integral high-column profiles. Low-word Verlinde integrality and nonnegative restricted products leave five profiles, of ranks $24, 30, 32, 35, 38$. The odd relative-Galois orbit criterion and exact twist-moment separating certificates reject all five profiles for one labelling and the ranks $24, 32$ for the other. Thus the input to the present computation is exhaustive: one labelling and three profiles, with ranks $30, 35, 38$.

The six low restriction columns, in the retained order, are

$$(1,0,0,0,0,0),\quad (1,1,1,1,1,1),\quad (1,1,2,2,2,3),$$
$$(1,1,2,2,2,2),\quad (1,1,1,2,2,2),\quad (1,1,2,1,2,2).$$

They are all self-dual, as follows from the real character table and its low-character correspondence. No self-duality assumption is made about the remaining central simples at this stage.

For each retained profile we rebuild the complete relative root-orbit domain and the cyclic-rotation trace equations. If a column has multiplicity m and a relative orbit has degree e, its orbit-count variable lies between 0 and $\lfloor m/e \rfloor$. Every rotation eigenmultiplicity lies between zero and the dimension of its rotation space. These are finite integer intervals. Before rational row reduction we put variables with larger upper bounds first. This is only a column

permutation, so it preserves the integer lattice and the solution set; it makes the subsequent exact search much smaller. The surviving free variables are enumerated with the inherited interval and integrality pruning, and every leaf is substituted into the original equations. The complete search results are

central rank	height	equations	variables	linear rank	nodes	solutions
30	6	4323	405	350	203	0
35	10	2476	587	523	4157	2
38	10	1942	527	470	2408	2

Consequently the rank-30 profile is excluded. In each other profile the two solutions are complex conjugate. Every occupied relative orbit has degree one, so all twists are explicitly in $\mathbb{Q}(\zeta_{15})$. The decoded twists have orders $3, 5, 15$, with least common multiple 15. This decoding is exact: a relative average of roots of unity having absolute value one can only average identical roots. The decoded lists are also checked to be conjugate as multisets of restriction–twist pairs. The low simples are exactly the twist-one simples.

Let $D = \mathrm{FPdim}(\mathcal{C})$ and write d_Z for a central simple dimension. The six normalized low rows are fixed by

$$S_{A_i,Z} = \lambda_i(Z) d_Z / D,$$

where $\lambda_i(Z)$ is the joint-character eigenvalue attached to the restriction column of Z. We apply Proposition 4.27. Since the low objects are self-dual and exhaust the twist-one objects, the low Galois signed permutations are determined by unique matches in the known low block. The congruence identities therefore supply the images of $T^a S e_i$ for every unit a modulo 15. Exact modular-word closure extends the known span from dimension 29 to dimension 30 in each profile. All recorded linear relations, symmetry and isometry identities are checked in $\mathbb{Q}(\zeta_{15})$.

For the rank-38 profile, the image of the individual basis vector e_6 becomes known. Its restriction is $(0, 0, 0, 0, 1, 0)$. The known rows give

$$[A_1 A_1 Z_6 : \mathbf{1}] = \sum_j \frac{S_{1j}^2 S_{6j}}{S_{0j}} = \frac{1}{3},$$

contradicting Verlinde integrality. The sum is recomputed independently using rational polynomial arithmetic modulo

$$\Phi_{15}(x) = x^8 - x^7 + x^5 - x^4 + x^3 - x + 1.$$

Conjugating preserves the rational value $1/3$, so both twist branches are excluded. Only the rank-35 profile remains.

For rank 35, let W be the resulting known 30-dimensional modular subspace. Its orthogonal complement is explicitly computed and verified to be T-stable. Its five distinct T-eigenvalues are

$$\{\zeta_{15}^2, \zeta_{15}^3, \zeta_{15}^8, \zeta_{15}^{10}, \zeta_{15}^{12}\}.$$

The center's anomaly-one representation factors through $SL_2(\mathbb{Z}/15\mathbb{Z})$ by the congruence theorem [47, Theorem II and Remark 4.5]. The Chinese remainder theorem identifies this finite group with $SL_2(\mathbb{F}_3) \times SL_2(\mathbb{F}_5)$. The computation constructs all 24 and 120 matrices in these groups, their conjugacy classes and the complete class-algebra multiplication tensors. A class-algebra element with simple spectrum yields every primitive central character; character orthogonality gives its degree. Discrete Fourier inversion on powers of the upper unipotent gives its exact T-spectrum. The computation verifies simple spectrum and all integer degrees and eigenmultiplicities, so no irreducible is omitted.

Tensoring these complete irreducible lists and comparing spectra leaves exactly two decompositions of $W^{\perp}$. Their dimensions are $1+4$ and $1+2+2$. For the first decomposition $-I$ acts trivially; for the second its negative eigenspace has dimension four. Charge conjugation can interchange only simples with the same restriction column and twist. The retained profile has exactly two such pairs, namely $\{21, 22\}$ and $\{23, 24\}$, and all other classes are singletons. An involution on these classes has negative eigenspace dimension at most two. Thus the $1+2+2$ representation is impossible.

For completeness, the remaining irreducible block is constructed directly, rather than inferred from approximate character values. Put $z_p = \zeta_{15}^{15/p}$ and $g_p = \sum_{x=0}^{p-1} z_p^{x^2}$. Two exact factors are

$$S_3 = \frac{1}{g_3}\begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}, \qquad T_3 = \operatorname{diag}(1, z_3),$$

and

$$(S_5)_{xy} = \frac{z_5^{-2xy} - z_5^{2xy}}{g_5} \quad (x, y \in \{1, 2\}), \qquad T_5 = \operatorname{diag}(z_5, z_5^4).$$

Enumerating every Cayley-graph edge checks that these matrices define representations of the stated finite groups. Their distinct T eigenvalues and nonzero off-diagonal S entries establish irreducibility. The four-dimensional tensor product has the required spectrum $\{2, 3, 8, 12\}$ in exponent notation. The remaining one-dimensional character has $S = 1$ and $T = \zeta_{15}^{10}$.

Each complementary T-eigenspace is one-dimensional. Hence any other realization of the surviving representation with this fixed T differs by a diagonal intertwiner. Symmetry of S fixes the relative squares of the four diagonal scalars, leaving precisely eight choices of three signs after removing a common scalar. The reconstruction checks $S = S^{\mathsf{T}}$, $S^2 = I$ and $STS = T^{-1}ST^{-1}$ for every choice. Four choices fail nonnegative integral Verlinde multiplicities. The other four pass all $\binom{36}{3} = 7140$ unordered nonunit triples, and unit triples follow from the unit row and $S^2 = I$. Exact simultaneous row–column permutations show that the four surviving matrices differ only by the two pair interchanges above. Complex conjugation supplies the other twist branch. This proves the stated exhaustive reduction. □

The complete 35×35 matrices and 35^3 fusion tensor are retained in `research/round5/m4_629/rank35_modular_completions.json` and `rank35_fusion_data.json`. Every nonunit basis object generates this entire necessary fusion ring. The supplied restriction matrix is a based-ring homomorphism, verified on all 3780 unordered product coordinates. These properties do not supply an associator.

The exact reduction certificate is `research/round5/m4_629/modular_reduction_certificate.json`. The script `replay_modular_chain.py` reconstructs all three bounded searches, the twist lists, modular closure, both Verlinde calculations, the complete finite-group character calculation and every modular completion. It runs with numerical optimizers disabled and is also replayed under Python's `-O` option. The replay records source and artifact hashes. Numerical infeasibility reports and earlier search interruptions play no part in the proof.

7.9 The multiplicity-five frontier

Proposition 7.5. *Every complex categorification of $R_{5,1204}^{(6)}$ is spherical and has a pseudo-unitary Galois conjugate. Its central induction matrix belongs, up to permutation of central simples, to an explicitly certified list of* 171 *restriction profiles. Their central ranks lie between* 22 *and* 51. *The list is a necessary condition; it supplies neither a categorification nor a complete fusion ring or modular datum for the center.*

Proof. The unrestricted signed-sphericalization test of Section 3.4 enumerates 512 associative signed quotient tables, partitioned into 16 sign-gauge orbits. One orbit consists of gauges of the original fusion tensor. The exact dimension-character tests reject all 46 character components on the other orbit representatives. As in the sphericality argument of Proposition 3.28, a categorification must therefore admit a spherical structure. Among the character components of the original ring, the only one without a zero simple dimension is the Frobenius–Perron orbit. Conjugating to its positive embedding gives pseudo-unitarity. The positive global dimension is $336 + 192\sqrt{3}$.

We may consequently apply the central-induction criteria with positive dimensions. Exhaustive low-column enumeration gives 39 frames. Only two admit the required character assignments. Both assignments on the first frame are excluded by exact full-character twist equations. The second frame has three identical low restriction columns; its six assignments are precisely the permutations of those columns. Relabelling the corresponding low simples identifies these six necessary-data problems, so one representative suffices.

For this representative, write the restriction Gram equations, joint Galois multiplicity equations, and low cubic-word equations as an integer nonnegative system $An = b$. Here n includes both high-column counts and low fusion multiplicities. The supplement records 165 exact bound certificates. Each upper bound $n_j \le B$ is proved by a separating functional for the augmented system

$$An = b, \qquad n_j - s = B + 1, \qquad n, s \ge 0;$$

its infeasibility and integrality exclude $n_j \ge B + 1$. Replacing n_j by $-n_j$ gives lower bounds in the same way. Every bound is checked against the preceding certified system. Bounded affine-lattice enumeration then visits 293972 nodes and produces exactly 39800 high restriction profiles. Requiring integral nonnegative low-word multiplicities through degree six and nonnegative restriction remainders leaves 1833 profiles.

For all unordered pairs of nonunit low simples, apply Theorem 4.17. The profiles determine 43 distinct joint restriction–word Gram matrices. The exact fixed-column Schur bounds of Lemma 4.18 enumerate every possible joint coefficient column. The resulting count, Gram, and restriction systems exclude 1568 profiles by exact separating functionals. The height-three full-character twist systems exclude another 91. Two additional joint count systems have inconsistent parity equations. Finally, adjoining all cubic words in the last three low simples gives an exact joint Gram obstruction for one further profile. Thus precisely

$$1833 - 1568 - 91 - 2 - 1 = 171$$

profiles remain in the certified necessary domain. None is discarded because a numerical optimizer reports infeasibility or exhausts its running-time budget.

The complete certificate is `research/round5/m5_1204/profile_certificate.json`; the sparse retained matrices are in `research/round5/m5_1204/remaining_profiles.json`. The replay program `software/round5_1204.py`, with its exact bounded integer enumerator `software/round5_1204.cpp`, reconstructs all low frames, the labelling quotient, every integral bound, the complete profile enumeration, and every joint-column domain before checking the separating and congruence certificates. Numerical optimization is disabled during replay, including under Python's optimized execution mode. □

7.10 The remaining multiplicity-eight ring

The ring $R_{6,8,1660}$ is self-dual and contains the Ising based subcategory with $g = b_1$ and $\sigma = b_4$. The unrestricted signed reduction recalled above reduces every categorification to the pseudo-unitary case. Its complete replay is `research/round5/m8_1660_conductor/replay_reduction.py`. We use positive dimensions and put $D = \mathrm{FPdim}(\mathcal{C})$.

Proposition 7.6 (Actual-twist support and individual capacity)**.** *For a central simple with restriction $v = (v_0, \dots, v_5)$, put*

$$A(v) = v_0 + v_1 + v_2 + v_3 + 2v_4 - 2v_5, \qquad B(v) = v_0 + v_1 - v_2 - v_3 + v_4 + v_5.$$

Then $A(v) \neq 0$ implies $\theta_v^{24} = 1$, and $B(v) \neq 0$ implies $\theta_v^{12} = 1$. Every actual central twist has order dividing 48.

Every high column containing σ satisfies

$$v_0 = v_1 = 0, \qquad v_4 = 1, \qquad v_2 + v_3 = 4, \qquad v_5 = 3.$$

There are exactly eight such central simples, each of dimension $D\sqrt{2}/8$. Each of the four allowed actual Ising twists occurs on exactly two of these simples.

Proof. The two saturated dimension-field characters have conductors 24 and 12. Evaluating their Fourier coefficients in (3.8) gives nonzero constant multiples of A and B, respectively. The character representatives are $1, 5, 7, 11$; their sign vectors are $(1, 1, -1, -1)$ and $(1, -1, -1, 1)$. The exact coefficient calculation is reconstructed by `software/dimension_block_support.py`. Proposition 3.21 gives the order bounds.

If a high column contains σ, its actual twist is primitive of order 16 and its g-coordinate is zero, by Corollary 4.32. Therefore both Fourier coefficients vanish. Solving these two integer linear equations gives $v_2 + v_3 = 4v_4$ and $v_5 = 3v_4$. Exact evaluation of the dimensions gives

$$\frac{v_4 \operatorname{FPdim}(v)}{D\sqrt{2}/4} = \frac{v_4^2}{2}.$$

The individual capacity bound (4.39) makes this ratio at most one. Hence $v_4 = 1$. The induction Gram entry at (σ, σ) is eight, and no low column contains σ. Thus there are eight such simples, each contributing half of one root mass, proving the stated allocation. This uses individual actual-root masses before relative trace projection.

For the exponent bound, consider a high column on which both A and B vanish. If $v_4 > 0$, the Ising support gives order 16. Otherwise $v_5 = 2(v_0 + v_1)$ and $v_2 + v_3 = 3(v_0 + v_1)$. Since $v_0 = 0$ and the column is nonzero, $v_1 > 0$, so its twist is -1. All other high columns have twist order dividing 24, and low twists are one. Their least common multiple divides 48. □

Proposition 7.7 (Complete necessary restriction profiles)**.** *For a categorification of $R_{6,8,1660}$, the pseudo-unitary Galois conjugate has one of two low frames, each with two admissible character labellings. After the actual-twist support and capacity bounds, the complete induction Gram equations, joint-character and normalized-dimension Galois balance, integer cubic Verlinde multiplicities, and nonnegative low-product restriction remainders, the numbers of high restriction profiles are as follows, per labelling:*

low frame	*cubic constraints*	*words through degree six*	*range of possible center ranks*
0	371	371	25–76
1	2011	1999	25–82

For a fixed low frame the two labellings give the same surviving restriction profiles. Every retained profile also passes entrywise nonnegativity of the residual Gram matrix of all fifteen quadratic words in the five nonunit low simples.

Proof. Enumerate the high columns using the induction Gram bounds, formal-codegree divisibility, complete character integrality and Galois closure, and the Ising projection constraints. Apply Proposition 7.6 and Lemma 4.16; the latter bounds the pure g multiplicity by one. There remain 20 high columns in the first frame and 52 in the second.

Their nonnegative integral multiplicities satisfy a bounded affine system. The Gram and Galois equations have ranks 11 and 30; exact row reduction leaves 9 and 22 free coordinates. Diagonal Gram entries bound every multiplicity. Cubic character formulas give integrality congruences and nonnegativity inequalities. Restricting each low product to the source based ring supplies coordinate bounds.

The bounded free coordinates are enumerated with exact interval and divisibility pruning. The complete searches visit 1009 and 63366 nodes, respectively, per character labelling. Higher-word tests use (4.27); degrees four through six remove twelve profiles in the second frame and none in the first. The quadratic-word residual Gram test subtracts the known low constituents as in Theorem 4.17.

All domains, rational row reductions, pruning decisions and retained profiles are reconstructed by `software/round5_1660.py`. The records are `research/round5/m8_1660/low0_profiles.json` and `low1_profiles.json` in the same directory. The source-bound receipts `low0_replay.json` and `low1_replay.json` record complete optimizer-free replays under `python -O`. Consequently neither numerical solver status nor Python assertions are used to prove completeness. □

The eight σ-supported simples have only two possible restriction multisets in these lists. In frame zero all eight have column $(0,0,2,2,1,3)$. In frame one, 1628 profiles have this same multiset; the other 371 have six copies of this column and one each of $(0,0,1,3,1,3)$ and $(0,0,3,1,1,3)$, per labelling. The exact first-trace cancellation condition on opposite actual Ising roots admits an allocation for both multisets, so it supplies no additional exclusion by itself. The exact control `research/round5/m8_1660/verify_sigma_partitions.py` reconstructs these multisets and their allocations, and checks the displayed Fourier forms on all six basis columns.

Proposition 7.8 (Complete actual-root profile reduction)**.** *The actual-root equations leave precisely the following necessary profile alternatives:*

low frame	*character labelling*	*input profiles*	*remaining profiles*
0	$(0,5,4,1,3,2)$	371	168
1	$(0,5,4,1,2,3)$	1999	798
1	$(0,5,4,1,3,2)$	1999	798

All three branches have second indicators $(1,1,1,-1,1)$*. Every retained profile has eight copies of the balanced* σ*-supported column* $(0,0,2,2,1,3)$*. Their possible center ranks are* $28,31,\dots,76$ *in the first frame and* $28,31,\dots,82$ *in the second.*

Proof. The proved exponent bound permits the use of the 48 actual roots as the complete twist domain. Impose the induction Gram equations, first and second indicator traces, cyclic-rotation multiplicities, joint-character reciprocity, normalized-dimension Galois balance, the four individual Ising-root masses, and the low-product restriction constraints. There are sixteen low-frame, character, and indicator branches. Exact Farkas vectors exclude thirteen of them. In the first retained branch, three shared separating vectors exclude 203 of the 371 restriction profiles.

For each second-frame branch the shared separating vectors leave 966 profiles. A six-row integer combination of the fixed-profile system has every coefficient divisible by four, but its right-hand side is not divisible by four for any asymmetric $(1,6,1)$ σ-pattern. This excludes all such patterns, including the 168 that remained after the separating-vector tests, leaving 798 profiles per labelling. The checker intersects these results with the complete degree-six survivor indices; it does not truncate the stored 2011-element cubic list.

The compact record `research/round5/m8_actual48/certificate.json` stores the shared integer vectors, congruences, and complete remaining index lists. Its replay program

`certificate.py` reconstructs all sixteen systems and checks every applicable profile exactly. The source-bound `replay_0.json` records a complete run with numerical optimizers and Python assertions disabled. □

Proposition 7.9 (A forced Ising-center subrepresentation)**.** *The canonical modular representation of a hypothetical center contains the full nine-dimensional representation of the Ising center as a subrepresentation. Its central projection matrix B satisfies $BB^\dagger = 2I_9$.*

Proof. For every occupied column in the complete lists of Proposition 7.7, its Ising projection is one of

$$(0,0,0), \quad (1,0,0), \quad (1,1,0), \quad (0,1,0), \quad (0,0,1).$$

At the allowed actual twist, each nonzero projection is a single simple of the Ising center, with multiplicity one. This finite support assertion is reconstructed for all 4740 labelled profiles by `research/round5/m8_modrep/ising_complement.py`. Corollary 4.30 therefore applies: the rows of B have disjoint supports, their individual positive dimension masses make them nonzero, and $BB^\dagger$ is a positive scalar matrix. The four σ rows each contain exactly two objects, so that scalar is two. □

The finite-quotient criterion of Subsection 4.10 is implemented for $\mathrm{SL}_2(\mathbb{Z}/48\mathbb{Z})$ using its two Chinese-remainder factors. The exact class algebras give 76 irreducible characters for $\mathrm{SL}_2(\mathbb{Z}/16\mathbb{Z})$ and seven for $\mathrm{SL}_2(\mathbb{Z}/3\mathbb{Z})$. Their 532 tensor products give 504 distinct signed T-spectra. The independent replay checks all class-algebra identities, character orthogonality, degrees, central signs and exact Fourier multiplicities. Two Ising-center spectra serve as positive controls. These records are in `research/round5/m8_modrep/`.

One may apply this spectrum cone to the complement of the forced Ising-center representation. Its known T-spectrum has multiplicity three at 1, two at -1, and one at each of the four Ising roots. The signed-trace lower bounds must also subtract these multiplicities: they become 3 at 1, -2 at -1, and -1 at the four Ising roots. The complement itself need not have permutation duality.

The full-spectrum and forced-complement searches produced no further exact exclusion. The finite-group character enumeration is complete; the coupled nonnegative integral completion problem for the remaining profiles is not solved. The ring remains unresolved. The remaining lists are necessary alternatives, not induction matrices or modular data asserted to come from fusion categories.

First retained certificate	Rings
Formal-codegree d-number obstruction	14753
Rank-two fusion subring	631
Complete oriented signed sphericalization	378
Involution-module representation obstruction	235
Commutative signed sphericalization	192
Complete lower-rank subring descent	144
Direct general-to-positive reduction	70
Further general-to-positive certificates	57
Subcategory dimension divisibility	20
Cyclotomicity	14
Multiplicity-weighted noncommutative induction	7
Biquadratic norm and trace obstruction	4
Stabilizer and cyclotomic traces	4
Solvability descent and involution-return indicators	4
Zero-spectrum obstruction	4
Stabilizer and averaged third rotation	3
Local signed-spectrum saturation	3
Noncommutative signed-spectrum saturation	3
Ramification and Casimir trace	3
Integral Morita and Clifford rotation	2
Order-six Morita dual and grading obstruction	2
Noncommuting dimension-field automorphisms	2
Signed reduction with the common induction test	2
Spectral involution-module obstruction	2
Cyclic associator and stabilizer rotation	1
Formal-codegree trace-ideal obstruction	1
Full-character cyclotomic rotation	1
general cyclic three projection	1
general fibonacci subfield rotation	1
general half e6 projection	1
general ising projection rotation	1
Low Verlinde and restriction products	1
Stabilizer and third-rotation obstruction	1
Ramification and signed reduction	1
Rank-three near-group subring descent	1
Different ideal and signed saturation	1
Trace ideal and signed saturation	1
Total	16552

Table 6: Disjoint allocation of general exclusions. Nested criteria are not counted again.

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